Plinio Innocenzi



What is an Electron?

An Accessible Guide to Its History, Theory, and Physical Properties

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ISBN 978-3-031-81983-4 ISBN 978-3-031-81984-1 (eBook)
https://doi.org/10.1007/978-3-031-81984-1

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I esteem more the finding of a truth, though of a light thing, than the long disputing of maximal questions without attaining any truth

Galileo Galilei

Preface

A researcher's career follows particular and largely unpredictable paths. As a young physics student at the University of Padua, the University of Galileo Galilei, I would not have imagined devoting my working career to science, let alone what kind of research I would follow. I was not very attracted to particle physics or theoretical physics, and I imagined physics in another way, which made up of experiments in the laboratory on a scale that was still "human" and not in large infrastructures like Conseil européen pour la recherche nucléaire (CERN), where the individual researcher is part of mega-experiments of which he or she is largely not in control. Fortunately, things turned out as I had hoped and I was able to follow my inclination; for the first few years I could totally concentrate on research in the lab, doing is much better than telling I also had the good fortune to start my career when nanoscience was in its early stages. At the nano scale, as Feynman had noted (*There's plenty of room at the bottom: An Invitation to Enter a New Field of Physics*), was where so many exciting discoveries and a world yet to be explored still lurked Nanosciences are also a very interesting field of research for another reason, they are really interdisciplinary, you have to be a bit of a physicist, chemist, materials scientist, and biological. The challenges of knowledge never end at the nano scale, very interesting phenomena are observed, and many of them are quantum in nature. Quantum dots are the bread and butter in nanoscience, for instance. My original training as a physicist has therefore been awakened; electrons at the nanoscale are responsible for many of the most interesting phenomena we know about. Another interesting aspect is that by controlling the properties of objects at the nanoscale, such as by confining electrons in a "box," results are obtained that are transferable to the real world to fabricate systems and devices with practical applications in several fields. Electrons and nanoscale phenomena are directly related and are the origin of this book on the electron. Another motivation behind this book arose from the observation that most people do not have a particularly clear idea of what an atom looks like, and even less of an elementary particle as elusive in nature as the electron. The electron is usually imagined as a ball of negative charge revolving around the nucleus. How come the electron is not attracted to the positively charged nucleus is a puzzling

question with no convincing answer. This and many other difficult questions have motivated generations of physicists to develop theories and carry out experiments that have gradually thinned the thick fog made of doubts and uncertainties about the nature of the electron.

In writing the book, I have used the original sources as much as possible, that is, the articles describing the experiments and the theories that were then developed from them. I have, therefore, included many original quotations in the text with the aim of giving voice, in some way, back to the protagonists of an extraordinary season that was the one that saw the study of the electron and its properties at the basis of the birth of modern physics. The bibliography at the end of each chapter is mainly composed of references to the original articles whose reading is strongly recommended. It is a reading that may yield pleasant surprises. The first of these surprises is the pleasure of rediscovering a way of communicating science that is now almost entirely lost. The articles are almost always written colloquially and simply and contain epistemological reflections that from the present perspective can even look obsolete. Some of these articles are for experienced physicists, but many parts remain easy to understand. Reading them will also hold another surprise because reading the original sources prevents information from being conveyed through intermediaries. These are later steps during which the content of the articles and the scientists' thinking can end up being misinterpreted if not substantially modified. The most important experiments are described in textbooks, which, in many cases, simply quote other sources, thus contributing to the distortion of the original. This is what happened in the case of J. J. Thomson's atomic model, which became in the popular vulgate, a plum model, something Thomson was careful not to describe. Ernest Rutherford suffered the same fate. His atomic model is now known as the planetary model. Again, Rutherford had been careful not to rotate electrons around the nucleus because he did not have an explanation of how they could describe stable orbits based on the knowledge of classical physics.

Chapters 2 and 3 are devoted to the discovery of the electron and its properties. To do this, I have preferred to follow the historical succession of theories and experiments that led from Dalton's atomic model to the Standard Model. The discovery of the electron goes hand in hand with the study of atomic structure; to study the electron is to delve into the interior of the atom to glean its secrets. This "historical" approach has the advantage of allowing one to understand the reasoning and problems that arose as knowledge about the electron progressed. I believe that without having a clear picture of the historical evolution of knowledge about the physics of the electron, it is difficult to understand its extraordinary and weird properties. One of the problems one usually encounters is that of the transition from the first quantum mechanics introduced by Niels Bohr in his atomic model, which is usually referred to as old quantum mechanics, to the second phase of the quantum revolution represented by the introduction of the Schrödinger wave equation. The combination of quantum concepts and classical physics methods in old quantum mechanics creates some confusion.

The next step was rather traumatic; in fact, having lost the reference to classical physics, it was necessary to navigate the conceptual difficulties of quantum mechanics. To consider the electron as a point particle, which nevertheless has a number of intrinsic properties such as spin, electric charge, and magnetic moment, it requires using entirely new mathematics and language. It was precisely the attempt to explain the properties of the electron that led to the construction of the first atomic models and then arrived by successive approximations at an accomplished model in which almost nothing remains of the electron imagined as a small negatively charged sphere except the concept of negative charge. The understanding of the structure of the atom and then quantum mechanics resulted from studies of the electron over a period of about twenty years at the beginning of the last century. We owe the development of quantum mechanics essentially to the efforts to understand the nature and properties of two particles, electrons and photons. The Chap. 4 is devoted to describing the experiments that served as the foundation for the construction of theories about the electron, from the wave and particle nature to the strange dual property of being in one state or another, represented by spin. These are very famous experiments, described in textbooks but represented in their historical sequence, which can be very useful for understanding the cognitive and logical processes that begins from theory and leads to experiments and vice versa.

In Chap. 5, we take a break from the electron and introduce some of the constants that are at the root of physics, such as Planck's constant. The constants of physics remain in some ways mysterious numbers that govern the universe, and the properties of elementary particles such as the electron cannot be described without using them.

In Chap. 6, we get better into the description of the quantum nature of the electron, where from the double slit experiment, unexpected properties such as quantum entanglement and tunneling will emerge.

The strange property of spin will be the focus of Chap. 7. Electron spin is probably the key to understanding the properties of matter itself and not just the electron as a particle. In Chap. 8, we will discover that the electron is not alone and has strange relatives, while the Chap. 9 is devoted to the properties of the electron under extreme conditions.

The individual chapters have been written with the idea that they can be read individually not necessarily following the logical order. However, it is recommended to follow the chapters as they are listed and to read the initial three chapters first, which are used to understand all the general properties of the electron and the intellectual pathway behind them. Describing the properties of the electron also means understanding, as much as possible, the fundamentals of quantum mechanics.

In the various chapters, I have reduced the use of mathematics to a minimum, leaving this onerous task to more specialized texts. Instead, I have resorted to a description of the phenomena by dwelling as much as possible on the reasoning behind the experimental results and the resulting theoretical models. This is an introductory book suitable for an informed amateur or high school and undergraduate students. The mathematics required for a detailed description is very complex and

requires a different approach. Instead, the book's purpose is to provide a general idea of the properties of the electron and to stimulate curiosity about the latest discoveries, such as the possibility of creating crystals from electrons. It is not a story about which the last word has been written yet, even though it began about two centuries ago.

Sassari, Italy Plinio Innocenzi

Contents

1 Introduction 1
References 9

2 Discovering the Electron. From Dalton to Bohr, Evolving Theories and Models 11
2.1 The First Atomic Model Has Electrons Missing 11
2.2 The Discovery of the Electron 15
2.2.1 The Experiment of J. J. Thomson. Solving the Mystery of Cathode Rays 16
2.2.2 Particles or Aether Fluctuations? 18
2.2.3 Giving a Name to the Negatively Charged Corpuscle 19
2.2.4 Electrons Are Not Raisins. Thomson Model and the Creation of a Scientific Myth 20
2.3 A Japanese Physicist Rotates Electrons Around the Nucleus 21
2.4 The Discovery of the Nucleus. Rutherford Experiment and Electrons in the Planet that Is Not There 23
2.5 Why Does an Electron Not Fall into the Nucleus? 25
2.6 The Model of Bohr. "A Genius Was Called for, and a Genius Appeared" 27
2.7 Beyond Bohr's Model, Sommerfeld, and the Elliptical Orbits 33
2.8 A Strange Model, Placing the Electrons in a Cube 36
2.9 The Bohr Atomic Model Is Not Yet a Full Quantum Model But Allows Calculating Some Important Constants Using Classical Physics 37
2.9.1 The Classical Electron Radius 37
2.9.2 The Bohr Radius 38
2.9.3 The Reduced Compton Wavelength 39
2.9.4 The Bohr Magneton 39
2.10 The Old Quantum Theory 40
2.11 A Brief Summary of the Discovery of the Electron 41
References 42

3 The Quantum Nature of the Electron 45
3.1 An Electron Is Not Just a Particle 45
3.2 The Mother of All Equations, Schrödinger's Equation 51
3.3 A Question of Principles: The Uncertainty Principle 54
3.3.1 Heisenberg's Microscope 55
3.4 A Question of Principles: The Pauli Exclusion Principle 58
3.5 A Question of Principles: The Correspondence Principle 61
3.6 The Standard Model 62
3.7 The Electron Would Be Never the Same Simple Charged Particle 65
3.8 A Brief Summary of the Development of Quantum Mechanics of Electrons 66
References 66

4 Knowledge Is the Son of Experience. From Theory to Experiments 69
4.1 Photons Creating Electrons, the Photoelectric Effect 70
4.2 The Charge of an Electron 75
4.3 The Experiment of Franck and Hertz 77
4.4 The Discovery of Spin. Stern and Gerlach's Experiment, a Fortunate Coincidence 80
4.4.1 The Sequential Experiment of Stern and Gerlach 83
4.5 The Electron Is (Also) a Wave. A Lucky Accident, Davisson and Germer's Experiment 87
4.6 It Is a Question of Mass 91
4.7 A Time Table of the Discovery of Electron Properties 91
References 93

5 The Mystery of the Constants 95
5.1 A Constant Measure of the Whole 95
5.2 The Constant of Mystery 101
5.3 Another Strange Constant 105
References 106

6 A Dual and Quantum Nature 109
6.1 The Mother of All Quantum Mechanical Experiments: The Double Slit Experiment 109
6.1.1 The Superposition Principle 112
6.1.2 Which Way? 114
6.1.3 Single Electron and a Double Slit 115
6.2 The Principle of Complementarity 116
6.3 The Copenhagen Interpretation 117
6.4 Quantum Tunneling, Electrons Cross Mountains 120
6.4.1 Faster Than Light? 123

6.5 Quantum Entanglement 124
6.5.1 The EPR Paradox 126
6.5.2 What Is the Quantum Entanglement of a Single Object? 129
6.5.3 How Do Two Particles Become One? 129
References 130

7 From the Single Electron to Spin 133
7.1 How Many Electrons Are There in the Universe? Perhaps Only One. The Theory of the Unique Electron 133
7.2 Do Electrons Also Age and Die? 136
7.3 The Electron Spin 139
7.4 The Spin-Statistics Theorem 144
7.4.1 Distribution Functions 145
7.5 The Ohanian Hypothesis on Spin 146
7.6 The Electron Magnetic Moment 146
References 148

8 Strange Relatives, Positrons, Muons, and Tau 149
8.1 The Discovery of the Electron's Antimatter Twin, the Positron 150
8.2 From Positron to Antihydrogen 153
8.3 Birth and Death of an Electron 154
8.4 Electron-Antielectrons, Joined for a Few Moments, the Positronium 158
8.5 The Heavy Brother of the Electron, the Muon 159
8.6 Positronium Also Has a Sibling, Muonium 160
8.7 The Most Unstable of All, the Tau Lepton 161
8.8 A Brief Schematic Summary 161
References 163

9 Virtual Particles, Quasiparticles, and Quantum Fields 165
9.1 The Last Frontier, Quantum Field Theory 165
9.2 But What Happens When Two Electrons Meet? 168
9.3 Virtual Particles, What Are They? 170
9.3.1 The Casimir Effect 171
9.4 Electrons Split, Spin-Charge Separation. But then Is the Electron Not so Elementary? 172
9.5 Quasiparticles, But What Are They? 175
9.5.1 An Example of Quasiparticle by Electrons, the Polarons 176
9.5.2 Other Electron-Based Quasiparticles, Plasmons, and Polaritons 177
9.5.3 Strange Demons Can Appear 178

9.6 Electron Crystals: The Wigner Crystals 179
9.7 When Electrons Couple, Cooper Pairs 180
9.8 The Electron in the Box and Strange Nanoparticles, the Quantum Dots 184
9.9 The End of the Story. What If the Electron Is Not an Elementary Particle? 186
References 187

Conclusions 189

Chapter 1
Introduction



The best that most of us can hope to achieve in physics is simply to misunderstand at a deeper level. Wolfgang Pauli.

What is an electron? A seemingly trivial question addressed to a physicist might, at best, generate an embarrassed smile and, at worst, a complex and lengthy explanation, probably almost incomprehensible to most. This question is also certainly not irrelevant to a chemist since the electron governs the reactions that form molecules. Probably from school physics notions, the most common representation of the electron is that of an electrically charged, very small spherical object, of a negative charge, to be precise, revolving like a planet around its sun, represented by the atomic nucleus made of protons and neutrons. This image is somehow representative of Bohr's atomic model. It is reassuring but quite far from the electron model to which contemporary physics has arrived. Indeed, the planetary model of the electron imagined as a small sphere revolving around its sun, that is, the atomic nucleus, has an immediate analogy with the solar system. The electrons are the planets, and the sun is the atomic nucleus. The earth's orbit around the sun, however, is not circular but elliptical, so one might wonder whether the electron around the nucleus will also assume an eccentric orbit. As we will find out, the answer to the question of electron orbits will take us very far.

In everyday life, we are surrounded by objects, people, animals, that is, reality as we perceive it that is measurable and definable in terms of size, shape, color. If this perception of well-sized and measurable objects shifts scale and we arrive at the subatomic one, this reassuring reality disappears. The only lifeline comes to us from the complicated and almost incomprehensible mathematics that describes the strange world made of subatomic particles. The image of the electrically charged ball revolving around the nucleus, a simple and easy to visualize model, is erased to be replaced by something indefinable. The certainty of finding this ball in a certain position and with a certain energy becomes a probability. The breakdown of this idea of the electron as an electrically charged sphere is not yet finished, and this

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P. Innocenzi, *What is an Electron?*, https://doi.org/10.1007/978-3-031-81984-1_1

Table 1.1 Four fundamental interactions

Force	Charge mediator	Relative magnitude	Range of action
Strong nuclear (holds quarks together)	Gluon	1	Short range (1.4×10^{-15} m)
Electromagnetic (acts between electrically charged particles)	Photon	1/137	Infinity
Weak nuclear (governs decay processes)	Bosons W^+, W^-, and Z	10^{-6}	Short range
Gravitational (acts on bodies endowed with mass)	Graviton (?)	6×10^{-39}	Infinity (10^{-18} m)

fundamental particle almost ends up becoming an elusive ghost with an evanescent nature. Even if we considered the electron a negatively charged ball, we would still have to explain why the electron does not fall into the nucleus, which has a positive charge given by protons, and why all electrons do not accumulate on the nucleus itself. Another fundamental factor to consider is the forces at play within the atom. We know that the fundamental forces of nature are four, at least for now and for the physical models we have been able to construct consistently:

1. *Gravitational force*. It is responsible for the interactions between bodies endowed with mass.
2. *Electromagnetic force*. It is responsible for interactions between charged particles.
3. *Strong nuclear force*. This is the force that keeps the nucleus stable by binding quarks together.
4. *Weak nuclear force*. It is responsible for radioactive decay.

The difference between the relative magnitude of the four forces listed in Table 1.1, taking the *Strong Nuclear Force* as a reference, shows how large the gap is with the *Gravitational Force*, which is over a trillion trillion trillion times weaker. However, the range of action of the *Strong Nuclear Force* is short, on the subatomic scale, while the *Gravitational Force* exerts its action on a theoretically infinite scale.

This classification of forces accounts for different physical phenomena and is the starting point for building the so-called *Standard Model*, which succeeds in explaining the subatomic world and provides the equivalent of the periodic table for the fundamental particles, the building blocks that make up the universe. The *Standard Model* is based on seventeen elementary particles, and one of them is precisely the electron [1]. Among the elementary particles, however, we do not find protons and neutrons; in fact, these elements of the atom are themselves made up of smaller particles that are elementary instead.

Some physicists are still a bit skeptical about the *Standard Model*; it would not all add up in detail but works excellently in describing the world of subatomic particles.

The perverse fascination exerted by the "beauty" of some equations has caused theoretical physicists to fall in love with the mathematics and symmetries used to model the natural phenomena. But it is not clear why nature and the laws that govern it should be elegant or "beautiful." They are what they are, complicated, in some ways chaotic and incomprehensible to most people and perhaps even physicists.

Returning to the four fundamental interactions, the force of gravity is certainly out of scale as far as subatomic objects are concerned, but this does not mean that relativity should not be considered. Indeed, we will see that one of the limitations of the first quantum models was the failure to integrate special relativity into the Schrödinger equation; this will be a key step in arriving at *Quantum Field Theory* (QFT). The fact remains that of the four fundamental forces, three of them follow the laws of quantum mechanics and one, the gravitational force, does not. Quite a puzzle for physics and for theoretical physicists who would like a theory of everything that can describe all phenomena according to a unified view.

Gravity is a Force?

In Table 1.1, forces are grouped into four fundamental interactions, one of which is gravity. Describing gravity is particularly challenging because extending beyond the limitations represented by Newton's well-known equations requires using Einstein's general relativity. Therefore, whether gravity is considered a "force" depends on the theoretical framework used to describe it.

Every mass attracts every other mass with a force proportional to the product of the two masses and inversely proportional to the square of the distance between them. This statement corresponds to Newton's law of universal gravitation. Gravity behaves just like other classical mechanical forces, such as stress and tension, and can be thought of as a "pull" between two objects with mass.

In Einstein's theory of General Relativity, however, gravity is not considered a force in the traditional sense. Instead, it is described as the curvature of space–time caused by mass and energy. Objects move along the paths of least resistance in the curved space–time. What we perceive as the "force" of gravity results from these objects following curved trajectories. This distinction is crucial because it changes our understanding of gravity from being a force between two objects to a geometric property of space–time itself.

It is not all because, on a large scale, General Relativity works very well, but in the scale of quantum objects, which is the scale of elementary particles such as electrons, gravity cannot be successfully integrated. Efforts to describe gravity as a quantum force involve the hypothetical exchange of particles called gravitons. If gravitons exist, gravity might be understood as a force mediated by these particles, similar to how photons mediate the electromagnetic force. However, a complete quantum theory of gravity does not yet exist.

So, gravity cannot be generally considered a force in the classical sense but rather a consequence of curved space–time. However, the classical Newtonian model of gravity as a force is still working approximately well.

The strong and weak nuclear forces, as defined, affect the atomic nucleus, so the electron must obey the laws of electromagnetism. Of these four forces, only two of them, the gravitational force and the electromagnetic force operate on a macroscopic scale and govern most of our daily activities, so thanks to the gravitational force we do not float suspended in space, like astronauts in space stations, and thanks to the electromagnetic force we communicate with cell phones.

Within the atom there are not only electrons that obey the electromagnetic force, because protons that are electrically charged must also submit to it. However, there is a hierarchy among the forces, this is given by distance and mass, so the gravitational force governs phenomena in space where there are planets and stars that have enormous masses and where the gravitational force can make its effects felt at great distances. On a much smaller scale, that of the atomic nucleus, however, the force that controls the processes is the strong nuclear force, which is so much more locally intense than the electromagnetic force that it can overcome the repulsion of particles with the same charge, such as protons, and keep them in the nucleus together with neutrons. The electron, therefore responds only to the electromagnetic force as opposed to the other charged particle in the atom, the proton, which has a positive charge and whose interactions in the nucleus are governed by the strong nuclear force. This is not the only characteristic that distinguishes the electron from the proton; in fact, the electron, to what is the state of our knowledge is a fundamental particle, that is, there are no other smaller ones that form its structure. Instead, the proton is made up of several fundamental bricks, the famous *quarks* [2]. It takes three in all to form a proton and as many to form a neutron.

The electron thus differs considerably from the proton and belongs to a particular family of subatomic particles called *leptons*. All these particles, of which there are six in total, have in common that they are not composed of smaller entities. The electron within this family of particles that we have defined as *leptons* has a near twin, the *muon*, which has negative charge, spin ½, and much larger mass. The third brother in the company is the *tau,* which besides being the most robust of the three is also the most unstable and difficult to observe. We will return later to deal in detail with the *lepton* family, which in addition to the three negatively charged particles also includes three different types of particles, the evanescent neutrinos. There are also a total of six *quarks*, like the *leptons*, just to respect an elegant symmetry.

Returning to the macroscopic world that we define as "real" this is made up of objects that we can touch and measure; they have dimensions familiar with our daily life, however, far removed from those of the subatomic world. If we go down the scale and move to the atomic range, the dimensions of the particles that make up an atom are very different. Not only the dimensions but also the forces at play are different. So a first question is how big is an electron and what mass does it have,

at least in comparison with the size of the atom and *nucleons* (protons or neutrons). The comparison with other subatomic particles serves to make the point. An electron has a size of about 10^{-18} m, which is 170,000 times smaller than the nucleus of the helium atom that has two protons and two electrons.

The fact that the electron has a measurable size, meanwhile, allows us to make some assumptions. The first is that one could consider it, at least mathematically, as a point *particle* in which all mass is concentrated at one point. However, this creates contradictions because if the electron is considered as a point particle it should not have a volume. However, the electron also has a mass that has been measured, about 9×10^{-31} kg, a really very small number even to imagine. So the definition of an elementary particle implies that the electron has some mass but no volume. Rather strange but for describing an electron and its properties it works quite well; it is the elegance of mathematics! The use of a term like *point particle* is in itself a nice trick. The definition, as we shall see, expresses well the difficulty in defining the nature of the electron.

We continue to play with these apparent contradictions, the electron is a particle that has mass and no volume, but it occupies some space. It is strange but perhaps true. However, the electron that has zero volume occupies a definite space, or rather its charge has a certain probability of being in that space from the geometry that is defined by the atomic orbitals. Figure 1.1 shows the orbitals of a hydrogen atom calculated from the Schrödinger equation. A warning, the term orbital as well as the term spin is a kind of interpretive trap, in fact the electron does not rotate in the orbital nor does it *spin*, that is, it rotates on itself. These are two residual definitions that remain from early interpretative models of the atom, when to simplify, the electron was simply represented as a charged ball rotating in precise orbits around its sun, the nucleus. From these early models the terms spin and orbital remain; quantum mechanics, however, uses these definitions in a very different way whereby the orbital represents a probability density of finding an electron in a certain position around the nucleus and the spin a specific state of the electron. More mathematics than physics, but the loss of cognizance of what we can intuit by referring to our perceptions of macroscopic reality is the cost of building an interpretive model that, however unintuitive, works very well.

All this strangeness of the electron is due to the fact that because of its intrinsic properties it is not possible to describe its behavior using classical physics but only by quantum mechanics. This entails a great complication and introduces us into a bizarre and counterintuitive world where the rules classical physics are completely turned upside down. Then again, even if we do not perceive it, we experience quantum conditions even in everyday life. Houses are divided into floors; we can inhabit and live in one floor or another, but not between two floors. Even when we walk our movement is discrete, our steps quantize the space we can travel. However, if we move on a vehicle with wheels our movement becomes continuous and no longer quantum. In much the same way as happens to an electron and its quantized orbits, so we should not be surprised that subatomic particles can find themselves in similar situations. The electron and its properties are, therefore, not understandable

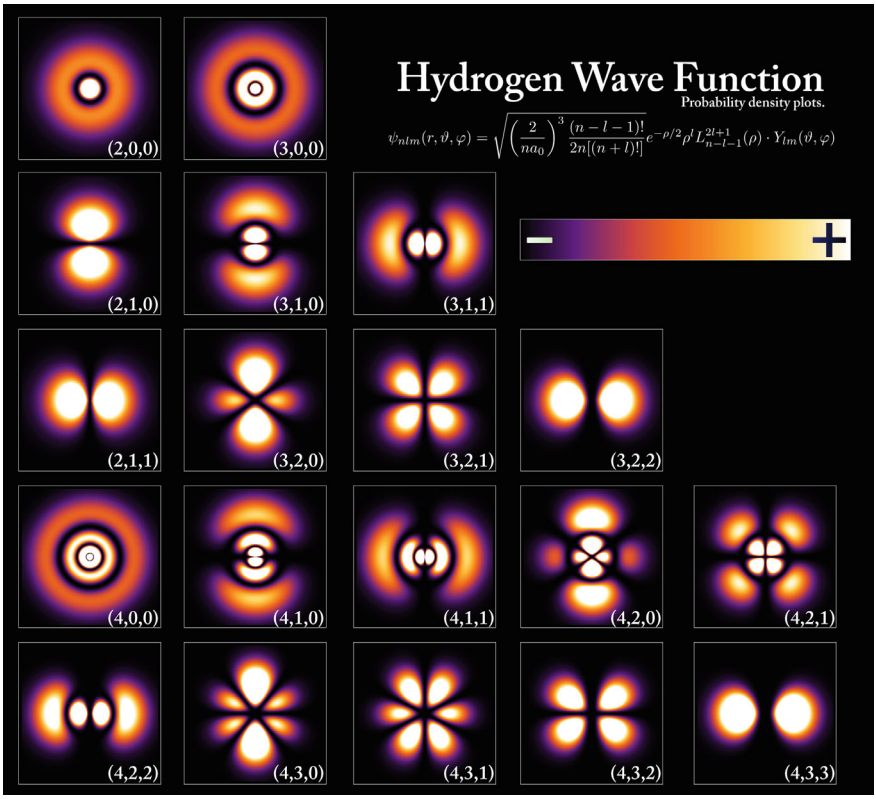


Fig. 1.1 Graphical representation of the probability density of electron location in the hydrogen atom for the principal quantum number n up to 4. Quantum mechanics cannot predict the exact location of a particle in space, only the probability of finding it at different locations. Brighter areas represent a higher probability of finding the electron [3]

without quantum mechanics, and in the coming chapters we will try to untangle this fascinating world in the simplest and most understandable way possible.

Back to atomic orbitals, once we have assessed that an electron can be in a given atomic orbital, another problem we face is to define how many electrons accommodate in an orbital. Here one of the most elusive and difficult to describe properties of an electron comes into play, which is the *spin*. The name gives the idea of spin, and indeed many people imagine that the electron rotates on itself according to a specific angular momentum; however, we have just said that the electron is considered a point particle, that is, a point. A point is a purely mathematical entity and cannot rotate. Therefore, the idea of spin, however abstract, derives from mathematical conditions: in an orbital at a certain instant only two electrons can be found, provided they are of opposite spin, ½ and −½. It is not possible for there to be two electrons with the same spin state, for example ½ and ½. The concept of electron spin is one of the most complicated in all of physics and extremely counterintuitive. The name itself,

Table 1.2 Properties of the electron

Intrinsic properties	Extrinsic properties
Mass	Position
Charge	Energy
Spin	Momentum
	Angular orbital momentum

Table 1.3 Identity card of the electron

Charge	1.602×10^{-19} Coulomb
Mass	9.109×10^{-31} kg
Spin	$\pm 1/2$

as seen, suggests that the electron rotates on its axis, hence the term spin. However, it can be said that never was the name more inappropriate… Suppose that the electron can be considered as a very small solid sphere of diameter 10^{-18} m that rotates with a spin of 1/2 $\hbar$ (with $\hbar$ the reduced Planck's constant, i.e., h divided by 2π: to which we will return later). This should be acceptable were it not that the edges of this particle would have to move at more than a million times the speed of light. Moreover, the smaller the diameter of this particle, the greater would be the speed with which it would have to rotate. So, the electron that has a specific spin simply cannot rotate; it cannot spin. Electrons will, therefore, have an undefined orientation but a spin angular momentum that has specific values.

Spin is an intrinsic property of subatomic particles, not a property exclusive to the electron, and defining the spin state of a particle is critical to understanding its properties. The neutron itself has spin, as does the proton. If spin were associated with the rotation of the electron, which would then give rise to the formation of a magnetic field, one could not explain why the neutron, which has neutral charge and a spin, does not generate magnetic fields. This confirms that electron does not rotate about its axis and that this rotation is not the origin of the spin property.

To bring some order to the matter, we could divide the properties of the electron between those that are *intrinsic* characteristics, that is, connected with the nature of the particle itself, and those that are *extrinsic,* related to its state in relation to interactions with the outside world. Simplified are two classes of properties connected with what the particle is and what it does. In the first category we find mass, charge, and spin, and in the second category we find position, energy, momentum, and orbital angular momentum (Table 1.2). Thus the spin property is one of the intrinsic properties of the electron.

At this point we can compile the "identity card of the electron" by summarizing its properties (Table 1.3), which are exactly the same for all electrons, an aspect that has led to a number of even rather paradoxical hypotheses, as we shall see.

Coming back to the ½ value of the spin, said like this, it does not mean much. Why this value and why do the electron spins have opposite signs? We will come to the explanation a little at a time; meanwhile the statement that only two electrons can

be accommodated in an orbital is nothing more than *Pauli Exclusion Principle*. As for the value of ½ we could try to imagine that if the first electron will take half the space, the other half will be occupied by the second electron. The ½ value, however, is closely related to one of the most intriguing properties of the electron, a spin of 1/2 in fact means that the electron must be rotated by two complete revolutions, thus a rotation of 720°, before it reassumes the same starting configuration. Rotation? But electrons do not rotate! This "rotation" is a mathematical operation that brings back the electron to its original spin state.

Thus electrons have a particular property, the spin, due to which they occupy a well-defined state that has a definite sign and value of spin state of opposite sign to that electron that should be in the same energy level around the atomic nucleus. Two electrons that have no mass thus find themselves living together in the same house sharing and occupying the same space but never overlapping because they are incompatible with each other. Strange but true.

Another question, but how can we be sure that this intrinsic property of the electron defined as spin exists? Every moving particle with a charge should, by the laws of electromagnetism create a magnetic field and thus be affected by a magnetic field itself. In 1922 two German physicists, Otto Stern and Walter Gerlach, performed an extraordinary experiment to verify the possible quantization of the angular momentum of electrons, i.e., stated more simply that electrons should possess two intrinsic spin states of opposite sign. In the next chapters we will have a chance to return to this important experiment and describe it in detail.

The spin state is not only observed in electrons; other particles share this property. All particles that have a half-odd-integer spin (spin 1/2, spin 3/2, etc.) and obey the *Pauli Exclusion Principle*. These particles are defined as *fermions*. However, *fermions* have a limitation, because of their quantized state there cannot be multiple fermions with the same energy. *Bosons*, another different class of particles characterized by having only spin 1, do not share this limitation.

Bosons play a fundamental role; we can call them the messengers of the fundamental forces. These messengers are all equally important, but one of them we have to deal with continuously in our everyday life, namely the *photon*. Both the photon and the electron can be described by the same equation but the photon in contrast to the electron has only one spin state, 1. The non-obvious consequence of this difference can be understood by looking at Fig. 1.2. The laser signal on the left is more intense and, therefore, generated by more photons than the signal on the right, which appears clearly less bright. Nothing strange, more photons, more light intensity, but looking closely at the space occupied by the two laser beams an anomaly can be noticed, the space turns out to be exactly the same. One thousand photons occupy the same space as one hundred. Quite different from the bizarre electrons that only because of spin state restrictions and thus because of *Pauli Exclusion Principle* cannot accumulate on top of each other and at the same energy. There is no limit to the number of photons that can occupy the same space. We could accumulate an almost unlimited number of photons in the same spot. Not quite unlimited, however, because beyond a certain limit the system could collapse and we would have created in principle a black hole. Let us think carefully, however, about the differences and consequences

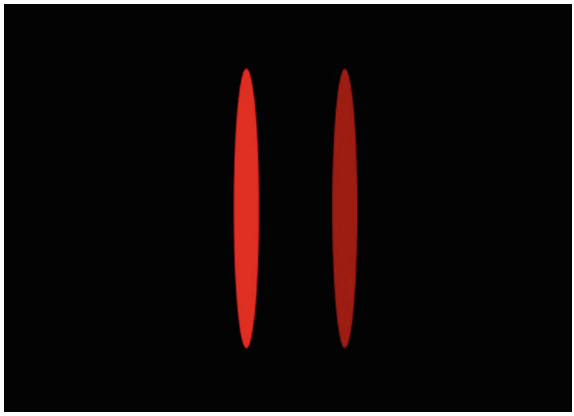


Fig. 1.2 Idealized image of two laser beams of different intensities

of the different nature of electrons and photons. Particles that have a spin state of one half occupy a volume, and those of spin one do not. Therefore, without electrons and without the exclusion given by the spin state, there would be no space-forming matter.

One final remark before closing this introduction. Two terms, force and field, have been employed several times that are in such common use that little thought is given to their physical meaning. If we throw a stone or shoot a projectile we are able to predict its trajectory with great accuracy simply by resorting to the equations of classical physics. The force exerted on the stone or projectile sets it in motion with a clear connection between cause and effect. When the orbit curves and the projectile falls back to the ground, we are not even surprised; we know it is the effect of the force of gravity. However, no one has touched the stone, as when it was thrown, and yet we say that its fall is due to a force exerted within a field, the gravitational field. The direct connection is no longer obvious, however; force and gravitational field are necessarily more abstract concepts, no apparent force acts, and defining non-mathematically what a field is becomes much more complicated. We could simply define the field as the limit of the range within which a force exerts its action. Things are a bit more complex, however, and *Quantum Field Theory* is the culmination of an extraordinary effort to frame particles, fields, and matter into a single theory. Mathematical space in quantum field theory returns to occupy a space, contains energy, and in fact excludes the possibility of a vacuum. It is in one of these fields, the electromagnetic field, that the electron finally found its home.

References

1. The updated particle registry is at the website of the particle data group, an international collaboration of physicists who using published data on particle properties keep the database updated. https://pdg.lbl.gov/

2. The word quark comes from a nonsense phrase in James Joyce's Finnegans Wake: three quarks for muster mark! U.S. physicist Murray Gell-Mann used it as a cue to christen the smallest, most indivisible bricks of elementary particles as quarks
3. https://commons.wikimedia.org/wiki/File:Hydrogen_Density_Plots.png

Chapter 2
Discovering the Electron. From Dalton to Bohr, Evolving Theories and Models



Understanding the intimate atomic structure and its component particles represents one of the most extraordinary adventures of the human intellect. Starting from the ideas of Greek philosophers such as Democritus, who first intuited that matter is formed by structural particles, the atoms, in about two thousand years, an accomplished experimental and theoretical construction of a comprehensive atomic model has been composed. This achievement was an incredible challenge since it was necessary to construct little by little a complex puzzle with only part of the pieces available from time to time. This research led to a revolutionary view of natural phenomena on an atomic scale thanks to quantum mechanics. At the same time, the rupture of the interpretive paradigms of classical physics has raised a series of questions that are still waiting to be answered, but whose culmination points are the *Standard Model* and *Quantum Field Theory.* The complex mathematics underlying these models and the quirks of the non-intuitive logic of quantum mechanics relegate the study of this topic to the narrow domain of specialists. Still, it is nevertheless possible to understand its foundational elements even using a simple language. This, too, is a challenge.

This chapter traces the development of the various atomic models up to the first conceptual achievement represented by the *Old Quantum Theory*. Although this chapter is part of the high school background, it is necessary to revisit or rediscover its contents critically and historically to understand the electron's many properties and oddities.

2.1 The First Atomic Model Has Electrons Missing

Until the discovery of the electron in 1897 by the English physicist Joseph John Thomson, the idea that matter was not continuous but consisted of fundamental units had gradually become the conceptual frame of reference. These particles were

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P. Innocenzi, *What is an Electron?*, https://doi.org/10.1007/978-3-031-81984-1_2

believed to be single entities and were referred to by the Greek name *atomos*, meaning not divisible. According to *atomism*, a philosophical current founded by Leucippus (early-first half of the fifth century B.C.-third quarter of the fifth century B.C.E.), atoms could take on different shapes and sizes [1]. For his student Democritus (460 B.C.–370 B.C.E.), these particles had no qualitative properties but only quantitative properties, such as shape and position, and the physical universe consisted of two fundamental principles: atoms and the vacuum. Democritus idea of the atom was based on the assumption that matter could not be divided infinitely; there had to be extremely small particles, unobservable by the human eye, that made it up. Although based on pure intuition, Democritus was basically right even though the ultimate units from which matter is formed are not atoms but the elementary particles of the Standard Model. The principles of atomism had been established solely on a philosophical and not an experimental basis. Although Democritus atomism lacked any experimental support, it was still the theoretical-philosophical basis for describing matter at the beginning of the nineteenth century. From the idea of the fundamental, indivisible elements, English chemist John Dalton [2] (1766–1844) began to develop his atomic theory, published in 1808 (*A new system of chemical philosophy*), which represents the first accomplished atomic model based on scientific considerations [3].

John Dalton made the assumption that matter was composed of indivisible fundamental elements, the atoms precisely, which can be represented as small filled spheres: *"These observations have tacitly led to the conclusion which seems universally adopted, that all bodies of sensible magnitude, whether liquid or solid, are constituted of a vast number of extremely small particles, or* ***atoms of matter bound together by a force of attraction****, which is more or less powerful according to circumstances, and which as it endeavours to prevent their separation, is very properly called in that view, attraction of cohesion; but as it collects them from a dispersed state (as from steam into water) it is called, attraction of aggregation, or more simply, affinity.*" [4].

The atoms of Dalton bond to form matter due to an intrinsic *force of attraction.* However, the nature of this affinity was not known or defined, but its existence had to be hypothesized to explain atomic aggregation and thus chemical reactions. What kind of force held atoms together was difficult to explain, and Dalton could only resort to a speculative explanation. Since the electron would be discovered ninety years later, it was virtually impossible for Dalton to understand the true nature of chemical reactions, and the only solution was to resort to the idea of the undefined concept of interatomic forces of attraction.

The model is a far cry from the quantum representation of the atom and its constituent particles now used; however, Dalton's theory represents the first attempt to arrive at a determination of the fundamental elements of matter and how they interact with each other. Dalton's viewpoint was primarily that of a chemist, and his interest was in explaining chemical reactions on the basis of the few notions then available. Dalton's model is thus an expression of the knowledge of his time when the electron and subatomic particles had not yet been discovered.

In his treatise, "*Foundations of the atomic theory*," he clearly described his idea of particles constituting matter: "*Whether the ultimate particles of a body, such as water,*

are all alike, that is, of the same figure, weight, is a question of some importance. From what is known, we have no reason to apprehend a diversity in the particulars: if it does exist in water, it must equally exist in the elements constituting water, namely, hydrogen and oxygen."

Dalton's atomic theory can be summarized in a few statements that define his idea of the atom and how it could lead through chemical reactions to the formation of compounds.

1. Matter consists of small units defined as *atoms* that cannot be created, divided into smaller units, or destroyed. These particles are "*solid, massive, hard, impenetrable and mobile.*"
2. All atoms of a specific element are identical in mass, size, and other properties. Atoms of different elements may have different properties, such as mass and size.
3. Atoms of different elements can be combined with each other according to a fixed ratio of integers.
4. Atoms can be combined, arranged, or separated during chemical reactions.

According to Dalton, chemical reactions do not destroy or create new atoms; every reaction is, in essence, a process by which atoms reorganize. A typical example is salt, which is a combination of sodium and chlorine atoms. When the compound forms, the sodium and chlorine atoms are still recognizable as such, even though they have reorganized to form salt.

How did Dalton arrive at his conclusions? His first simple observation was based on the *Law of Conservation of Mass,* a principle that states that the total mass of the starting products is equal to the mass of the substance formed after the reaction.

But how did he apply this principle to his work? If you take 32 g of oxygen, O_2, these can react with 4 g of hydrogen, $2H_2$, to form two water molecules, $2H_2O$, weighing a total of 36 g, the same as the starting molecules.

If we instead react the oxygen molecule, O_2, with a single hydrogen molecule, H_2, taking 32 g of oxygen and two of hydrogen we will have a weight of 32 g + 2 g = 34 g before the reaction and still the same weight after the formation of hydrogen peroxide, H_2O_2:

$$O_2 + H_2 \rightarrow H_2O_2$$

Therefore, the mass in both reactions is always conserved. Dalton also observed that in the first case, the ratio of the starting components, one oxygen molecule, and two hydrogen molecules, was 32 g/4 g = 8; in the case of hydrogen peroxide, however, there is a ratio of 32 g/2 g = 16 g. The number of oxygen atoms that combine with the same number of hydrogen atoms to form oxygen peroxide is twice the number needed to form water.

Water $H_2O \rightarrow$ 2 hydrogen atoms−1 oxygen atom

Hydrogen peroxide $H_2O_2 \rightarrow$ 2 hydrogen atoms−2 oxygen atoms

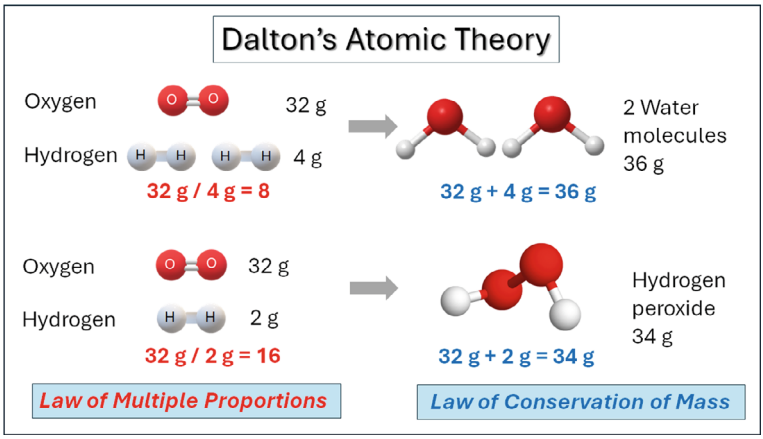


Fig. 2.1 Dalton's atomic theory and the laws of conservation of mass and multiple proportions

This is a multiple ratio (2: 1), which allowed, for the first time, to deduce the fixed stoichiometric proportions in which chemicals reacted according to the *Law of Multiple Proportions* (also known as *Dalton's Law*) (Fig. 2.1). These observations led Dalton to deduce that there must be indivisible constituent particles that, combined together in simple ratios, can form matter.

Dalton's model explains most of the observations underlying the chemistry of the late eighteenth and early nineteenth centuries by reducing it to a series of processes that can be described through a simple combinatorial theory. Dalton, in one of his most famous sentences, clearly expresses the idea that: "*Matter, though divisible in an extreme degree, is nevertheless not infinitely divisible. That is, there must be some point beyond which we cannot go in the division of matter. … I have chosen the word 'atom' to signify these ultimate particles*." [5].

Indeed, atoms are the smallest particles that retain specific properties. Still, these depend, however, on the building blocks that form the atom: electrons, protons, and neutrons, the latter two in turn formed by other even smaller particles, the quarks. The ultimate constituent unit of the nucleon particles, protons, and neutrons are the quarks, elementary particles like electrons. It is a bit more complicated than Dalton might have imagined, although he has the great merit of having applied the scientific method to work out his observations in an attempt to describe the intimate structure of matter and explain its connection with chemical reactions.

A Dalton atom is an atom without electrons, an indivisible entity that can ideally be represented as a small rigid sphere; it is thus a physical object that interacts with other atoms without changing its nature. Atoms are thus purely physical entities for Dalton, an idea probably derived from his study of the physical properties of the atmosphere and other gases. Dalton intuits that when atoms form molecules, some deeper interactions must take place, but he did not have the experimental tools to go beyond this initial intuition. Matter is composed of atoms. However, the single atom in nature, except noble gases, does not exist. Each atom will tend to interact to

achieve a stable energy configuration, and to do so, it must donate or accept electrons. At the time of Dalton, however, electrons were still to be discovered.

The road traced by John Dalton (1766–1844) will lead from his first atomic theory to the Standard Model. A long journey that lasted two centuries. Dalton, according to his biographers, had a shy and modest personality (*R. A. Smith. Memoir of Dr. Dalton, and History of The Atomic Theory up to his time. Memoirs of The Literary and Philosophical Society of Manchester. 1856. Vo. 13. London*). His family did not have much financial means, and to continue his studies at 12, he began teaching in his village school. This employment enabled him to study and support himself until he moved to Manchester to become a tutor at New College. There, he could access laboratories and devote himself to his great scientific passion, meteorology. This passion became a veritable mania; throughout his life, he diligently jotted down his detailed weather observations every day. More than 200,000 weather annotations made over 57 years remain; the last one followed shortly before he died on July 27, 1844, at the age of 78. Dalton is also famous for the disability that bears his name, color blindness. He developed a theory to explain color vision deficiency that, however, was later demonstrated to be wrong. Dalton's life was completely absorbed in science, and he lived modestly without marrying: "For Dalton, science was the occupation of life, of a life spent in the most laborious manner." At his death, over 40,000 people attended his funeral. His scientific legacy was taken by another outstanding scientist who was his student, James Prescott Joule (1818–89).

2.2 The Discovery of the Electron

The discovery of the electron, a pivotal moment in the understanding of atomic structure, represented a significant leap from Dalton's theory of indivisible atoms. This breakthrough, which took over a century after Dalton's time, revealed the existence of particles smaller than the atom. It unveiled the complex inner structure of matter, shattering the notion of the atom as the ultimate physical boundary.

The discovery of the electron dates back to the late nineteenth century, in 1897 to be precise, and was made by the English physicist Joseph John Thompson (1856–1940). Hand in hand with the discovery of the electron began reflections on the structure of what Dalton had correctly identified as the constituent element of matter, the atom. Subsequent experiments made it possible to identify as early as 1919, in addition to electrons, the presence of other charged particles, the protons. Once it had been experimentally observed that these two particles had opposite charges, it remained to be understood how electrons and protons could form an atomic structure apparently

in contradiction to the physical laws then known. The neutron would be discovered later in 1932 and would close the first stage of knowledge about atomic particles. As knowledge about the particles that made up the atom advanced, more sophisticated models of its structure were developed. It is a very fascinating story because it is primarily a story of ideas and how, starting from the earliest discoveries through a conceptually evolutionary process, we arrived at the model currently used to explain atomic structure and the set of elementary particles that is the *Standard Model*.

2.2.1 The Experiment of J. J. Thomson. Solving the Mystery of Cathode Rays

In 1896, Joseph John Thomson, a professor of physics (*Cavendish Professor of Physics*) at Cambridge University, began conducting experiments with cathode-ray tubes in his laboratory [6, 7]. These devices consist of sealed glass tubes inside which a vacuum is created; at the ends of the tube are placed an anode and a cathode that emit electrons even though before Thomson the nature of the particles generated from the cathode was not clear at all [8]. The system was called a cathode-ray tube precisely because the cathode produces the particle beam. Cathode-ray tubes had been known for some time and studied for their property of ideally functioning as simple, small particle accelerators whose nature was still widely debated.

It is Thomson himself at the beginning of his article in 1897 who takes stock of the heated discussion on cathode rays: "*The experiments discussed in this paper were undertaken in the hope of gaining some information as to the nature of the Cathode Rays. The most diverse opinions are held as to these rays; according to the almost unanimous opinion of German physicists they are due to some process in the aether to which, inasmuch as in a uniform magnetic field their course is circular and not rectilinear, no phenomenon hitherto observed is analogous: another view of these rays is that, so far from being wholly aetherial, they are in fact wholly material, and that they mark the paths of particles of mutter charged with negative electricity.*" [9]. The strange nature of this discussion should not be surprising; the world of subatomic particles was still quite foreign to the scientific community. Thomson experiment would mark a fundamental turning point with his demonstration of the existence of the electron [10].

Cathode-ray tubes, which have now fallen into disuse, have been widely used for many applications, particularly for televisions, until the advent of the new generation of LCD flat screens that employ completely different technologies. The old televisions were large in size because they used cathode-ray tubes to generate the images.

The cathode-ray tube needs a vacuum inside to operate, and when a difference of potential is applied between the cathode and anode, a particle flux is generated (Fig. 2.2) whose nature was finally clarified by Thomson's experiment.

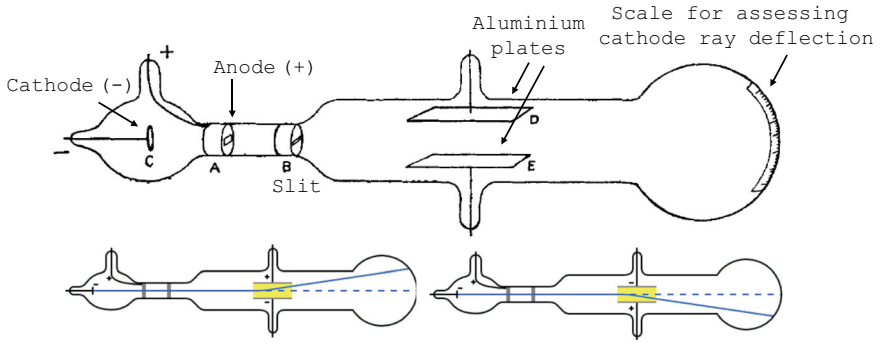


Fig. 2.2 Diagram of the experimental apparatus used by Thomson for electron studies (upper part, original scheme) [9]. Bottom figures. Deflection of the cathode rays by the electric field. The blue line represents the cathode ray, and the yellow represents the electric field [11]

It is possible to detect the particle beam reaching the far end of the cathode-ray tube using phosphorescent materials, such as zinc sulfide. Once the beam passes through the slit positioned after the anode and hits the phosphors, they emit light. Thompson used this experimental setup to test the properties of the particles generated inside the cathode-ray tube. For this purpose, he placed two oppositely charged metal plates. Once the cathode beam was generated and focused by passing through the slit, he observed a significant deflection. The beam moved away from the negatively charged plate while being attracted toward the positively charged one. Thomson's description of the experiment in his article is very clear. The original diagram used to show the experimental apparatus is shown in Fig. 2.2: "*The rays from the cathode C pass through a slit in the anode A, which is a metal plug fitting tightly into the tube and connected with the earth; after passing through a second slit in another earth-connected metal plug B, they travel between two parallel aluminium plates about 5 cm. long by 2 broad and at a distance of 1.5 cm. apart; they then fall on the end of the tube and produce a narrow well-defined phosphorescent patch. A scale pasted on the outside of the tube serves to measure the deflexion of this patch. At high exhaustions the rays were deflected when the two aluminium plates were connected with the terminals of a battery of small storage-cells; the rays were depressed when the upper plate was connected with the negative pole of the battery, the lower with the positive, and raised when the upper plate was connected with the positive, the lower with the negative pole. The deflexion was proportional to the difference of potential between the plates, and I could detect the deflexion when the potential-difference was as small as two volts. It was only when the vacuum was a good one that the deflexion and I could detect the deflexion when the potential-difference was as small as two volts.*" [9].

This result experimentally demonstrated that negatively charged particles formed the cathode ray. Once Thompson discovered the charge of the particles, he asked what mass they had. To answer this question, he placed two magnets on the two sides of the cathode-ray tube and observed that the magnetic field deflected the cathode rays.

This experiment enabled him to determine the relationship between the mass and charge of this particle, and he realized, to his amazement, that they had a mass much smaller than the atom. Thomson was also the first to demonstrate that an electric charge in motion possesses mass independently of the charge and how it is set in motion.

It remained to verify whether what he had just discovered was a general property or not. To this end, Thompson repeated the experiment using different metals as electrode materials and found that the properties of the cathode rays were constant, regardless of which material he used for the cathode.

Thomson's experiment led to significant conclusions. It not only shed light on the nature of cathode rays but also confirmed the existence of subatomic particles. His discovery that these particles, initially called *corpuscles*, are a common constituent of all atoms was groundbreaking. The results of the experiment allowed drawing some conclusions:

- Negatively charged particles form cathode rays.
- The particles observed in cathode rays have a very small mass, 1000 times smaller than the hydrogen atom, and therefore must necessarily be part of the atomic structure.
- Corpuscles are an element present within all atoms.

Thomson's association of negative charge with a *corpuscle*, i.e., a particle, was a significant observation. This deduction, based on the English physicist's experiments and knowledge, was brilliant. However, subsequent studies revealed that the nature of the electron is much more elusive than could have been imagined just after its discovery; it was just the beginning of a long journey. On this journey another milestone to the knowledge of the electron will be reached by Joseph Thomson's own son, George Paget Thomson.

2.2.2 Particles or Aether Fluctuations?

J. J. Thomson published his article in 1897 when the German school still considered the physical theory of the aether as the simplest explanation for many physical phenomena. Regarding the nature of cathode rays, two schools of thought had formed, the German school considering them to be a phenomenon due to some process in the aether, the other that they were instead material in nature, negatively charged particles. The aether theory of the German physicists promoted the existence of a medium that fills empty space. This medium would be essential for the propagation of electromagnetic and gravitational forces. The nature of the aether was still being experimented and pondered by the scientific community in the early 1900s. In 1902, Dutch physicist Hendrik A. Lorentz, in his Nobel lecture, observed: "*Permit me now to draw your attention to the ether. Since we learnt to consider this as the transmitter not only of optical but also of electromagnetic phenomena, the problem of its nature became more pressing than ever. Must we imagine the ether as an elastic medium of*

very low density, composed of atoms which are very small compared with ordinary ones? Is it perhaps an incompressible, frictionless fluid, which moves in accordance with the equations of hydrodynamics, and in which therefore there may be various turbulent motions? Or must we think of it as a kind of jelly, half liquid, half solid? […] Unfortunately, all the experiments undertaken on these lines have been unsuccessful; the ether always slips through our fingers." [12].

Aether theory is reminiscent of field theory, which we will describe in Chap. 8, but with one major difference; aether would have a physical nature and some properties such as density or velocity. In quantum field theory, the fields are essentially mathematical entities with no physical interpretation. Instead, the *luminiferous* (luminiferous = light-bearing) aether theory regarded the aether as the medium through which the transmission of electromagnetic waves such as light or X-rays took place in a manner similar to the transmission of sound in air. Instead, the Michelson-Morley experiment of 1887, which was specifically designed to measure the motion of the earth in the ether, proved conclusively that no such effect existed. Another blow to the aether theory came from Albert Einstein's theory of special relativity, which eliminated its necessity by proving that the speed of light is constant in all inertial reference systems. This theory radically changed the understanding of space and time, making the concept of the aether obsolete.

It is Thompson himself who explains why the aether theory is inconsistent, simply because it does not work and does not allow for calculations that predict outcomes: "*The electrified-particle theory has for purpose of research a great advantage over the aetherial theory, since it is definite and its consequences can be predicted; with the aetherial theory it is impossible to predict what will happen under any given circumstances, as on this theory we are dealing with hitherto unobserved phenomena in the rather, of whose laws we are ignorant*." [9]. This observation is the basis for the development of modern physics; a theory, however fascinating, must be consistent with experimental results and must allow predictions to be made.

2.2.3 Giving a Name to the Negatively Charged Corpuscle

Having discovered the electron by Thomson in 1897, it was necessary to use a proper name for these negatively charged corpuscles, as the English physicist had defined them. In 1894 an Irish physicist, George Johnstone Stoney (1826–1911), in his paper entitled "Of the electron or atom of electricity," wrote: "*In this paper an estimate was made of the actual amount of this most remarkable fundamental unit of electricity for which I have since ventured to suggest the name* ***electron***."

In reality, Stoney referred to something different than the actual electron, namely, a charge unit responsible for the passage of current in electrolytes. Stoney's electrons were similar to negatively charged atoms; another Irish physicist, George Francis Fitzgerald (1851–1901), Stoney's own grandson, would suggest using the term electron for the "corpuscles" discovered by Thomson. Fitzgerald's electrons, however, were still corpuscles separate from the atom, and according to a theory proposed by

Joseph Larmor, Thomson's fellow student at Cambridge, they were a structure of the aether. Although physicists meant different things by the particle, the name electron would later be adopted by the scientific community, in *nomen omen.*

Despite the long road ahead, the science made remarkable progress in a surprisingly short period. In less than thirty years, a complete atomic model based on quantum mechanics was developed, an achievement that was difficult to predict in the late 1800s.

2.2.4 Electrons Are Not Raisins. Thomson Model and the Creation of a Scientific Myth

Thomson's "corpuscles" accounted well for the negatively charged particles generated by cathode rays. As we have seen, the next step of correlating corpuscles with a subatomic particle was not obvious, however. It was Thomson himself, despite criticism, who associated corpuscles, that is, electrons, with atomic structure. Dalton's indivisible sphere with which atoms had hitherto been imagined finally appeared as something much more complex. What would the atomic structure look like if electrons were part of an atom? Thomson began to reason about this problem in a series of papers until he arrived at a model that appeared quite reasonable for the experimental data acquired up to then. His model, which he gradually modified over the years, represents the starting point of a journey to understand the atomic structure and the strange properties of the particle he had just discovered.

Unfortunately, Thomson's model has undergone a strange process that has led to the creation of a scientific myth, that of the *plum pudding* atom model [13, 14]. In this model, electrons would be immersed and randomly distributed, like raisins in a pudding, in a positively charged sphere. It is too bad that Thomson had a completely different idea in mind, and no traces of this model can be found in his articles. What model Thomson had in mind is accounted for by his student Bohr: "*According to the theory of the latter* (*Thomson*) *the atom consists of a sphere of uniform positive electrification, inside which the electrons move in circular orbits.*" [15].

Thomson did indeed have a positively charged sphere in mind, but his corpuscles are not raisins but negatively charged particles traveling in circular orbits, as he clearly specified already in the title of his paper (On the structure of the atom: an investigation of the stability and periods of oscillation of several corpuscles arranged at equal intervals around the circumference of a circle; with application of the results to the theory of atomic structure): "*The view that the atoms of the elements consist of a number of negatively electrified corpuscles enclosed in a sphere of uniform positive, electrification, suggests, among other interesting mathematical problems, the one discussed ill this paper, that of the motion of a ring of n negatively electrified particles placed inside a uniformly electrified sphere.*" [16].

The other problem faced by Thomson was to give mechanical stability to his atom, particularly to assess how many electrons could fit into circular orbits and at

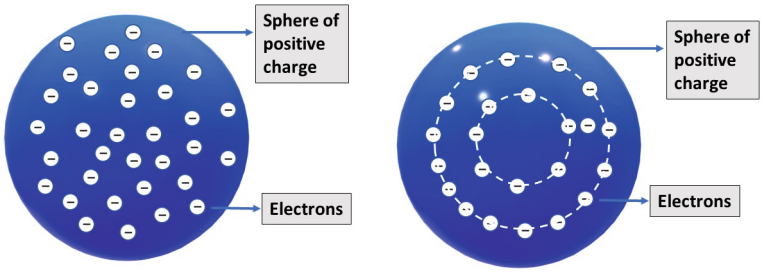


Fig. 2.3 The "plum pudding atomic model of Thomson" fruit of a scientific myth (left) and Thomson's model (right)

what distance: "*The number of corpuscles in a ring vary from ring to ring; each corpuscle is travelling at a high speed around the circumference of the ring in which it is situated, and the rings are so arranged that those which contain a large number of corpuscles are near to the surface of the sphere, while those in which there are a smaller number of corpuscles are more in the inside.*"

Figure 2.3 shows the "plum pudding" model, the result of a scientific myth, and the one that may more closely represent the model Thomson had in mind. Where the idea of plum pudding originated is not easy to reconstruct, perhaps from a journalist who, listening to Thomson, wanted to simplify his idea to popularize it. Almost every textbook on basic physics and internet sites propose this strange idea, which has no basis in Thomson's work.

Another important Thomson deduction was that the electrons did not have to be hundreds but relatively few; moreover, the emission of light from atoms resulted from the vibrations of electrons, provided that the number of electrons was of the same order as the number of spectral lines. However, the model could not explain the regularities of the spectral lines.

2.3 A Japanese Physicist Rotates Electrons Around the Nucleus

Beginning with J. J. Thomson's discovery of the electron, theories and models to explain the properties of electrons in an atom followed one another rapidly. Many experimental elements are still missing; evidence of a heavy nucleus made of protons will, in fact, come later with Ernest Rutherford. In the early 1900s, close to the discovery of the electron, the existence of a positive charge was only assumed based on the hypothesis that negatively charged electrons must be balanced by a charge of the opposite sign to ensure the neutrality of the atom. In 1904, a Japanese physicist, Hantaro Nagaoka (1865–1950), probably inspired by his astronomy studies, proposed a "quasi-planetary" model in which electrons rotate in circular orbits around a heavy nucleus [17]. A step forward from Thomson's electrons in concentric circular

orbits because the idea of a positively charged nucleus emerges for the first time. Nagaoka's hypothesis comes very close to Rutherford's idea of a positive nucleus, who, in fact, will cite the work of Hantaro Nagaoka as one of the sources.

Nagaoka's model, over time has been sidelined and vulgarized with the image of Saturn's rings according to which electrons moving around the nucleus like the fragments that make up the rings: "*The system, which I am going to discuss, consists of a large number of particles of equal mass arranged in a circle at equal angular intervals and repelling each other with forces inversely proportional to the square of distance; at the centre of the circle, place a particle of large mass attracting the other particles according to the same law of force. If these repelling particles be revolving with nearly the same velocity about the attracting centre, the system will generally remain stable, for small disturbances provided the attracting force be sufficiently great. The system differs from the Saturnian system considered by Maxwell in having repelling particles instead of attracting satellites. The present case will evidently be approximately realized if we replace these satellites by negative electrons and the attracting centre by a positively charged particle.*"

Although this model later became known as "Saturnian," it is likely that Nagaoka had in mind a system of three-dimensional orbits for electrons: "*We must consider the same number of rings of particles (electrons), which may or may not be in the same plane*" (Fig. 2.4).

Nagaoka, in his model, solved what he thought was one of the major problems with the Thomson's atomic model, namely, that opposite charges are impenetrable [18]. So, to imagine electrons inside a positively charged sphere contravened this principle, a correct intuition, by the way. The positive charge is concentrated in the center of the atom and the electrons outside. This is the first conceptualization of the atomic nucleus, a fundamental step, at least from a theoretical point of view, in the development of atomic models.

Nagaoka's model, however, failed to explain the lack of stability of electrons with respect to oscillations orthogonal to the plane of rotation, as observed by J. J. Thomson himself. This led Nagaoka himself to abandon his model, which nevertheless served to develop future models, as noticed also by Rutherford [19]: "*It is of interest to note that Nagaoka has mathematically considered the properties*

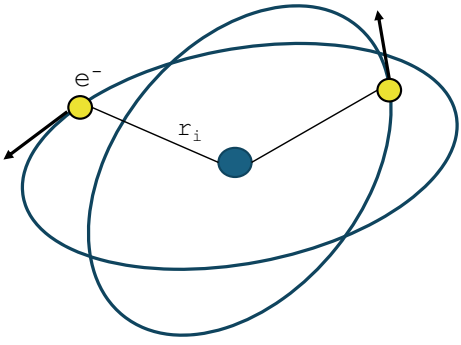


Fig. 2.4 Hantaro Nagaoka's atomic model. Individual electrons describe orbits of radius, r_i, around the large-mass atomic nucleus. The orbits may not be in the same plane

of a 'Saturnian' atom which he supposed to consist of a central attracting mass surrounded by rings of rotating electrons."

2.4 The Discovery of the Nucleus. Rutherford Experiment and Electrons in the Planet that Is Not There

In 1909, two assistants of physicist Ernest Rutherford (1871–1937), a former student of J. J. Thomson's, performed under their professor's guidance an experiment that would become one of the milestones of physics. It was the famous alpha particle experiment carried out by Hans Geiger and Ernest Marsden, following the directions of Rutherford, who had theorized it.

The experiment was based on observing the exit angle of a beam of alpha particles (nuclei of helium atoms, thus positively charged) fired at a thin sheet of gold. Based on Thomson's model in which the atoms consisted of a positively charged *continuum* the expectation was to observe only deflections at relatively high angles. Things, to the surprise of Rutherford himself, did not quite work out that way. As he describes: "*The observations, however, of Geiger and Marsden on the scattering of a rays indicate that some of the a particles must suffer a deflection of more than a right angle at a single encounter. They found, for example, that a small fraction of the incident a particles, about 1 in 20,000, were turned through an average angle of 90° in passing through a layer of gold-foil about 0.00004 cm. thick, which was equivalent in stopping-power of the a particle to 1.6 mm of air. […] A simple calculation based on the theory of probability shows that the chance of an a particle being deflected through 90 ~ is vanishingly small*." [19].

The majority of the alpha particles were observed to pass through the gold foil with minimal or no deflection in the experiment, suggesting that the atoms were mostly made up of empty space. Only a small fraction of alpha particles, as described by Rutherford, were deflected at large angles. The fact that most of the particles were still being transmitted without being deflected was indicative of an important aspect of atomic structure, namely that the atom was, in essence, empty. Otherwise, it would not explain how alpha particles could traverse it undisturbed. Rutherford's conclusions were published in a 1911 paper (*The scattering of alpha and beta particles by matter and the structure of the atom*) in which he explained the results of Geiger and Marsden's experiment. Deflections at high angles are justifiable if there is a nucleus that occupies a very small volume within the atom: "*…it has been supposed that the atom consists of a central charge supposed concentrated at a point, and that the large single deflexions of the a and b particles are mainly due to their passage through the strong central field*." [19].

In his 1913 book *Radioactive substances and their radiations* he still reiterates that: "*In order to account for this large scattering of particles, I supposed that the atom consisted of a positively charged nucleus of small dimensions in which practically all the mass of the atom was concentrated. The nucleus was supposed to*

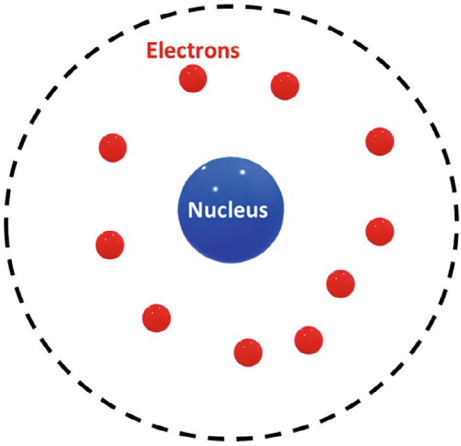


Fig. 2.5 Atom of Rutherford

be surrounded by a distribution of electrons to make the atom electrically neutral and extending to distance from the nuclear comparable with the ordinary accepted radius of the atom."

As we have previously seen, the only reference to possible circular orbits of electrons around the nucleus is taken from Nagaoka. The graphical representation that most closely corresponds to Rutherford's idea of the atom is that of a small positively charged nucleus surrounded by electrons (Fig. 2.5). The model is different from the school model of planetary orbits but closer to what Rutherford described in his article. Rutherford's idea marks a significant advancement in our understanding of the atom's structure. It refutes Dalton's notion of an indivisible particle and Thomson's concept of a positively charged sphere with rotating electrons. Instead, it introduces the idea of the atom as a composition of two distinct charges, the positive nucleus and the negative electrons. While Rutherford's model does not delve deeper, it establishes a crucial distinction between electrons and the nucleus.

Rutherford's idea of the atom would be the starting point for the first quantum model theorized by Bohr, who in his famous paper published in 1913 (*On the Constitution of Atoms and Molecules*),[15] wrote: "*In order to explain the results of experiments on scattering of a rays by matter Prof. Rutherford has given a theory of the structure of atoms. According to this theory, the atoms consist of a positively charged nucleus surrounded by a system of electrons kept together by attractive forces from the nucleus; the total negative charge of the electrons is equal to the positive charge of the nucleus. Further, the nucleus is assumed to be the seat of the essential part of the mass of the atom, and to have linear dimensions exceedingly small compared with the linear dimensions of the whole atom.*"

Fundamental elements for understanding atomic structure still need to be included; the nucleus is, in fact, not just a small massive sphere at the center of the atom but has a much more complex structure, and there are two different particles, protons and neutrons, that form it. However, the discovery of the proton and neutron (which took place in 1932) would lead to many other questions, such as how charges of the

same sign remain permanently attached in the nucleus. The answer will come only after many years when the *Standard Model* is finally worked out.

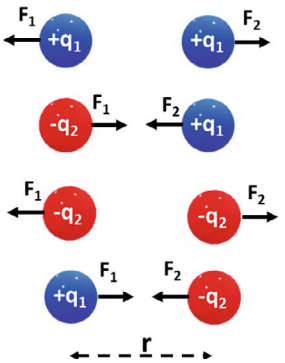
2.5 Why Does an Electron Not Fall into the Nucleus?

The models of Nagaoka and then Rutherford, imagining the electrons as negatively charged particles around the nucleus, fail to answer a fundamental question: How come the electrons do not fall into the nucleus? In his model, Rutherford is careful not to rotate the electrons around the nucleus. The analogy with a planetary model, which also inspired Nagaoka, does not hold in the case of electrons and nucleus because they are electrically charged particles, and their eventual motion could be calculated based on the laws of classical physics. By applying the classical equations, the contradictions about the possible motion of electrons around the nucleus emerge. Coulomb's law governs static interactions between charged particles:

$$|\mathrm{F}| = k_e \frac{|q_1||q_2|}{r^2}. \tag{2.1}$$

F is the electric Force between charges q_1 and q_2, expressed in the equation as moduli, so they can be positive and negative charges with four combinations, as shown in Fig. 2.6. Coulomb's law expresses how much the Force will depend on the distance between the two charges. The Force is also inversely proportional to the square of the charge distance according to Coulomb's constant, $k_e \approx 8.988 \times 109\ \mathrm{N{\cdot}m^2{\cdot}C^{-2}}$. The moduli indicate, as we have seen, that the charges can have both positive and negative signs, and the possible combinations show that the Force between the charges can be repulsive when they have equal value or attractive when the charges are different.

According to classical mechanics, if electrons were to rotate in circular orbits around the nucleus, only Coulomb-type attractive forces could ensure their stability. The problem with the stability of the electron in a circular rotation and the Coulomb



$$|F_1| = |F_2| = k_e \frac{|q_1||q_2|}{r^2}$$

Fig. 2.6 Magnitude of the electrostatic force F between two point charges, q_1 and q_2, is directly proportional to the product of the magnitudes of the charges and inversely proportional to the square of the distance between them. Similar charges repel each other, and opposite charges attract each other

force to which it is subjected is that according to classical physics, every charged particle in rotation should emit electromagnetic radiation, that is, lose energy. The electron should then fall on the nucleus following a spiral orbit in a very short time. It has been calculated that the time required for this to happen is one thousandth of a billionth of a second (0.000000000001 s); an electron could, therefore, not exist for more than 10 to 12 s. Fortunately, this does not happen, and the electron, wave, or particle it may be, occupies, albeit in probabilistic terms, a specific position within the atom without precipitating into the nucleus.

Regarding Coulomb's law, it is also good to remember that its application has definite limitations:

- Electric charges need to have a symmetrical distribution in all directions, such as in point charges or charged metal spheres.
- The charges should be separate from each other and not overlap, meaning they must be distinct point charges.
- The charges must remain stationary about each other, following the electrostatic approximation.

Returning to the models of Thomson, Nagaoka, and Rutherford, and the question of electron stability, this did not actually arise for Thomson's atom because, in the case of many electrons moving in circular orbits, their motion could be considered similar to that of a direct current capable of generating only static electromagnetic fields. Rutherford circumvented the problem by avoiding describing how the electrons eventually had to move around the nucleus. Nagaoka's model, instead, had stability problems and was quickly discarded. Bohr perfectly illustrates the problem: "*In an attempt to explain some of the properties of matter on the basis of this atom-model we meet, however, with difficulties of a serious nature arising from the apparent instability of the system of electrons: difficulties purposely avoided in atom-models previously considered, for instance, in the one proposed by Sir J. J. Thomson. According to the theory of the latter the atom consists of a sphere of uniform positive electrification, inside which the electrons move in circular orbits. The principal difference between the atom-models proposed by Thomson and Rutherford consists in the circumstance that the forces acting on the electrons in the atom-model of Thomson allow of certain configurations and motions of the electrons for which the system is in a stable equilibrium; such configurations, however, apparently do not exist for the second atom-model*."

Thus, when Bohr began to address the study of the atom's structure, the small positive nucleus model surrounded by electrons was well established. However, the position occupied by the electrons within the atom and their eventual movement relative to the nucleus appeared to be a complex puzzle to solve.

2.6 The Model of Bohr. "A Genius Was Called for, and a Genius Appeared"

We must be clear that when it comes to atoms, language can only be used as in poetry. Niels Bohr

Rutherford's atomic model was based on the experimental evidence for the existence of a nucleus where most of the atom's mass was concentrated. Many questions, however, were still to be resolved; the first issue to explain was the eventual motion of the electron, and the other concerned the emission of spectral lines. In fact, an important experimental finding was already acquired as early as the end of the nineteenth century: when an atom is excited by light, it emits characteristic spectral lines, that is, energy at specific values rather than a continuous spectrum. This was a piece of established experimental evidence, even though it could not be correlated with specific properties of the atomic structure. It would be Bohr who understood that discrete spectral lines were what should be expected if the electrons were distributed in some specific circular orbits around the nucleus.

In 1885, Swiss mathematician Johann Jakob Balmer (1825–1898) proposed an equation that described the emission wavelengths in the visible range of the hydrogen atom quite well. Balmer observed that there was an empirical relationship between a specific wavelength in the visible, 364.50682 nm, and the four wavelengths of the hydrogen emission lines in the visible spectrum (between 400 and 700 nm): 410 (violet), 434 (blue), 486 (cyan) and 656 nm (red). Taking this reference wavelength as a constant, B = 364.50682 nm, a simple equation could be written to derive the wavelengths of hydrogen for any integer greater than two:

$$\lambda = B\left(\frac{n^2}{n^2 - 2^2}\right), \tag{2.2}$$

where λ is the wavelength, B is the Balmer limit, and n is an integer greater than two. The empirical formula shows that there is a certain regularity in the emission of spectral lines, and this phenomenon must correspond to a specific physical property of the atoms. The series continues with infinite lines whose wavelengths asymptotically approach the limit of Balmer $B = 364.5$ nm in the ultraviolet.

Balmer with his formula had identified all the possible transitions of an electron from $n \geq 3$ to $n = 2$, where n is the *principal quantum number* that identifies the energy level of the electron (Table 2.1) in the Bohr model, as we will see.

For transitions from the lowest energy level, $n = 1$, and closer to the nucleus, more energetic radiation was needed, and spectral emissions were observed in the ultraviolet range. This is the spectral series called the *Lyman series*, after the American physicist Theodore Lyman (1874–1954), who first identified them. It was later possible to define the spectral lines corresponding to the other energy levels, with each series named after its discoverer (Fig. 2.7). Each series corresponds to transitions that produce spectral lines in different regions of the electromagnetic spectrum: Lyman

Table 2.1 Transitions of the hydrogen atoms in the visible spectrum region (400–700 nm)

Balmer spectral line	Wavelength	Color	Electronic transition
H_α	654.4 nm	Red	$2 \rightarrow 3$
H_β	466.1 nm	Cyan	$2 \rightarrow 4$
H_γ	434.2 nm	Blue	$2 \rightarrow 5$
H_δ	410.2 nm	Violet	$2 \rightarrow 6$

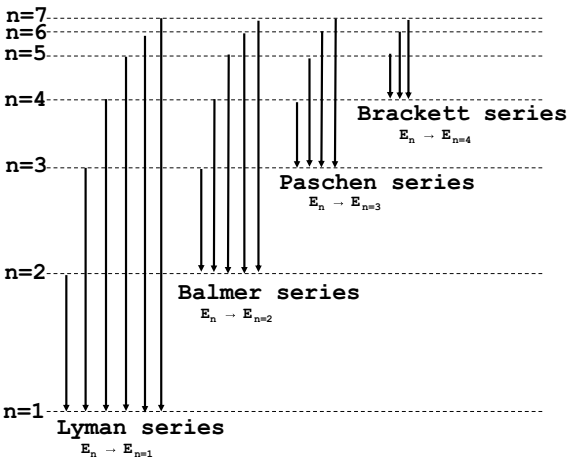


Fig. 2.7 Electronic transitions in the Lyman (UV), Balmer (Visible), Paschen (Infrared), Brackett (Infrared) series

(UV), Balmer (Visible), Paschen (Infrared), Brackett (Infrared), Pfund (Infrared), and Humpreys (Infrared). The series take the name of the scientists that identified them.

For the hydrogen atom, the spectral lines can be, therefore, categorized into different series each corresponding to transitions to a specific lower energy level. Figure 2.8 illustrates Bohr's representation of the transition process between different electronic energy levels [20]. H_α ($3 \rightarrow 2$) and H_β ($4 \rightarrow 2$) indicate the Balmer series transitions in the visible corresponding to the red and green spectral lines of hydrogen, respectively. The transition processes of the first three spectral lines of Lyman's (1914) ultraviolet series are also shown ($4 \rightarrow 1$, $3 \rightarrow 1$, $2 \rightarrow 1$).

Few years later, in 1888, the Swedish physicist Johannes Rydberg (1854–1819), succeeded in generalizing Balmer's formula for all transitions of the hydrogen atom by introducing the wavenumber, which is the reciprocal of the wavelength:

$$\frac{1}{\lambda} = R_H \left(\frac{1}{n_1^2} - \frac{1}{n_h^2} \right) \tag{2.3}$$

with: λ the wavelength of electromagnetic radiation emitted in vacuum, R_H, the Rydberg constant for the hydrogen atom being approximately 1.09677583×10^7 m^{-1}, n_1 the lowest energy level, and n_h the energy level of the electronic transition

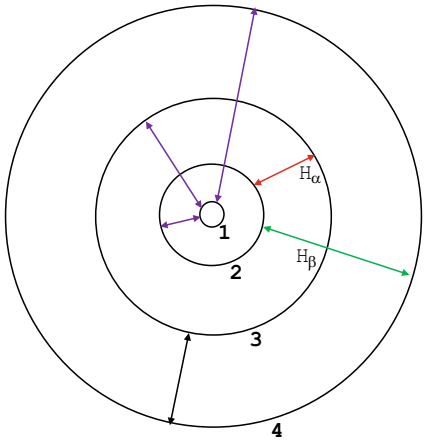


Fig. 2.8 Scheme used by Bohr to illustrate the electronic transitions during his Nobel Lecture [20]

[21]. In practice, the Rydberg constant corresponds to 4/B in Balmer's formula. Rydberg's formula applies to the hydrogen atom but can also be generalized to *hydrogen-like* atoms, i.e., all atoms with only one valence electron, such as alkali metals, or ionized alkaline earth metals.

$$\frac{1}{\lambda} = \mathrm{RZ}^2\left(\frac{1}{n_1^2} - \frac{1}{n_h^2}\right). \tag{2.4}$$

In the formula, Z is the atomic number, and R is the Rydberg constant for the specific atom with atomic number Z. In formulating his expression, Rydberg made no assumptions about the correlation of the equation with a corresponding physical phenomenon, and R was a fitting parameter. However, the use of the wavenumber, simply the reciprocal of the wavelength λ, represented a happy insight according to Bohr himself. The frequency of light, ν, is indeed proportional to the wave number (there is the speed of light):

$$\frac{1}{\lambda} = \frac{\nu}{c} \tag{2.5}$$

and thus also to the quantum of light with energy $\mathrm{E} = \mathrm{h}\nu$:

$$\frac{1}{\lambda} = \frac{E}{hc}. \tag{2.6}$$

We will return to this fundamental relationship later, for now it is important to recall as a conceptual step that experimental evidence had been found between excitation of an atom and emission wavelength and that this relationship was expressed by empirical equations correlated with progressive integers. The idea of discrete

electron energy levels once the existence of the electron was known was a logical consequence.

This logical connection was made in 1913 by the Danish physicist Niels Bohr [22] (1885–1962), the man of destiny for quantum mechanics. Between July and November 1913 Bohr published a trilogy of articles in the *Philosophical Magazine* on the structure of atoms and molecules. The first part dealing with the origin of the spectrum of hydrogen was truly revolutionary. To solve the mystery of spectral emissions Bohr moved from Denmark to England just to work with J. J. Thomson and Ernest Rutherford. He did not get any direct support from them but his stay in Cambridge first and Manchester later gave him the opportunity to assist to inspiring lectures and to reflect on the problem. In 1912 he wrote some notes where he realized that the idea of Max Planck that light is composed by quanta of energy could be applied to the atomic model of Thomson. This was a fundamental conceptual step as we are going to see.

According to Bohr, the integers n found in the Balmer (and Rydberg) series had to correspond to the electronic orbitals located at discrete distances from the center of the atom. In this way Bohr not only explains the origin of spectral lines with specific electronic transitions, but introduces a quantum view of atomic structure. It is thus from the observation of discrete spectral lines that Bohr proposed his model in which electrons can only occupy orbits that have particular energy levels around the atomic nucleus. The analogy with the solar system is obvious; planets orbit the sun according to the laws of gravitational force. Similarly, electrons revolve around the atomic nucleus according to circular orbits, although the motion is governed by electromagnetic interaction forces. However, the analogy between electrons and planets around the sun stops at orbits since planets, even if they describe only well-defined routes, cannot jump from one orbit to another and then return to the starting orbit. And *mutatis mutandis* electrons do not revolve around the nucleus at all.

Although rather distant from how the atomic structure will be described according to the Schrödinger equation and finally the *Standard Model*, Bohr's model for the first time introduces the concept of quantization of electron orbits, even if the idea of a particle in circular motion around a nucleus will soon be replaced by much more complex and counterintuitive models for which the physical representation of the atom will become virtually indefinable except in purely mathematical terms.

Bohr builds his atomic model using as a reference the hydrogen atom, which is the simplest one since it consists of an electron and a proton. According to Bohr's model an electron can only be found in a definite energy state, each level or orbital is denoted by the letter n and a progressive integer number starting with 1, for n = 1, 2, 3, 4…. The lowest orbital, thus the one closest to the atomic nucleus, will be n = 1. An electron cannot occupy an intermediate state between n = 1 and n = 2 or n = 3 but only the energy levels corresponding to the discrete value of the orbital's energy. The number n corresponds to the *Principal Quantum Number*, n, or also Bohr's number.

This idea explains the presence of the discrete, non-continuous spectral lines described by Balmer and Rydberg with their empirical series. An electron cannot

occupy all possible energy states corresponding to different orbits placed at continuous distances from the center of the atom. It is the first truly introduction of a quantum concept to describe the atomic model.

The atoms will have energy levels in which electrons rotate that are at different distances from the nucleus because each atom has a different number of protons and neutrons. The emission of a spectral line is determined by how much light is emitted by the excited electron in returning to its fundamental level.

The model is generally summarized using Bohr's postulates, which are basic assumptions that are in any case unproven or experimentally unprovable. These are:

1. The electron rotates around the central nucleus of the atom describing circular orbits. These orbits follow the laws of classical mechanics, and the Coulomb force generates the electron motion.
2. Not all orbits are allowed, but only those that have specific values of angular momentum, which must be an integer multiple of $\hbar$.

$$\mathrm{L} = nh/2\pi = n\hbar \tag{2.7}$$

with n = 1, 2, 3 ….. and $\hbar = h/2\pi$.

3. The third postulate states that the electron emits no electromagnetic radiation as it orbits the nucleus. The electrons are in a stationary state.
4. If an electron passes from one orbit with a certain energy value, $\mathrm{E_i}$, to another with final energy $\mathrm{E_f}$, a quantum of light with energy will be emitted:

$$\nu = E_i - E_f\,. \tag{2.8}$$

When electrons jump from a lower to a higher energy level they absorb energy, which conversely is lost in the opposite process (Fig. 2.9). An important observation is that in postulating allowed angular moments, Bohr uses a constant we have not yet discussed, Planck's constant. We will devote much attention to this constant later; for now it can be regarded as a value that defines the minimum possible energy, the quantum of light. Then the electron will be able to take a value of angular momentum, defined by its velocity and mass (the momentum or momentum) and its position during rotation with respect to the nucleus (the angle and distance), which is a multiple of a constant value.

We can now go back to Rydberg's constant and find out why Bohr was so enthusiastic about it being expressed in terms of 1/λ. Bohr derives the energy levels of an electron, $\mathrm{E_n}$, in the hydrogen atom on the basis of the assumptions made in his model and applying the classical physics laws:

$$E_n = -\frac{e^4 m_e}{8\varepsilon_0^2 h^2 n^2}. \tag{2.9}$$

Thus, the energy of a photon emitted during a transition from a higher energy level ($\mathrm{n_2}$) to a lower energy level ($\mathrm{n_1}$) is:

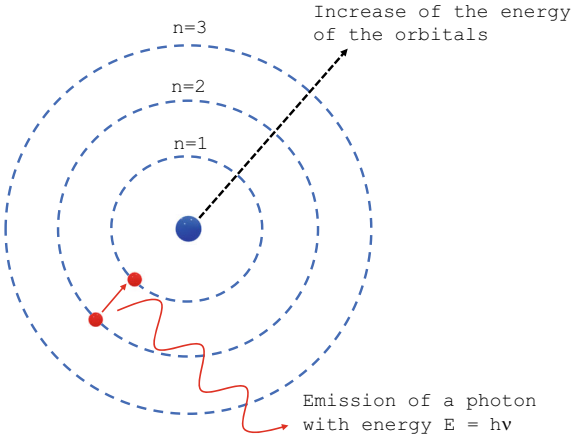


Fig. 2.9 Bohr's atom model. The nucleus in the center houses positively charged protons and neutrons, while negatively charged electrons revolve around it on certain orbits in which they are confined. A quantum leap of the electron from one given orbit to another is accompanied by absorption or emission of electromagnetic radiation

$$E_{photon} = E_{n_2} - E_{n_1} = -\frac{e^4 m_e}{8\varepsilon_0^2 h^2}\left(\frac{1}{n_2^2} - \frac{1}{n_1^2}\right). \tag{2.10}$$

Recalling the previous relationships (2.5) and (2.6), which link energy to frequency and thus wavelength:

$$E_{photon} = h\nu = \frac{hc}{\lambda}. \tag{2.11}$$

It can be written:

$$\frac{hc}{\lambda} = -\frac{e^4 m_e}{8\varepsilon_0^2 h^2}\left(\frac{1}{n_2^2} - \frac{1}{n_1^2}\right). \tag{2.12}$$

The equation begins to suggest something and becomes very similar to Rydberg's expression, in fact solving for $1/\lambda$:

$$\frac{1}{\lambda} = \frac{e^4 m_e}{8\varepsilon_0^2 h^2}\left(\frac{1}{n_1^2} - \frac{1}{n_2^2}\right), \tag{2.13}$$

the two expressions are virtually identical if we pose:

$$R_H = \frac{e^4 m_e}{8\varepsilon_0^2 h^2}, \tag{2.14}$$

and therefore we obtain exactly the Rydberg series:

$$\frac{1}{\lambda} = R_H\left(\frac{1}{n_1^2} - \frac{1}{n_h^2}\right). \tag{2.15}$$

The calculation of Bohr gives the value of the Rydberg constant in physical terms, not as a fitting parameter. Rydberg constant is, in fact, expressed from fundamental constants:

- Electron charge: e.
- Electron mass: m_e.
- Permittivity of the free space: ε_0.
- Planck's constant: h.
- Speed of light: c.

The expression obtained by Bohr provides a theoretical framework for the Rydberg formula and connects the experimental finding of the spectral emission lines with the electron transition in an atom. The Bohr model successfully explains the spectral lines of the hydrogen atom and shows how these lines connect to the quantized levels of the electrons in atoms. The Rydberg constant becomes a physical constant representing the value of the maximum wavenumber (inverse of wavelength) of the photon that can be emitted from a hydrogen atom or the wavenumber of the photon with the minimum energy required to ionize that atom.

It is important to underline that the electron in the model of Bohr is described using a classical approach and the quantum mechanics is applied only to define the possible orbits around the nucleus that can be described by the revolving electron. It is the same Bohr that underlines this point: "(1) … *the dynamical equilibrium of the systems in the stationary states can be discussed by help of the ordinary mechanics, while the passing of the systems between different stationary states cannot be treated on that basis.* (2) *That the latter process is followed by the emission of a homogeneous radiation, for which the relation between the frequency and the amount of energy emitted is the one given by Planck's theory.*" [20].

The Bohr model is still a mix of classical and quantum mechanics, it is only with Schrodinger that a full quantum model will be described, but the electron would become a much more undefined entity that is still puzzling for its complexity.

2.7 Beyond Bohr's Model, Sommerfeld, and the Elliptical Orbits

Bohr's model worked well enough for a simple atom like hydrogen that has only one electron; attempts to extend the model to atoms with more than one electron, even just helium, a two electron atoms, led to frustrating attempts. Heisenberg and Born had calculated all the possible orbits that one of the two electrons of helium could describe when the atom was in an excited state. The method pursued by the two physicists was to apply the computational techniques used for celestial mechanics with the perturbation theory developed for the motion of planets around a central star. However, all attempts by the two physicists led to failure: "*The result of our calculation is negative.*"

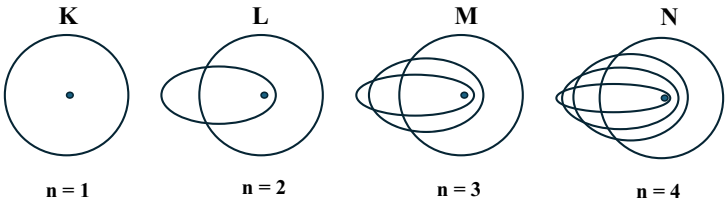


Fig. 2.10 Sommerfeld's extensions of Bohr's 1913 solar system model of the hydrogen atom show the addition of elliptical orbits to explain the fine structure. The picture reproduces the original one from Sommerfeld for n = 1, 2, 3, and 4. The orbits are drawn confocally

The Sommerfeld atomic model, also known as the Bohr-Sommerfeld model, is a modification of Bohr's model of the atom proposed by German physicist Arnold Sommerfeld (1868–1951) [23] in 1919 with the aim of overcoming the limitations of the circular orbit model [24]. The model introduced elliptical orbits for the electrons of an atom, instead of just the circular orbits proposed by Bohr. To explain the formation of the fine spectral lines in the hydrogen atom Sommerfeld proposed that the electron travels also elliptical orbits besides than circular ones (Fig. 2.10). An electron revolving around the nucleus is affected by nuclear charge in such a way that it travels elliptical orbits with the nucleus positioned at one of the two foci. Sommerfeld suggested that it is not necessary that all electrons in the same orbit have the same energy so he divided the main energy levels into subenergy levels. Again, the experimental observation of fine emission structures, i.e., clearly identifiable lines but with very close wavelengths, was a crucial element in the development of the theory.

According to Sommerfeld, not all electrons in the same orbit must rotate on the same circular path, but some of them may move on elliptical trajectories. The nucleus of an atom is located in one of the two foci of elliptical orbitals. During circular motion the distance of the electron from the nucleus obviously remains constant, what varies is only the angle of rotation. In an elliptical orbit, on the other hand, both the angle of rotation, defined as the azimuthal angle, and the distance from the nucleus vary.

Sommerfeld then introduced the concept of quantum numbers that define angular momentum. These discrete numbers describe the shape of the electron's orbit so that in Sommerfeld's model, the energy levels of an atom are determined by both the *principal quantum number* (n) and the *angular momentum quantum number* (l). The principal quantum number corresponds to the energy level of the electron; higher values correspond to higher energy levels, in agreement with Bohr's model. The *angular momentum quantum number,* on the other hand, describes the shape of the electron's orbit, always keeping in mind that elliptical orbits must have the nucleus at one of the two foci.

Sommerfeld's atomic model successfully explains the fine structure of the rows obtained in the hydrogen emission spectrum. But this model, like Bohr's atomic model, fails to explain the emission spectrum of atoms or ions with more than one

electron and the Zeeman effect. Moreover, in Sommerfeld's atomic model, electrons are still regarded as particles whose velocity and position can be determined completely. Following the discovery of the dual nature of electrons and the definition of *Heisenberg's indeterminacy principle*, it became clear that even Sommerfeld's atomic model was unable to fully describe atomic structure.

Scientists and Extraordinary Mentors

A fascinating aspect of J. J. Thomson's personality was his ability to create a scientific school with some of the most brilliant scientists among its students. Thomson's biographies convey to us the image of a brilliant scientist along with that of an engaging professor with a great passion for imparting knowledge. The result was extraordinary; his pupils included no fewer than seven Nobel laureates, beginning with Ernest Rutherford himself (Chemistry, 1908). He was joined by Lawrence Bragg (Physics, 1915), Charles Barkla (Physics, 1917), Francis Aston (Chemistry, 1922), Charles Thomson Rees Wilson (Physics, 1927), Owen Richardson (Physics 1928), and Edward Victor Appleton (Physics, 1947). J. J. Thomson himself had first received the Nobel Prize in 1906. Even more remarkable is that J. J. Thomson's son, George Paget Thomson, also received the Nobel Prize in Physics in 1937.

Another outstanding mentor was Arnold Sommerfeld, who had among his Ph.D. students four Nobel Prize winners, Werner Heisenberg (Physics, 1932), Wolfgang Pauli (Physics, 1945), Peter Debye (Chemistry, 1946), and Hans Bethe (Physics, 1967) and other three winners between his postdocs, Linus Pauling (1954, Chemistry), Isidor I. Rabi (Physics, 1944) and Max von Laue (Physics, 1914). He also supervised more than thirty other physicists and chemists that gave a fundamental contribution to scientific research. Despite his fundamental contributions to atomic physics, to whom we owe the introduction of the second and third quantum numbers, the *azimuthal quantum number,* and the *magnetic quantum number,* J. J. Thomson did not receive the Nobel Prize. He is, however, the record holder of the number of Nobel Prize nominations, an impressive 84 times.

Einstein recognized Sommerfeld's exceptional capability as a mentor to young scientists. Although Einstein and Sommerfeld did not always agree on scientific matters, especially concerning quantum theory, but Einstein acknowledged Sommerfeld for his teaching and nurturing of young talent skills. Sommerfeld had a rigorous, systematic teaching style and always encouraged his students to explore new ideas in physics.

There are many ways to contribute to the advancement of science, but the transmission of the knowledge remains a key pillar for progress.

2.8 A Strange Model, Placing the Electrons in a Cube

The so-called octet rule is one of the best-known rules of thumb used in chemistry. It states that elements in the main group tend to form chemical bonds in a way that each atom has eight electrons in its outermost shell, which is the same electron configuration as a noble gas. This rule, known as *Lewis's rule*, comes from one of the earliest proposed atomic models, that of the *cubic atom*. Strange as this model may seem now, it is of great importance historically because Pauli, in defining the Exclusion Principle, was inspired by Lewis's idea. The idea of the cubic atom came to Gilbert Newton Lewis (1875–1946) as early as 1902, but he left the article in the drawer without publishing it until 1916 [25]. The idea was then taken up by Irving Langmuir in 1919 as the *cubical octet atom* [26].

The cubic model is based on the idea that electrons are arranged in fixed positions around the nucleus at the corners of a cube (Fig. 2.11). This arrangement gives the atom a defined geometric shape, and electrons are considered static in their positions. Following the model the atoms could have several concentric cubes, with electrons arranged in successive layers around the nucleus. Langmuir described the Lewis model in this way: "*These electrons arrange themselves in a series of concentric shells, the first shell containing two electrons, while all other shells tend to hold eight. The outermost shell, however, may hold 2, 4, or 6, instead of 8. The 8 electrons in a shell are supposed to be placed symmetrically at the corners of a cube or in pairs at the corners of a regular tetrahedron. Then atoms combine they usually hold some of their outer electrons in common, two electrons being thus held for each chemical bond. These electrons may form parts of both atomic shells of 8 electrons.*"

Unlike later atomic models, where electrons occupy orbitals with various probabilities of being found in different locations, the cubic model treats electrons as being in fixed positions. This concept does not consider the actual motion or wave-like property of electrons; quantum models for the electrons will be introduced only later by Bohr. The number of electrons in each cube (or shell) increases as the atom became larger, with more electrons filling the corners of successive cubes. For example, the outermost cube could hold up to 8 electrons, as there are 8 corners on a cube. Lewis developed the cubic model to explain the *chemical bonding* of atoms, atoms could share or transfer electrons at the corners of their cubes to achieve stable configurations, thus explaining how chemical bonds form.

Figure 2.12 shows how bonds form following the cubic atom model. If two atoms share an edge they form single covalent bonds (Structure C), resulting in the exchange

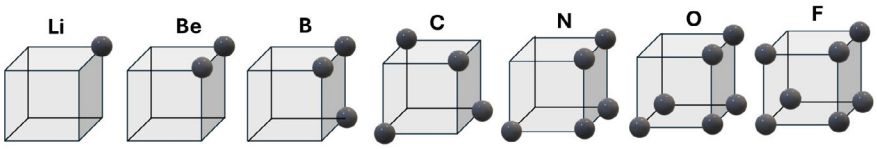


Fig. 2.11 Atoms according to cubical atom model, the second row of the atomic table

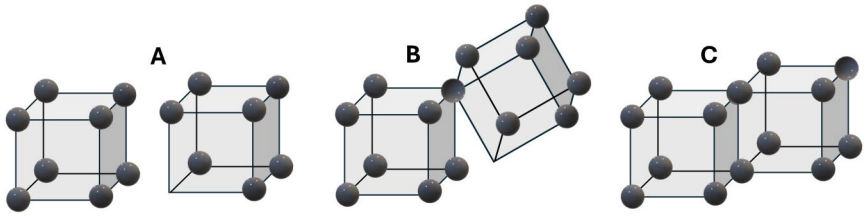


Fig. 2.12 Double bonds are formed by sharing a face between two cubic atoms

of two electrons. Ionic bonds are formed by the transfer of an electron from one atom to another without sharing an edge (Structure A). Lewis also supposed the possible formation of an intermediate state with only one corner shared (structure B).

The cubic atomic model did not account for the dynamic nature of electrons, as they were considered fixed in place rather than in constant motion around the nucleus. The model could not explain the fine details of atomic spectra or the properties of electrons in different energy levels, which were explained by the Bohr model and quantum mechanics. It did not include any wave-like properties of electrons, which were discovered later. However, the octet rule is thus widely used to explain chemical bonding.

2.9 The Bohr Atomic Model Is Not Yet a Full Quantum Model But Allows Calculating Some Important Constants Using Classical Physics

This section introduces some constants derived from Bohr's atomic model. Although the Bohr model is not yet a quantum model and does not use the concept of a wave function, by applying the classical laws of electromagnetism it is possible to derive with good accuracy some constants that are still used in simplified calculations: the *classical electron radius*, the *Bohr radius*, the *reduced Compton constant*, and the *Bohr magneton.* As can be easily deduced from the names of the constants, they are precisely derived using Bohr's semiclassical atomic model.

2.9.1 The Classical Electron Radius

Quantum mechanics and the *Standard Model* of particle physics describe the electron as a point-like particle with no measurable spatial extent. A strange particle, therefore, that has no structure, has charge, mass, but does not occupy a space unless defined as point-like. However, the electron size has been derived from classical

electrodynamics and is still defined as the *Classical Electron Radius,* r_e. It represents a characteristic length scale associated with the electron, though it is not the actual physical size of the electron. The concept arises from the classical theory of the electron, particularly when calculating the energy stored in the electron's electric field. It is related to early models that attempted to reconcile the electron's charge and mass before the development of quantum mechanics. The classical electron radius represents the scale at which classical electrodynamics would predict an electron's electric field to exert a certain energy equivalent to the electron's rest mass.

The *Classical Electron Radius* is obtained by equating the electrostatic potential energy of a sphere of charge *e* and radius r_e with the relativistic mass–energy of the electron ($m_e c^2$), on the hypothesis, therefore, that the mass-energy is fully contained within the sphere:

$$r_e = \frac{e^2}{4\pi\epsilon_0 m_e c^2}, \tag{2.16}$$

where e is the elementary charge (the charge of the electron), ϵ_0 is the permittivity of free space, m_e is the mass of the electron, and c is the speed of light in a vacuum.

The classical electron radius is approximately:

$$r_e \approx 2.817\ldots 10^{-15}\text{m(or 2.817 femtometers)}.$$

Although the classical electron radius has limited direct physical application in modern physics, it still appears in various contexts, such as in the scattering of light by electrons (*Thomson scattering*) and in classical electromagnetism calculations.

2.9.2 The Bohr Radius

In Bohr's model, the electron is still represented as a charged sphere rotating around the nucleus and is "forced" to occupy only very specific energy levels. How far in this model will the electron be from the nucleus? The answer is given by the so-called *Bohr radius*, which represents the most probable distance between an electron in the fundamental state (i.e., that of lowest energy) and the nucleus. The *Bohr radius* is a constant, denoted by the symbol a_0 and is defined by the formula:

$$a_0 = \frac{4\pi\varepsilon_0\hbar^2}{m_e e^2}, \tag{2.17}$$

con ϵ_0 the permittivity of free space, $\hbar$ the reduced Planck's constant, m_e the mass of the electron, and e the elementary charge of the electron. The value of the Bohr radius is:

$$a_0 \approx 5.29177210902\ldots \times 10^{-11}\text{m}.$$

2.9.3 The Reduced Compton Wavelength

The previous two constants are closely related to another constant, the *Reduced Compton Wavelength* (or reduced Compton constant). This is a quantum mechanical property of a particle and represents a fundamental limit to the precision with which the position of a particle position can be known, given its mass. It is defined as reduced because such as the reduced Planck's constant is given by dividing per 2π for simplifying some quantum mechanical calculations. The reduced Compton wavelength, $\lambda\!\!\!^{-} = \lambda/2\pi$ is given by the formula:

$$\lambda\!\!\!^{-} = \frac{\lambda}{2\pi} = \frac{\hbar}{mc}, \tag{2.18}$$

where $\hbar$ is the reduced Planck's constant, m is the mass of the particle, and c is the speed of light in a vacuum.

This value defines a fundamental scale below which the wave-like properties of matter become significant for a particle of a given mass. It also shows the inverse relationship that exists between the mass of a particle and its reduced Compton wavelength: 0054he more massive is the particle, the smaller its reduced Compton wavelength.

For an electron, the reduced Compton wavelength is approximately [27]:

$$\lambda\!\!\!^{-} \approx 2.42631023867\ldots \times 10^{-12}\text{m}.$$

This length is much smaller than the Bohr radius and reflects the quantum mechanical effects at very short distances.

In Chap. 5, devoted to describe the role of some fundamental constants, such as the Planck's constant, we will see that the three constants, r_e a_0, $\lambda\!\!\!^{-}$ are all correlated with another strange number, the fine structure constant, α.

2.9.4 The Bohr Magneton

Another physical constant derived from the Bohr atomic model is the *Bohr magneton* [28] (μ_B). It arises from a combination of classical and quantum mechanical principles. It is a physical constant representing the natural unit of the magnetic moment of an electron caused by its orbital or spin motion in an atom. *Bohr magneton*, therefore, links the electron magnetic momentum to its momentum (angular momentum). Physically, it represents *a quantum of the electron magnetic moment*, which corresponds to its smallest magnetic moment, exactly as the quantum of the electronic angular momentum is given by $L = \hbar$. The *Bohr magneton* measures the strength of an electron magnetic moment due to its orbital or spin motion in an atom. The spin of an electron generates a magnetic moment proportional to the *Bohr magneton.*

The *Bohr magneton* is defined as:

$$\mu_B = e\hbar/2m_e, \tag{2.19}$$

where e is the elementary charge (the charge of an electron), $\hbar$ is the reduced Planck's constant, and m_e is the electron mass.

The value of the *Bohr magneton* is approximately:

$$\mu_B \approx 9.274 \times 10^{-24} \mathrm{J/T}\left(\mathrm{m}^3/\mathrm{s}\right).$$

Here, J/T (joules per tesla) is a unit of magnetic moment, representing the energy associated with a magnetic field.

The formula for the *Bohr magneton* contains terms like the electron mass m_e, the elementary charge e, and concepts related to classical electromagnetism, such as the electron's magnetic moment due to its charge. At the same time, the presence of the reduced Planck's constant $\hbar$, which is a purely quantum mechanical quantity, introduces the quantum mechanical aspect. The *Bohr magneton* comes from early quantum theory, based on Bohr model of the atom, which was semiclassical. Bohr's model combines classical orbits of electrons with quantum ideas (quantized angular momentum). Hence, the *Bohr magneton* fits into a framework where classical physics begins to merge with quantum mechanics.

2.10 The Old Quantum Theory

The Bohr-Sommerfeld model describing atomic structure as a function of the motion and position of electrons in an atom closes a phase of fundamental theoretical discoveries and insights. This progress happened in a relatively short span of time, from Thomson's discovery of the electron in 1897, Rutherford's discovery of the atomic nucleus in 1911, to Bohr's model of 1913 and its extension by Sommerfeld in 1919. Experimental discoveries went hand in hand with the development of theoretical models, which increasingly broke with classical physics until Bohr and his atomic model used quantum concepts for the first time. This phase is now referred to as "old quantum theory." Thomson's discovery of the electron explains the nature of cathode rays, and what these new corpuscles were was initially unclear. Were they separate entities, or did they make up part of the atomic nucleus? Thomson gave the answer, opening a whole new perspective on atomic structure. There remained the problem of how many electrons there were in an atom, how they were distributed, and what, if any, their motion was. The planetary model of electrons revolving around the nucleus was Bohr's idea, who introduced a series of postulates, such as that of stationary orbits, to stabilize the rotation of electrons. Bohr's real revolution, however, is the idea that the angular momentum of the orbits is quantized, as well as the energy levels. His model, as we have seen, is still based on the use of the equations of classical physics, but quantum mechanics has now made its way into the description of the electron.

A particle that was a fundamental part of the atomic structure but extremely elusive, so in order to describe its behavior, it had been necessary to introduce entirely new concepts. This will only be the first step.

The old quantum theory has many limitations, and beyond a system as simple as the hydrogen atom, the calculations do not work; it also fails to account beyond the wavelength of the emission lines for their intensities. The following fundamental step will be taken by Erwin Schrödinger in 1925, and the electron will become a completely different entity with an elusive and multifaceted nature. Schrodinger's equation will introduce coherence to the theoretical model now based on entirely new concepts inspired by quantum mechanics. The new quantum model will make it possible to explain physical phenomena, such as the fine structure of emissions, that were not solved using the old model. The price to be paid will be high, however, because the electron will never be the same again because from physical particles it will become a quantum object. It will be Bohr himself who will try to find a correspondence between classical and quantum physics and will develop two important principles in this regard: *The Correspondence Principle* and the *Complementarity Principle*. We will return to these principles after describing the double-slit experiments when we have more information about what should be a quantum electron.

2.11 A Brief Summary of the Discovery of the Electron

First Atomic Theory (1808). The atom is a rigid sphere and the smallest structure existing in nature.

John Dalton (Eaglesfield, 6 September 1766–Manchester, 27 July 1844).

A new system of chemical philosophy, 1808 [3].

Discovery of the Electron (1897). Cathode rays are composed of negatively charged corpuscles. The value of the electron mass-to-charge ratio is calculated.

Thomson Atomic Model (1904–1906). Electrons rotate rapidly in concentric orbits in a positively charged sphere. Electrons are recognized as subatomic particles.

Joseph John Thomson (Manchester, 18 December 1856–Cambridge, 30 August 1940).

On the Structure of the Atom: an Investigation of the Stability and Periods of Oscillation of a number of Corpuscles arranged at equal intervals around the Circumference of a Circle; with Application of the Results to the Theory of Atomic Structure. 1904 [16].

Quasiplanetary Model (1904). Electrons rotate in circular orbits around a nucleus. It is the first model in which the atom is represented as a positively charged nucleus and electrons revolving around it.

Hantaro Nagaoka (Ōmura, 15 August 1865–Tokyo, 11 December 1950).

Kinetics of a system of particles illustrating the line and the band spectrum and the phenomena of radioactivity, 1904 [17].

Discovery of the Atomic Nucleus, Electrons Are Located Around the Nucleus (1911). The atom consists of a positively charged nucleus surrounded by a distribution of electrons such that the atom is electrically neutral.

Ernest Rutherford (Brightwater, New Zealand, 30 August 1871–Cambridge, 19 October 1937).

The scattering of alpha and beta particles by matter and the structure of the atom, 1911 [19].

First Quantum Model of Atoms (1913). The electron describes stationary orbits around the nucleus whose energy levels can have only discrete values.

Niels Bohr (Copenaghen, 7 October 1885–Copenaghen, 18 November 1962).

I. On the constitution of atoms and molecules, 1913 [15].

Cubic Atomic Model (1916–1919). The electrons occupy the corners of a cube and can be shared by other atoms explaining in this way the formation of atomic bonds.

Gilbert Newton Lewis (Weymouth, 23 October 1875–Berkeley, 23 March 1946)–Irving Langmuir (Brooklyn, 31 January 1881–Woods Hole, 16 August 1957).

The Atom and the Molecule, 1916 [25].

The Arrangement of Electrons in Atoms and Molecules, 1919 [26].

The Electrons Describe Elliptical Orbits (1919). Electrons in the same orbit do not need to have the same energy and can occupy subenergy levels.

Arnold Sommerfeld (Könisberg, 5 December 1868–Münich, 26 April 1951).

Atombau und Spektrallinien (*Atomic Structure and Spectral Lines) 1919* [24].

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Chapter 3
The Quantum Nature of the Electron



Bohr's hypothesis that electrons could occupy only certain energy states within the atom solved the problem of discrete emission of spectral lines, but other fundamental questions remained unanswered. As we saw in the previous chapter, according to Bohr, electrons have discrete angular momentum; they can only be at specific distances from the nucleus, describing definite circular orbits. Sommerfeld then refined the model by including elliptical orbits and energy sublevels. The success of the model, although restricted to experimental evidence from the hydrogen atom alone, was due to the possibility of correlating spectral emission with the energy transitions of an electron in the atom, which is no small step. The excellent match with experimental evidence indicated that the application of the equations of classical physics to the electron worked quite well, provided that discrete conditions were then imposed on the angular moments and energy levels.

The electron in the old quantum model was still a negatively charged ball revolving around the nucleus following quantized orbits. However, this model, besides failing to explain some phenomena, was severely limited by its lack of coherence. The electron is a charged particle that rotates but must follow specific laws that emerge using Planck's concept of quantum of light. The reasons why Bohr's atomic model was welcomed as a tremendous theoretical achievement but also skepticism are pretty clear. It was a matter of making a further effort to build a coherent theory, but this would require an imaginative attempt to take the limit of knowledge beyond common sense. Physics would never be the same again.

3.1 An Electron Is Not Just a Particle

Louis de Broglie (1892–1987) took a fundamental conceptual step forward. The French physicist began by reconsidering the nature of the electron beyond the strictly corpuscular model, seeking an answer to two fundamental questions. The first issue

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P. Innocenzi, *What is an Electron?*, https://doi.org/10.1007/978-3-031-81984-1_3

concerned the very nature of the electron itself. If light has a dual nature of wave and particle, is it not possible that matter itself could have this same property and thus have a wave-like property typical of a wave? Einstein had already demonstrated the wave nature of light in 1905 by explaining the photoelectric effect, which allowed him to determine that light consists of discrete, localized quanta of energy [1]. Thus, de Broglie's considerations were based on experimental observations that could give rise to reasonable doubts that the wave nature was a property also shared by electrons.

Luis de Broglie described later his thoughts as follows: *"In our present state of knowledge* [1926], *it is no longer disputed by any physicist, that matter is not formed by the combination of a few primordial elements (probably only two, the proton and the electron), and that the simple bodies of chemistry differ from one another because their atoms are different assemblages of these primordial elements.* [...] *On the other hand, the theory of relativity has brought us a notion of extreme importance, that of the idea of the identity of mass and energy or, if you like, since mass is the essential characteristic of matter, the identity of matter and energy."* [2]. In 1924, in his Ph.D. thesis at the Faculty of Science at the University of Paris, de Broglie introduced the hypothesis that also material particles could have a wave-like nature [3].

Since the electron is matter but also energy, it must also behave as a wave. Thanks to Davisson and Germer's 1927 experiment (Chap. 4), this hypothesis would be confirmed shortly afterward. The next step, a logical consequence of the theoretical models and experiments, would be to theorize the dual wave-particle nature of elementary particles such as electrons.

De Broglie also posed the question of how electrons confined in discrete orbits did not emit energy, which, as we have seen, would condemn them to a very fast decay in a spiral orbit toward the nucleus.

Instead, De Broglie shows that every moving object with some momentum, p = mv (m mass and v velocity) according to classical mechanics, has an associated wavelength from which the dual wave-particle nature can be deduced:

$$\lambda = h/p \tag{3.1}$$

with h the Planck's constant and p the momentum.

This simple equation has very important conceptual implications. Once it was understood the dual nature of wave and particle light, as shown by the photoelectric effect, the next logical step was questioning whether this nature was also a property of matter. A particle such as an electron has mass, however small, and as demonstrated by several experiments that we will describe in detail in Chap. 4, it can also exhibit wave-like properties. The dual nature of wave and particle was thus beginning to be well supported by experimental evidence; however, understanding its physical meaning was still a long way.

De Broglie's equation connects a property of a wave, wavelength λ, with a property of mass, momentum, p = mv. All matter, in principle, has a wavelength, but according to de Broglie's equation, the larger the mass, the smaller the wavelength, so an object of large size, far larger than that of a particle such as an electron, a ball for instance, will have a negligible wavelength. In the case of the electron, on the other hand,

the mass is so small that its l becomes important. This consideration also makes it possible to establish the limit of applicability of the laws of quantum mechanics; size makes all the difference.

To proceed, let us define a wave in the physical sense. A wave can be defined as a disturbance in propagation that transports energy and information from one position to another without the transfer of matter. Classical waves can be classified as either mechanical or electromagnetic waves. The former require a medium to propagate, while the latter do not.

The next logical step made by de Broglie is a consequence of the assumption that electron orbits are stationary. One explanation for the stationarity of orbitals could be related to the possible wave-like property of electrons, such as that of photons. Therefore, de Broglie assumed that Bohr's stable orbitals correspond to orbitals that become standing waves around the nucleus, assuming a specific wavelength that would later be defined as *de Broglie wavelength.*

We can now briefly recall what a standing wave is; it is a wave whose oscillations do not propagate along a specific direction, so the wave oscillates in time but within a defined space without propagating. The simplest example is that of the strings of a guitar. When a string is plucked, stationary waves are produced because they do not travel along it but oscillate in amplitude with respect to fixed points. These particular string positions are called nodes and do not move at all. Therefore, the amplitude at the nodes of the standing waves will be zero; the nodes can be located at the extremes of the string or in between. The higher the vibration frequency of the string and the smaller the wavelength, the greater the number of nodes. However, there can only be certain wavelengths in each of the allowed vibrations of the string (Fig. 3.1).

De Broglie's idea is easier to understand if we consider three stationary waves (Fig. 3.2) made up of one, two, and three wavelengths in that order. These values are discrete integers with no intermediate values. We can cut out sections of the plotted sheet and create continuous circular strips. By arranging these strips around a center, we notice that each strip has a specific radius value, such as r_1, r_2, and r_3, which defines a set of stationary waves around the center of the three strips. We can imagine that the center of the three stripes represents the atomic nucleus, and the stripes themselves the stationary orbits of the electrons around the nucleus. The fact that the electrons exist around the nucleus as stationary waves explains why they do not lose energy. It's a simple and elegant solution.

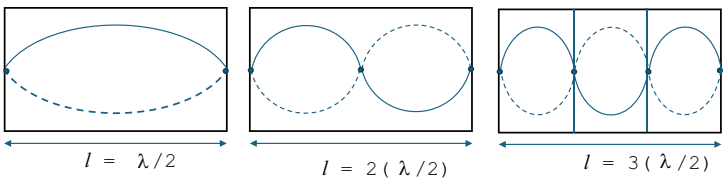


Fig. 3.1 Stationary waves are generated by plucking a guitar string. Each point represents a node. The length of the string, *l*, must equal a whole number multiplied by half the wavelength

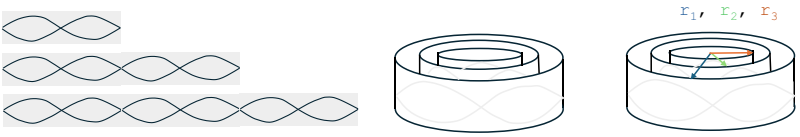


Fig. 3.2 Graphical representation of three stationary waves of one, two, and three wavelengths **a**. The three stripes are connected to form circular stripes **b**. The circular stripes have radii r_1, r_2, and r_3 with the center representing an ideal nucleus and stripes the electron stationary orbits

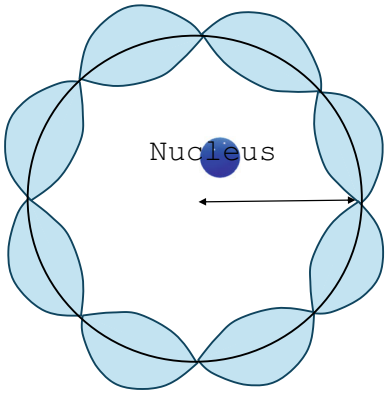


Fig. 3.3 Schematic representation of a stationary de Broglie orbit around the nucleus

Furthermore, if an electron behaves as a stationary wave, its wavelength must coincide perfectly with the orbit of the hydrogen atom (Fig. 3.3). If it did not, each successive orbit would gradually reduce its amplitude until it reached zero, and the wave would no longer exist.

Thus, for de Broglie, the electrons around the nucleus move as stationary waves, and the circumference of the orbits is quantized; that is, these orbits are at specific distances from the nucleus. De Broglie's model is based on two assumptions:

- The energy of the electron is not lost as electromagnetic radiation. The electron has a wave-like behavior.
- The electron remains in its specific orbits without spiraling into the nucleus.

According to de Broglie, electrons should be regarded as stationary waves around permitted orbits rather than particles. Taking Bohr's circular orbits as a reference, de Broglie's stationary waves develop precisely around these. For a stationary wave to form around the circular orbit, the wavelength of the electron must be an integer multiple of the wavelengths, that is, it can take the values: $\lambda = 1$, $\lambda = 2$, $\lambda = 3$….

Recalling that the circumference of the orbit of the electron in the Bohr atom is given by:

$$2\pi r = n\lambda \text{ con } n = 1, 2, 3, 4 \ldots \tag{3.2}$$

For the first orbitals, it is quite easy to visualize how electron stationary waves develop:

$$n = 1 \rightarrow 1\lambda \quad n = 2 \rightarrow 2\lambda \quad n = 3 \rightarrow 3\lambda$$

therefore because: $\lambda = h/mv$, we get:

$$2\pi r = \frac{nh}{mv} \rightarrow mvr = \frac{nh}{2\pi} \rightarrow L = \frac{nh}{2\pi} = n\hbar \tag{3.3}$$

with L the angular momentum of the orbit of the electron (Fig. 3.4).

At this point, we can associate the wave-like or particle-like nature of particles, such as electrons and photons, to their specific properties (Fig. 3.5).

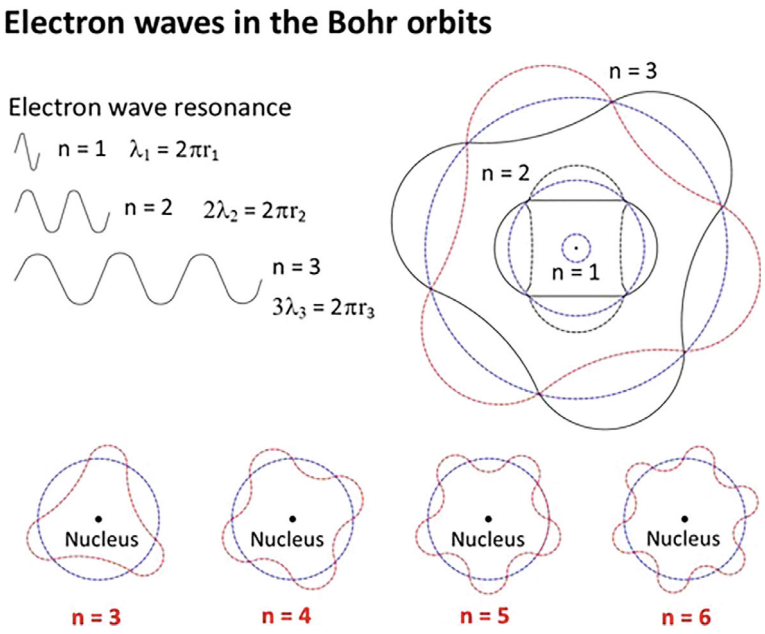


Fig. 3.4 Visualization of electron waves for the first three Bohr orbits. (Courtesy of Massimo Garzotti)

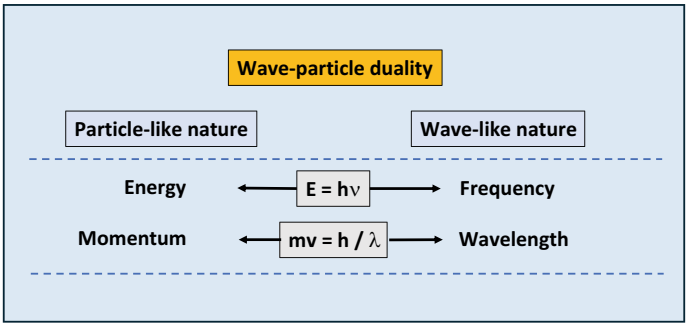


Fig. 3.5 Particle-wave duality is expressed by the equations that correlate energy with frequency and momentum with wavelength

Let us digress a little because we use many terms from classical physics that are then confused with those from quantum mechanics. One very important term is *momentum*, **p**, which in classical mechanics is simply the product of a body's mass, m, times its velocity, **v**:

$$\mathbf{p} = \mathrm{m}\mathbf{v}.$$

It is a vector quantity because the body is moving in a three-dimensional classical space, and therefore, the velocity will have a specific direction and above all, it is a conserved quantity. The concept of *momentum* is closely related to the development of Newtonian mechanics. Before Newton, it was believed that a body moved only if a force was continuously applied. It was Newton himself who introduced the concept of *momentum*, which in some ways was a revolutionary idea. A body, if no other forces intervene, maintains its state of motion or rest indefinitely, and the definition of *momentum* is exactly what it indicates, in fact, in the simple mathematical relationship, p = mv, force does not appear. Furthermore, *momentum* is a fundamental physical quantity because in isolated systems, on the whole, it is conserved (*Law of conservation of momentum).* In quantum mechanics, instead, position and momentum are conjugate variables and the duality creates an uncertainty relation that is expressed through the *Heisenberg Uncertainty Principle.* The momentum is expressed through an operator of the wave function:

$$\boldsymbol{p} = \frac{\hbar}{i}\nabla = -i\hbar\nabla$$

with ∇ the gradient operator, $\hbar$ the reduced Planck constant, and i the imaginary unit.

To denote *momentum* that is a Latin word (the plural is, therefore, *momenta*), the letter, **p**, is used. Where does the letter **p** come from? It is not very clear, probably from the Latin *impetus*, *im-* + *pellere*, with *pellere* meaning to push forcefully. Since the letter i is used for imaginary numbers and m for mass, perhaps p was most convenient.

3.2 The Mother of All Equations, Schrödinger's Equation

Beginning with the Bohr-Sommerfeld model and ending with the de Broglie hypothesis, the electron's elusive and complex nature was becoming apparent. However, despite the strides made, experimental evidence and a comprehensive theory were lacking. Only coupling experiments and theory would pave the way for a true quantum electron model. The next chapter will delve into these experiments.

The equation named after the Austrian physicist Erwin Schrödinger (1887–1961) represents one of the most important intellectual steps in the history of physics and a fundamental mathematical tool for the quantum description of atomic structure and the electron in the atom. The equation makes it possible to represent with a quantum wave function all particles, such as electrons or photons, which can be described by wave functions.

De Broglie's hypothesis about the wave nature of matter had fascinated Schrödinger, however, the French physicist's idea still lacked a mathematical description to support it; a wave without a wave equation was difficult to justify. In any case, finding an equation that could describe how a particle that behaves like a wave evolves in time and space was not an easy problem to solve.

In classical physics, each wave can be described by a corresponding wave function. Still, de Broglie had not even tried to find this function, leaving the door open to criticism from physicists such as Pieter Debye (1884–1966), who were skeptical of his theory. Debye himself invited Erwin Schrödinger in 1925 to give a seminar on de Broglie's theory. At the end of Schrödinger brilliant presentation, Debye objected that he lacked an equation describing the wave behavior of matter. Schrödinger accepted the challenge, and after several months of intensive work, he published a series of articles in the *Annalen der Physik*, *"Quantisierung als Eigenwertproblem"* (Quantization as a problem at eigenvalues), from March 1926, in which he presented what is now known as the Schrödinger equation.

There is no rigorous derivation of Schrödinger's equation; the starting point is to consider the possible connections between waves and photons and construct a similar relationship for waves and electrons. When Richard Feynman was asked how Schrödinger derived his equation, his answer was: "*Where did we get that [Schrödinger's equation] from? It's not possible to derive it from anything you know. It came out of the mind of Schrödinger.*" [4].

In practice, to construct the Schrödinger equation, it is necessary to start from postulates, which are precisely those of de Broglie-Einstein. In the case of a free particle, moving along a direction x, the wave function that accounts for the wave behavior of this particle is described as proposed by Schrödinger by such an Eq. (3.4):

$$\boxed{-\frac{\hbar^2}{2\mathrm{m}}\frac{\partial^2\psi(x,t)}{\partial x^2}+V(x,t)\psi(x,t)=i\hbar\frac{\partial\psi(x,t)}{\partial t},} \tag{3.4}$$

where $\hbar$ is the reduced Planck, m the mass of the particle, x the distance, t the time, ψ the wavefunction, V the potential, and i the imaginary unit.

The equation can obviously be generalized from two-dimensional to three-dimensional space for an electron or a generic particle. The equation shown is the *time-dependent* and non-relativistic equation. It is rather complicated, and in fact, it is since it is a second-order partial derivative differential equation and has a solution only if initial and boundary conditions are defined. The fact that it is a differential equation implies that the quantity being sought is not a variable but a function. This function is exactly the wave function that is identified with the Greek letter ψ, and in the equation, there are the derivatives of this function. The partial derivative indicates that the function contains several multiple variables, such as position x and time t, in the one-dimensional version for space; in three-dimensional space, we would have x, y, z, and t. In the left-hand term of the equation, the partial derivative is of the second order of the function with respect to space (in the one-dimensional version, only x), and in the right-hand term of the first order.

The equation derived by Schrödinger can be considered the quantum equivalent of the second principle of Newton dynamics, $\mathbf{F}=\mathrm{m}\mathbf{a}$, of "classical" physics. Newton's equation states that there is a linear relationship between the force applied, $\mathbf{F}$, to a body of mass m, and the resulting acceleration $\mathbf{a}$. If the applied force increases in intensity so will the acceleration; moreover, force and acceleration are vector quantities in three-dimensional space.

In the case of the Schrödinger equation, it is through the three-dimensional wave function (ψ) that the interactions governing the state of the system are correlated. Although not obvious at first glance, the equation describes the motion of the particle by taking into account its kinetic energy (the first term on the left) and the potential energy (the second term on the left) acting on it. The difference from a classical approach is that the motion of the particle is described by a wave function, ψ. The first two terms can be expressed mathematically by a mathematical operator called a Hamiltonian that describes the total energy of the particle. Another important property of the Schrödinger equation is linearity. This choice also makes it possible to describe interference between wave functions.

Once an equation describing the motion of a particle as a wave function has been constructed and works well from a mathematical point of view, a spontaneous question arises: what does it mean from a physical point of view? The answer is not unambiguous, and the difficulty in giving a convincing explanation is the basis of the different interpretations of quantum mechanics. Initially, Erwin Schrödinger

started with the idea, based on de Broglie's model, that the electron in the hydrogen atom was a wave with its own physical nature. As we have seen, the wave nature of the electron for de Broglie was represented through circular stationary waves; Schrödinger extended this idea by using three-dimensional standing waves, namely wave functions.

Is the Schrödinger equation just a mathematical construction that works well for calculations, the famous "*shut up and calculate*" approach, or is there some physical reality behind this formula that only mathematics can fully represent? There are many interpretations of the equation, which we will describe in Chap. 6 when we will discuss the two-slit problem. However, we can note that the Schrödinger equation contains no probabilistic elements; before the observation, the wave function changes only deterministically. It was Max Born (1882–1970) who suggested interpreting the wave function as a probability amplitude, where the square of the wave function, $|\psi|^2$, describes the probability that an electron can be found at a certain position x at an instant t. The electron is thus still a particle, and the wave functions representing it are not physical waves but simply probability amplitudes.

$$P(x,t)dx = |\psi(x,t)|^2 dx. \tag{3.5}$$

Atomic orbitals describe the areas where electrons are most likely to be found. The Schrödinger equation, therefore, calculates the wave function deterministically, but the information it carries is defined only in probabilistic terms. This is not an easy step to accept because it conceptually breaks the classical approach in physics that univocally associates a spatial position with an object as a function of its mass and the forces acting on it. As it is well known, Einstein himself, in this regard, was extremely skeptical, and in a letter he wrote to Max Born on December 4, 1926 he wrote about his doubts: "*Quantum mechanics is certainly imposing. But an inner voice tells me that it is not yet the real thing. The theory says a lot, but does not really bring us any closer to the secret of the 'Old One'. I, at any rate, am convinced that He is not playing at dice.*" [5].

So, if a sound wave is like a mechanical wave and a light wave can be described in terms of electromagnetic field oscillations, the electron is a charge density (or a particle?) whose position and shape are definable only in probabilistic terms. This is not a small step forward.

The equation, though extraordinarily effective, contains elements of incompleteness that undermine the foundations of quantum mechanics itself. Indeed, the equation allows one to know the evolution of the state of a system if one knows the current state, that is, the initial conditions necessary for the solution of the equation itself. The problem, however, is that the measurement process does not follow the Schrödinger equation, giving rise to one of the biggest open questions in quantum theory: the *Measurement Problem.* The equation, in fact, describes the state of the system in probabilistic terms, but the measurement determines the state non-probabilistically.

According to British physicist Roger Penrose: "*That's not the way the world operates when you make a measurement on the system, it does not follow the Schrödinger equation.*..." [6].

This problem seems to undermine the whole theory of quantum mechanics and has given rise to several schools of interpretation; Schrödinger himself had caught the contradiction he pointed out with his famous thought experiment to be known as *Schrödinger's Cat* (1935) [7]. However, regardless of the measurement problem and different interpretations, the equation works remarkably well and makes it possible to explain atomic structure even in many-electron atoms beyond the hydrogen atom. The tremendous scientific development since the beginning of the twentieth century is due precisely to quantum mechanics and the fact that despite everything, calculations manage to describe physical phenomena at the nanoscale and atomic scale accurately, *primum vivere deinde philosophari* (first live, then do philosophy).

3.3 A Question of Principles: The Uncertainty Principle

The consequences of quantum mechanics and the profound paradigm shift from classical mechanics are inherent in the Schrödinger equation. Knowing an electron's exact position and momentum is impossible, regardless of what an observer initially knows about the state of the particle. Position and momentum can only be known in probabilistic terms. At no instant will it be possible to know both variables precisely; the information gained about one will be at the expense of knowing the state of the other variable. This statement is nothing less than *Heisenberg's Indeterminacy Principle* (*The Uncertainty Principle*), another of the pillars on which quantum mechanics rests [8].

Mathematically, the *Uncertainty Principle* is expressed through a simple and very familiar relationship known from high school:

$$\Delta x \Delta \mathrm{p} \geq \frac{\hbar}{2}. \tag{3.6}$$

The Greek capital delta symbol, Δ, indicates precisely the indeterminacy of the measure of position, x, and momentum p (i.e., the product of mass m by velocity v). Thus, the equation indicates that the product of the indeterminacy of position with the indeterminacy of momentum must be greater than or at most equal to some constant, $\hbar$ (h/2π), Planck's constant divided by two. As previously anticipated, we will go into more detail on the meaning of this constant in one of the next chapters; for now, it is enough to note that it has an extremely small value, $h = 6.62607015 \times 10^{-34}$ J second. The value of h indicates that the uncertainty principle applies well to objects whose dimensions have an atomic or subatomic scale. This observation is very important because it suggests that the principle is closely related to the dual, wave-particle nature that can be observed in systems at that scale.

Heisenberg's Indeterminacy Principle applies not only to *position-momentum* but also to the *time-energy* binomial. Measuring the quantum state of a particle in a very short interval of time carries uncertainty about its energy state. On the other hand, a precise measurement of energy requires a longer time, which implies the impossibility of identifying the exact moment when the energy measurement was made. In this case, the relation becomes:

$$\Delta t \Delta E \geq \frac{\hbar}{2}. \tag{3.7}$$

Schrödinger's equation indicates that the probability of finding a particle at a certain position is given by the wave function or rather its square, as suggested by Born. The wave function is expressed using a sine function that has a specific wavelength. This wavelength is inversely proportional to the momentum of the particle ($p = h/\lambda$). To obtain more information about the location of the particle, it is necessary to use several wave functions of different wavelengths, which are added together. As more and more wavelengths are added together, the probability of locating the particle increases, but at the same time momentum information is lost, which is not now associated with a single wavelength. Therefore, as the probability of locating the particle increases, given by the summation of the wave functions, the particle will be more and more easily located at a certain position but its velocity, or momentum, will be less and less known.

According to Heisenberg, if we know the momentum, we do not know the position, so the sinusoidal wave function associated with the wave nature of the particle remains undefined and stretches infinitely in space in all directions. If we have a single wavelength, which we know from de Broglie's relation, is related to momentum, we have no information about the position of the wave function. To get more information on both position and momentum, it is necessary to transform the infinite sine function into something more localized. To localize the wave function, it is necessary to add up as many wavelengths as possible.

Once accepted, the dual particle-wave nature results in the end of determinism and the realization of the limitations of classical physics, a process well known in scientific thought. The progress of theory generally never erases the validity of a previous model but, if anything, identifies its limitations. The relativistic theory does not cancel the validity of Newtonian mechanics but limits its application to bodies moving at speeds much slower than the speed of light. The enunciation of the *Uncertainty Principle* heralds the beginning of the quantum age and the age of probability.

3.3.1 Heisenberg's Microscope

The *Uncertainty Principle* raises a number of physical and philosophical questions that are still being debated. One of the questions Heisenberg tried to answer was

whether the relation was just a mathematical relation resulting from calculations or expressed an intrinsic physical reality. Heisenberg was convinced that every definition in physics should have an experimental basis. At the beginning of the article in which Heisenberg introduced the Uncertainty Principle, he wrote: "*We believe we understand the physical content of a theory when we can see its qualitative experimental consequences in all simple cases and when at the same time we have checked that the application of the theory never contains inner contradictions.*" [8]. Immediately afterward, he introduced a famous thought (*gedanken*) experiment to support his claim, now known as the *Heisenberg Microscope.*

The origin of this idea goes back to the period of undergraduate studies and has curious background. Heisenberg was an extremely bright physics student who had attracted the attention of Arnold Sommerfeld during his doctoral studies at Ludwig Maximilian University in Münich. Heisenberg's interest was mainly in theoretical physics, but during his doctoral examination in 1923 he was nevertheless to take part of the exam on topics in experimental physics, the so-called *Prakticum*. His examiner was the physicist Wilhem Weiss (from whom Wiess law, which we will introduce in the next chapter, comes). It was not a particularly complex examination, but Heisenberg failed to answer even simple questions on microscopy, much to Weiss's disappointment. It was thanks to Sommerfeld's good offices that Heisenberg was able to pass the examination, albeit with a low mark. However, Heisenberg's poor performance spurred him on to deepen his knowledge of what he lacked, in particular how a microscope worked. The study so excited him that he wondered whether there was a theoretical limit to the minimum size that could be observed. This reflection led him to formulate a thought experiment, known as the *Heisenberg Microscope*, whose conclusions, although based on assumptions that later turned out not to be entirely correct, led to the formulation of the uncertainty principle. Heisenberg's reasoning was based on the assumption that the photon is a particle, and therefore, when it hits an electron, a particle-to-particle collision with momentum transfer should occur [9].

The smallest distance that can be resolved with a microscope, Δx, according to the laws of optics, depends on two different parameters: the wavelength of the light used, λ, and the aperture angle of the microscope, θ, that is, the size of the lens:

$$\Delta x = \frac{\lambda}{2 \sin \theta}. \tag{3.8}$$

Therefore, Δx will be directly proportional to the wavelength, so the smaller λ is, the smaller the distance that can be resolved. In his thought experiment, Heisenberg, therefore, took as a reference the γ-rays, which have a very small wavelength.

Δx, on the other hand, is inversely proportional to the sine of the aperture, $\sin\theta$, so reducing it would decrease the resolution, as a consequence of the diffraction of light. When the photon-particle collides with the electron-particle it will transfer some momentum to the electron (Fig. 3.6).

After colliding with the electron, which is supposed to be at the center of the lens, for the photon to be visible it must be scattered by the electron within the cone of

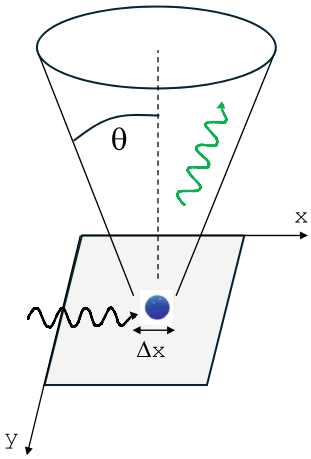


Fig. 3.6 Schematic representation of the Heisenberg Microscope. A photon moving along the x-direction with respect to the electron is scattered toward the microscope lens at an angle 2θ

amplitude 2θ. The opening angle of the microscope lens introduces an indeterminacy about the direction of origin of the photon. The act of measurement involves a particle–particle interaction, during which the photon will transfer to the electron some momentum proportional to the photon's energy.

Assuming for simplicity that the photon impacts the electron along the x-axis with momentum p, the photon that is elastically scattered toward the lens acquires a transverse momentum component between $+\,\mathrm{p}\sin\theta$ and $-\mathrm{p}\sin\theta$, where p is the momentum after scattering. Consider the two extreme cases, in which the photon will be scattered toward either the left or the right end of the cone identified by the lens. In this case, the total momentum of the electron-photon system after collision will be given by (Fig. 3.1):

$$\mathrm{p}'_{\mathrm{ex}} + \mathrm{p}\sin\theta\,(\text{right}) \quad \mathrm{p}''_{\mathrm{ex}} - \mathrm{p}\sin\theta\,(\text{left})$$

with $\mathrm{p}'_{\mathrm{ex}}$ and $\mathrm{p}''_{\mathrm{ex}}$ the momentum of the electron after the collision in the two edges of the cone. By the principle of conservation of momentum, the final momentum along x must be equal to the initial momentum always along x:

So the indeterminacy of the momentum of the photon will be given by:

$$\left(\mathrm{p}'_{\mathrm{ex}} + \mathrm{p}\sin\theta\right) = \left(\mathrm{p}''_{\mathrm{ex}} - \mathrm{p}\sin\theta\right) \rightarrow \mathrm{p}''_{\mathrm{ex}} - \mathrm{p}'_{\mathrm{ex}} = \triangle \mathrm{p}_x = 2\mathrm{p}\ \sin\theta. \tag{3.8}$$

If the measurement with the photon g causes the electron to move, then the certainty of knowing its position will also decrease. The act of measurement involves a particle–particle interaction, during which the photon will transfer momentum p, proportional to the photon's energy, to the electron, so if the particles also possess wave nature, $\lambda = h/\mathrm{p}$, so:

$$\Delta \mathrm{p} = 2\frac{h}{\lambda}\sin\theta. \tag{3.9}$$

At this point multiplying the two indeterminacies yields precisely Heisenberg's relation. In fact, regardless of the wavelength of the photon and the size of the aperture, the product of the uncertainty in the measured position and the measured momentum is greater than or equal to a lower limit, which is (up to a small numerical factor) equal to Planck's constant:

$$\Delta x \Delta \mathrm{p} = \frac{\lambda}{2 \sin \theta} \frac{2h \sin \theta}{2\lambda} = h. \tag{3.10}$$

For the German physicist, the thought experiment plays an important role in the construction of quantum mechanics to bring it out of conceptual abstraction and back into a physical reality: "*…as we can think through qualitatively the experimental consequences of the theory in all simple cases, we will no longer have to look at quantum mechanics as unphysical and abstract*."

The Heisenberg microscope thought experiment provides a sound heuristic interpretation for understanding the uncertainty principle, but it is a classical analogy with significant shortcomings. The errors in the thought experiment arise from the oversimplification of quantum interactions, the neglect of wave-particle duality, and a classical interpretation of measurement that doesn't fully align with the principles of quantum mechanics. Despite these issues, the thought experiment remains a valuable tool for conceptualizing the fundamental limits imposed by quantum theory, even if it doesn't perfectly represent the underlying physics.

3.4 A Question of Principles: The Pauli Exclusion Principle

Bohr's atom model, with the introduction of quantized orbitals, left numerous open questions; among them was the question of why all the electrons of an atom in its fundamental state were not all in the innermost shell. Bohr postulated that this should not happen, but there was no clear explanation as to why: "*The question as to why all electrons for an atom in its ground state were not bound in the innermost shell had already been emphasized by Bohr as a fundamental problem in his earlier works…. However, no convincing explanation of this phenomenon could be given on the basis of classical mechanics. It made a strong impression on me that Bohr at that time and in later discussions was looking for a general explanation.*"

We also recall that in Lewis-Langmuir's cubic model, the fundamental number was 8, and many physicists, including Sommerfeld, were convinced that the answer lay in this number. Pauli, however, tells us how, instead, Bohr followed another idea: "*It made a strong impression on me that Bohr at that time and in later discussions was looking for a general explanation which should hold for the closing of every electron shell and in which the number 2 was considered to be as essential as 8 in contrast to Sommerfeld's approach.*" [10]. The sequence of even numbers 8, 6, 4, and 2 seemed to be the key to understanding the distribution of electrons in the different atomic levels.

Another unanswered problem was the anomalous Zeeman effect, which is closely related to the introduction of the concept of spin. This problem also represented quite a puzzle for Pauli. Until 1924, three quantum numbers were used to describe the orbits of an electron in the atom; Pauli came to the conclusion that it was necessary to add a fourth. In January 1925, he published an article in German in which he proposed the introduction of a fourth quantum number to define a specific property of the electron [11], which for the time being he defined as: "*two-valuedness not describable classically*," which even now is probably the best definition of spin possible. Pauli's definition of the principle is as follows: "*In an atom there cannot be two or more equivalent electrons, for which in strong fields the values of all four quantum numbers coincide. If an electron exists in an atom for which all of these numbers have definite values, then this state is* '*occupied*'."

This idea could explain why the electrons in a multiple-electron atom do not all collapse into the lowest-energy orbital; at most, the orbital could accommodate two. Pauli had, therefore, introduced an exclusion principle that in an atom, there cannot be two electrons with all four quantum numbers in common. Although there was no explanation to derive the principle from the laws of quantum mechanics, as Pauli himself admitted, it actually explained well the properties of the electron in the atom: "*Already in my initial paper, I especially emphasized the fact that I could not find a logical substantiation for the exclusion principle nor derive it from more general assumptions. I always had a feeling, which remains until this day, that this is the fault of some flaw in the theory."* In the same year two physicists Samuel Goudsmit and George Uhlenbeck proposed the idea of spin, as we will discuss in more detail in Chap. 6. The association between spin and Pauli Exclusion Principle was an inevitable consequence. The principle has general validity and applies not only to electrons but to all particles with half-integer spin, the fermions; it to does not work, instead, for bosons that have an integer spin. By the Exclusion Principle bosons can share the same quantum state while fermions cannot.

The explanation of Pauli Exclusion Principle commonly shown is connected with the concept of symmetry and antisymmetry: "*Among the different classes of symmetry, the most important ones (which moreover for two particles are the only ones) are the symmetrical class, in which the wave function does not change its value when the space and spin coordinates of two particles are permuted, and the antisymmetrical class, in which for such a permutation the wave function changes its sign*…."

Consider two generic particles (1) and (2) that are in the a and b states, respectively. The wave function ψ, of the two-particle system will be given by:

$$\psi = \psi_1(a)\ \psi_2(b),$$

where ψ corresponds to the probability that particle 1 is in state a, $\psi_1(a)$, and that particle 2 is in state b, $\psi_2(b)$. If, however, the two particles are identical and indistinguishable, as in the case of two electrons, then a linear combination of the two wave functions describing the two states must be used, $\psi_1(a)$ *and* $\psi_1(b)$. The states, however, cannot be identified precisely because the particles are identical:

$$\psi = \psi_1(a)\ \psi_2(b) \pm \psi_1(b)\ \psi_2(a)$$

with ψ that represents the possibility that states a and b are occupied by the identical particle (electron). Hence:

$$\psi = \psi_1(a)\ \psi_2(b) + \psi_1(b)\ \psi_2(a) \quad \text{Symmetry} \rightarrow \text{bosons}$$
$$\psi = \psi_1(a)\ \psi_2(b) - \psi_1(b)\ \psi_2(a) \quad \text{Antisymmetry} \rightarrow \text{fermions}$$

The minus sign implies that the wave function goes to zero if the states e and b are identical, if $a = b \rightarrow \psi = 0$. Thus two identical particles cannot occupy the same state.

Pauli Exclusion Principle opens up profound implications of a philosophical and physical nature. As we saw in the introduction, in classical terms, electrons are identifiable and different only in terms of *extrinsic properties*, such as position and momentum. From the quantum point of view, all electrons, taken individually, are absolutely identical and indistinguishable in terms of *intrinsic properties*. Their individuality arises only through the relationship of the electrons to the system they are in, such as their differing quantum states. According to the Pauli Exclusion Principle, an electron within an atom takes on "unique" and identifiable properties, given that no other electrons can take on those specific values given by the quantum numbers that identify the orbital and energy.

This raises questions about what it means for physical objects, such as elementary particles, to be "individual" entities. Are the particles inherently unique, or is their distinctiveness context-dependent? The principle suggests that each particle expresses unique properties through spatial or energetic differentiation in a universe governed by the same physical laws. This implies that individuality may not be intrinsic but rather emergent from the relationship between properties, challenging our preconceptions about the nature of particles.

The Pauli Effect

Scientists, particularly physicists, have a widespread reputation, whether real or undeserved, for being eccentric and paradoxically very superstitious. Wolfgang Pauli was much feared by his colleagues for the caustic comments of which he was capable. One fierce expression with which he demolished the work of others that has gone down in history is "not even wrong." Apart from his difficult character and extraordinary scientific contribution, Pauli also is remembered for an effect named after him: the *Pauli effect*.

Pauli's presence in the laboratory often produced seemingly mysterious or mischievous accidents in which technical equipment break down or experiments would malfunction. The effect has become part of the folklore in the physics community and is often mentioned humorously when something goes wrong with laboratory instrumentation or during experiments without a clear

explanation. Although Pauli was a theoretical physicist and rarely conducted experiments himself, it became a running joke among his colleagues that experimental setups would fail when he was nearby. One famous story involving the *Pauli Effect* took place when an expensive instrumentation in a laboratory in Göttingen unexpectedly broke down during Pauli's visit. Pauli himself was not directly involved in the experiment, yet the equipment malfunctioned for no apparent reason, but incidents like this fed into the belief that Pauli had a "psychic" or jinxing influence over machinery, even though there was no evidence for such an effect.

One of the most famous instances attributed to the *Pauli Effect* occurred at a laboratory in Stuttgart in the 1930s. While Pauli was traveling by train, an experimental apparatus exploded in a nearby laboratory, despite the fact that Pauli was not physically present. The incident was considered so remarkable that it contributed to the mystique of the Pauli Effect.

3.5 A Question of Principles: The Correspondence Principle

The breakdown of the paradigms of classical physics introduced by quantum mechanics has, since the introduction of Bohr's model, raised the question of whether a correspondence existed between the two worlds. To recompose this rupture, Bohr introduced the *Principle of Correspondence* in 1920 [12]. It states that the properties of quantum systems must resemble classical physics in the limit of large quantum numbers. Quantum mechanics, which works for systems like atoms and electrons, should converge to classical mechanics, which describes macroscopic systems, when applied to large systems or high energies. The principle bridges the gap between classic and quantum physics, ensuring that quantum predictions align with classical results for large systems. As quantum numbers, which characterize the quantum states of a system, increase, the properties of a quantum system increasingly approximate that of classical systems. Bohr applied this principle in his model of the atom to explain why classical physics was successful at explaining large-scale phenomena while failing at atomic and subatomic scales. In the context of atomic systems, for very high energy levels, i.e., large quantum numbers, the difference between successive energy levels becomes so small that the quantum system behaves almost indistinguishably from a classical one. For instance, the orbits of electrons in an atom at very high energy levels will resemble the continuous orbits described in classical mechanics, as opposed to the quantized orbits in quantum mechanics. Wolfgang Pauli in his Nobel Lecture observed that the principle stated by Bohr: "*One was an effort to bring abstract order to the new ideas by looking for a key to translate classical mechanics and electrodynamics into quantum language which would form*

a logical generalization of these. This was the direction which was taken by Bohr's correspondence principle."

3.6 The Standard Model

After identifying the atomic structure and defining the equations that describe the properties of electrons within atoms, a period of extraordinary discoveries and the development of the fundamental theoretical model of quantum mechanics came almost to an end. The adventure, however, did not close with the discovery of electrons, protons, and neutrons. Thanks to the development of much more sophisticated detection instruments, much experimental evidence pointed to the existence of even smaller elementary particles.

The atomic models we have described so far are based, as we have seen, on three subatomic particles: the negatively charged electrons, the positively charged protons, and the neutrons that have no charge. Of these, only one particle is now considered fundamental, i.e., it is not made up of other even smaller units; this is the electron. The electron, however, is not alone. It is part of a family of particles sharing a negative charge, known as *leptons*. On the other hand, the protons and neutrons are not fundamental particles and are made up of other smaller particles, the famous *quarks*. The use of particle accelerators since the 1950s has made it possible to detect many more subatomic particles, leading to a rapid evolution of our knowledge about the atom. It became clear that it is the combination of three quarks that forms the proton and neutron. The quarks are the constituent units of the nucleus particles, protons and electrons, and of hadrons in general; six of them have been identified and have rather strange names: *up, down, charm, strange, top,* and *bottom.*

Of these six *quarks*, two in particular, the *up* and *down quarks*, are found within the proton and neutron, while the others are used to explain the formation of other particles that have been discovered using accelerator physics. Between the 1950s and 1970s, a paradoxical circumstance arose as new particles continued to be discovered thanks to the availability of ever more powerful accelerators, so much so that they were colloquially referred to as the *particle zoo.* The situation was so confusing that the particle physicists despaired of being able to bring order to such a complex system. In 1955, the physicist Willis E. Lamb (1913–2008), Nobel Prize winner for his discoveries about the fine structure of the hydrogen spectrum, even went so far as to propose during his Nobel Lecture a punishment for those who discovered new particles: "*A deluge of other elementary particles appeared after 1930: neutron, neutrino, p meson, m meson, heavier mesons and various hyperons. I have heard it said that the finder of a new elementary particle used to be rewarded by a noble prize, but such discovery now ought to be punished by a $10,000 fine.*" [14].

One of the most intriguing properties of these zoo particles was their mass, which was much greater than that of the electron, so they came to be known as *hadrons*, the Greek root of which indicates that they are heavy. The attempt to rationalize and define in a complete model how these were formed was the basis of the theory of

quarks and the *Standard Model*, which comprised only three groups of elementary particles: *leptons, bosons,* and *quarks.*

The negatively charged particles, the *electron*, the *muon*, and the *tau*, play a significant role in the *lepton* family. In addition to these, there are three corresponding *neutrinos* that are electrically neutral: the *electron neutrino*, the *muon neutrino*, and the *tau neutrino*. All leptons are fermions and are not affected by strong nuclear interactions.

The *up* and *down quarks* together with the electron and electron neutrino form the first column of the table summarizing the *Standard Model* and are referred to as first-generation fermions (Fig. 3.7). This generation of particles is the most common in the universe and the combination of *up* and *down quarks* forms protons and neutrons. The electron neutrinos are the most common particles in the cosmos; every second, our bodies are traversed by one trillion neutrinos, but fortunately, they do not interact with matter, which is good for us but makes them extremely elusive and difficult to detect.

There are three generations of particles in the *Standard Model*; each is associated with a pair of *quarks* and a pair of *leptons*. In the second generation, the particles are larger than in the first, but the muons are much less stable than the electrons. Finally,

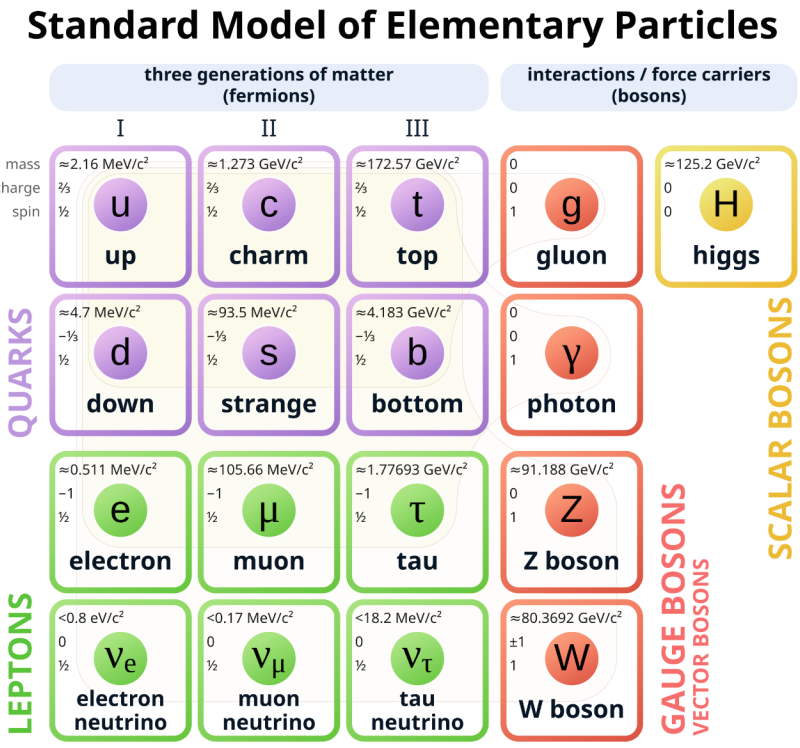


Fig. 3.7 Standard model of elementary particles [13]

the third generation consists of the *tau* particle with its corresponding neutrino, the tau neutrino, and the *top* and *bottom quark* pair. The *tau lepton* and the third-generation *quarks* are the heaviest but also the rarest in the universe, which is made up almost exclusively of the first-generation particles. The three generations, therefore, differ substantially only in their masses, which are increasing.

Quarks and *leptons* can, therefore, describe virtually all the particles we know; *quarks up*, *quarks down*, and *electrons* are the building blocks of matter in the cosmos. The twelve *leptons* are all fermions, i.e., they have no integer spin, unlike bosons. It must also be remembered that, like *leptons*, there are *antileptons*, so that next to the table of matter particles in the *Standard Model*, there is a similar one for antimatter. The dimension of the elementary particles in the *Standard Model* is assumed to be point-like, with no internal structure or measurable size. They are modeled as having no spatial dimension, existing only as a single point in space. Although they are described as point-like, the usual concepts of "size" or "dimension" do not apply in this context. Unlike composite particles such as protons or neutrons, which have measurable sizes due to their internal structures, elementary particles have no substructure, and no physical size can be assigned to them. Experiments in high-energy particle physics have measured tight upper limits on the size of elementary particles. For example, the electron has been probed on scales smaller than 10^{-18} m without showing any internal structure, indicating that if it has any size, it must be smaller than this scale. The same is true for quarks and the other elementary particles; their effective "size" is smaller than any currently measurable distance. According to *Quantum Field Theory* (QFT), particles are excitations of underlying fields. These excitations are point-like, and interactions are described at specific points in space–time without the need to attribute any physical size to the particles.

One important question still remains to be defined: there is a strong short-range "nuclear" force inside the nucleus that holds protons and neutrons together. It must be strong enough to overcome the repulsion of particles such as protons that have the same charge. We know that in the case of electromagnetic forces, there are charge mediators, which are the photons that carry information like messengers. At the atomic level, it works in a similar way, and there must be particles that exchange information. So just as there are photons for the electromagnetic force, there are particles for the strong nuclear force called *gluons*, and just as many particles for the *weak nuclear force*, the W^+, W^-, and Z^0 *bosons*. The latter are the strangest of all because the other bosons we know, *photons* and *gluons,* have no mass and no charge, while these particular bosons instead have both mass and charge (Z^0 has mass but no charge).

So, at this point, we have described six fundamental particles, the *leptons*, of which three have a negative electric charge, and three are neutral, to these must be added six *quarks* and four *bosons*, the charge mediators of the three known forces with the exception of the gravitational force. The collection of these particles represents the so-called *Standard Model*, which can be regarded as the analogue of the periodic table for elementary particles. The fundamental particles of the *Standard Model*, as the building blocks of matter, demonstrate the simplicity and yet the incredible complexity from which the entire universe can, in fact, be constructed.

Another boson appears in the *Standard Model* that stands quite alone. This boson is the last one to have been discovered in 2012, and it is the *Higgs boson*. It is the only scalar boson, and it plays a key role in closing the *Standard Model* for the state of current knowledge. The *Higgs boson* allows other particles to have mass, which is why its discovery was greeted with such great excitement. In order for the *Higgs boson* to give mass to other particles, it would have to permeate the entire universe somehow, forming a sea of virtual *Higgs bosons*, the *Higgs field*. While massless photons and gluons ignore the *Higgs field*, quarks, electrons, and other particles interact with it, which is why these particles have mass. Massless particles, such as photons and gluons, do not interact with the *Higgs field* and are therefore massless. Only particles that interact with the *Higgs field*, such as quarks and electrons, acquire mass. In this way, the *Higgs boson* is responsible for the existence of mass in the universe.

Although the *Standard Model* has a name that is anything but evocative, it is probably one of the most comprehensive theories in physics. It succeeds in coherently ordering every particle in the universe even though there is still a missing stone in this system: the gravitational force. There is likely another boson, the *graviton*, and this should open a new window toward a unified theory of forces, but so far, there is no trace of the *graviton*, which is not to say that it does not exist, but we have not yet been able to find it.

3.7 The Electron Would Be Never the Same Simple Charged Particle

The *old quantum mechanics* had fundamentally changed the perception of the electron. In a very short time, it had gone from wondering whether the electron was a particle of the ether to realizing that it was an essential element of the atomic structure, responsible for fundamental physical and chemical phenomena. Bohr's introduction of the concept of quantization undoubtedly opened up a whole new way of looking at the electron within the atom. In this first phase, following J. J. Thomson's discovery of the electron and the study of its properties, the focus was on the role of the particle within the atomic structure, which suddenly became much more complex than had been assumed. De Broglie's next step was to extend the wave nature of the photon to the electron. In this way, the electron takes on a dual wave-particle nature, which will form the interpretative basis of quantum mechanics. From a theoretical point of view, this transition would not be easy because the deterministic laws of classical physics can no longer provide a reassuring framework for representing physical phenomena. The intrinsic indeterminacy introduced by Heisenberg becomes an insurmountable limit to knowledge. How to interpret the non-causality and indeterminacy introduced by quantum mechanics became the subject of an important scientific and philosophical debate that is still ongoing.

3.8 A Brief Summary of the Development of Quantum Mechanics of Electrons

Introduction of the Concept of Wave-Particle Dualism (1924).

Louis-Victor Pierre Raymond de Broglie (Dieppe, 15 August 1892–Louveciennes, 19 March 1987).

Recherches sur la Théorie des Quanta (*Researches on the Theory of Quanta*). [1].

Pauli Exclusion Principle (1925). Pauli states that no electrons can have the same four quantum numbers.

Wolfgang Pauli (Wien, 25 April 1900–Zurich, 15 December 1958).

Über den Zusammenhang des Abschlusses der Elektronengruppen im Atom mit der Komplexstruktur der Spektren (on the connection between the configuration of the electron groups in the atom and the complex structure of the spectra). 1925. [11].

Introduction of the Wave Function (1926). The motion of electrons is described by the equation of Schrödinger.

Erwin Rudolf Josef Alexander Schrödinger Wien, 12 August 1887–Wien, 4 January 1961).

Quantisierung als Eigenwertproblem (Quantization as an Eigenvalue Problem). 1926 [15].

Probabilistic Foundation of Quantum Mechanics (1926). The square of the wave function, $|\Psi|^2$, should be interpreted as the probability density for finding an electron in a particular region of space.

Max Born (Breslau, 11 December 1882–Göttingen, 5 January 1970).

Zur Quantenmechanik. II. 1926. [16].

Uncertainty Principle (1927). The position and momentum of a particle cannot be known exactly at the same time.

Werner Karl Heisenberg (Würzburg, 5 December 1901–München, 1 February 1976).

Über den anschaulichen Inhalt der quantentheoretischen Kinematik und Mechanik (*The Physical Content of Quantum Kinematics and Mechanics*). 1927. [8].

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primordiaux. […] D'autre part, la théorie de la relativité nous a apporté une notion d'une extrême importance celle de l'idée l'identité de la masse et de l'énergie ou, si l'on veut, puisque la masse est la caractéristique essentielle de la matière l'identité de la matière et de l'énergie
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Chapter 4
Knowledge Is the Son of Experience. From Theory to Experiments



(*Leonardo da Vinci*)

Schrödinger's equation and the quantum model of the atom are the results, as we have seen in the previous chapter, of the intellectual contributions of various scientists who, as the knowledge of the atomic structure and its constituent elements progressed, sought to develop an accomplished theoretical model well supported by experimental evidence. Theory and experiments have gone hand in hand as necessary elements. Robert A. Millikan well highlighted this close connection in his 1923 Nobel lecture: "*…Science walks forward on two feet, namely theory and experiment…Sometimes it is one foot which is put forward first, sometimes the other, but continuous progress is only made by the use of both—by theorizing and then testing, or by finding new relations in the process of experimenting and then bringing the theoretical foot up and pushing it on beyond, and so on in unending alternations.*" [1].

It has been a path in which theory has often anticipated experimental results, but any model to break out of the realm of pure hypothesis necessarily requires the support of experiments: "*Experiments have a life of their own. They may suggest new theoretical advances. They may be based on false theoretical conceptions and produce valid findings. What is more, they may constitute crucial experiments in the sense of providing conclusive evidence against one theory, without, however, necessarily validating a rival theory.*" [2].

The experiments that will be described in this chapter provided an indispensable basis for the development of quantum mechanics and knowledge of the properties of the electron [3]. Quantum theories undoubtedly exert great fascination, but not less do the experiments that have allowed to reconstruct the subatomic universe fragment by fragment up to the *Standard Model*. This chapter briefly describes some of the "classic" experiments that have helped us understand the electron's nature and physical properties. These experiments are very famous and are taught in high school physics courses because they represent a fundamental step in the extraordinary journey to disclose the mysteries of the intimate structure of matter. Thomson's experiment which led to the discovery of the electron has already been described in Chap. 1.

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P. Innocenzi, *What is an Electron?*, https://doi.org/10.1007/978-3-031-81984-1_4

From this first fundamental discovery, a series of brilliant experiments followed in the early part of the last century to unravel the properties of the electron piece by piece. Many of these experiments take the name of the scientists who performed them: *Millikan oil-drop experiment*, *Davisson-Germer experiment, Franck-Hertz experiment*, and *Stern-Gerlach experiment*. This series of experiments made it possible to determine the charge of the electron, its wave nature, and the existence of a spin state, in addition to confirming Bohr's quantum theory. These were fundamental steps without which the entire theoretical scaffolding of quantum mechanics would not have been consistent. Five experiments have been selected. They are reported in chronological order to follow the development of the theoretical models of quantum mechanics over time.

4.1 Photons Creating Electrons, the Photoelectric Effect

The first experiment we remind in this chapter connects light and electrons through quantum mechanics. This experiment concerns the photoelectric effect. It was so significant in the development of quantum mechanics that Alber Einstein, who provided the theoretical explanation for it, was awarded the Nobel Prize in Physics in 1921. The photoelectric effect has several fathers and Nobel Prize winners and has played a pivotal role in shaping our understanding of quantum mechanics. The discovery and explanation of the phenomenon are the result of the work of several scientists who, over several years at the turn of the 1800s and early 1900s, contributed through their experiments and theories to the understanding of the effect.

The photoelectric effect was first observed in 1887 by German physicist Heinrich Rudolf Hertz (1856–1894) [4]. Hertz became famous for his research on radio waves, to the extent that the frequency of electromagnetic wave oscillation has been named after him. In his experiments on cathode tubes, Hertz observed that when ultraviolet light illuminates two metal electrodes to which a voltage is applied, the voltage value at which sparking occurs between the cathode and anode changes (Fig. 4.1). This simple observation implied that light and electricity were somehow interconnected. However, it was not yet clear to Hertz what kind of particles could generate this voltage variation in the cathode-ray tube. Hertz's observations suggested that light could somehow influence the electric properties of materials, but he did not delve deeply into the underlying cause [5].

It was a student of Hertz, the physicist Philipp von Lenard (1862–1947) [6], a few years later, in 1902 [7], who showed that the effect of light on metal was connected with those electrically charged particles discovered by Joseph John Thomson in 1897. Lenard observed that the intensity of incident radiation does not affect the kinetic energy of photoelectrons and defined the particles emitted by the metal as: *mobile carriers of negative electricity.* The experiment carried out by von Lenard, which earned him the Nobel Prize in 1905, the very year Einstein published his article explaining the photoelectric effect, allowed him to assess that the intensity of incident radiation does not affect the ability of light to produce photoelectrons [8]. The energy

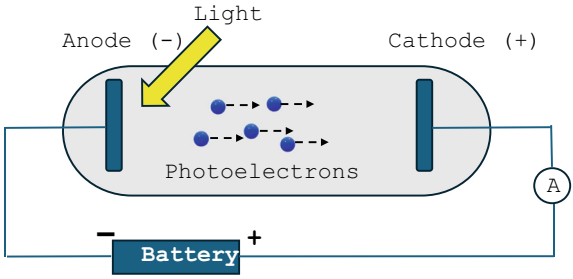


Fig. 4.1 Schematic diagram of the apparatus used for the experiment on electron emission from the surface of a metal exposed to UV radiation

of emitted electrons depends only on the wavelength (frequency) of the incident light, not its intensity. The fundamental property of light in producing the photoelectric effect is not its intensity, but its frequency, which represents its energy. Low-frequency light, even with very high intensity, could not generate photoelectrons, which were emitted only by increasing the frequency to a certain threshold. Intensity is related to the number of photons striking the metal surface, and frequency, on the other hand, to energy. Even by increasing the number of photons, thus the intensity of the incident radiation, if it does not exceed a specific value, it is unable to make electrons emit. In the process of photoemission, an electron within the material absorbs the energy of a photon, and if this is greater than its binding energy, the electron can be emitted from the atom. If, on the other hand, the photon's energy is too low, it will not be able to generate the corresponding photoelectron (Fig. 4.2). Von Lenard's observations were critical because they highlighted the limitations of classical physics in explaining the photoelectric effect. However, while he made important experimental findings, he did not provide a theoretical explanation for them.

According to classical theory, the intensity of the incident light determines whether or not it can generate photoelectrons. More intense light should be able to produce electrons "liberated" from the cathode surface more easily. Conversely, decreasing the intensity of the radiation should decrease the ability of the light to make the metal emit electrons.

Instead, the intensity of light, given by the number of photons arriving at the surface in the unit of time, determined the number of electrons released from the metal (measured as an electric current). Electron emission in classical theory is always possible as long as the time required to transfer the energy needed to "release" is sufficient. The classical interpretation of light as an electromagnetic wave fails to explain the correlation between light and electron emission from a metal.

Einstein's reflections in this regard, when he critically considers the limits of the classical concept of electromagnetic radiation as a *continuum* in space, are highly enlightening: "*According to the Maxwellian theory, energy is to be considered a continuous spatial function in the case of all purely electromagnetic phenomena including light, while the energy of a ponderable object should, according to the present conceptions of physicists, be represented as a sum carried over the atoms and electrons. The energy of a ponderable body cannot be subdivided into arbitrarily many or arbitrarily small parts, while the energy of a beam of light from a point source*

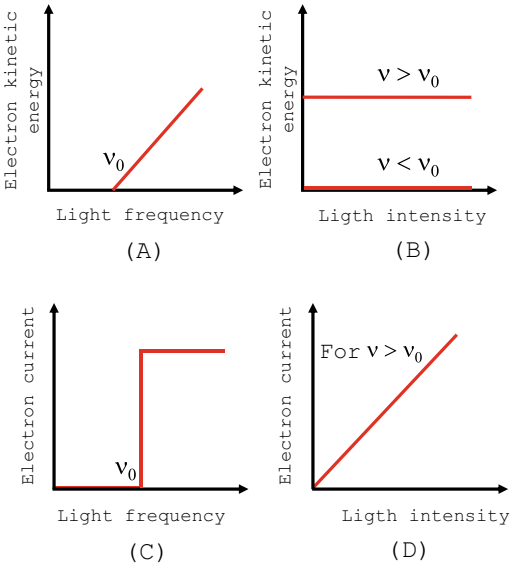


Fig. 4.2 Schematic drawings illustrating the characteristics of the photoelectric effect, derived from Lenard and Millikan's experiments. **a** The kinetic energy of each emitted electron increases linearly with frequency above a particular threshold value, n_0. **b** The electron kinetic energy is independent of light intensity above the threshold frequency n_0 and zero below. **C** The number of electrons emitted per second (i.e., the electric current) is independent of the light frequency above the threshold frequency n_0 and null below. **d** The number of electrons increases linearly with light intensity for $n > n_0$

(according to the Maxwellian theory of light or, more generally, according to any wave theory) is continuously spread an ever increasing volume.

The wave theory of light, which operates with continuous spatial functions, has worked well in the representation of purely optical phenomena and will probably never be replaced by another theory. It should be kept in mind, however, that the optical observations refer to time averages rather than instantaneous values. In spite of the complete experimental confirmation of the theory as applied to diffraction, reflection, refraction, dispersion, etc.*, it is still conceivable that the theory of light which operates with continuous spatial functions may lead to contradictions with experience when it is applied to the phenomena of emission and transformation of light.*

It seems to me that the observations associated with blackbody radiation, fluorescence, the production of cathode rays by ultraviolet light, and other related phenomena connected with the emission or transformation of light are more readily understood if one assumes that the energy of light is discontinuously distributed in space. In accordance with the assumption to be considered here, the energy of a light ray spreading out from a point source is not continuously distributed over an increasing space but consists of a finite number of energy quanta which are localized

at points in space, which move without dividing, and which can only be produced and absorbed as complete units." [9]. This is a long quote but very important because Einstein's observations are in fact the recognition of the quantum nature of light.

Another experimental evidence that did not add up with the classical electromagnetic theory was the value of the kinetic energy, $E_c = ½\ mv^2$, with which electrons were emitted from the metal used as a target. The kinetic energy of the photoelectron should have depended on the intensity of the electromagnetic radiation; a higher intensity should have corresponded to a higher kinetic energy of the electron. In reality, the kinetic energy depends on the frequency of the incident electromagnetic radiation, not its intensity (Fig. 4.2). A later experiment performed in 1916 by Robert Millikan, the same scientist who measured the charge of the electron, showed that the kinetic energy of photoelectrons also depended on the energy of the incident radiation, as predicted by Einstein's theory of the photoelectric effect [10]. Although Millikan was initially skeptical of Einstein's hypothesis that light could be quantized, his experiments ultimately provided conclusive support for the theory: "*It was in 1905 that Einstein' made the first coupling of photo effects and with any form of quantum theory by bringing forward the bold, not to say the reckless, hypothesis of an electro-magnetic light corpuscle of energy hv, which energy was transferred upon absorption to an electron. This hypothesis may well be called reckless first because an electromagnetic disturbance which remains localized in space seems a violation of the very conception of an electromagnetic disturbance, and second because it flies in the face of the thoroughly established facts of interference. The hypothesis was apparently made solely because it furnished a ready explanation of one of the most remarkable facts brought to light by recent investigations,* viz*., that the energy with which an electron is thrown out of a metal by ultra-violet light or X-rays is independent of the intensity of the light while it depends on its frequency. This fact alone seems to demand some modification of classical theory or, at any rate, it has not yet been interpreted satisfactorily in terms of classical theory*." [10].

Einstein deduced in 1905, well before the Millikan experiment, that the particular response of photoexcited electron was closely related to the property of light. The energy of light was not continuous but localized in *"quanta,"* or packets of energy, as defined by Planck's constant, and known as photons. A single quantum of light, provided it had sufficient energy, was capable of generating a single electron. Therefore, intensity, i.e., the number of photons, had no effect; frequency appeared to be the determining factor, not intensity.

The photons of a light beam have a characteristic energy that is proportional to the frequency of the light. Since increasing the intensity of low-frequency light only increases the number of low-energy photons, this change in intensity will not produce a single photon with sufficient energy to release an electron. Furthermore, the energy of the emitted electrons does not depend on the intensity of the incoming light of a given frequency, but only on the energy of the individual photons (Fig. 4.3).

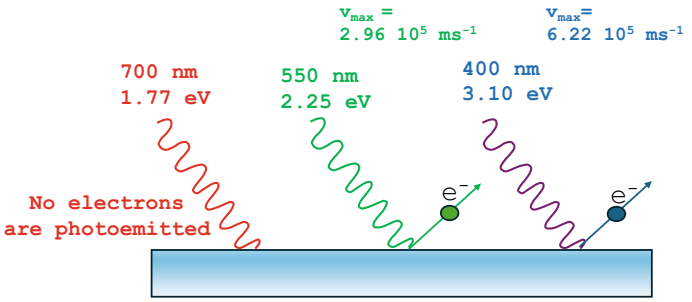


Fig. 4.3 Photoelectric effect. A photon with an energy of 1.77 eV, corresponding to red light (700 nm), does not have enough energy to induce an emission of an electron from the metal surface. However, photons with shorter wavelengths, in the green (550 nm) and blue (400 nm) ranges, can generate a photoelectron. The kinetic energy of the photoelectron increases with the energy of the incident photon

Ugly Scientists

However brilliant and talented, scientists are not always agreeable people on the human level and correct from the ethical one. Philipp Eduard Anton von Lenard undoubtedly made significant contributions to science, but he is also remembered for his quarrels with other scientists, including Einstein, and the field choices he made in Nazi Germany. In 1905, he was awarded the Nobel Prize in Physics: "*for his work on cathode rays*". His biographical note, however, is not particularly flattering: "*Lenard was an experimentalist of genius, but more doubtful as a theorist. Some of his discoveries were great ones and others were very important, but he claimed for them more than their true value. Although he was given many honours […] he believed that he was disregarded and this probably explains why he attacked other physicists in many countries. He became a convinced member of Hitler's National Socialist Party and maintained unreserved adherence to it. The party responded by making him the Chief of Aryan or German Physics*." (Philipp Lenard—Biographical. NobelPrize.org. Nobel Prize. www.nobelprize.org/prizes/physics/1905/lenard/biographical.

Despite initial mutual esteem, von Lenard became prey to strong anti-Semitic prejudices that led him to an aversion to Einstein both personally and scientifically, calling him the leader of Jewish physics. The confrontation between the two scientists was also over method, von Lenard believed that only experiments constituted the foundation of science not theory. Einstein was a theoretical physicist and this, in addition to the fact that he was Jewish, deeply irritated von Lenard. After World War II, von Lenard was expelled by the University of Heidelberg and deprived of the title of professor emeritus; he was also arrested but, given his advanced age, confined for life in a small German village. (Bruce J. Hillman, Birgit Ertl-Wagner, Bernd C. Wagner. The

Man Who Stalked Einstein: How Nazi Scientist Philipp Lenard Changed the Course of History. Lyons Pr. 2015).

4.2 The Charge of an Electron

J. J. Thomson's 1897 cathode-ray experiment marks a pivotal moment in the history of physics; electrons are finally recognized as negatively charged subatomic particles. The atom is no longer the rigid and impenetrable sphere described by Dalton but a new frontier to be explored. However, there was still a long way to go in understanding atomic structure and the intimate nature of the electron.

Another fundamental experiment in the journey to discover the atom and the properties of the electron was the measurement of the electron charge. The first measurement was made in 1909 by Robert Millikan (1868–1953) and Harvey Fletcher (1884–1981), who performed a relatively simple but highly informative experiment [11]. The experiment is now known as "*Millikan's oil drop*" and provided the first measurement of the charge of the electron, the fundamental unit of charge, one of the universal constants.

Before Millikan's oil-drop experiment, the value of the electron charge was known only approximately and there was much debate about its precise value. The experiment helped resolve this debate and established that the *electron charge is a fundamental constant of nature.*

The experiment, in its simplicity, is astonishing, and one must take into account that it was carried out in the early 1900s with a relatively rudimentary instrumentation [12]. The experiment was accurately conceived according to what was possible at the time, and the design of the experimental apparatus still amazes today. The experiment was realized by suspending tiny drops of oil in an electric field and measuring the force exerted on the drops by an electromagnetic field. The name by which the experiment is known is precisely the "*Millikan oil drop*" because it is based on the nebulization of oil into tiny droplets that are then used for measurements.

By measuring the size of the droplets and the strength of the electric field, Millikan was able to determine the charge of a single electron. It was a simple but extraordinary experiment that allowed for a relatively accurate measurement of electric charge. The experiment consisted of passing a beam of small oil droplets created with an atomizer through an electric field and measuring the deflection of the droplets due to the field (Fig. 4.4). The droplets, once sprayed in the upper part of the experimental chamber, fell through a small circular opening in the lower part by the force of gravity. Millikan and Fletcher used a microscope to observe the behavior of the falling oil droplets as a function of the applied electric field. The electric field was produced by applying a voltage between two metal plates, and the intensity of the field could be adjusted by changing the voltage. Another trick used was to negatively charge the oil

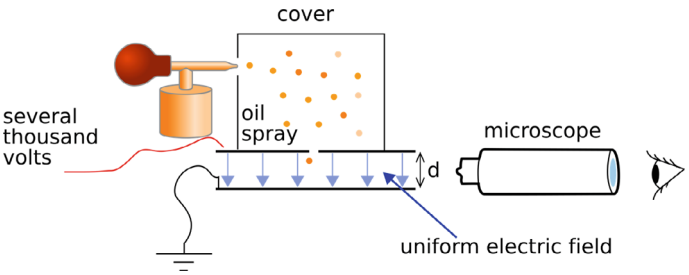


Fig. 4.4 Simplified diagram of Millikan's oil-drop experiment [13]

droplets so that they could be moved inside the lower part of the measuring chamber simply by varying the voltage of the electromagnetic field. For this, the gas inside the measuring chamber was ionized using X-rays; the X-ray generated electrons sticking to the surface of the oil droplets, which made them negatively charged.

To measure the charge of the droplets, Millikan and Fletcher used a technique called the balanced field method. In this way, the electric field was adjusted until the force on the drops due to the field was balanced by the force of gravity acting on the drops themselves. Hence:

$$q\,E = m_{(\text{oil droplet})}G \quad \text{and solving for the charge}$$
$$q \rightarrow q = m_{(\text{oil droplet})}G/E$$

where q is the charge, E the applied electric field, m the mass of the oil droplet, and G the force of gravity, which is responsible for the droplets falling to the bottom of the chamber (Fig. 4.5). Knowing the values of m, G and E, which are experimental data, it is then possible to determine the charge of the droplets and, by extension, the charge of a single electron.

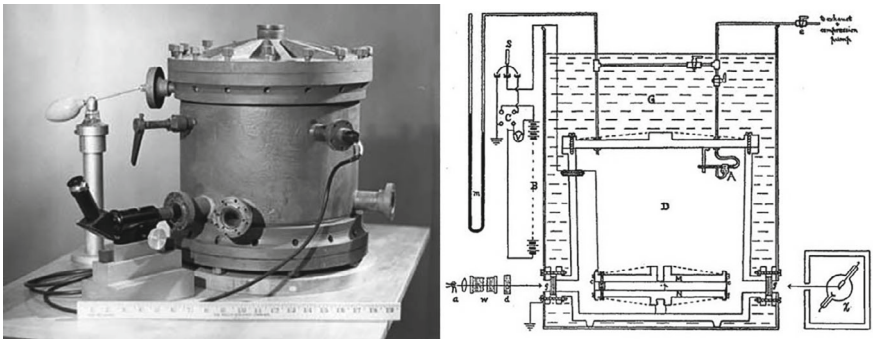


Fig. 4.5 Millikan and Fletcher's original apparatus for the oil-drop experiment (left). Diagram of the original apparatus (right) [14]

Millikan's oil droplet experiment represented a major breakthrough in understanding the fundamental nature of matter that later led to the development of the modern theory of quantum mechanics. It remains a classic example of the application of the scientific method and is still taught in physics courses around the world. However, Millikan's oil-drop experiment was the subject of controversy when it was first published in 1913, and some of the results of the experiment were later found to be erroneous. In fact, the error on the electron measurement was greater than the 1% reported by Millikan. One problem with the experiment was that it was based on the hypothesis that the oil droplets were all spherical in shape. However, it was later discovered by performing more precise measurements that the droplets were not perfectly spherical, which led to some errors in the results.

Another problem with the experiment is the assumption that the droplets were of homogeneous composition. However, it was then verified that they contained impurities that affected the charge and contributed to the errors in the results; as usual, the devil is in the details. Despite the errors during the execution and evaluation of the results, the conceptual value of the experiment remains intact. Robert Andrews Millikan was awarded of the Nobel Prize in Physics in 1923: "*for his work on the elementary charge of electricity and on the photoelectric effect.*"

The charge of the electron is now known with great accuracy and the currently accepted value of the electron charge (May 2019) is:

$$e = 1.602176634 \times 10^{-19}\text{coulomb}(C).$$

4.3 The Experiment of Franck and Hertz

Niels Bohr's atomic model, the first quantum model of an atom, was proposed by the Danish physicist in 1913. As we saw in the previous chapters, the model explained well the emission of the spectral lines in the hydrogen atom that were observed only at certain wavelengths. These emissions were considered an indication that electrons in an atom could only take on specific energy values that Bohr made correspond to atomic orbitals. Only a year later, in 1914, a pair of German physicists, James Franck (1882–1964) and Gustav Hertz (1887–1975), performed an important experiment whose results were used by Bohr himself to confirm the quantum nature of the atomic structure [15].

The experiment carried out by the two physicists, earned them the Nobel Prize in 1925: "*for their discovery of the laws governing the impact of an electron upon an atom*" and the title of James Franck Nobel lecture was: "*Transformation of Kinetic Energy of Free Electrons into Excitation Energy of Atoms by Impacts.*" The idea behind the experiment was to study what happens when an electron bumps into an atom; the results will be another important building block in support of the quantum atomic model.

When a particle collides with another particle, two different types of collisions can occur: elastic or inelastic. In an elastic collision, the kinetic energy must be conserved, so in a collision between an electron and an atom that has a much larger mass, due to the conservation of momentum and energy, the electron will only change direction of motion but transferring only a negligible amount of energy to the atom. The kinetic energy of the electron before and after the collision will be the same. In an inelastic collision, on the other hand, the energy lost by the electron in the collision is not converted into kinetic energy but rather may cause ionization, excitation, or particle production. If the electron's kinetic energy is high, it can transfer enough energy in case of collision with an atom to succeed in exciting it. If in an electron-atom collision, the energy that can be transferred by the colliding electron is less than that required to excite an atomic electron from its fundamental state to the first excited level, the collision remains elastic. Otherwise, if the kinetic energy of the electron exceeds the energy difference between the two atomic states, energy transfer and inelastic collision will occur.

The transfer of kinetic energy during a collision is not the only way to excite an atom; in fact, it can also be done by using light of an appropriate wavelength, but this must correspond exactly to the difference in energy between the fundamental and excited states. If electrons are used instead of photons and the energy transfer occurs through an inelastic collision, this limitation, in theory, does not exist since the kinetic energy is not quantized either before or after the collision. To realize the experiment, therefore, it was necessary to create the conditions in which free electrons could collide inelastically with single atoms.

Franck and Hertz used a vacuum tube to create a low-pressure gas of mercury atoms. Inside the tube, they emitted electrons using three electrodes: a cathode for emission, a grid to accelerate them, and an anode (see Fig. 4.6a). The purpose of the experiment was to test whether the kinetic energy of the electrons changed during collisions with mercury atoms. They measured the current between the electrodes as a function of the applied potential difference to observe how the kinetic energy varied during the collision between the accelerated electrons and the gas atoms.

The experiment held some surprises, for Franck and Hertz observed that when an electron collided with a mercury atom, it lost only a specific amount of energy before "bouncing back" after colliding with the atom. This result suggested that the energy absorbed by atoms had to equal the difference between two energy levels of atomic electrons. If the electron had collided with the atom at a higher energy level, it would have lost the amount of kinetic energy corresponding to the energy between the levels. If the electrons had been slower, they would not have lost energy since the collisions were elastic.

Figure 4.6b shows precisely how a current decrease is observed at the energy value corresponding to 4.9 eV and its multiples, which corresponds to the energy difference between the first two levels of the hydrogen atom. Otherwise, as long as the electron kinetic energy is less than the value of 4.9 eV, this continues to increase as the applied potential increases.

The experiment shows that before a collision, an electron within the mercury atom occupies the lowest available energy level. There can be no intermediate levels or a

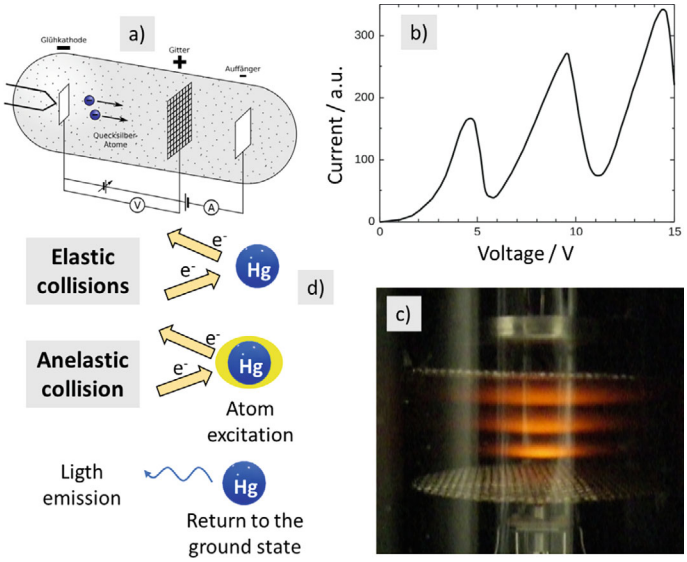


Fig. 4.6 **a** Schematic diagram of the experimental apparatus used by Franck and Hertz [16]. **b** The change in current between electrodes as a function of the accelerating potential [17]. **c** Franck-Hertz experiment with neon gas: three glowing regions are observed [18]. **d** Elastic and inelastic electron collisions with mercury atoms. Slowly traveling electrons change direction after elastic collisions, but do not change their velocity. Faster electrons lose most of their velocity in inelastic collisions and the lost kinetic energy is transferred to the mercury atom. The excited atom then emits light and returns to its original state [19]

continuous band of energy separating the nucleus and electrons. After the collision, the incident electron provides enough energy to excite the electron in the first orbital of the mercury atom at the top level, 4.9 eV. After the collision, the atom emits a photon of corresponding energy, and the electron returns to its fundamental state. This result showed that an electron within an atom must occupy discrete energy levels, as indicated by Bohr, although, at the time the experiment was performed, the two physicists were unaware of Bohr's theory. It was Bohr himself who used Franck and Hertz's experiment to lend further experimental support to his theory, as described in the Award ceremony speech for the Nobel Prize of Franck and Hertz [20]: "*The pioneer in this field is Lenard. But Franck and Hertz have developed and refined Lenard's method so that it has become a tool for studying the structure of atoms, ions, molecules and groups of molecules. By means of this method and not least through the work of Franck and Hertz themselves, a great deal of material has been obtained concerning collisions between electrons and matter of different types. Although this material is important, even more important at the present time is the general finding that Bohr's hypotheses concerning the different states of the atom and the connexion between these states and radiation, have been shown to agree completely with reality.*"

4.4 The Discovery of Spin. Stern and Gerlach's Experiment, a Fortunate Coincidence

Bohr's atomic model, represented until 1922, when the experiment was performed by the two German physicists Otto Stern (1888–1969) and Walther Gerlach (1889–1979), a good theory to explain the atomic structure. Electrons were electrically charged particles revolving around a positively charged dense nucleus, with the orbits assuming only discrete energy values. This model is part, as seen, of what we now identify as the "*old quantum theory.*"

Stern described later the dilemma in this way: "*At this time according to Bohr's theory one assumed that the magnetic moment of the hydrogen atom is produced by the movement of the electron around the nucleus in the same way as a circular current in a wire is equivalent to a magnet. Then the statement of the quantum theory means that the electrons of all hydrogen atoms move in planes perpendicular to the direction of the magnetic field. In this case one should find optically a strong double refraction which was certainly not true. So there was a serious dilemma.*"

Stern and Gerlach designed, therefore, their experiment to test the Bohr-Sommerfeld hypothesis of space quantization of angular momentum. According to the old quantum theory, space quantization in the case of a single electron orbiting the atomic nucleus, with a single quantum of angular momentum in the case of the hydrogen atom, can be visualized relatively simply (Fig. 4.7). The angular momentum vector L (and the atomic magnetic moment proportional to it) has only two permissible orientations, parallel ($m = +1$) or antiparallel ($m = -1$), corresponding to a clockwise or counterclockwise motion of the electron in the circular orbit.

Both Stern and Gerlach were quite skeptical of quantum mechanics and their experiment was intended to show that the theory was false [21]: "*What was really interesting was the experiment that I did together with Gerlach on space quantization. I had thought that [quantum theory] couldn't be right … I was still very skeptical about quantum theory and thought that a hydrogen or alkali atom must exhibit birefringence in a magnetic field … At that time I had thought about [space quantization] and realized that one could test it experimentally.*" [22].

Therefore, they expected their experiment to show that there was no quantization of angular momentum. If there were such a quantization, an electron, or a hydrogen atom, when passing through a magnetic field, should show a peculiar behavior and two different paths due to the deflection induced by the magnetic field. Otherwise, a continuous distribution of particles subjected to the action of a non-uniform magnetic field should be observed. For experimental matters, as we shall see, they used silver

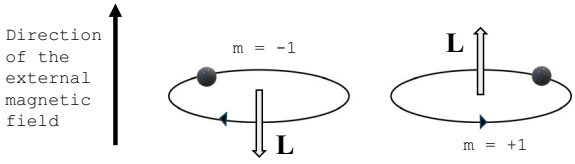


Fig. 4.7 Quantization of space in the case of a single-electron orbiting the atomic nucleus according to the *old quantum theory*

atoms, and the result was that they observed the splitting in two of the beams of silver atoms, which, although very small, provided clear evidence of the violation of the laws of classical physics.

Their experiment is a crucial piece in the intricate puzzle of conceptual enigmas underlying quantum mechanics, from the uncertainty principle to entanglement. The experiment, whose initial interpretation was later proven wrong, still amazes today, and together with the two-slit experiment, it represents one of the experimental foundations of quantum mechanics. What Stern and Gerlach demonstrated was not the spatial quantization of angular momentum but the existence of a different quantum state that of the electron spin. The fourth quantum number, which is defined by the spin state, closes the descriptive model of the electron in the atomic structure. The magnetic field-induced splitting of the beam of silver atoms in apparent agreement with classical quantum theory later turned out to be a lucky coincidence. Although the Stern-Gerlach experiment is associated with the discovery of electron spin, the two physicists and the scientific community had no idea of the true significance of their results. It was not until a few years later, in 1927, after the spin of the electron was theorized by George Uhlenbeck [23] and Samuel Goudsmit in 1926 [24], that it began to be understood that the Stern-Gerlach experiment had, in fact, demonstrated the existence of another quantum number, that one associated with the electron spin. As we have already pointed out, spin is a property of all particles, even though bosons and fermions have integer or half-integer values, respectively. The spin of an electron corresponds to an intrinsic angular momentum and takes only values of $+1/2$ or $-1/2$, while the orbital angular momentum is characterized only by integer numbers.

Stern and Gerlach carried out the experiment in February 1922 using silver as the precursor material. Silver was evaporated in a furnace to obtain atoms of the same element (Fig. 4.8). The atoms produced by the evaporation process were then collimated through a series of apertures to form a beam. Another essential element of the experiment was the use of an inhomogeneous magnetic field to pass through the beam of silver atoms. The fact that these were not charged particles, but neutral atoms was particularly intriguing. Silver atoms in their fundamental state contain the same number of protons and electrons, 47; therefore, the classical Lorentzian-type forces should not act. The 47 electrons are located in different energy levels, and according to *Pauli Exclusion Principle*, two electrons at the same energy level cannot have the same quantum numbers. The principal quantum number indicates the *shell* in which the electron is located, that is, the distance of the electron from the atomic nucleus. The lowest *shell* is the one denoted by the letter K. The secondary quantum numbers (s, p, d, f) indicate the shape of the orbital in which the electron is located, representing a probability density of finding the electron with a 90% probability. The magnetic quantum number, m, indicates precisely the orientation of the orbital shape. There is one last quantum number the fourth that differentiates precisely the two electrons in the same shell, it is the intrinsic angular momentum, the one identified by Stern and Gerlach's experiment.

Returning to the silver atoms, the pair of electrons with the same quantum numbers except for the spin are identified as paired electrons. So, if we look closely at all the different shells of the silver atom, we can see that all of them are occupied by

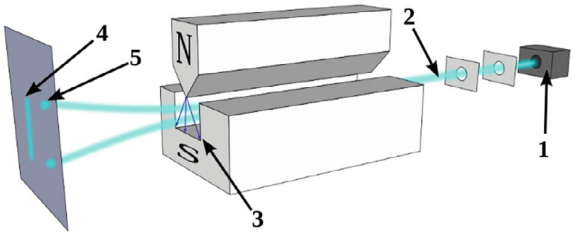


Fig. 4.8 Schematic diagram of the experimental apparatus used by Stern and Gerlach. A beam of silver atoms produced by a furnace (1) is collimated (2) and made to pass through an inhomogeneous magnetic field with a gradient along the direction orthogonal to the direction of propagation of the atomic beam (3). When leaving the magnetic field, the silver atoms impact a detector screen perpendicular to its propagation direction. According to predictions based on classical physics, a continuous signal should have been observed (4), while experimental results showed only two distinct signals (5). The silver atom after the measurement can only be in two spin states $|+1/2\rangle$ or $|-1/2\rangle$ [25, 26]

a pair of electrons, except for the last level, where there is an unpaired electron. This electron can assume both spin states, up and down, without violating *Pauli Exclusion Principle*. The silver atom with only one valence electron can be considered equivalent to the hydrogen atom with only one electron; experimentally, however, it is much easier to use silver atoms instead of hydrogen.

In Stern and Gerlach experiment, when the silver atoms pass through the magnetic field, which is inhomogeneous, the atoms do not distribute themselves on the detector continuously but are deflected to two positions, one up and one down, with respect to the axis of the detector placed after the magnetic field. The splitting of the beam of silver atoms observed on the detector is caused by the spin of the unpaired electron in the valence *shell.* According to classical physics, a homogeneous distribution of silver atoms on the detector screen should have been expected instead. The direct Stern's description of the experiment and his commentary is clear and precise: "*We sent a beam of silver atoms through a very inhomogeneous magnetic field. In such a field the magnets are deflected because the field strength on the place of one pole of the magnet is a little different from the field strength acting on the other pole. So in the balance a force is acting on the atom and it is deflected. A simple calculation shows that from the classical theory follows that we should find a broadening of the beam with the maximum intensity on the place of the beam without field. However, from the quantum theory follows that we should find there no intensity at all, and deflected molecules on both sides. The beam should split up in two beams corresponding to the two orientations of the magnet. The experiment decided in favor of the quantum theory.*" [27].

Although this experiment was at the time mistakenly regarded as a proof of the "quantization of space," what was observed by the two physicists were the two quantum states of the electron spin. Another important observation is that can be measured one component of the silver atom spin at a time, meaning that measuring the spin along one axis destroys all information along the others.

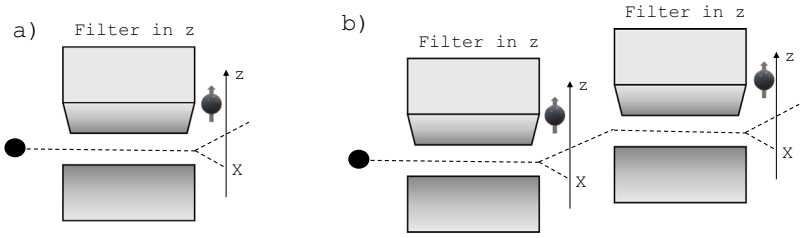


Fig. 4.9 Sequential experiment of Stern and Gerlach. **a** The system with the inhomogeneous magnetic field along the z-axis, acts as a filter in z, only one of the two beams is allowed to pass while the other is blocked with a filter. **b** The experiment is extended sequentially, even in the second detector only electrons with spin up are made to filter

4.4.1 The Sequential Experiment of Stern and Gerlach

Let us now look at the result of the experiment from another perspective. The experimental device could, in fact, be seen as a kind of filter capable of selecting electrons with a certain spin state. Before passing through the magnetic field, the spin is randomly oriented, however, immediately afterward, it takes two specific orientations that we can call for simplicity *up* and *down* (Fig. 4.9a). If one of the two electron beams, *up* or *down*, is blocked, the electron beam would have only one spin species. Stern and Gerlach's experiment can, at this point, be extended sequentially; this allows understanding of how an electron beam that has a specific spin state is affected when it is again subjected to a perturbation represented by the inhomogeneous magnetic field of the experimental system.

Let us then put two devices in sequence; the second one will be placed at the output of the spin *up* electrons oriented by the magnetic field axis so only spin *up* electrons will cross the field of the second device (Fig. 4.9b). At this point, one can check at the output of the second device what happens if the spin *down* detector is blocked. Electrons with spin *up* are still observed, and this means that the electrons with spin *up* orientation, so along the z-axis we have chosen as a reference, that come out of the first magnetic field are not perturbed by a magnetic field with the same orientation. There are no surprises in this result. However, it remains to be verified that only spin *up* and not spin *down* are observed; by recrossing the magnetic field, the spin *up* electrons selected by the first system could reorient themselves *up* and *down* again. However, if this time we block the slit with the exit of the *up* electrons, the detector in the area where the *down* electrons should be revealed measures zero (Fig. 4.10). The magnetic field itself does not perturb the state of the electrons.

However, the same experiment can be repeated using a magnetic field that is no longer oriented along the z-axis but along the x-axis. The first device is used to select electrons in the up state along the z-axis. These electrons are then passed through a second device whose magnetic field, however, has a different orientation, orthogonal to the previous one.

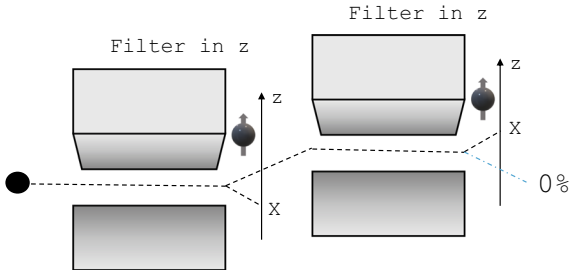


Fig. 4.10 Sequential experiment of Stern and Gerlach. In the second detector, the slit where spin up electrons pass through is blocked. No electrons are revealed

If an electron behaved like a classical magnet and was deflected upward in the first device, it should not experience any deflection in the second. Its poles would be in the same condition. The inhomogeneity of the external magnetic field would not affect the electron, which should hit the center of the detector screen. However, the electron is never detected at the center of the screen because being endowed with spin it is a two-state system and thus deflects to two specific directions.

What we observe is that about 50% of the electrons that have passed the first filter will deflect to the right along the x-axis and 50% to the left (taking the x-axis in the drawing to the right or left) (Fig. 4.11). We can define this state as right spin and left spin. Therefore, an electron that has spin up along the z-axis also has a 50% probability of having spin oriented along a specific x-axis direction, left or right.

In this situation, even quantum mechanics cannot predict which way a given electron will deviate; it is impossible. The mathematical formulation of quantum mechanics, thus the Schrödinger equation, allows us to calculate the probability of an event but not to predict the outcome of a single measurement. The experimental data could be interpreted as a limitation of quantum mechanics itself, a limitation in our ability to experiment and work out a consistent consequent model. Alternatively, we may think that Stern and Gerlach's experiment and quantum mechanics are nothing more than a fundamental property of nature. This example also shows that it is impossible to visualize spin as a classical vector. Spin is not determined before measurement; it only has probabilities of pointing in one direction or another.

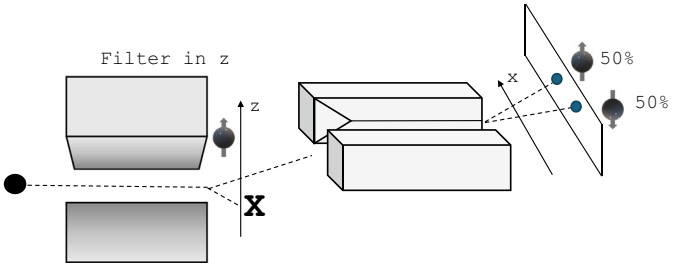


Fig. 4.11 Sequential experiment of Stern and Gerlach. The second system of magnetics is placed vertically to the first

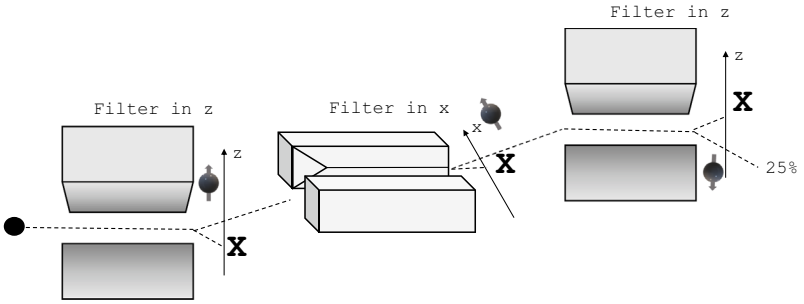


Fig. 4.12 Sequential experiment of Stern and Gerlach with three magnetic fields. The second magnet system is placed vertically to the first, the third has the same configuration as the first

Let us add a third system with the magnetic field still oriented along the z-axis like the first one. So sequentially, there will be a first system that orients electrons *up* and *down*, a second with a magnetic field along the x-axis where only *up* electrons are passed and which orients electrons with spin left or right, and a third with the same field as the first. A filter is still placed after the second device; only the left electrons will pass through the third magnetic field. The effect of the magnetic field of the third device, however, is to still produce 50% electrons with spin *up* and 50% with spin *down* along the z-axis (Fig. 4.12).

This behavior can only be explained by assuming that after passing through the second filter, an electron forgets that its spin was directed upward along the z-axis. The measurement of the spin along the x-axis makes the previous information about its state with respect to the z-axis irrelevant. Thus, the Stern-Gerlach apparatus does not behave as a simple filter, as we defined it in the description of the sequential experiment.

This situation occurs whenever two quantities cannot be measured simultaneously. In this case, the two quantities are the directions of spin orientation with respect to different axes. Before the actual measurement, none of the classical characteristics exists objectively. We cannot assume that the spin has a specific state before we observe the deflection of the electron. We can probabilistically predict that in a beam of electrons, one half will have a certain spin state and the other half the opposite. However, it is impossible to know in advance whether a specific electron will have one or the other configuration; it will be like tossing a coin, heads, or tails. A coin with heads and tails can be considered a quantum system. Just as an electron has two spin states, the coin has only two possible configurations, heads and tails, which are an integral part of the coin's properties. When we toss it in the air and then let it fall back to the ground or the hand, we have performed a measurement, and the coin assumes a definite configuration, e.g., heads. However, the coin will still simultaneously be both heads and tails, even though our measurement has forced it to show a specific configuration. If we withdraw the coin into the air, we only know that there is a 50% probability that the coin will either be heads or tails, much like what happens to an electron with its spin state.

This consideration applies to all observable quantities for which it is generally impossible to predict the outcome of a single measurement. Quantum mechanics allows only the probability of an event to be calculated, and no physical quantities can be known simultaneously. Measuring one makes previous information about the other irrelevant.

Two Scientists, One Nobel. The Fate of Stern and Gerlach

The Stern and Gerlach experiment is named after the two German physicists who conceived and carried it out. Their lives, however, had different fates and only Stern was awarded the Nobel Prize in 1943 (delivered in 1944) but not Gerlach, although the motivation for the prize was more general and was not concerning only the quantization of angular momentum (For his contribution to the development of the molecular ray method and his discovery of the magnetic moment of the proton). The original experiment was designed by Stern with the aim of dismantling quantum theory, even though the result was later diametrically opposed. However, the practical implementation of the experiment was not trivial, so Stern sought an experimental physicist who could actually carry it out. His choice fell squarely on Walter Gerlach then aged 33. Max Born who strongly supported the realization of the experiment exclaimed enthusiastically: "*Thank God, now we have someone who knows how to do experiments. Come on, man, give us a hand*."

There was another problem to be solved; however, the time was not the most favourable in Germany to raise funds for the experiment because the Weimer Republic was, in fact, plagued by hyperinflation. Stern then turned to his friend Max Born, who mobilized to find him the necessary funding in different ways; Einstein himself, who esteemed Stern and was his mentor, contributed with personal funds. The experiment was then performed in February 1922, but Gerlach alone carried it out; the paper was published in the same year, with the two scientists as co-authors. Otto Stern 1933 moved to the United States, where he became a Research Professor of Physics at the Carnegie Institute of Technology in Pittsburgh, while Walter Gerlach remained in Germany. The careers and destinies of the two physicists had permanently separated. During World War II, Gerlach took part in the Nazi nuclear program and, for this reason, was captured after the war as part of *Operation Epsilon. Operation Epsilon* was the code name by which the program to capture German scientists suspected of working on the German nuclear program was referred to. There were ten in all, and Gerlach found himself in distinguished company with Max von Laue and Werner Heisenberg. In 1946 he was released and resumed his academic career in Germany, but he was never awarded the Nobel Prize, although his experiment remains a milestone in the history of physics.

4.5 The Electron Is (Also) a Wave. A Lucky Accident, Davisson and Germer's Experiment

Louis de Broglie's hypothesis of the wave nature of the electron was very elegant, but at the time it was formulated it lacked empirical evidence. Experiments by J. J. Thomson and R. Millikan had shown that the electron had charge and mass, and thus a corpuscular nature. Although de Broglie had theoretically predicted the wave nature of the electron, it remained to be demonstrated, but how? It had to be shown that the electron had properties typical of a wave, such as interference and diffraction.

De Broglie published his theory in 1924, and fortunately for him, in 1927, two young researchers in the United States working at *Western Electric Company Laboratories,* which would later become the famous *Bell Laboratories*, performed an experiment that paved the way for understanding the wave nature of the electron. The experimental proof of the validity of his theory enabled de Broglie to receive the Nobel Prize in Physics in 1929.

The Davisson-Germer experiment [28], also known as the electron diffraction experiment, was an important step in the evolution of the theory of the wave-particle duality of electrons. It was performed in 1927 by Clinton Davisson (1881–1958) and Lester Germer (1896–1971) and was based on the study of the scattering of electrons hitting the surface of a nickel crystal. Previously, in 1913, W. H. Bragg and his son W. L. Bragg had performed an experiment with X-rays that demonstrated the wave-like nature of this radiation, which has a wavelength similar to that of crystalline atomic planes.

The two Braggs discovered that when X-rays interact with the regular atomic structure of a crystal lattice, they undergo diffraction due to the periodic arrangement of atoms. When waves encounter an obstacle or pass through an aperture, they exhibit a characteristic behavior called diffraction. Diffraction occurs when waves bend around obstacles and spread out, creating interference figures. This phenomenon is a clear indicator of wave behavior. Waves diffracted from the crystal lattice exhibit a particular diffraction spectrum with characteristic peaks that are used in diffractometry to determine crystal phases. Since the wavelength of electrons varies from a few tenths of a nanometer (at 10 eV) to a few picometers (at 100 keV), it is comparable to the interatomic distances in solids. Therefore, if electrons were also wave-like, a diffraction experiment such as that used for X-rays should give similar results by showing specific diffraction figures. This insight came to Walter Maurice Elsasser (1904–1991), a German physicist, and was the basis of the Davisson-Germer experiment. According to Elsasser, an experimental demonstration of electron diffraction would be proof of the wave nature of the electron.

The experiment was based on observing the angle at which an electron beam generated by a tungsten filament and accelerated in a vacuum chamber was scattered toward a target consisting of a nickel crystal (Fig. 4.13). A moving detector made it possible to observe the electrons scattered by the nickel crystal as a function of the angle, θ, and to vary the speed at which they arrived at the target by modulating the values of the acceleration voltage, V.

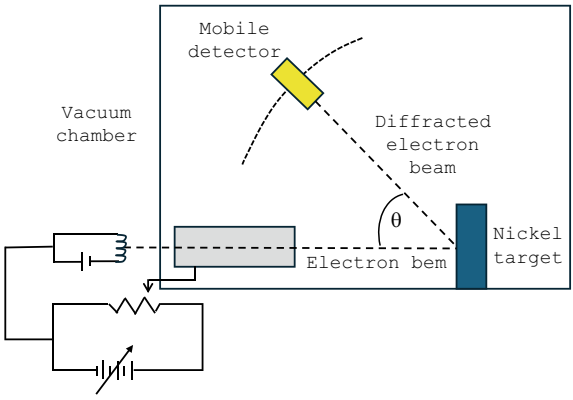


Fig. 4.13 Schematic diagram of the experimental apparatus used in Davisson and Germer's experiment. The electron beam normally strikes a nickel crystal, the diffracted electrons are detected using a moving detector that measures the electron beam emerging at different angles

The experiment was actually the result of a laboratory accident, as Davisson and Germer described in their 1927 *Physical Review* article: "*The investigation reported in this paper was begun as the result of an accident which occurred in April 1925. At that time we were continuing an investigation, first reported in 1921, of the distribution-in-angle of electrons scattered by a target of ordinary (poly-crystalline) nickel. During the course of this work a liquid–air bottle exploded at a time when the target heavily oxidized by the inrushing air. The oxide was eventually reduced and a layer of the target removed by vaporization, but only after prolonged heating at various high temperatures in hydrogen and in vacuum. When the experiments were continued it was found that the distribution-in-angle of the scattered electrons had been completely changed. […] Before the accident in previous experiments we had been bombarding many small crystals, but in the tests subsequent to the accident we were bombarding only a few large ones.*" [29].

The laboratory accident had transformed the nickel target into a material more similar to a single crystal. In a polycrystalline material in which the many small crystals have different orientations, the electron scattering figure will appear as a continuous curve otherwise diffraction curves will be observed as shown in Fig. 4.13 taken from the original experiment.

The result of the experiment actually showed that the intensity of the signal due to the electrons was not fixed but varied depending on the angle. As we have already pointed out, if the electron had wave-like properties, exactly like that of light, then it should have exhibited the phenomenon of constructive and destructive wave interference and a figure similar to that of X-rays diffracted from a crystal. Everything then seemed to indicate that the electron also had the properties of a wave, according to Davisson and Germer: "*Because of these similarities between the scattering of electrons by the crystals and the scattering of waves by the three- and two-dimensional gratings a description of the occurrence and behaviour of the electron diffraction beams in terms of the scattering of an equivalent wave radiation by the atoms of the crystal, and its subsequent interference, is not only possible but most simple and natural. This involves the association of wavelength within the incident electron*

beam, and this wavelength turns out to be in acceptable agreement with the value h/mv of the undulatory mechanics, Planck's action constant divided by the momentum of the electrons." [22].

Further precise confirmation of the wave nature of the electron was obtained just a few years later, in 1921, by George Paget Thomson (1892–1975) [30]. The Davisson-Germer experiment showed that the regular arrangement of atomic planes in a crystal was capable of generating reflections at specific angles in the electrons striking it. However, by using an electron beam of higher energy and a target formed by a thin film of effectively monocrystalline nickel, it was possible to penetrate deeper into the crystal and clearly observe the diffractions generated by the planes. Indeed, under these experimental conditions, the diffracted wave is generated by a large number of atomic planes, and the diffraction figure takes on a characteristic structure with well-defined peaks (Figs. 4.14 and 4.15).

The final demonstration of electron diffraction and thus of the wave nature of the electron earned G. P. Thomson the Nobel Prize in 1937 along with Davisson. The homonymy with J. J. Thomson, who discovered the electron in 1987, is not accidental; he was in fact his father. The comment of the physicist Max Jammer (1915–2010) on Thomson's family captures the paradoxical aspect of the work of father and son, both Nobel laureates: "*One may feel inclined to say that Thomson, the father, was awarded the Nobel Prize for having shown that the electron is a particle, and Thomson, the son, for having shown that the electron is a wave.*" [32].

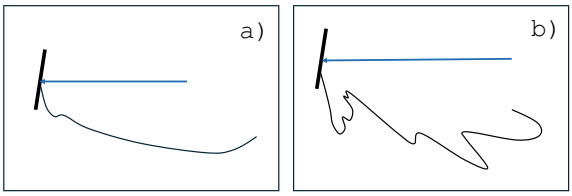


Fig. 4.14 **a** Diffusion of an electron beam accelerated to 75 V from a nickel crystal block (many small nickel crystals). **b** Diffusion of an electron beam accelerated to 75 V from several large nickel crystals

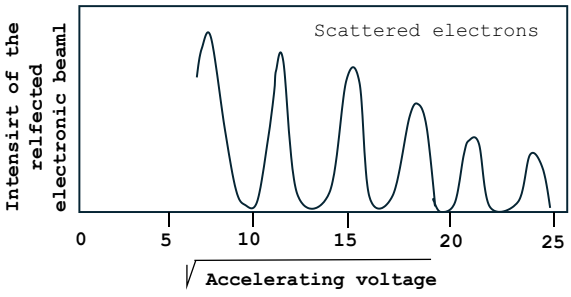


Fig. 4.15 Electrons striking a crystal lattice are reflected by showing constructive interference in agreement with Bragg conditions [31]

Thus X-rays and electrons share the wave nature and are diffracted by a nickel crystal showing similar diffraction patterns. One can use at this point de Broglie's relation relating wavelength to momentum and calculate the wavelength of an electron moving at a speed of 10^7 m s^{-1}, similar to that of the experiment.

$$\lambda = \frac{h}{p} = \frac{h}{m_e v} = \frac{6.626 \times 10^{-34}}{9.11 \times 10^{-31} \times 10^7} \frac{\text{J s}}{\text{kg m s}^{-1}} = 0.073 \text{ nm}$$

This wavelength is similar to that of X-rays, which is in the range of 0.01^{-10} nm, and to that of the nickel crystal lattice spacing, which is $d = 0.215$ nm, the distance between two adjacent atomic planes. This value thus justifies the electron diffraction phenomenon and the X-ray diffraction analogy.

One last question: How are diffraction patterns produced? By the interference of the electronic waves of different electrons colliding on the crystal, or is a single electron responsible for the phenomenon? The answer, well supported by experimental observations, is that the interference is produced by the parts of a wave associated with a single electron that has been scattered by the crystal. Because the single electron diffraction produces the same kind of figure, this finding well supports the explanation.

The Davisson-Germer experiment was a major milestone in the development of quantum mechanics. It helped establish the concept of wave-particle duality, which states, as we already described, that particles can exhibit both wave and particle properties depending on how they are observed. The experimentally demonstrated wave properties of the electron, however, opened up new interrogatives about its true nature, and the wave-particle discussion, with all its unanswered questions, would become part of the foundation of quantum mechanics.

Although the experiment was performed by Davisson in collaboration with his student Germer, hence the name by which the experiment is known, it was Davisson alone who received the Nobel Prize, perhaps because he came up with the idea first. Davisson shared the prize 50/50 with G. P. Thomson, who had actually made a major contribution in determining electron diffraction but had essentially perfected Davisson and Germer's experiment, for whom only the consolation of a minor prize remained.

The two had, among other things, very different characters; Lenard Germer had been a fighter pilot during World War I, and in addition to physics, he devoted himself intensively to his other passion, mountaineering, and died of a heart attack during a mountain hike. In addition to the experiment with Germer, he has the great merit of having realized the potential of electron diffraction as an analytical tool. Germer, on the other hand, was known to be very shy and reflective, so much so that he published few papers per year because he absolutely wanted to be sure of the results. His career seemed to be over after only one year of graduate studies due to lack of funds but another

of the scientists mentioned in this chapter, Robert Millikan, came to his aid for their experiment. Millikan, who had recognized Davisson's great talent, recommended him as a physics teacher at Purdue University. From there, his career took off even though he preferred to work at Bell Laboratories rather than at the University because it ensured that he could concentrate on his basic physics experiments. After retiring from his work at Bell Laboratories, he devoted himself to university teaching and his favourite hobbies, walking and ice skating, always remaining a shy and reserved person.

4.6 It Is a Question of Mass

This group of experiments was done in the first two decades of the 1920s. They can still be considered part of the "old theory of quantum mechanics." However, they played a crucial role in unambiguously demonstrating the dual wave-particle nature of the electron. These properties, however, are intrinsic to matter and are independent of whether it has a charge or not. The ability to observe the wave nature of matter is closely related to its mass. Again, it is Planck's constant h that dictates the rules of the game, as we will discuss in detail in the next chapter. If the constant h, which is a very small number anyway, was zero, then the de Broglie wavelength, $\lambda = h/p$, would also be zero. However, h is not zero, even though it has a very small value. To obtain measurable wavelengths, it is necessary that the value of the momentum in the denominator of the de Broglie relation, $p = \text{mv}$, be very small, which means that the mass of the object in question must also be very small. In general, in the world of subatomic particles, even if the velocity v is high, the mass m is small enough to give a measurable de Broglie wavelength. As soon as we move up in scale and consider macroscopic objects, the mass becomes so large that the resulting de Broglie wavelength is so small that it cannot be detected experimentally. For example, if we consider a one-kilogram baseball moving at 10 m sec^{-1}, the de Broglie wavelength is 0.66×10^{-25} nm. It is too small to be detectable, so we can safely continue to apply the rules of classical physics to baseballs while ignoring their possible quantum properties.

4.7 A Time Table of the Discovery of Electron Properties

- **The photoelectric effect**

Experimental discovery of the photoelectric effect (1887). The phenomenon was observed for the first time but not explanation was given.

Heinrich Rudolf Hertz (Hamburg, 22 February 1857–Bonn, 1 January, 1894).

Ueber einen Einfluss des ultravioletten Lichtes auf die electrische Entladung (On the influence of ultraviolet light on the electrical discharge), 1897 [5].

The maximum energy of the emitted photoelectrons depends on the wavelength of incident light (1902). The intensity of incident radiation does not affect the kinetic energy of photoelectrons; the particles emitted by the metal are defined as mobile carriers of negative electricity.

Philipp von Lenard (7 June 1862, Pressburg-Messelhausen, 20 May 1947).

Ueber die lichtelektrische Wirkung (About the photoelectric effect). 1902 [7].

Theoretical explanation of the photoelectric effect (1905). The energy of individual photons must be high enough to overcome the minimum energy needed to release an electron of the metal.

Albert Einstein (Ulm, 14 March 1879–Princeton, 18 April 1955).

Concerning an Heuristic Point of View Toward the Emission and Transformation of Light. 1905 [9].

It exists a minimum frequency of light below which no electrons are ejected (1916). The photoelectrons are emitted regardless of the light's intensity.

Robert Andrews Millikan (Morrison, March 22, 1868–San Marino, December 19, 1953).

A Direct Photoelectric Determination of Planck's "h". 1916 [10].

- **The spin**

***Discovery of the electronic spin* (1922)**. The experiment of quantization of angular momentum demonstrates the existence of electronic spin.

Otto Stern (Sohrau, 17 February 1888–Berkeley, 17 August 1969) and Walther Gerlach (Biebrich, 1 August 1889–München, 10 August 1979).

Der experimentelle Nachweis der Richtungsquantelung im Magnetfeld. (*Experimental evidence of space quantization in the magnetic field)* 1922 [33].

***Pauli exclusion principle* (1925)**. Pauli states that no electrons can have the same four quantum numbers.

Wolfgang Pauli (Wien, 25 April 1900–Zurich, 15 December 1958).

Über den Zusammenhang des Abschlusses der Elektronengruppen im Atom mit der *Komplexstruktur der Spektren (On the connection between the configuration of the electron groups in the atom and the complex structure of the spectra).* 1925 [34].

***Theorization of electronic spin* (1926)**. The experiment of Stern and Gerlach is explained on the basis of another quantum number associated with the electron spin.

George Eugene Uhlenbeck (Batavia, 6 December 1900–Boulder, 31 October 1988) and Samuel Abraham Goudsmit (Den Haag, 11 July 1902–Reno, 4 December 1978).

Spinning Electrons and the Structure of Spectra. 1926 [24].

- **Wave-like nature of electrons**

Experiment of electron diffraction **(1927)**. Demonstration of the wave nature of electrons.

Clinton Joseph Davisson (Bloomington, 22 October 1881–Charlottesville, 1 February 1958) and Lester Halbert Germer (Chicago, 10 October 1896–Gardiner, 3 October 1971).

The Scattering of Electrons by a Single Crystal of Nickel. 1927 [28].

Demonstration of electron diffraction from atomic planes **(1931)**. The electron diffraction is finally demonstrated as general properties of the wave-like properties of electrons.

George Paget Thomson (Cambridge, 3 May 1892–Cambridge, 10 September 1975).

The diffraction of electrons by single crystals. 1931 [30].

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Chapter 5
The Mystery of the Constants



In physical theories, both classical, quantum, and relativistic versions, there is extensive use of physical constants. However, the meaning of many of these constants is difficult to understand in all its implications, are they an expression of a mathematical modeling process or a physical reality? One of these constants, Planck's constant, is the basis of quantum mechanics and is encountered in all the main equations describing the properties of an electron. Another of these constants is the electron charge, which implies that all electrons in the universe are absolutely identical. The charge of an electron is one of the universal constants, and every electron, as we have seen, has exactly the same charge with exactly the same value. Moreover as far as we can currently measure it does not decay, at least in the timescale of the life of the universe. The charge of the electron is found in other constants whose meaning is yet to be deciphered, at least as to what physical sense they may have. One of these, the fine structure constant, is so strange that it has occupied generations of physicists in trying to understand its meaning.

This chapter is devoted to first describing the Planck's constant and its enunciation, which is the basis for all future development of quantum mechanics, and then two constants directly related to the intrinsic nature of the electron.

5.1 A Constant Measure of the Whole

One of the fundamental formulas of quantum mechanics that we have already met is $E = h\nu$. The relationship indicates that energy is directly proportional to the frequency of oscillation of the wave function according to a constant h, known as *Planck's constant*. It is a constant that is almost ubiquitous in the subatomic world but whose nature and implications perhaps few truly understand. Its introduction by Planck is in some ways a historical event that marks the beginning of a new scientific paradigm, that of quantum mechanics. Planck's constant, in fact, appears in all the

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P. Innocenzi, *What is an Electron?*, https://doi.org/10.1007/978-3-031-81984-1_5

basic equations of quantum mechanics, from de Broglie wavelength to *Heisenberg's indeterminacy principle* (Uncertainty principle), in the *Schrödinger equation* and in the quantum numbers that define the energy levels of electrons in an atom.

$E = h\nu$	Relationship between energy and photon oscillation frequency.
$\lambda = h/m\nu$	De Broglie wavelength.
$\ell = nh/2\pi$	Energy levels of electrons in an atom.
$\Delta x \Delta p \geq h/2\pi$	Heisenberg's indeterminacy principle.
$-\frac{\hbar^2}{2m}\frac{\partial^2\psi(x,t)}{\partial x^2} + V(x,t)\psi(x,t) = i\hbar\frac{\partial\psi(x,t)}{\partial t}$	Schrödinger equation.

It is a ubiquitous and fundamental constant, but where was it derived from and when was it introduced? The constant is, of course, named after the German physicist Max Planck (1858–1947) and derives from his explanation for a physical phenomenon that had long left the scientific community rather puzzled. The problem is now known as the *Ultraviolet Catastrophe* but this definition was used by Paul Ehrenfest only in 1911 [1], several years after the introduction in 1900 of the Planck's constant [2] and his explanation of the black body radiation. The definition does not bode well, but it says a lot about how much this phenomenon has intrigued physicists before finding an explanation. It is not a catastrophic event in the real sense, but well describes the frustration of the physicists that before Planck tried to explain it using the concepts of classical physics.

The physical phenomenon that attracted so much interest was the so-called black-body radiation. Matter, as is easily observed experimentally, emits light radiation with a wavelength that is dependent on its temperature. Beginning at absolute zero, $T = 0$ in Kelvin degrees, the absolute temperature scale corresponding to $-273.5\,^{\circ}\mathrm{C}$, a body begins to emit electromagnetic radiation. The relative spectrum with the frequency distribution of radiation produced by glowing objects had been well mapped out by the late 1800s. The curves, known as blackbody radiation, had a peculiar shape that seemed very difficult to explain. The solution was not trivial and lay precisely in the modeled mathematical description of the blackbody spectrum, but to do this it was necessary to break through a conceptual barrier on which classical physics, that of continuity, had been built.

A blackbody is an idealized physical body that is capable of absorbing all incident electromagnetic radiation, regardless of frequency or angle of incidence [3]. The term "blackbody" comes precisely from its ability to absorb all colors of light. A blackbody not only absorbs but also emits blackbody radiation. The radiation produced by a blackbody results in an emission spectrum whose intensity depends only on the temperature of the body. Several scientists tried to find an explanation and a physical law that could fit the experimental results: Gustav Kirchhoff in 1860, followed by Josef Stefan in 1879 (*Stefan law*) [4] and Ludwig Boltzmann in 1884 (*Stefan–Boltzmann law*) [5], Wilhelm Wien in 1897 (*Wien approximation*) [6].

A further explanation for blackbody electromagnetic emissions was proposed by a pair of British physicists, John William Strutt Rayleigh (1842–1919) and James

Jeans (1877–1946). The resulting formula is known as the *Rayleigh–Jeans law*. Lord Rayleigh first laid the groundwork for the law in an article published in 1900 [7]. He derived the formula for the spectral energy density of blackbody radiation using classical mechanics. Rayleigh attempted to explain the distribution of energy in blackbody radiation using classical wave theory, Jeans expanded upon Rayleigh's work in 1905, incorporating additional considerations about the distribution of energy in blackbody radiation and deriving the same result independently [8, 9]. He provided a more detailed treatment of the problem, cementing the classical prediction for radiation at longer wavelengths.

Rayleigh and Jeans observed that in an ideal blackbody under thermal equilibrium conditions, energy is uniformly distributed among all possible energy states, and thus the frequencies of the photons produced could be well described by their thermal energy. The *Rayleigh-Jeans law* models the shape of the blackbody spectrum quite well for high wavelengths (low frequencies), up to infrared light. The problem, however, arises at shorter wavelengths. In the visible and ultraviolet regions, in fact, the intensities that were predicted, applying Maxwell's equations, were too high. Indeed even the intensity should eventually go to infinity with increasing frequency (Fig. 5.1). This is precisely what Ehrenfest defined as *Ultraviolet Catastrophe,* not a real catastrophic event but the result of models and interpretations built on the basis of classical physics.

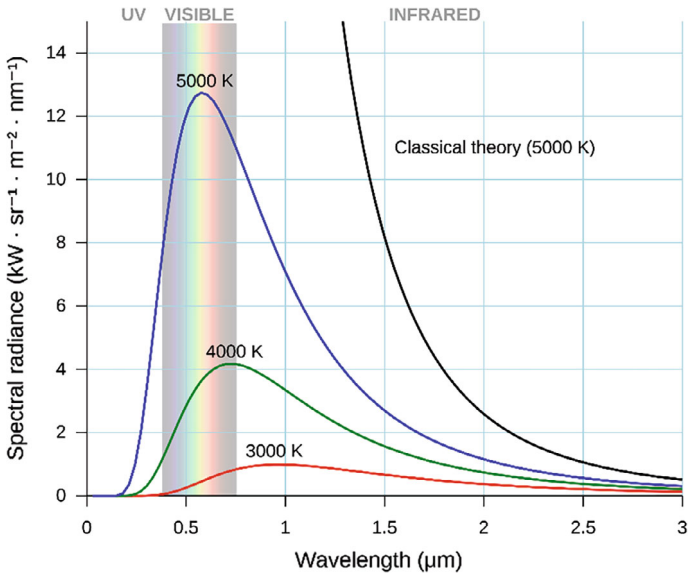


Fig. 5.1 Rayleigh-Jeans law, represented as "classical theory" in the graph, fails to describe for the energy emitted by an ideal blackbody. The error, which is much more pronounced for short wavelengths, is the difference between the black curve (as classically predicted by Rayleigh-Jeans law) and the blue curve (the measured curve as predicted by Planck's law) [10]

Experimental blackbody spectra have a peculiar pattern, the intensity of the emission increases toward higher frequencies and then decreases again; the maximum of the emission also shifts toward higher wavelengths with temperature. The problem was precisely inherent in the classical approach because according to classical physics any quantity can be divided practically infinitely without a defined upper limit. One can proceed by applying successive division processes always reaching a smaller magnitude. So according to Rayleigh-Jeans calculations, particles can vibrate any amount of energy down to an infinitesimal value. The attempt to mathematically distribute thermal energy equally through all possible energy states results in too much energy being condensed in a very small space, especially at higher frequencies, such as those of ultraviolet radiation.

A solution to the *Ultraviolet Catastrophe* was proposed by Max Planck in an article in 1900 [11]. If we look at the dates of the publications of Rayleigh [7], Planck [2], and Jeans [8], we see that Jeans continued to work on a classical model even after the article of Planck. He refused the quantum idea and was never convinced of the implications of the theory: "*so little progress has been made in this direction* (quantum mechanics) *that an attempt to report on it reduces mainly to a chronicle of conflicting opinions*."

It would, however, Planck's solution that will lead to one of the greatest revolutions in physics and a new worldview. Planck was, however, quite unaware of the implications of his model. Planck used the constant has a fit parameter for matching the experimental data of blackbody radiation and did not suggest any connection of quanta to the constant.

Planck started from a simple observation. The curves expressing the variation of energy as a function of temperature and wavelength have a variable maximum but a similar trend. The similar trend of the curves at different temperatures, therefore, seemed to suggest that there was a common factor. Planck tried to hypothesize that the vibrating particles could only do so at energies that were the multiple of a minimum energy. It is interesting to note how critical observation of experimental data was used by Planck to arrive at his deduction. What he had intuited was precisely the quantization of energy states by establishing a priori that these states could take on only certain values [12]. Planck then decided to take as a reference a minimum energy, that is, a very small number yet to be measured that would later become known as Planck's constant: "*…the distribution of a given amount of energy over the different colours of the normal spectrum using one constant of nature only* (Planck's constant h) *and after that the value of the temperature of this energy radiation using a second constant of nature* (the Boltzmann constant)." [2]. Planck's trick worked, and his equation, the *Planck equation,* was able to describe the shape of the blackbody spectrum very accurately at all frequencies of light, even at the highest frequencies. Planck's formula of radiance once the constants, Boltzmann constant and the new constant h, were inserted perfectly simulates the pattern of blackbody emission at various temperatures.[1]

[1] The formula identified by Planck makes it possible to calculate the spectral radiance $B(\nu, T)$ of a blackbody (W m^{-2} s r^{-1} mm^{-1}) as a function of frequency, ν, and absolute temperature, T, with

Having found the formula, Planck's own constant was still missing to be measured. This was accomplished by varying the constant until the mathematical law derived by Planck agreed perfectly with the observed spectrum. Planck used the letter h from *hilfsgrösse*, *auxiliary variable*, which suggests how he did not initially fully understand the implications and meaning of the constant he identified [4]. Indeed, it would later fall to Einstein to verify and extend Planck's insights into the quantization of energy states.

Returning to the blackbody emission curves, which express how radiance varies as a function of wavelength, these have a particular shape that is not so intuitive to explain and that depends on the temperature of the body. This specific physical property can be better understood if we consider the blackbody as a cavity within which light will bounce back losing energy with each reflection through successive absorptions. We can imagine the cavity as a box that has a small hole through which light can enter and will be trapped inside. The energy transmitted by the light to the walls will heat them and thus in turn they will emit electromagnetic radiation. If we consider the wavelike nature of light, the dimensions of the cavity impose specific boundary conditions on it given precisely by its size. Only certain wavelengths will be allowed, those whose displacement will be equal to zero with respect to the walls of the cavity. Such waves will be the ones responsible for blackbody emission. A blackbody, regardless of size, can be considered a system that limits the possible energy levels, which can take on only discrete values.

The solution Planck found was much more than a mathematical artifice to simulate an experimental curve. It was Einstein himself, as we anticipated, who fully understood what the actual implications of Planck's discovery were. His idea implied that light was quantized, that is, that vibrating particles of light could take on only certain values of energy. Planck's discovery was the basis for Einstein's hypothesis of the existence of a photon that was part wave and part particle and carried a quantum of energy equal to that of the frequency of a wave times Planck's constant, $E = h\nu$. Experimental proof was provided by the photoelectric effect and with its explanation provided by Einstein. Despite the growing evidence, both theoretical and experimental, in favor of the quantum mechanical model, Planck tenaciously tried to reconcile it with classical physics: "*My futile attempts to fit the elementary quantum of action somehow into the classical theory continued for a number of years, and they cost me a great deal of effort. Many of my colleagues saw in this something bordering on a tragedy.*" A nice definition given of Planck is that of "reluctant revolutionary." [13].

The value of Planck's constant in units of the international system is expressed in Joule seconds:

$$h = 6.62607015 \times 10^{-34} \text{J s}$$

h the constant introduced by Planck that will take its name from here, c the speed of light, and k_B the Boltzmann constant. $B(\nu, T) = \frac{2h\nu^3}{c^2} \frac{1}{\exp\left(\frac{h\nu}{k_B T}\right)-1}$.

And it is extremely small. When compared with the scale of the world described by the laws of classical physics, the one we use to describe phenomena on a "human" scale, it is an incredible small number. On the "human" scale, energies are of the order of kilojoules and time we calculate in seconds, a value of 10^{-34} appears infinitely smaller. When the product energy (J) and time (s) approach the value of Planck's constant, quantum effects govern the physical phenomena.

Planck's constant also sets the *Planck distance*, a truly special length because according to quantum mechanics and relativity it would also represent the minimum measurable distance:

$$\text{Planck distance: } l_{\mathrm{P}} = \sqrt{\frac{\hbar \mathrm{G}}{c^3}}$$
$$\ell_p = 1.616199256 \times 10^{-35}\mathrm{m} \tag{5.1}$$

this quantity is universal, since it is expressed in terms of the gravitational constant, G, which describes the magnitude of the gravitational force exerted by two masses, the speed of light c, which expresses the maximum speed at which communication can take place in space, and ħ the reduced Planck's constant, $\mathrm{h}/(2\pi)$, which as we have written relates instead the energy of light to the electromagnetic frequency.

Speed of light in vacuum:	c	Units: (distance)/(time)
Gravitational constant:	G	Units: $(\text{distance})^3/[(\text{mass})(\text{time})]^2$
Planck's constant:	$\hbar = h/(2\pi)$	Units: $\left[(\text{distance})(\text{length})^2\right]/(\text{time})$

Substituting the units of the three universal constants yields in the relation for ℓ_{p} the result is a length:

$$l_{\mathrm{P}} = \sqrt{\frac{\frac{\text{mass}\times\text{length}^2}{\text{time}} \times \frac{\text{length}^2}{\text{mass}} \times \frac{1}{\text{time}^2}}{\frac{\text{length}^2}{\text{time}^3}}} = \sqrt{\frac{\text{length}^4}{\text{length}^2}} = \text{length} \tag{5.2}$$

In practice, with all the limitations of the physics we know now, it is impossible, under the known laws of quantum mechanics and gravity, to make a measurement of a length with an accuracy less than the Planck length. It is the smallest scale in which the force of gravity can have an effect.

In addition to the Planck length, it is possible to define a complete set of quantities that define the so-called *Planck scale*, so you have a Planck *time* and a *Planck energy*.

$$\text{Planck time: } t_{\mathrm{P}} \equiv \sqrt{\frac{\hbar \mathrm{G}}{c^5}} \approx 5.391 \times 10^{44}\mathrm{s} \tag{5.3}$$

This time is taken by a photon traveling at the speed of light to travel a Planck length in vacuum, in practice the shortest measurable time.

$$\text{Planck energy: } E_P = \sqrt{\frac{\hbar c^5}{G}} \approx 1.956 \times 10^9 \text{J} \approx 1.22 \times 10^{19} \text{GeV} \tag{5.4}$$

Below these quantities of space, time, and energy, the physics defining the *Standard Model*, Quantum Field Theory, and general relativity lose their meaning. This would be the world of the Planck epoch, the initial moment of the Big Bang, about 10^{-43} s, where even the concept of time loses physical meaning and where there are no theories to describe what happened in the initial instants of the formation of the universe, before the advent of time thus below Planck time.

Below the Planck distance at such a small point size an enormous amount of energy could be stored so that gravity would become fundamental, a gravity that works on a scale that is quantum mechanical, this would be the famous quantum gravity.

5.2 The Constant of Mystery

Let us now leave Planck's constant to describe another of the physical constants that have intrigued generations of physicists. Constants pose a number of questions to which in most cases one must end up admitting that there is no answer, except in philosophical terms. The constants one encounters in the physical description of the universe are so many, understanding exactly what they mean and their nature is not simple and perhaps not even possible. Why do they take on that particular value, what is their meaning? These questions apply to all constants, but there is one in particular that has been and still is at the center of the doubts of generations of physicists, the *Fine Structure Constant*, denoted by the Greek letter α and worth about 1/137; a strange number with no apparent special meaning [14]. At first glance, this constant does not seem very different from the most important constants used in relativity and quantum mechanics, namely the speed of light, *c*, Planck's constant, h, and the gravitational constant, G.

There is something inherently different about the *Fine Structure Constant*, however, so much so that Paul Dirac called it "*The most fundamental unsolved problem in physics*," while Wolfgang Pauli put it at the top of the questions he would ask the devil: "*When I die the first question to the Devil will be: What is the meaning of the fine structure constant?*".

What is the origin of the fine structure constant? We have seen that the emission of an atom being excited occurs at very specific wavelengths that correspond to the energy levels of the electrons in the atoms. As measurement techniques progressed, however, it was verified experimentally that by going to observe the emission lines more precisely, finer structures could be defined. Each individual emission line appeared to be composed of two almost overlapping lines, this effect being due, apart from relativistic corrections, to the presence of electron spin. Sommerfeld discovered something else, that the difference in energy between these fine structure lines is always the multiple of a very special number, the square of the electron charge,

divided by two times the Plank constant, h, the electrical permittivity in vacuum, ε_0, and the speed of light, c.

$$\alpha = \frac{e^2}{2\varepsilon_0 hc} = \frac{e^2}{4\pi\,\varepsilon_0 \hbar c} \tag{5.5}$$

where

e: the electric charge ($1.602176634 \times 10^{-19}$ C).
h: Planck's constant ($6.62607015 \times 10^{-34}$ J Hz)$^{-1}$
$\hbar$: the reduced Planck's constant, $\hbar = h/2\pi$.
c: the speed of light (299,792,458 m s)$^{-1}$
ε_0: the electrical permittivity in vacuum ($8.8541878128\ldots \times 10^{-12}$ F m)$^{-1}$

If we consider the units of all the constants we can observe something peculiar, they all cancel out so α is a dimensionless constant whose numerical value is:

$$0.0072973525693 \simeq 1/137.035999084\ldots$$

$$\text{and finally:} \quad \frac{1}{\alpha} \simeq 137$$

The mathematical symbol $\simeq$ is used to indicate that the value 137 is not quite accurate; it comes close but does not correspond to a whole number. For now, the most accurate measurement of the fine structure constant performed in 2021 gives the value of 137.035999206 [15] with the accuracy of 81 parts per trillion [16].

Another debated question is how much this strange value has actually remained constant since the formation of the universe, because it is not even a given that the constants have not changed since the beginning of the history of the universe. According to some hypotheses, it could be that there was an instant when the value of the fine structure constant was exactly 137.

The *Fine Structure Constant* cannot be calculated but only measured, and it is one of the constants whose value is known most accurately. The implications of the dimensionless character are fascinating; the example that is most often used is that of communication with an alien civilization. In fact, if we wanted to converse at least scientifically with aliens we could start precisely from a mathematical constant such as π and a physical one such as α, both of which are purely numbers and as such do not depend on the different units of measurement used by earthlings and aliens.

The *Fine Structure Constant*, however, is not confined to atomic spectra and is encountered in other cases. Consider two electrons and measure the repulsive energy between them; they are in fact two particles with the same charge and will tend to repel each other with some force. We then take a photon with the same wavelength as the distance between the two electrons. Here the constant α reappears, in fact, the energy of the photon with that length will turn out to be precisely 137 times greater than the repulsive energy between the electrons.

There is a very elegant way to visualize the *Fine Structure Constant,* and it is through Feynman diagrams, used in *quantum electrodynamics* (*QED*). In the

diagram, the arrows represent two charged particles and the wavy lines represent photons (Fig. 5.2). When two electrons move following a trajectory that brings them closer, they will arrive at a distance at which they will exchange a photon and then repel each other. According to the QED model, it is precisely the exchange of the photon that causes attraction or repulsion. The two vertices of the Feynman lines represent the point at which the charged particles emit or absorb a photon. It is at this point that the constant α plays a key role, because it is the coupling constant that exactly determines the intensity of the interaction. If we rewrite the previous formula for α as:

$$4\pi\varepsilon_0\hbar c\alpha = e^2 \tag{5.6}$$

the correlation between the electron and the electromagnetic field whose intensity is given precisely by α becomes clearer. Another peculiar coincidence is that if we calculate the rotational speed of the electron in the first energy level in Bohr's model, the speed turns out to be 1/137 that of light.

Let us now take up the formula for Bohr's radius, which, as we described in Chap. 2, is a physical constant (Eq. 2.17):

$$a_0 = \frac{4\pi\epsilon_0\hbar^2}{\mathrm{m_e e^2}} \tag{2.17}$$

If we use Eq. (5.5) of the fine structure constant to rewrite Eq. (5.2.17), we get:

$$a_0 = \frac{\hbar}{\mathrm{m_e}c\alpha} \tag{5.7}$$

This shows that the Bohr radius is inversely proportional to both the fine structure constant α and the electron mass $\mathrm{m_e}$. Since the fine structure constant quantifies the strength of the electromagnetic interaction, a larger α would correspond to a stronger

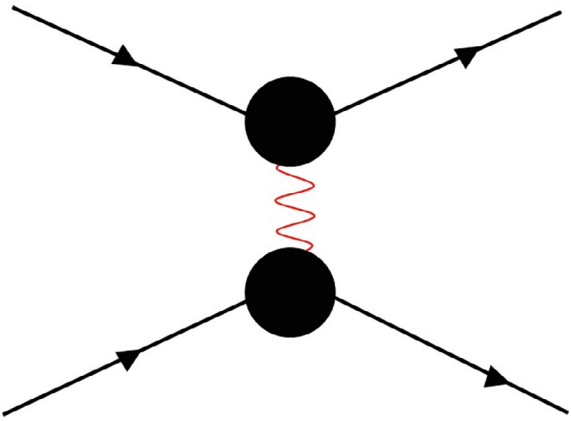


Fig. 5.2 Feynman diagram showing two electrons approaching, exchanging a photon, and moving apart

electromagnetic force, resulting in a smaller Bohr radius. On the other hand, a smaller α leads to a larger Bohr radius.

The surprises do not stop here, because if we recall the other two physical constants introduced in Sect. 2.8, the *Classical Electron Radius,* r_e, and the *Reduced Compton Wavelength,*$\lambda\!\!\!^{-}$, they are all related to the fine structure constant α:

$$r_e = \alpha\lambda\!\!\!^{-} = \alpha^2 a_0 \tag{5.8}$$

When considering length scales, Eq. (5.8) shows that the *Bohr radius* (a_0) is the largest of the three constants and is inversely proportional to α. The *classical electron radius* (r_e) is smaller than the Bohr radius by a factor of α and is directly proportional to the fine structure constant. Additionally, the *reduced Compton wavelength* is smaller than the Bohr radius, and the ratio between the Bohr radius and the reduced Compton wavelength is exactly α. Likewise, the classical electron radius is smaller than the reduced Compton wavelength by a factor of α.

There is no, or at least no known, reason why these specific properties can be expressed in terms of the constant α, and this has obviously stimulated much conjecture and questions, such as those posed by Feynman: "*There is a most profound and beautiful question associated with the observed coupling constant, e—the amplitude for a real electron to emit or absorb a real photon. It is a simple number that has been experimentally determined to be close to 0.08542455. (My physicist friends won't recognize this number, because they like to remember it as the inverse of its square: about 137.03597 with an uncertainty of about 2 in the last decimal place. It has been a mystery ever since it was discovered more than fifty years ago, and all good theoretical physicists put this number up on their wall and worry about it.) Immediately you would like to know where this number for a coupling comes from: is it related to pi or perhaps to the base of natural logarithms? Nobody knows. It's one of the greatest damn mysteries of physics: a magic number that comes to us with no understanding by humans. You might say the "hand of God" wrote that number, and "we don't know how He pushed His pencil. "We know what kind of a dance to do experimentally to measure this number very accurately, but we don't know what kind of dance to do on the computer to make this number come out—without putting it in secretly*!" [17].

So what lies behind this constant we do not know, but like most physical constants they seem to have set themselves to a particular value precisely because a small variation in this would make the universe impossible. Are they then the constants that determined the universe as we know it or are they instead a consequence of the formation of the universe itself that set its values? It looks like the classic chicken or the egg problem… If we consider the magnitude of the constant α, a slightly larger value would imply that electrons can get closer to the nucleus and thus being more bound would be less available for atomic bonds. Conversely, a smaller value would make atoms less stable and the formation of matter in the universe impossible. A small percentage change in α would make the formation of carbon atoms and thus carbon-based forms of life on earth impossible.

However, the danger of using these constants to construct a numerology even by theoretical physicists who should be well versed is always lurking. The number α as Planck's constant emerges from the need to find mathematical formulations that correspond to actual measurements. Looking in constants such as α and π for special meaning is an effort most likely destined to lead to fruitless results [18].

5.3 Another Strange Constant

The value of the *Fine Structure Constant* has been and still is, as seen, the torment and delight of generations of physicists who have tried and still try to grasp its meaning. There is yet another intriguing constant and that is the ratio of the mass of the proton to that of the electron, usually denoted by the Greek letter μ or β.

$$\mu = m_p/m_e = 1836.15267343\ldots$$

The value of this ratio cannot be derived through theoretical calculations but can only be measured. This constant, like the fine structure constant, is also dimensionless and could be used in a possible encounter with alien civilizations to establish contact. The mass ratio between protons and electrons, μ, is a fundamental constant, but several theories, such as the supersymmetry (SUSY), which hypothesizes the existence of symmetry between particles with integer spin (bosons) and particles with semi-integer spin (fermions), predict that the value of μ in the first moments of the creation of the universe was different, as is hypothesized in the case of the constant α.

A special aspect of this relationship is that it is obtained by using the values of the masses of an elementary particle, the electron, and a baryon such as the proton. The proton is not an elementary particle like the electron and is formed as we have seen by three quarks, just like the neutron. However, the proton and electron share an important property of being stable. A free neutron, in fact, has a half-life of only 613.9 s, while no decays have yet been experimentally observed for electrons and protons although according to theoretical calculations the proton should have a half-life of the order of 10^{23} years. It is probably their stability that is the property that makes a lepton like the electron and a baryon like the proton the particles that determine the chemical properties of the elements.

Even in the case of the ratio μ, the question of the constancy of that value is still being studied [19]. The *Standard Model* does not explain why some physical constants take on certain specific values, nor does it require that these remain constant in space and time. A change in the value of μ would have a direct effect on the strong nuclear force and would be no small effect.

If we consider the three constants we have discussed in this chapter, we can see that Planck's constant, h, like the gravitational constant, G, and the speed of light, *c*, are defined by dimensions. Can they be defined as *fundamental physical constants*? Probably strictly speaking this definition should be reserved precisely

for dimensionless constants such as α and μ, which cannot be derived. Therefore, a *dimensionless physical constant* is a pure number without units and with a numerical value independent of the unit system used.

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Chapter 6
A Dual and Quantum Nature



The discovery of the electron and its strange properties is the cornerstone of quantum mechanics. Based on their spin properties, photons and electrons, now categorized as fermions and bosons under the *Standard Model*, exhibit an enigmatic dual nature of wave and particle. This property, the interpretation of which is still debated today, is supported by various experiments, as we saw in Chap. 4. One of these, in particular, can be considered the basis of any interpretative model on the dual nature of photons and electrons, the double-slit experiment. This experiment will lead to the famous Copenhagen interpretation, which is still a key to explain quantum mechanics today. With all its incredible weirdness, the double-slit experiment is thus the basis of quantum mechanics, as we shall see in this chapter. Experiments and interpretations of the strange quantum properties of the electron have also led to the discovery of other physical phenomena seemingly impossible according to classical physics, such as the tunnel effect and quantum entanglement. These phenomena have no classical equivalent and can be explained only by resorting to quantum properties of electrons.

6.1 The Mother of All Quantum Mechanical Experiments: The Double Slit Experiment

If one had to single out one experiment to be defined as the mother of all quantum mechanical experiments, the *Double-Slit Experiment* (DSE) is probably the one. Feynman even called it *the greatest mystery in physics*. The DSE can help us better understand the electron's nature without the illusion of reaching definitive conclusions.

DSE was first performed by Thomas Young (1773–1829) in the early nineteenth century using sunlight. It was the first in a long series of experiments that are still performed in many laboratories worldwide for education and research purposes.

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P. Innocenzi, *What is an Electron?*, https://doi.org/10.1007/978-3-031-81984-1_6

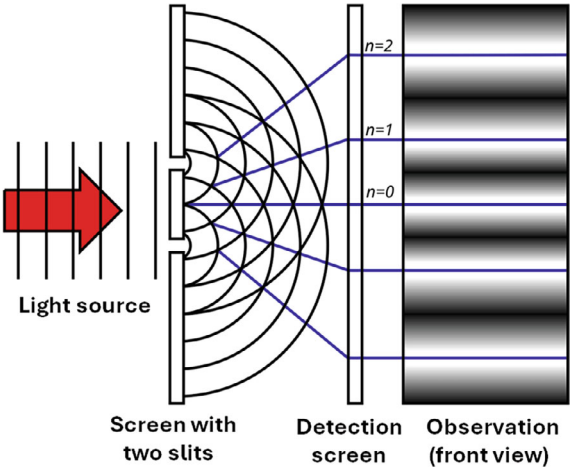


Fig. 6.1 Schematic of double-slit experiment carried out by Thomas Young [2]. The light source illuminates a screen with two small slits; the light passing through the slits creates on the second detection screen an interference pattern of dark and white bands

Since then, it has been repeated and perfected many times using not only light but particles, atoms, and molecules.

In Young's original experiment, a light beam is directed toward a screen equipped with two small slits, actually two small holes in the original version [1]. The light passing through the slits illuminates a second screen on the opposite side, creating a precise pattern of alternating light and dark bands (Fig. 6.1). The result is a demonstration of the wave nature of light; only constructive and destructive interference of waves can, in fact, explain the results.

By the time of Young's experiment, the nature of light had already been the subject of several experimental observations. Newton's idea that light consisted of small particles emitted by a luminous body and had no wave nature was still widely accepted. However, Newton's theory clashed with a growing body of experimental evidence that the particle model could not explain, such as light diffraction in correspondence of apertures and edges of materials. The diatribe over the nature of light and many other particles, including electrons, was just beginning.

The first double-slit experiment using an electron beam was performed by Claus Jönsson of the University of Tübingen in 1961, 160 years after the 1801 experiment by Thomas Young. According to Feynman: "*We choose to examine a phenomenon which is impossible, absolutely impossible, to explain in any classical way, and which has in it the heart of quantum mechanics. In reality, it contains the only mystery.*" [3].

In the DSE, a beam of electrons is fired toward a metal foil that has a parallel double slit (Fig. 6.2). The slits are very close to each other, the distance depending on the wavelength of the incident radiation and the position of the detector after the slits. The detector is placed immediately after the slits so that it can record what happens when electrons pass through the two apertures. If the electrons are, in fact, small particles, one should observe on the detector an image that mimics that of the slits. This is what can be expected to happen under the assumption that electrons, by their

nature, are something like a small charged sphere; that is, they have a corpuscular nature. However, the image that is produced on the detector shows something quite different and in some ways surprising, having in mind the idea of the electron as a particle (Fig. 6.3).

Instead of observing images corresponding to the specular projection of the slits on the detector (Fig. 6.3d), interference images form, which is an indication that the electron must be wave-like. If electrons are assumed to be waves, when they reach the slits, they will be transmitted as circular waves that can form constructive or destructive interference depending on whether they are in phase or not. When they are not in phase, they will lead to the mutual cancelation of the propagating waves; therefore, the image reproduced on the screen will correspond to the destructive interference of the dark troughs. The areas where the waves have summed constructively appear as white stripes. The two zones show that there are areas of the detector where more

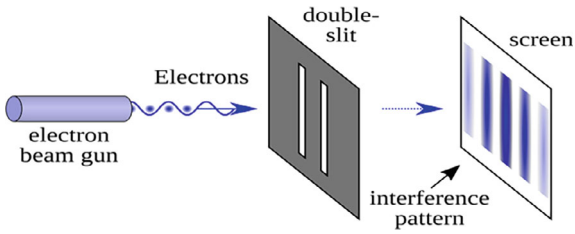


Fig. 6.2 Scheme of the double-slit experiment with electrons that produce an interference pattern [4]

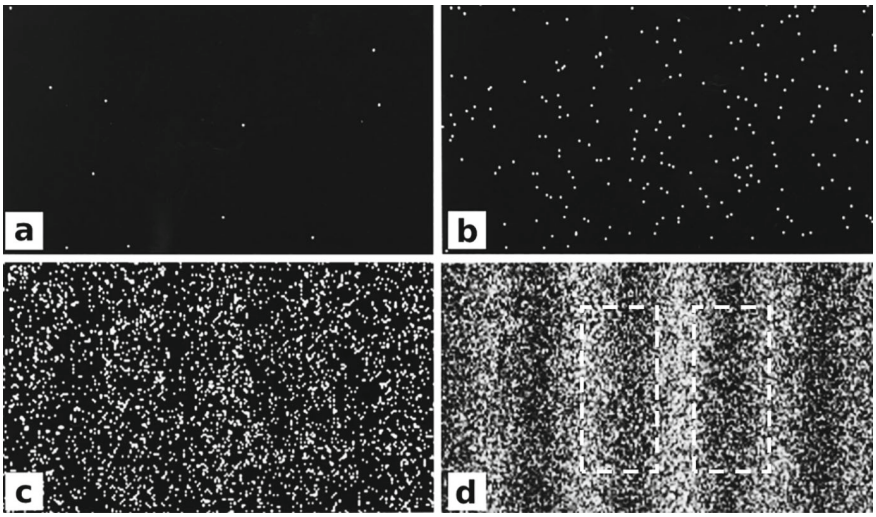


Fig. 6.3 Double-slit experiment. At the beginning only scattered electrons are observed on the detector: **a** 10 electrons. **b** 200 electrons. **c** Some patterns start to emerge as the number of scattered electrons increases (6000). **d** Stripe-like patterns are clearly observed after 140,000 electrons hit the detector; dashed lines show where the slit images would fall [5]

or fewer electrons have arrived; this effect is impossible to explain unless a wave-like nature of the electrons is assumed. The results perfectly coincide with Thomas Young's experiment in which a light source, hence photons, was used instead of electrons.

It all seems to add up, but … if you look closely at the interference images and enlarge them it seems that they are formed by a collection of small dots close to each other. So electrons are not really waves but particles? This observation has sparked one of the most intellectually stimulating debates among physicists about the nature of the electron.

One explanation is based on one of the counterintuitive anomalies of quantum mechanics. Until the electrons reach the screen, they behave as waves and thus can interfere, giving rise to the interference images. However, when they approach the screen and are measured, they change state and collapse to a specific point. The intriguing part is that the measurement of the wave changes the wave itself. This collapse does not happen randomly, as shown by the interference images, and electrons are more likely to be observed where the waves are weaker.

The explanation typically given for this experiment involves the dual wave-particle nature of the electron. However, this explanation might seem like a meta-magical illustration of the nature of the electron, which would have the property of choosing whether to behave as a wave or as a particle. May be the electron could *always* be a wave. What is observed on the detector would be a highly localized wave but still a wave. Anyway, all the interpretations, even if they sound extremely reasonable, should be taken with care, electrons are quantum objects…

How does it happen that the wave that describes the electron and is mathematically expressed by a probability density through the Schrödinger equation could have this strange property? The explanation given for this quantum phenomenon is the so-called *Collapse of the wave function*, which is closely related to what is referred to as *The Measurement Problem*. The impossibility of direct observation of this collapse has generated different interpretations of quantum mechanics. Each of these interpretations, however, leaves open several questions to be answered by each theory.

6.1.1 The Superposition Principle

According to the Schrödinger equation, the wave function describing the electron, or a generic quantum particle, evolves as a linear superposition of different states. This observation is the basis for another of the postulates of quantum mechanics, the so-called *Superposition Principle* (*Quantum superposition*), according to which a particle can be found in a superposition of different states. The Schrödinger wave equation provides a complete description of the state of a quantum system at any given time. Mathematically, the *Superposition Principle* follows from the property of the Schrödinger equation to be linear. This means that if we take two solutions of the wave equation, ψ_1 and ψ_2, any linear combination of this solution, ψ, is also a solution.

$$\psi = c_1\psi_1 + c_2\psi_2 \tag{6.1}$$

with c_1 and c_2 arbitrary complex coefficients that describe the probability amplitudes, which determine the likelihood of measuring the system in one state or the other.

This is also true if the wave equation has more than two solutions, any combination of these will be again a solution. Therefore, several states coexist at the same instant, yet when a measurement is made, the physical system assumes a well-defined state. The measurement plays a unique and crucial role in determining the state of the system, and any future wave function evolution will depend on this measurement. This means that the measurement is an event that "changes" the system, but what this "change" is, it is difficult to define. A system described by a wave function as a superposition of different states suddenly assumes only one well-defined state when a measurement is made. When the electron is measured on the detector, which "something" happens that causes the wave function to collapse, and on the detector, the electron, therefore, appears as a dot that would correspond to a highly localized wave. That "something" can be a photon emitted by the detector or any other entity that interacts with the electron for the measure.

The superposition and the probability of the resulting measures determined by the observation were described by Paul Dirac in a very clear and simple way: "*The non-classical nature of the Superposition process is brought out clearly if we consider the Superposition of two states, A and B, such that there exists an Observation which, when made on the System in state A, is certain to lead to one particular result, a say, and when made on the System in state B is certain to lead to some different result, b say. What will be the result of the Observation when made on the System in the superposed state? The answer is that the result will be sometimes a and sometimes b, according to a probability law depending on the relative weights of A and B in the Superposition process. It will never be different from both a and b. The intermediate character of the state formed by superposition thus expresses itself through the probability of a particular result for an observation being intermediate between the corresponding probabilities for the original states, not through the result itself being intermediate between the corresponding results for the original states. In this way we see that such a drastic departure from ordinary ideas as the assumption of Superposition relationships between the states is possible only on account of the recognition of the importance of the disturbance accompanying an Observation and of the consequent indeterminacy in the result of the Observation.*" [6].

From what we have seen, may be, electrons could in principle always be waves, or a cloud of negative charge, if we prefer, but what then is their nature? What are these waves made of? We know waves of different natures, sound waves, for example, or the waves we observe on the surface of the sea. This last example can actually help us visualize the concept of a field. It is precisely from the electromagnetic field, a kind of *continuum* that characterizes the universe, which we can imagine as a large liquid surface on which ripples form, that electrons are generated. That is why electrons are all the same, they are waves that propagate in the electric field and have identical nature. Each of the seventeen particles in the *Standard Model* has its own specific field; therefore, all of them are identical to each other for the same reason, according

to which electrons are identical. This is the *Quantum Field Theory*, which we will return to in Chap. 9.

6.1.2 Which Way?

Let us return to the DSE since there are many other interesting questions to be addressed. If we observe carefully at the experiment, it is missing an important piece of data that has not been defined, namely, in which slit the electron actually passed through. In fact, the observation of the interference figure is guaranteed only when no measuring apparatus is added to determine the path of the particle. If a sensor is used to follow the route of the electron through the slits, the experiment is defined as "which way." [7]. The result is very strange, it is the disappearance of the interference fringes. The wave-like property changes in favor of the corpuscular one. This effect motivated Niels Bohr to introduce the *Complementarity Principle* [8], according to which the two aspects, corpuscular and wave-like, cannot be observed simultaneously since they are mutually exclusive. The experiment determines the properties of the particles involved in it.

The *which way* experiment is extremely intriguing; if we try to measure where the electron goes, it behaves like a particle, and the interference fringes on the detector disappear. This happens regardless of where the detector is located, before or after the two slits. The electron "notices" that something is observing it, and the wave function collapses.

What does the detector look like that determines which pathway the electron passes through? It could be a photodiode that, however, must necessarily emit a photon to determine the passage of the electron, somehow interfering with it. The measurement node is again crucial here, and the discussion is far from exhausted. In this regard, Feynman's observation is as follows: "*In our experiment [which way double slit experiment] we find that it is impossible to arrange the light in such a way that one can tell which hole the electron went through, and at the same time not disturb the pattern. It was suggested by Heisenberg that the then new laws of nature could only be consistent if there were some basic limitation on our experimental capabilities not previously recognized. He proposed, as a general principle, his uncertainty principle, which we can state in terms of our experiment as follows: "It is impossible to design an apparatus to determine which hole the electron passes through, that will not at the same time disturb the electrons enough to destroy the interference pattern." If an apparatus is capable of determining which hole the electron goes through, it cannot be so delicate that it does not disturb the pattern in an essential way. No one has ever found (or even thought of) a way around the uncertainty principle. So we must assume that it describes a basic characteristic of nature.*" [3].

The "which way" double-slit experiment represents a challenge to our classical idea of particle as a physical entity. The experiment indicates that the trajectory or position of a particle remains undefined until they are observed. The result of the

measure depends on the experimental physical setup but no matter how accurate the measurement system is the determination of the particle's path is limited by the uncertainty principle.

6.1.3 Single Electron and a Double Slit

So if we were able to shoot one electron at a time through the slit what would we observe? We have the answer, because the experiment performed with single electrons shows us unequivocal results.

The single electron shows up as a dot on the detector, and as long as the number of electrons is relatively small, say around two hundred, they are still randomly distributed on the screen. However, if more single electrons continue to be fired in succession, the interference fringes slowly begin to appear. Once again, we can think of the electron as a wave; in fact, regardless of whether it can make constructive or destructive interference, it always shows interference fringes. The result of the experiment with the single electron shows unequivocally the dual nature of the electron. Since only one electron passes through the slit at a time, electron–electron interactions can be ruled out. When the single electron, as a wave, arrives on the target, we again have the *Measurement Problem* and the collapse of the wave function for which the point is observed. Provided that this remains only one of the possible explanations. Let us, therefore return to the question of the dual wave-particle nature and let us be helped once again by Feynman, who, in this regard, had also formed an opinion: "*Newton thought that light was made up of particles, but then it was discovered that it behaves like a wave. Later, however (in the beginning of the twentieth century), it was found that light did indeed sometimes behave like a particle. Historically, the electron, for example, was thought to behave like a particle, and then it was found that in many respects it behaved like a wave. So it really behaves like neither. Now we have given up. We say: "It is like neither. There is one lucky break, however—electrons behave just like light. The quantum behavior of atomic objects (electrons, protons, neutrons, photons, and so on) is the same for all, they are all "particle waves," or whatever you want to call them. So what we learn about the properties of electrons (which we shall use for our examples) will apply also to all "particles," including photons of light.*" [9].

An interesting aspect of the double-slit experimental setup is that this can be used not only for leptons as electrons or bosons as photons but also to test quantum properties using molecules, which are relatively much larger in size. The experiment was replicated using fullerenes, which are molecules formed by 60 carbon atoms that bond together to form a hollow spherical structure [10]. In addition to fullerenes, the double-slit experiment was thus repeated not only with electron [11, 12], but also atoms [13], molecules [8], and neutrons [14]. In 2019 a new record was achieved by demonstrating the dual nature of a molecule formed by 2000 atoms [15]. In all these cases, the formation of interference figures was observed as a further demonstration of the dual nature of these systems.

6.2 The Principle of Complementarity

The double-slit experiment still poses a great challenge to our ability to understand quantum mechanics and its implications. One of the interpretative models was provided by Bohr by the *Principle of Complementarity*. The principle was formulated by Niels Bohr in 1927 and represents a cornerstone of quantum mechanics. Bohr introduced the general idea that particles, such as electrons or photons, have complementary properties that cannot be observed or measured simultaneously. These properties include the wave- and particle-like nature, which are mutually exclusive but equally essential for fully describing quantum phenomena.

Complementarity implies that the way a system is observed determines the type of property that will manifest. For instance, in the DSE as we have seen, when a photon or electron passes through the slits, it behaves like a particle, appearing at a definite location on a detector. However, when no observation is made, the same system behaves like a wave, creating an interference pattern. Bohr argued that both wave and particle descriptions are valid but not simultaneously. Depending on the experimental setup, the complete nature of quantum systems can only be understood by acknowledging these complementary perspectives. The complementary principle extends beyond wave-particle duality, applying to other pairs of quantum properties, like position and momentum, described by Heisenberg's uncertainty principle. The hypothesis of complementarity introduced by Bohr thus reflects the limits of classical concepts in fully describing quantum systems. The principle has profound philosophical implications, particularly in how it challenges classical notions of reality, objectivity, and the limits of human knowledge. The classical idea of an independent, objective reality, where objects always possess well-defined properties, may not apply to quantum systems because their properties depend on how they are observed. Reality is not absolute but is shaped by the context of observation. This consideration raises questions about the nature of reality itself: Is there a "reality" independent of our measurements, or is reality only defined through interaction with an observer?

The principle also challenges the notion that we can fully know or describe a system in all aspects simultaneously. The properties of waves and particles in quantum mechanics are complementary but mutually exclusive, meaning we cannot simultaneously have complete knowledge of both. This limitation highlights the boundaries of knowledge and suggests that our comprehension of nature is inherently partial and context-dependent. The classical view of science as progressing toward an all-encompassing, complete description of reality, is in strong contrast to this interpretation of the quantum systems. In classical physics, determinism rules, meaning that given complete knowledge of a system, one could predict its future evolution. The principle of complementarity, along with quantum mechanics as a whole, questions this deterministic view. The act of measurement affects the system, and we can never predict both a particle's position and momentum with perfect accuracy, as described by Heisenberg's uncertainty principle. The inherent uncertainty introduces an element of fundamental indeterminism in nature that is still widely

debated, raising questions about free will, causality, and the predictability of the universe. It is the fascination and strangeness of quantum mechanics.

6.3 The Copenhagen Interpretation

Having come to this point in describing the properties of an electron and the different interpretations of its nature, which is closely related to quantum mechanics, it is important to summarize the famous *Copenhagen interpretation* (CI), which provides a key to understanding the paradoxes of quantum mechanics. The theory was proposed in 1925 by two of the founders of quantum mechanics, Niels Bohr and Werner Heisenberg. Bohr, formulated his ideas that would become central to this interpretation while working at the *University of Copenhagen's Institute for Theoretical Physics*, which he founded in 1920, and for this reason is named Copenhagen interpretation (Fig. 6.4). Bohr's theoretical approach is at the ground of one of the earliest interpretations of quantum mechanics and still one of the most popular.

Bohr believed that the act of observation plays a critical role in defining physical reality at the quantum level. According to Bohr's interpretation, a quantum system does not have definite properties, such as position or momentum, until they are measured. The dual nature wave-particle is context-dependent; the quantum properties depend on how they are observed or measured. Particles can exist in a state of superposition, meaning they can be in multiple possible states at the same time. Before measurement, a quantum system is described by a wave function that contains all the possible outcomes. When a measurement is made, the wave function collapses into one specific state, and the other possibilities vanish.

One of the most controversial aspects of the CI is the role played by the observer. The act of observation or measurement causes the collapse of the wave function; therefore, quantum systems do not have definite properties, such as position or momentum, until they are measured. Prior to observation, quantum objects exist in a range of probabilities, not in defined states. Niels Bohr emphasized that there is no meaningful distinction between the observer and the system in quantum mechanics. Reality, as it is intended in quantum phenomena, cannot be separated from the measurement.

The *Principle of Complementarity*, as we have previously described, states that the wave and particle aspects of quantum objects are complementary but mutually exclusive. The system can be interpreted as either a wave or a particle as a function of the experimental setup, but both properties cannot be observed simultaneously. This principle goes beyond particles and waves, indicating that different descriptions of quantum phenomena are complementary rather than contradictory. The CI emphasizes the probabilistic nature of quantum mechanics. Rather than predicting exact outcomes, it predicts probabilities of various possible outcomes. This is reflected in Heisenberg's uncertainty principle, which asserts that certain pairs of properties, like position and momentum, cannot both be precisely known at the same time. This is not due to measurement limitations but is a fundamental aspect of quantum systems.

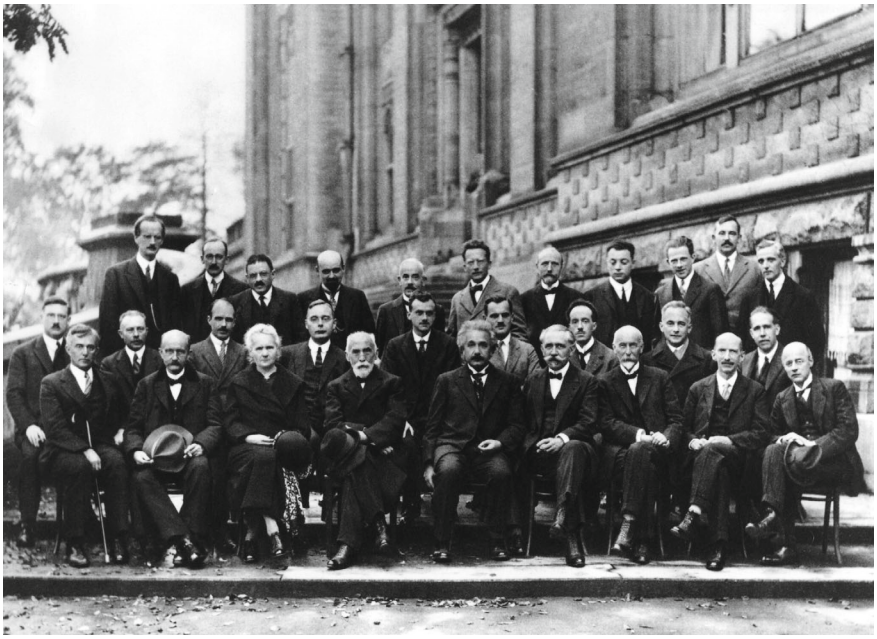


Fig. 6.4 The Solvay conference 1927 picture of the participants. This is the most amazing picture of the story of the physics. Most of the greatest scientists of all time are in the image. Front row from left to right: Irving Langmuir, Max Planck, Marie Curie, Hendrik Lorentz, Albert Einstein, Paul Langevin, Charles-Eugene Guye, Charles Wilson, Owen Richardson. Second row: Petrus Debye, Martin Knudsen, William Lawrence Bragg, Hendrik Kramers, Paul Dirac, Arthur H. Compton, Louis de Broglie, Max Born, Niels Bohr. Back row: Auguste Piccard, E. Henriot, Paul Ehrenfest, Edouard Herzen, Theophile de Donder, Erwin Schrodinger, J. E. Verschaffelt, Wolfgang Pauli, Werner Heisenberg, Ralph Howard Fowler, Leon Brillouin

It is a point that has repeatedly described in the book but it is really important to stress.

The CI draws a distinction between the quantum world and the classical world. Quantum mechanics applies to systems like particles and atoms, while classical mechanics is used to describe the macroscopic world. Bohr, again, with his *Principle of Correspondence* states, however, that the classical physics and quantum mechanics are only tools to describe in a different way the physical world, and within these dimensional limits there is a correspondence. The wave function collapse occurs when a quantum system interacts with a classical measuring device, which is why everyday objects don't exhibit quantum superpositions.

It must be underlined that Bohr and Heisenberg both supported an epistemological approach rather than an ontological one. In other words, quantum mechanics is about what we can know and measure, not necessarily about the true nature of reality. They were cautious about claiming what quantum objects "really are" when unobserved. In this sense, the CI is agnostic about whether quantum systems have definite properties before measurement. The CI assumes that during observation there is *something* away

that causes the wave function to collapse. The nature of this something has been much debated and still the debate is open. In general the success of the CI is based on the fact it works well mathematically so it is best to avoid lengthy discussion about the nature of this something, *Shut up and calculate!*, [16] as Feynman used to say about it.

The CI is not without its critics. Some physicists and philosophers have found the idea of wave function collapse and the role of the observer unsettling, as it suggests that reality is incomplete or undefined without measurement. On the other hand, the CI is not the only theoretical tool to describe the quantum mechanical world. Other interpretations have been built, such as Many-Worlds Interpretation, Pilot Wave Theory, and Objective Collapse Theories. They have emerged as an alternative to address the puzzling mysteries of quantum mechanics.

Philosophy and Physics

The development of quantum mechanics and the scientific revolution that followed occurred in Europe's first three decades of the 1900s. The European scientists who were key players in this new scientific approach had been trained in the European academic environment still strongly dominated by a theoretical-philosophical approach. This led to accompanying experiments with profound reflections on their nature and philosophical implications. Quantum mechanics is based on a set of principles, mainly the result of theoretical reflections that require a logical deductive ability capable of taking the analysis of the problem to a higher level, which only a humanistic philosophical culture could provide. Another extraordinary aspect of those years was the emergence of scientific schools where scientists were trained and, at the same time, the great circulation of ideas, leading to even bitter but fruitful debate. All this gradually faded away as the ominous shadows of the racial laws and World War II thickened. Many of Europe's brightest minds, such as Enrico Fermi, Albert Einstein, Max Born, emigrated to the United States. Largely then, they were involved in the Manhattan Project, a real race against time without room for philosophical musings. A nice article by David Kaiser in Nature well describes this new approach (Kaiser, D. History: Shut up and calculate! *Nature,* 2014, **505**, 153–155): "*This war-forged pragmatism produced enormously impressive research and influenced a generation of leading scientists. Their approach to basic research, and the institutions in which they pursued it, assumed an aura of inevitability. But the approach came with some trade-offs, largely unnoticed at the time. Important questions that resisted the powerful, phenomenological methods tended to get eclipsed. Anything that smacked of 'interpretation', or worse, 'philosophy', began to carry a taint for many scientists who had come through the wartime projects. Conceptual scrutiny of foundations struck many as a luxury. The wartime style was reinforced in the United States by exponentially rising university enrolments after the war. The new classroom realities left little space for informal*

discussion of philosophy or foundations. The Rad Lab rallying cry of "Get the numbers out" shaded into "Shut up and calculate!" The history of physics, however, did not end with the generation of pragmatism, the exploration of the many still unknown phenomena of nature requires an open mind capable of imagination and the ability to solve complex problems in which speculative thinking can also play a key role.

6.4 Quantum Tunneling, Electrons Cross Mountains

Quantum physics is characterized, as seen, by a number of oddities that have no analogue in classical mechanics. The winners of a hypothetical championship of quantum weirdness's would have to be two absolutely amazing phenomena: *entanglement* and *quantum tunneling*. Neither of these effects has a classical analogue, and although they are counterintuitive and difficult to explain, like much of quantum mechanics, they have been used to develop many practical applications.

The dual nature of the electron and the need to use quantum mechanics to describe its properties are closely related to the impossibility of making predictions about the outcome of an electron's measurement and behavior. What happens if an electron encounters a barrier? Given its quantum nature, some strangenesses can be expected. Let us start with the "classical" case, a particle, but we can also imagine a very common macroscopic object, such as a ball, moving at a constant speed and not interacting with the surrounding world (Fig. 6.5). Since we are dealing with a macroscopic object, we have no problem in determining position and momentum at every instant; we can use Newtonian mechanics without any problems. In classical physics, in fact, particles follow "deterministic" trajectories that are described by the classical laws of motion. The position of the particle is precisely defined as a function of time.

If, however, we consider a particle of such a small dimension that can be described as a quantum object, things become more complicated in this case. Suppose that

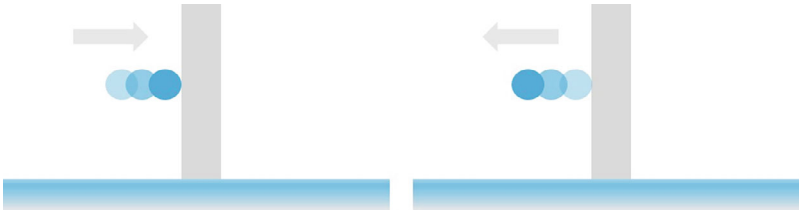


Fig. 6.5 A ball hits a physical barrier and is repelled by it. The phenomenon is described using the equations of motion of classical physics

the position of this particle is initially known fairly well. The probability distribution containing the information about its position will be maximal. So, in the case of quantum mechanics, there is a maximum probability distribution for locating the particle, whereas, in classical mechanics, the information specifies the precise position.

If the particle is in motion, in the classical case, its motion corresponds to a change in position that can always be uniquely determined. In contrast, for the quantum particle, according to the Schrödinger equation, it will be the maximum of the probability distribution of the wave function that moves at a constant speed as it does for the classical particle. One effect of the quantum particle's motion, however, is also to cause the probability distribution that defines its position to widen. That is, we always know with less certainty what the particle's position might be. This is nothing strange; it is an effect of *Heisenberg's indeterminacy principle*. If we initially knew with very good probability of the position of the particle, the probability of determining its momentum was much less. As we instead know with increasing precision the particle's momentum, we lose information about its position.

Let us now delve into the details of what happens to the two particles, the classical and the quantum one, when they encounter an obstacle. The first important step is to define what is meant by an obstacle; let us say right away that it will have a different nature for the two particles. In the case of the quantum particle, the obstacle is thought of as a potential barrier, which is a barrier that is energetically impossible to overcome. For the classical one, we can safely think of a wall. So, imagine shooting a particle like a ball toward a wall at a fixed angle (Fig. 6.5). The ball will bounce and return unless it has enough energy to destroy the wall like a projectile, but it will certainly not be able to pass through it, leaving it physically intact. Still remaining with the case of the macroscopic particle, our ball, let us now imagine the physical barrier as a mound (Fig. 6.6).

If the kinetic energy is large enough, the ball will be able to go over the mound and get to the other side otherwise it will bounce and go back. So there is a threshold energy that separates the two possibilities, the ball either goes over the barrier represented by the mound or is repelled by it, there are no other options and it will all depend on the height of the mound and the energy of the ball.

What if, instead of a "classical" ball, we consider a particle whose dimensions are such that it behaves as a quantum object? The first consideration is that the quantum particle would be, by its definition, dual, i.e., wave and particle, and some strange

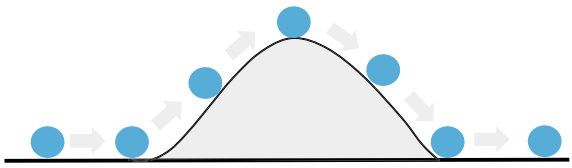


Fig. 6.6 A ball encounters a physical barrier, and if it has enough kinetic energy it can climb over it. According to classical mechanics, particles that do not have enough energy to climb over the obstacle cannot reach the other side

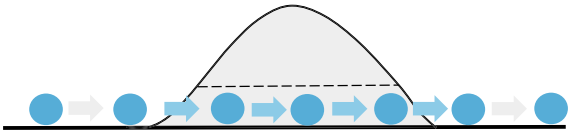


Fig. 6.7 Quantum tunneling allows an electron with an energy lower than that of the barrier to go through and appear on the other side

properties would potentially be connected with that property. In simulating the effect of a quantum particle colliding against a barrier, we use the concept of a potential barrier, which can be considered for all intents and purposes as a virtual energy wall defined by a mathematical step function that is zero everywhere except in a specific finite interval.

In the case of the quantum particle we can take the electron itself as an example. As we know in the realm of quantum mechanics, particles such as electrons do not behave like classical particles and exhibit wave properties that are described by wave functions. The interaction of the electron wave-particle with the barrier, taking into account precisely the quantum properties, holds some surprises.

Consider again the probability distribution of the moving electron; if its energy is much higher than the potential energy wall, the electron will pass the barrier, similar to the classical particle. Like in the classical case, if the electron's energy is too much smaller than that of the barrier, it will turn back. The intermediate case is the one that holds a surprise for us, and it is the one in which the electron's energy is smaller than the potential barrier, which, however, is not too high. In this case, a classical particle would simply bounce back, but in the quantum case, a part of the wave function manages to pass to the other side, even though this would be energetically forbidden. In fact, the electron collides with the barrier at a lower energy than the maximum of the barrier itself. Imagine having to climb over a wall; if it were four meters high, we would surely give up. Otherwise, if the wall were one meter high, we would attempt to climb it, but simply walking through it as if it weren't there would be a different story. Things go exactly this way for the electron: quantum effects allow it to get to the other side when, in reality, it would be impossible (Fig. 6.7).

The key point is that when the electron encounters an energy barrier, its wave function does not abruptly collapse to zero at the edge of the barrier, as classical physics might suggest. Instead, the wave function gradually decreases, and there is a very small but finite probability of finding the particle on the other side of the barrier. This phenomenon is another consequence of *Heisenberg's indeterminacy principle.* Let us assume that we know the particle's position, with the probability of placing it equal to one. According to Heisenberg's relation, this implies that its velocity should be infinite, an eventuality that we can discard as physically impossible. Therefore, the chance that the particle could be beyond the potential barrier will be very small but not zero (Fig. 6.7). There will always be some uncertainty in the particle's position, which may be "scattered" in space as described by its wave function.

During the motion of the electron, the wave function describing it evolves and expands. Its amplitude at any point in time and space makes it possible to calculate the probability of finding the particle at that time, if a measurement is made. By definition, this probability can be nonzero at many points at once. If the particle

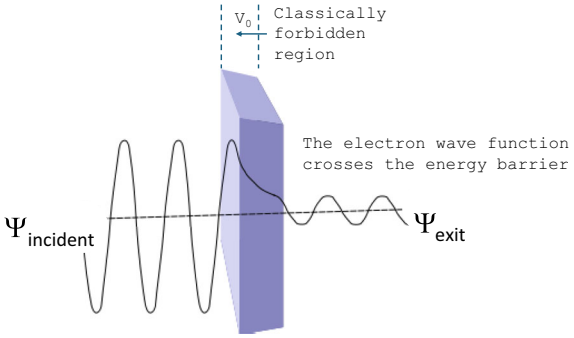


Fig. 6.8 A particle that obeys the laws of classical mechanics and has an energy E, which is less than the maximum energy of the potential barrier, V_0, cannot penetrate or pass the barrier that is classically forbidden. In the case of a quantum particle described by a continuous wave function Ψ_{incident}, it will undergo exponential decay at the barrier and on the opposite side the Ψ_{exit} function will be very small but nonzero. As long as the probability is nonzero there will be a possibility of finding the electron even beyond the barrier

collides with an energy barrier this encounter changes the amplitude of the wave function, which begins to decay exponentially within the barrier. However, a part of it can pass it, and its amplitude does not zero on the opposite side of the barrier. Thus, however small, there remains a finite probability of detecting the particle beyond the barrier.

Quantum tunneling, therefore, occurs when the wave function of a particle extends into a region beyond an energy barrier (Fig. 6.8). Even if the particle does not have enough energy to cross the barrier according to classical physics, there is a finite probability of finding it on the other side. This probability decreases exponentially with thickness, particle size, and barrier height, which means that the thicker and higher the energy barrier, the less likely tunneling becomes. In addition, smaller particles are more likely to pass the barrier. The phenomenon has an inherently probabilistic nature, the probability of which decreases exponentially as the height of the potential barrier increases.

In the process of tunneling through the potential barrier whose height exceeds its total energy, the electron behaves exactly like a wave. In the region just after the barrier the electron is again a physical particle that can be located at a certain position. Before and after the barrier is crossed, therefore, the electron can be regarded as a particle whose total energy is less than the barrier itself before and after it is crossed. Particle-wave-particle, the magic of the electron (Fig. 6.9).

6.4.1 Faster Than Light?

The consequences of the tunneling effect may go beyond the mysterious crossing of energy barriers by an electron. One intriguing aspect is that if one compares the wave

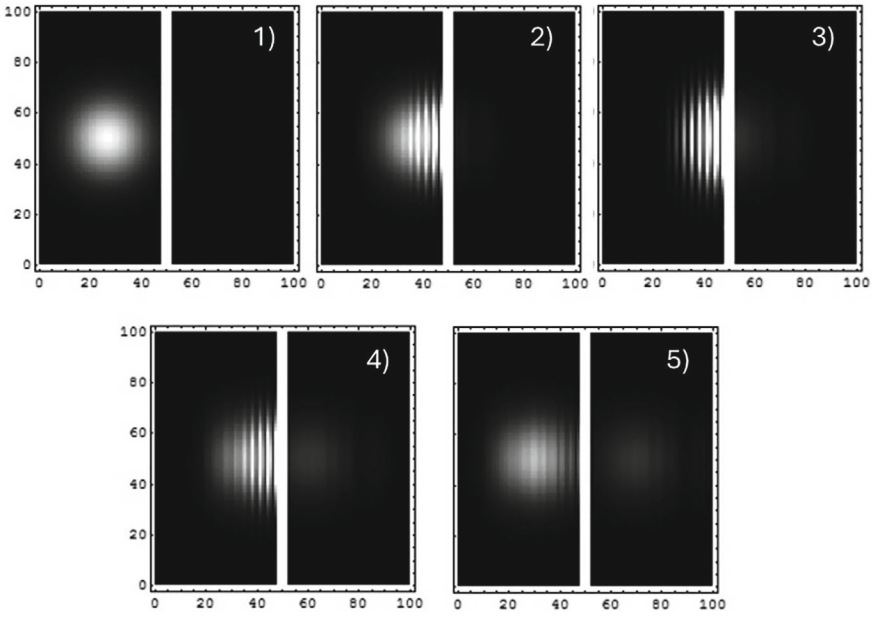


Fig. 6.9 Picture shows the evolution of the wave function through a potential barrier, represented by the vertical white line. A nonzero part of the wave function passes through the barrier so there is a nonzero probability that the electron is located beyond the barrier. Calculated with Mathematica, using the Crank-Nicolson method (finite difference method). Author Jean-Christophe Benoist [17]

function of an electron that has crossed a potential barrier with that of an electron that has traveled undisturbed, one observes something strange.

The peak of the part of the wave function that passed through the tunnel covered a greater distance at the same time. If the electron traveled at or very close to the speed of light, then the peak of the part of the wave function that crossed the barrier by the tunneling effect should have moved faster than the speed of light [18]. An experiment performed with photons showed that the initial limits of the wave functions of the photon that traveled unimpeded and the tunneling photon reached their respective detectors at the same time; therefore, there is no violation of Einstein relativity. The experiment showed that it is the peak of the wave function that can be faster than light but this does not imply that information or energy travels faster than light.

6.5 Quantum Entanglement

One of the strangest and most perplexing to intuitively understand phenomena of quantum mechanics is the famous quantum entanglement or quantum correlation. According to Erwin Schrödinger, this strange property is the hallmark of quantum mechanics that sets it apart substantially from the thinking of classical physics. This

phenomenon is also the basis for possible developments in quantum computing and quantum cryptography.

Quantum entanglement was first hypothesized during the famous Solvay Conferences in the 1920s and 1930s and was the subject of great discussion between Bohr and Einstein. In 2022 finally, the Nobel Prize in Physics was awarded to three physicists, Alain Aspect, Joseph F. Clauser, Anton Zeilinger, for: "*For experiments with entangled photons, establishing the violation of Bell inequalities and pioneering quantum information science.*"

As we have seen, one of the fundamental properties of the electron is its spin. When we measure an electron's spin, it will have a specific value, and if we correlate two electrons, their spins will always have opposite directions. If the first electron has a spin *up*, the second one will necessarily have to be *down*. However, remember that up and down have no particular meaning and are only linguistic artifacts that indicate two different electronic states indicated as spin.

A fairly representative example used to depict the specificity of spin properties is that of the pair of gloves. One of the gloves is right-handed, and the other is left-handed. We cannot think of using two left or right-handed gloves; in fact, our hands are oriented in a mirror-image manner. Once two electrons are correlated so that the spin of one is *up* and the other *down*, they can be separated until they are a great distance from each other, but without being able to determine their spin, not at least until a measurement is made. In this case, if the spin of one electron will be *up*, the spin of the other will be *down*. Apparently, there is nothing strange in this. Suppose two astronauts were given two coins, one dollar, and one euro, before leaving on an interstellar journey. The astronauts, however, can only look at the coin once a special message is sent to them, so they do not know which coin they received at departure. During their journey, at one point, they simultaneously receive the same message telling them that they can finally verify which coin is theirs. At the same instant, when they discover their own coin, they will also immediately know what the other astronaut's coin is. Analogous example can be made using the left and right pair of gloves.

It is important to note that the measurement does not change the state of the coin or glove pair; it simply allows one to know which is the other instantly. However, there is an important difference in the case of electronic spin. When we consider the coin or glove type, this is always the same from the instant they are separated; the measurement does not change its state, it merely allows us to know it. In the case of the electron the situation is different, in the initial state it is not defined whether the electron is with spin up or down, it is in fact in both states, it is only the measurement that defines it, thus simultaneously determining the state of the other electron as well. This occurs because the electron, as shown by the double-slit experiments, is in a superposition of states. The quantum state of entangled particles is described by the same wave function.

If the electron behaves like a wave, it is the measurement that causes the wave function to collapse and locate it at a defined position. In the case of the electron pair, the situation is even stranger since the measurement would seem to simultaneously cause the state of both to be defined. The measurement determines the *up* or *down*

state of one electron and forces the other electron to assume a correlated state with opposite spin.

The distant correlation of the two electrons, a consequence of the quantum superposition of states, has a profound implication because it is closely related to a peculiar property namely, the *nonlocal* character of quantum physics. This property is not easy to explain and in fact has raised quite a few doubts since the beginning of the development of quantum mechanics.

6.5.1 The EPR Paradox

The difficult interpretation of *quantum entanglement* raised many perplexities even in Albert Einstein, who in 1935, together with two young scientists Boris Podolsky and Nathan Rosen proposed what would come to be known as the *EPR Paradox,* from the first letter of the last name of the three scholars. For *Einstein, quantum entanglement* was a "*spooky action at a distance.*" [19]. In the abstract of their article the three authors clearly refuse the idea that the formulation of quantum mechanics that was emerging was complete: "*In a complete theory there is an element corresponding to each element of reality. A sufficient condition for the reality of a physical quantity is the possibility of predicting it with certainty, without disturbing the system. In quantum mechanics in the case of two physical quantities described by non-commuting operators, the knowledge of one precludes the knowledge of the other. Then either (1) the description of reality given by the wave function in quantum mechanics is not complete or (2) these two quantities cannot have simultaneous reality. Consideration of the problem of making predictions concerning a system on the basis of measurements made on another system that had previously interacted with it leads to the result that if (1) is false then (2) is also false. One is thus led to conclude that the description of reality as given by a wave function is not complete.*"

The *EPR Paradox* was formulated not so much to show how strange *entanglement* was but rather precisely to demonstrate the fundamental incompleteness of quantum mechanics. The fact that the two correlated particles are affected by the measurement simultaneously, without a measurable time interval, would imply the violation of the postulate of special relativity, for which no communication beyond the speed of light is possible. Einstein's explanation was that there must be hidden variables that fixed spin orientations. The impossibility of measuring such variables limits the possibility of knowing the state of the spin, but such variables determine it anyway.

This idea was rejected by Niels Bohr, who asserted that hidden variables did not exist and that one simply had to accept *quantum entanglement* as a real physical property; it is as it is. The diatribe raised by the EPR Paradox was then resolved in 1964 by Irish physicist John Bell with his *Bell's theorem* [20]. Bell's theorem is also known as Bell's inequality because he derived a mathematical constraint that must be satisfied by any local hidden variable theory. This inequality sets precise limits on the correlations between measurements made on entangled particles, assuming that local realism, no faster-than-light communication, is satisfied. Several experiments have

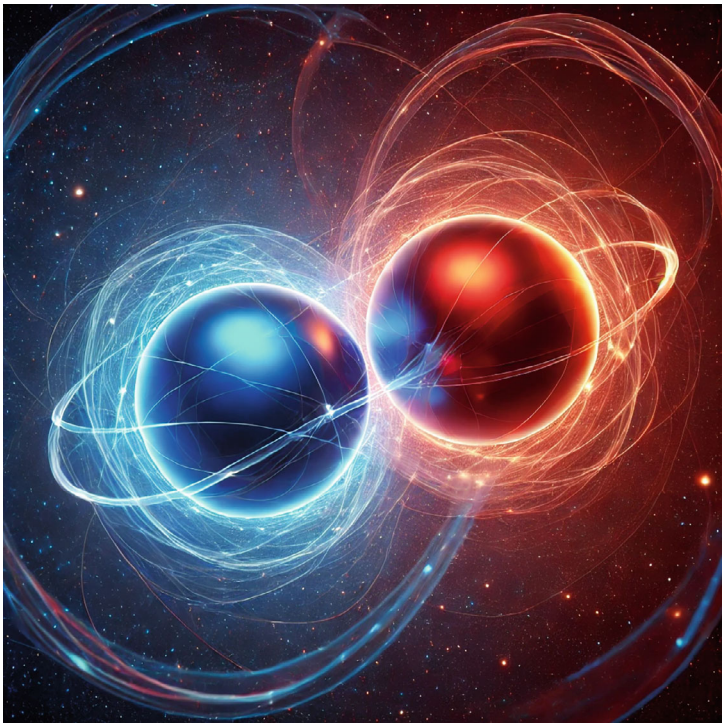


Fig. 6.10 An artistic view of quantum entanglement between particles. Any graphical representation is generally failing when quantum phenomena are involved and only art images can be used but with much caution. It is an intrinsic limitation of the quantum world

violated Bell's inequality; the one carried out beginning in 1970 by physicists John Clauser and Stuart Friedman at the University of California Berkeley proved that hidden variable theories were inconsistent. Definite demonstration has been given by Alain Aspect, John F. Clauser, and Anton Zeilinger the 2022 Nobel Prize winners (Fig. 6.10).

The results are determined only by the measurement itself, which was verified again and again in the course of other subsequent experiments. Following Bell's theorem, no theory of nature that respects both locality and realism can reproduce all the predictions. Bohr was right, and the EPR theory was incorrect; however, the doubts about the incompleteness of quantum mechanics are still far from being clarified.

The Philosophical Thinking of Niels Bohr

Niels Bohr was the first to introduce the concept of quantization into the atomic model named after him. The idea of a discontinuity of physical properties

opened up a number of interpretive issues that were not easy to resolve. Bohr's approach to resolving the apparent opposition between classical and quantum physics was based, as seen above, on the Principle of Complementarity. This idea formed the basis of his interpretation of quantum theory and, thus, the resulting Copenhagen Interpretation. Bohr intended to resolve the profound implications of quantum mechanics on the nature of reality, the limits of knowledge, and the relationship between the observer and the observed. The properties of a quantum system, according to Bohr, are defined by measurement. Unlike classical physics, where the properties of a system are thought to exist independently of observation, quantum properties are context-dependent and only acquire specific values when a measurement is made. A consequence of this idea is that the observer cannot be separated from the observed. The act of measurement forces the quantum system to adopt a particular state, which only becomes real in the context of that specific experiment. Bohr believed that objective reality in the classical sense does not exist at the quantum level; instead, reality is only defined in the context of interactions between systems and measuring devices.

Bohr's philosophical approach sought to reconcile the wave-particle duality of light and matter, which was one of the major paradoxes of early quantum theory. According to Bohr, duality is a fundamental property; light and electrons do not have a definite wave-like or particle-like nature per se; the specific property emerges only from measurement. Understanding quantum mechanics is only possible if one accepts that the classical concepts of wave and particle cannot be accepted simultaneously. The use of the terms must be contextualized according to experimental conditions. This is eminently epistemological thinking, which considers both the foundations of nature and the limits of human knowledge. He argued that classical concepts, such as position, momentum, wave, particle, etc., are indispensable tools for describing and communicating about experiments, even though they do not fully capture the nature of quantum phenomena. Bohr maintained that all knowledge must be expressed in terms of classical language, as this is the only language we have for describing experiments and observations.

Bohr insisted that any description of a quantum system must be contextual, meaning that the experimental setup and the questions asked determine the properties that can be meaningfully assigned to the system. Classical physics serves as a necessary reference point for defining quantum phenomena, but only within the context of a given experimental arrangement. A complete understanding of quantum systems requires considering the system and the measuring apparatus as a single, indivisible entity. Bohr's philosophy rejects the classical realism that underpins much of classical physics and that assumes that physical properties exist in a well-defined way, independent of observation. Physical system properties do not have an independent existence; rather, they emerge through the act of measurement. For Bohr, questions about what

"really exists" outside of observation are meaningless in the quantum context. He famously stated: "There is no quantum world. There is only an abstract quantum physical description."

6.5.2 What Is the Quantum Entanglement of a Single Object?

In quantum mechanics, particles are objects mathematically described by wave functions, that is, mathematical expressions that incorporate all the information that can be known about the particle.

The wave function can be distributed in space, which is why particles can act as if they were waves, yet if two particles are *entangled*, they are described by a single wave function and are mathematically speaking the same object. Two particles and only one wave function to describe them. That is the real meaning of *entanglement* and that is why the properties of the particles are interdependent, if you do something to one of the particles such as measuring it, you change the wave function and therefore you also alter the other particle because it too is described by the same wave function.

6.5.3 How Do Two Particles Become One?

The observation of *entanglement* raises the interesting question of how it could be possible for two electrons or, more generally, initially distinct particles to combine their two wave functions into one. In principle, it is possible to separate the wave functions of the entire system consisting of the two particles into a part describing particle one and a part describing particle two; however, after the particles have interacted, this is no longer true. It is impossible to identify any part of the total wave function and say which part corresponds to particle one or particle two. This *entanglement* happens for basically any kind of interaction between them. So once two particles are *entangled*, you cannot know anything about one of them without considering the whole wave function; it is as if their properties are now distributed over both of them.

That is why no matter how far apart the *entangled* particles are, they are still entangled by their joint wave function. This entanglement is the really strange and counterintuitive property. We are used to considering the properties of an object as being in or on the object itself. The color of an object, for example, is a property that has a precise location and defines a quality of a specific object. A blue cup and a green cup have a property that is well-defined by their color; if they were small enough to behave as *entangled* quantum objects, the two cups would be both blue

and green; the color property is spread non-locally between the two cups. The non-locality property is a specific feature of quantum mechanics that is a consequence of *entanglement* precisely. Things in a place do not depend only on what happens there, as is the case in the classical world; there must be an instantaneous connection between different regions and particles regardless of their distance. Quantum space is what emerges from *quantum entanglements* that connect objects in a vast network of interactions.

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Chapter 7
From the Single Electron to Spin



The nature of the electron, a point particle according to the *Standard Model*, presents a paradox that challenges our ability to comprehend the physical reality. This paradox lies in the apparent contradiction between the complexity represented according to mathematical models that fully describe experimental results and the world around us perceived as a representation of an intuitive process. For example, how come all electrons are the same? As we shall see in this chapter, a seemingly simple question stimulated original reflections that later led to significant theoretical developments. Stern and Gerlach's experiments demonstrated that the electron possesses an intrinsic property, the spin, whose generalization led to the distinction between fermions and bosons. The spin property, however, is apparently not reconcilable with the point properties of the electron. It has, in fact, not any analogue in classical physics. This incongruence also applies to the electron's magnetic dipole moment. The key to everything is perhaps the spin property, which has Pauli's Exclusion Principle as its consequence, to which we owe the existence of matter as we perceive it.

7.1 How Many Electrons Are There in the Universe? Perhaps Only One. The Theory of the Unique Electron

Richard Feynman (1918–1988), the famous American physicist who won the Nobel Prize in Physics in 1965, received a strange phone call one day in the spring of 1940 while at Princeton. It was his old teacher, physicist John Archibald Wheeler (1911–2008), who proposed a theory of his own that he thought explained why the electrons in the universe all have the same charge and mass. All electrons have exactly the same properties, such as electric charge, so it can be argued that they are all indistinguishable. This characteristic feature implies that if an experiment involves a large number of electrons, it is impossible to identify one electron specifically at the end of the experiment [1].

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P. Innocenzi, *What is an Electron?*, https://doi.org/10.1007/978-3-031-81984-1_7

Feynman recounted the phone call himself in the lecture he gave at the Stockholm Academy of Sciences when he was awarded the Nobel Prize in Physics: "*… I received a telephone call one day at the graduate college at Princeton from Professor Wheeler, in which he said, "Feynman, I know why all electrons have the same charge and the same mass" "Why?" "Because, they are all the same electron!" And, then he explained on the telephone, "suppose that the world lines which we were ordinarily considering before in time and space—instead of only going up in time were a tremendous knot, and then, when we cut through the knot, by the plane corresponding to a fixed time, we would see many, many world lines and that would represent many electrons, except for one thing. If in one section this is an ordinary electron world line, in the section in which it reversed itself and is coming back from the future we have the wrong sign to the proper time—to the proper four velocities—and that's equivalent to changing the sign of the charge, and, therefore, that part of a path would act like a positron." "But, Professor", I said, "there aren't as many positrons as electrons." "Well, maybe they are hidden in the protons or something", he said*" [2].

This call marked a special and in some ways bizarre moment, that of the enunciation of the *One Electron Theory*. According to this theory there would be only one electron in the entire universe moving up and down in time. According to the *One Electron Theory* when an electron is hit by light it can be repelled back into space but likewise this photon-electron collision could lead to a jump back in time. The electrons involved in this time travel would be none other than positrons, the antimatter analogue of the electron. So not only would we have the same electron but also as a consequence that all positrons are nothing but the one single electron moving through time.

The weak point of the single electron theory is the positron; if the single electron travels forward in time as an electron and returns as a positron, there should be an equal number of electrons and positrons. However, this equality is not experimentally confirmed, and for the moment, much of the antimatter is not found; we live in a universe made of matter. The missing part, assuming it exists and is not just the result of an incomplete or wrong theory, could be *Dark Matter*, and this is currently the very focus of research and studies by theoretical and experimental physicists. However, at present despite all efforts no experimental evidence for the existence of dark matter and dark energy has been found, and these failures are beginning to undermine the castle of theories built on them.

However, how would a single electron move through space–time? According to Wheeler, each electron would draw lines in the universe through space–time. Rather than having a myriad of such lines, Wheeler suggested that they might all be parts of a single line, like a huge tangled knot drawn by a single electron. A given moment in time is represented by a section of space–time that meets the knotted line many times. Each point of encounter with this specific section of space–time represents an electron that can be considered as real at that precise instant. At those points, half of the lines will be directed forward in time, and half will have made a turn and be directed backward. Wheeler suggested that these backward sections would appear

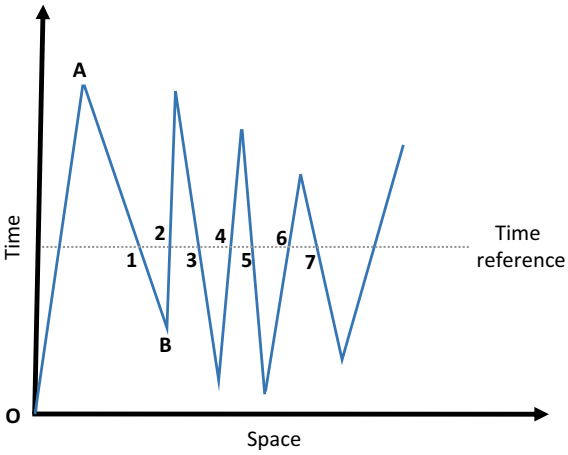


Fig. 7.1 Unique electron in space–time

as the antiparticle of the electron, the *positron*. It is a fascinating idea that also gives a deeper meaning to the existence of antimatter.

The following graph can help to understand Wheeler's idea that an electron could be in an infinite number of positions simultaneously (Fig. 7.1). The graph has space in the abscissa and time in the ordinate; the blue curve shows the trajectory of the single electron in space–time. If one takes as a reference in the time axis, that of the ordinates, a certain instant, an electron moving in space–time up and down with respect to the chosen time would appear as a multiplicity of particles even though it is nothing but the same electron.

The electron travels from point O to A, following a normal trajectory whereby one displacement in space corresponds to one in time. Things change when the electron reaches point A, where the electron reverses its direction of travel, to reach point B, where it again reverses its displacement in time. This reversal of direction in time, whereby the electron, once it reaches point A in space–time, actually begins to travel backward in time, means that the electron can be in two places simultaneously. The electron is simultaneously at point 1 and point 2. At that precise instant, the electron can exist in multiple places, and points 1, 2, 3, 4, 5, 6, and 7 in the diagram can be extended in a virtually infinite sequence. The single electron can, therefore, be observed in countless different places at any one time taken as a reference. At the positions where the electron travels to the future, it behaves like a normal electron. If the electron were in a way that would continue forever on this journey, up and down time effectively, the entire universe could have a single electron.

However, the problem remains that one has to come to terms with the experimental reality that many more electrons than positrons have been observed. When Feynman raised this question with Wheeler the latter replied that the missing positrons might be hidden in the protons. However, this hypothesis is also incorrect because inside the protons are quarks.

In the end Feynman did not take Wheeler's one-electron theory too seriously but still something borrowed from it (although he used the term "steal"): "*I did*

not take the idea that all the electrons were the same one from him as seriously as I took the observation that positrons could simply be represented as electrons going from the future to the past in a back section of their world lines. That, I stole!" Feynman, therefore, considered the idea of the mathematical equivalence of antimatter as "*time-reversed matter*" [2].

Feynman later proposed the interpretation that the positron was nothing more than an electron moving backward in time in his 1949 article: *The Theory of Positrons* [3]. Japanese physicist Yoichiro Nambu (1921–2015), Nobel Laureate in Physics in 2008, applied it to all particle-antiparticle pair formations and annihilations, stating: "*The time itself loses sense as the indicator of the development of phenomena; there are particles which flow down as well as up the stream of time; the- eventual creation and annihilation of pairs that may occur now and then, is no creation nor annihilation, but only a change of directions of moving particles, from past to future, or from future to past; a virtual pair, which, according to the ordinary view, is foredoomed to exist only for a limited interval of time, may also be regarded as a single particle that is circulating round a closed orbit in the four-dimensional theatre; a real particle is then a particle whose orbit is not closed but reaches to infinity …*" [4].

As fascinating as it was and inspired Feynman to construct his diagrams, the one-electron theory never caught on. The wave nature of the electron underlying Quantum Field Theory considers electrons as oscillations of the fundamental electromagnetic field. It is not a single electron but still the physical expression of a single field.

7.2 Do Electrons Also Age and Die?

The electron is the negatively charged particle with the smallest mass among all those in the *Standard Model* that similarly have a negative charge, such as its big brother, the muon. As we know, the electron is an elementary particle, so it should not be possible to observe its decay in the form of other particles. If the electron could decay, it should decompose by the *law of conservation of energy* (LCE) into particles of smaller mass. In fact, by breaking down into smaller particles, some of the energy would be released as photons or kinetic energy, and this would not violate the principle. The muon, μ^-, which has a greater mass than the electron, can eventually decay into electrons without violating the LCE. Muon decay occurs with remarkable speed, in 2.20 μs, , and generates an electron and two neutrinos, a muon neutrino and an electron antineutrino (Fig. 7.2).

If the electron is to decay into smaller particles, as opposed to the muon, the choice is not very wide; this leaves neutrinos, photons, gluons, and gravitons, which, however, have no electric charge and are only hypothetical particles for now. Even when combining several of these particles together, the net charge would always be zero. This is quite a problem because if the electron decayed, its charge would simply vanish into thin air. For this reason, the electron cannot decay and is considered a fundamental particle, at least until proven otherwise. We will see later, however, that not all physicists are totally convinced that there is nothing underneath leptons.

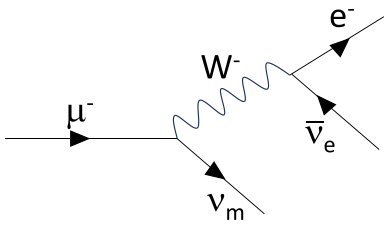


Fig. 7.2 Muon decay. The muon, μ^-, decays into an electron, e^-, a muon neutrino, ν_m, and an electron antineutrino $\overline{\nu}_e$. The W^- boson is the mediator of the weak nuclear force

The hypothetical decay of the electron would represent a violation of the *Standard Model*. Nevertheless, it is worth understanding whether or not there is experimental evidence to confirm the eternal life of the electron. For this purpose, at the Gran Sasso National Laboratories in Italy, the Borexino experiment, designed to hunt neutrinos, has been used to study a hypothetical decay process in which an electron decays into an electron neutrino, ν_e, and into a photon of energy 256 keV.

$$\mathrm{e} \rightarrow \nu_\mathrm{e} + \gamma\,(256\ \mathrm{keV})$$
$$\text{(hypothetical electron decay reaction)} \tag{7.1}$$

If this photon were actually produced, it would have to be revealed through a bright flash in one of the 2212 photomultipliers used in the experiment. The experiment was carried out in 2015, but after 408 days of observation, no decay was observed. This result made it possible to estimate what can be considered the minimum value of the average lifetime of an electron, 6.6×10^{28} years. However, perhaps that could be a conservative estimate. To translate, the decay time of an electron would be at least a quintillion times larger than the age of the universe, with a lifetime of at least 66,000,000,000,000,000,000,000,000,000 years. Despite the unimaginable number of years of life calculated for an electron, it does not necessarily have eternal life. This is where Penrose's *Conformal Cyclic Cosmology* (CCC) [5, 6] comes in, whereby everything in the universe should eventually decay into photons. No matter how long it takes, if nothing remains in the universe other than photons, this state would lead directly to a new *Big Bang* and the beginning of a new universe.

The electron, therefore, would have not quite an eternal life but at least a life longer than that of the universe. Its decay would violate the LCE and represent a break from the *Standard Model.* Decay is a natural process for heavy particles but not for light particles such as the electron. A free neutron is very unstable and decays into a proton, an electron, and an electron antineutrino within minutes (Fig. 7.3). This process is possible because the neutron is heavier than all three particles into which it decays.

$$\mathrm{n}^0 \rightarrow \mathrm{p}^+ + \mathrm{e}^- + \overline{\nu}_\mathrm{e}. \tag{7.2}$$

However, the very principle of charge conservation means that other decays have never been observed.

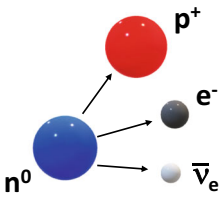


Fig. 7.3 Decay of a neutron, n^0, into a proton, p^+, an electron, e^-, and an electron antineutrino, $\overline{\nu}_e$

In particle physics, there are two other associated quantities that are strictly conserved, the *baryonic number* and the *leptonic number*. These two numbers are additive, that is, given by the sum of different quantities, and express a quantum number associated with a system (Table 7.1).

The *baryonic number* is defined as:

$$B = \frac{1}{3}\left(\mathrm{n}_q - \mathrm{n}_{\bar{q}}\right) \tag{7.3}$$

where $\mathrm{n_q}$ and $\mathrm{n}_{\bar{q}}$ represent the number of *quarks* and *antiquarks*, respectively. *Baryons* (protons and neutrons) are composed of three quarks and have a baryon number + 1, given by the 3 quarks. In fact it will be 0 while $\mathrm{n_q} = 3$ so B is worth 1. *Antibaryons*, which have three antiquarks, will have an opposite baryon number, − 1, while for mesons, particles formed by one quark and one antiquark, the baryon number becomes zero.

The *law of conservation of baryon number* is also the reason why protons, which have a mass 1836 times greater than that of electrons, do not decay and are stable. In fact, protons are the lightest baryons and their decay would violate one of the conservation laws.

The other conserved quantity is the *leptonic number*:

$$L = \mathrm{n_l} - n_{\bar{\mathrm{e}}} \tag{7.4}$$

with $\mathrm{n_l}$ and $\mathrm{n}_{\bar{\mathrm{e}}}$ the number of leptons and antileptons, respectively.

According to the *Standard Model*, the leptonic number must be constant; if a hypothetical violation of it were observed experimentally, this would lead to new physics,

Table 7.1 Leptonic numbers. The bosons all have zero leptonic charge

Leptonic number + 1 (leptons)	Leptonic number −1 (antileptons)
Electron	Positron
Muon	Antimuon
Tau	Antitau
Electron neutrino	Electron antineutrino
Muon neutrino	Muon antineutrino
Tau neutrino	Tau antineutrino

all to be rewritten. Physicists who do not believe that the *Standard Model* is complete are looking in this direction. In the meantime, the near-eternal life of the electron and the baryonic and leptonic constants provide a fairly consistent framework to support the *Standard Model*.

The stability of the electron may lead one to extend some considerations; however, it is the only one of the three negatively charged elementary particles that do not decay. Unlike the muon and the tau particle, it is virtually eternal. If we look at the periodic table of elementary particles (Fig. 1.16) depicted in the *Standard Model*, the electron falls in the first-generation column, where there is also the electron neutrino and the pair of up and down quarks. The proton is formed by the first-generation quarks, as is the neutron. In practice, the matter of the universe is formed only by the particles of the first generation, and the others that appear and disappear so quickly what role do they play? [7, 8].

7.3 The Electron Spin

If the double-slit experiment can be considered the mother of all experiments, the definition of electron spin is, in its own way, one of the most complex concepts in physics. The quirks of quantum mechanics make the description of spin counterintuitive yet fascinating [9].

Spin is a quantum property of the electron, one of its fundamental properties. As we have seen, it can take only two discrete values that are defined mathematically by the fractions $+ 1/2$ or $-1/2$. In the introduction, it was described how spin is a property on which the state of matter depends; if electrons did not have the spin property, the universe would not exist. Spin is closely associated with the Pauli Exclusion Principle, as we have seen.

The idea of electron spin, one of the most elusive quantum properties to define, began to develop in the early 1920s in an attempt to explain some otherwise incomprehensible experimental results. In 1926, a pair of Dutch physicists from Leiden University, G. E. Uhlenbeck and S. Goudsmit, published an article in *Nature* entitled: "*Spinning Electrons and the Structure of Spectra*" [10]. In the abstract they wrote: "*So far as we know, the idea of a quantised spinning of the electron was put forward for the first time by A. K. Compton (Journ. Frankl. Inst., Aug. 1921, p. 145), who pointed out the possible bearing of this idea on the origin of the natural unit of magnetism. Without being aware of Compton's suggestion, we have directed attention in a recent note (Naturwissenschaften, Nov. 20, 1925) to the possibility of applying the spinning electron to interpret a number of features of the quantum theory of the Zeeman effect, […]. In this letter we shall try to show how our hypothesis enables us to overcome certain fundamental difficulties which have hitherto hindered the interpretation of the results arrived at by those authors.*"

Dirac himself in 1928 started from the idea of electron spin to develop his quantum theory of the electron (*The Quantum Theory of Electron*) [11]: "*The new quantum mechanics, when applied to the problem of the structure of the atom with point-charge*

electrons, does not give results in agreement with experiment. The discrepancies consist of "duplexity" phenomena, the observed number of stationary states for an electron in an atom being twice the number given by the theory. To meet the difficulty, Goudsmit and Uhlenbeck have introduced the idea of an electron with a spin angular momentum of half a quantum and a magnetic moment of one Bohr magneton." The idea of a binary physical property applied to a point particle raised some puzzlement, but the theory's disagreement with special relativity necessarily required the assumption that the electron had two states: "*The question remains as to why Nature should have chosen this particular model for the electron instead of being satisfied with the point-charge. One would like to find some incompleteness in the previous methods of applying quantum mechanics to the point-charge electron such that, when removed, the whole of the duplexity phenomena follow without arbitrary assumptions.*"

An early clue to electron spin was provided by an experiment performed in 1896 by Dutch physicist Peter Zeeman (1865–1843) (Fig. 7.4), who studied the effect of a magnetic field on the atoms of a sodium gas produced by a Bunsen burner: "*When we study the well-known Bunsen sodium flame by means of Rowland's grating, we see a spectrum consisting mainly of two separate sharp yellow lines, which in our grating lie about 1 mm from each other*" [12].

Zeeman had experimentally observed the effect of a magnetic field on the wavelengths at which a photon was emitted from the atom when an electron jumps from different energy levels. These energy levels, corresponding to specific spectral lines, split into multiple lines when the atom is placed inside a magnetic field. This effect is



Fig. 7.4 Albert Einstein visiting Pieter Zeeman (left) in Amsterdam, with his friend Paul Ehrenfest (right), ca 1920. Public domain via Wikimedia Commons Photographer unknown [13]

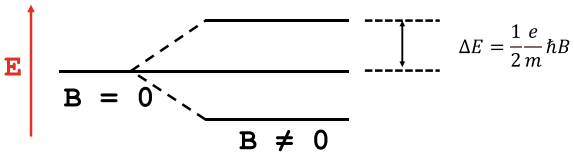


Fig. 7.5 Spectral line splits into three in the presence of a magnetic field, B. The difference in energy between the levels, ΔE, is directly proportional to the intensity of the magnetic field, *B*, through the constants e (electron charge), m (electron mass), and $\hbar$ (the reduced Planck's constant)

produced by the interaction between the electron magnetic moment and the external magnetic field. This interaction slightly changes the energies of the different electronic states, causing the separation of spectral lines observed in the emission or absorption of light (Fig. 7.5).

This effect is known as the *Zeeman Effect*. The discovery of the effect occurred twenty years before the beginning of the development of quantum mechanics, and although it earned Zeeman and Hendrik Lorentz (1853–1928) the Nobel Prize as early as 1902, it remained a difficult phenomenon to interpret for a long time. Lorentz, who shared the Nobel Prize for explaining the Zeeman effect, had associated electrons and their wave nature with the observation. However, a comprehensive explanation would only be achieved with the full development of quantum mechanics.

An initial explanation of Zeeman's experimental finding was developed on the basis of semi-classical arguments in physics. If we assume that the electron is a small charged ball moving in a circle around the atom, the motion leads to the generation of a dipole magnetic moment and a magnetic field exactly as if it were a small magnetic bar with a negative and positive pole. The different alignments of the orbital magnetic field with respect to the external field cause the energy levels to be divided into three. It is a reasonable explanation, but one that runs up against another effect, the *anomalous Zeeman effect,* discovered just a year later, in 1897, by physicist Thomas Preston (1860–1900) [14]. Indeed, in some cases, the magnetic field causes energy levels to split further apart for reasons that were completely mysterious at the time. In fact, the definition of anomalous comes from the fact that at the time the effect was discovered, we are still in the late 1800s, the concept of spin had not yet been defined so that a comprehensive explanation for the phenomenon could not be given using classical physics.

The semi-classical explanation is plausible, but it is necessary to think of the electron as a small spinning charged ball. The ball idea, however, raises another major problem because producing the observed magnetic field moment would require that the electron rotation occurs faster than the speed of light, as we have seen in the **Introduction**.

This limitation was first understood by the Austrian physicist Wolfgang Pauli. He showed that if one assumes that electrons have a certain size, their surface would have to move faster than light to generate the required angular momentum, an observation that has already been discussed in Chap. 2. Therefore, the size of the electrons should

be zero, which shows that the classical idea of angular momentum cannot be matched from a physical point of view.

Pauli rejected the idea of associating a classical property such as rotation with the electron, insisting instead of calling it, *classical undescribable two-valuedness*: "*On the other hand, my earlier doubts as well as the cautions expression "classically non-describable two-valuedness" experienced a certain verification during later developments, since Bohr was able to show on the basis of wave mechanics that the electron spin cannot be measured by classically describable experiments (as, for instance, deflection of molecular beams in external electromagnetic fields) and must therefore be considered as an essential quantum–mechanical property of the electron.*"

This is an important conceptual step, a measurable physical property that has a solid experimental basis given by Stern and Gerlach's experiment demonstrating the quantization of the spin angular momentum is defined by the specific property of taking on discrete non-describable states, at least with respect to classical deterministic physics.

Electron spin is a quantum property and has all the quirks one might expect from quantum mechanics. Spin can be defined as "*intrinsic angular momentum*" because it is a form of angular momentum unrelated to the motion of the particle in space. It is, unlike the *orbital angular momentum* associated with a hypothetical rotation of the electron around the nucleus in the semi-classical model, an intrinsic property of the electron, just as is its mass and charge.

In 1920, as we have seen, the wave nature of particles such as the electron was finally disclosed by the Schrödinger equation. The equation describes how quantum objects behave by treating wave functions in terms of probability distributions. In the Schrödinger equation, however, there is no spin. It was Pauli to introduce spin from a mathematical point of view into the wave function by introducing the two quantum spin components. This extension of the Schrödinger equation becomes a highly complex function that requires the introduction of the concept of *spinor*.

Spinors are used to describe particles, such as electrons, which have strange rotational properties (even if they are point-like particles that in principle do not rotate). In three-dimensional objects described using classical physics, a rotation of 360° returns the same object exactly to the starting point. This property is also true for vectors that point to specific directions in three-dimensional space. In the case of spinors, however, to return to the starting point a double rotation is required, 360° + 360° so 720°. For a *spinor*, a 360° rotation changes the sign of the *spinor* and transforms it into its negative.

These strange mathematical objects encode the quantum mechanical spin information and the state of an electron spin is not represented by ordinary vectors, but by *spinors*.

One way to visualize the behavior of a spinor is to use a Möbius band (Fig. 7.6). The spinor is visualized as a vector that changes sign when it makes a 360° rotation along the band.

Electrons, as far as has been understood so far, can have only two possible spin configurations, as shown by Stern and Gerlach's experiment. Let us return for

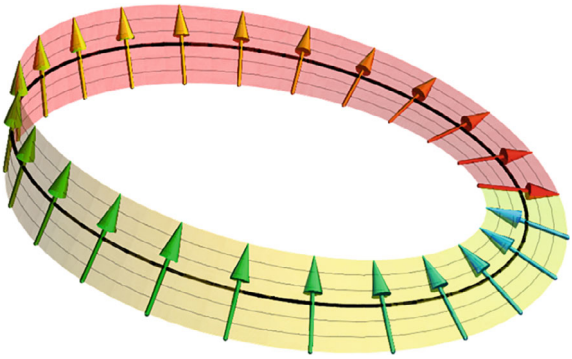


Fig. 7.6 Spinor and the Mobius strip [15]

simplicity to calling them up and down. If we consider a number of electrons, it is reasonable to think that 50% will have spin *up* and 50% spin *down*. In reality this is true only at the time when a spin measurement is made. Before the measurement, the electrons do not take on any defined configuration and are in a superposition of states; they are both up and down. It is the measurement that will define their state at that instant.

To understand spin a little better, we need to think not of angular momentum but of classical momentum, p, which is defined as the product of mass times velocity, p = mv, and is a vector quantity. The momentum of a particle is fundamentally related to its position. For similar reasons, in quantum mechanics, position and momentum are conjugate variables, which means that it is possible to represent a wave function of a particle in terms of either of these properties. The conjugation of the variables is at the ground of *Heisenberg's indeterminacy principle*, whereby increasing the knowledge of one property is at the expense of knowledge of the related property. If position is related to momentum, the twin property of orbital angular momentum of a particle in rotational motion is the orientation of the particle, the spin. One can, therefore, think of the angular momentum of an electron not in terms of the classical property connected with rotation but rather as the rotational degree of freedom that leads to the conservation of its associated quantity, the intrinsic angular momentum, i.e., the spin.

The quantum model, with its unintuitive definition of spin, tells us that particles described by *spinors* have quantum moments with quantum spin numbers that are half of odd numbers, 1/2, 3/2, and 5/2. The electron itself has a spin 1/2, exactly like protons and neutrons. Their intrinsic angular momentum can only be observed with the minus or plus sign of half of Planck's reduced constant, ħ, projected in whatever direction you want to measure it. And this, for now, is the result of experimental measurements that validate the model.

We know that particles with semi-integer spin are *fermions*. They differ from the bosons that have integer spin and are described not by *spinors* but by vectors, and the original state is restored following a 360° "rotation." This disparity in the properties of fermions and bosons is reflected in a profound difference in their physical nature. These properties define how bosons interact with each other, which, for example,

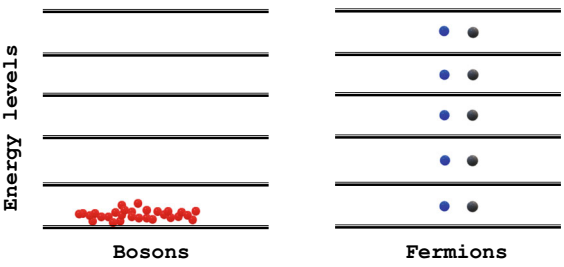


Fig. 7.7 An unlimited number of bosons can occupy the same energy level without violating the laws of quantum mechanics, only two fermions of opposite spin can occupy the same state instead

can occupy the same quantum state while fermions can never occupy the same state (Fig. 7.7).

The properties of fermions are expressed by *Pauli Exclusion Principle* and determine the composition of the periodic table, the discrete energy levels of electrons and for matter its structure as we know it.

So even though electrons do not spin, the property we define as spin is an intrinsic property of matter that goes to structure and define reality itself through *spinors,* i.e., the strange nodes in the space–time of subatomic particles.

7.4 The Spin-Statistics Theorem

The half or integer value of spin divides the elementary particles into the categories of bosons and fermions, as we have just described. Generally, a system of many particles needs statistical treatment to be described. In classical systems, the Maxwell–Boltzmann statistics is applied. Classical particles are distinguishable and do not obey quantum mechanical restrictions so that multiple particles can occupy the same state. Two molecules can be in the same position and have the same energy without any exclusion or symmetry considerations. The probability of finding particles in a given state decreases exponentially with energy.

In quantum systems, however, two different statistics are used, the *Fermi–Dirac statistics* for fermions, named after Enrico Fermi and Paul Dirac, and the *Bose–Einstein statistic* for bosons, named after the Indian physicist Satyendra Nath Bose and Albert Einstein. The application of specific statistics is defined in a theorem, the *Spin-Statistics Theorem*:

- All integer spin particles are necessarily *bosons* since they obey the Bose–Einstein statistic.
- All particles with spin half-integer are necessarily *fermions* since they obey the Fermi–Dirac statistic.

This means that the nature of a particle spin dictates how it behaves under particle exchange and how its wave function is structured. For bosons, the wave function is symmetric under particle exchange. This symmetry allows multiple bosons to occupy

the same quantum state simultaneously. This property leads to unique quantum phenomena, such as Bose–Einstein condensation, where a large number of bosons collapse into the lowest energy state at very low temperatures.

On the other hand, for the Pauli Exclusion Principle, two identical fermions cannot occupy the same quantum state. For fermions, the wave function is antisymmetric under particle exchange, meaning that swapping two fermions flips the sign of the wave function. At low temperatures, fermions fill up the available energy levels up to a maximum energy called the Fermi energy. This creates a step-function distribution, with all states below the Fermi energy filled and those above it empty.

7.4.1 Distribution Functions

Each type of statistics has a different distribution function that describes the average number of particles $\langle n_i \rangle$ in a given energy state i:

- *Maxwell–Boltzmann Distribution*

$$\langle n_i \rangle = \frac{e^{-(\varepsilon_i - \mu)/k_B T}}{Z} \tag{7.5}$$

with ε_i is the energy of state i, μ, the chemical potential, k_B Boltzmann's constant, T the absolute temperature, and Z the partition function[1] for the system.

- *Bose–Einstein Distribution*

$$\langle n_i \rangle = \frac{1}{e^{(\varepsilon_i - \mu)/k_B} - 1} \tag{7.6}$$

In the Bose–Einstein distribution, the denominator $e^{(\varepsilon_i - \mu)/k_B T}$ indicates that as ε_i approaches μ, $\langle n_i \rangle$ can become very large; i.e., many particles can occupy the same state.

- *Fermi–Dirac Distribution*

$$\langle n_i \rangle = \frac{1}{e^{(\varepsilon_i - \mu)/k_B} + 1} \tag{7.7}$$

In the Fermi–Dirac distribution, the presence of $+1$ in the denominator ensures that $\langle n_i \rangle \leq 1$, is consistent with the Pauli Exclusion Principle, meaning a maximum of one fermion per state.

[1] The *partition function* describes the statistical characteristics of a system at thermodynamic equilibrium. It depends on temperature and other factors, like the volume that contains the gas. Several thermodynamic properties of the system, such as total energy, free energy, entropy, and pressure, can be defined using the partition function or its derivatives.

7.5 The Ohanian Hypothesis on Spin

Are these the final words on spin? Almost certainly not. For the American physicist Hans Ohanian, spin has a much more physical sense. In one of his 1986 papers, he wrote [16]: *"According to the prevailing belief, the spin of the electron or of some other particle is a mysterious internal angular momentum for which no concrete physical picture is available and for which there is no classical analogue. However, on the basis of an old calculation by Belinfante [Physica 6, 887 (1939)], it can be shown that the spin may be regarded as an angular momentum generated by a circulating flow of energy in the wave field of the electron. Likewise, the magnetic moment may be regarded as generated by a calculating flow or charge in the wave field. This provide an intuitively appealing picture and establishes that neither the spin nor the magnetic moment are" internal", they are not associated with the internal structure of the electron, but rather with the structure of its wave field. Furthermore, a comparison between calculations of angular momentum in the Dirac and electromagnetic fields show that the spin of an electron is entirely analogous to the angular momentum carried by a classical circularly polarized that wave."*

According to Ohanian, therefore, it is possible to derive exact values of the angular momentum of the electron spin and magnetic moment by going to look at the energy of the charge in the so-called Dirac field. So even though the electron is a point object its field is capable of generating angular momentum.

7.6 The Electron Magnetic Moment

The magnetic moment of an electron is a fundamental property that arises from its intrinsic spin and charge. It determines how the electron interacts with external magnetic fields and plays a crucial role in several quantum phenomena. The magnetic moment of an electron is a vector quantity that represents the strength and orientation of its magnetic field. It is related to the spin and charge of the electron. A charged particle with angular momentum, whether due to orbital motion or intrinsic spin, generates a magnetic moment. When an electron is in a magnetic field, it responds to the applied field as if it were a dipole, and many experiments, such as the Zeeman experiment well support this property. Indeed, the definition of a point particle poses the usual problem of interpretation; classical physics is used to describe the properties, but this must be consistent with the quantum model of the system.

Spin is associated with a magnetic moment. Just as a classical charged object that spins around its axis generates a magnetic field (like a spinning charged sphere), the intrinsic spin of the electron gives rise to a magnetic moment. The *electron spin magnetic moment* is expressed by the formula:

$$\boldsymbol{\mu}_s = -g_s \mu_B \frac{\boldsymbol{S}}{\hbar} \tag{7.8}$$

with $\boldsymbol{\mu}_s$, the spin magnetic moment vector; g_s the *electron spin g-factor*, a dimensionless constant that determines the strength of the magnetic moment and $\hbar$ the reduced Planck's constant. For a free electron, $g_s \approx 2.002319$; μ_B is the *Bohr magneton*, which is the fundamental unit of the electron's magnetic moment and that we already described in Chap. 2 (Eq. 2.19); **S**, is the spin angular momentum vector.

It is also possible to determine the *magnitude of the electron spin magnetic moment* using the following equation that is written introducing the two constants, g_s and μ_B and taking into account the spin value:

$$|\mu_s| = g_s \mu_B \sqrt{s(s+1)} = g_s \mu_B \frac{\sqrt{3}}{2} \tag{7.9}$$

for a particle with spin ½ like an electron, s = 1/2, and this simplifies Eq. (7.9) to approximately $\mu_s \approx \mu_B$ (taking $g_s \approx 2$)

An important property is the sign and direction of the magnetic moment. It is antiparallel to the electron's spin because the electron has a negative charge. Hence, the direction of the magnetic moment is opposite to what it would be for a positively charged particle. If an electron's spin is "up" (pointing along the +z-direction), its magnetic moment points along the −z-direction.

In addition to the *spin magnetic moment*, an electron can have an *orbital magnetic moment,* $\boldsymbol{\mu_L}$, if it is in an atom and moving in a circular orbit around the nucleus (classical view). The *orbital magnetic moment* is given by:

$$\boldsymbol{\mu}_L = -\mu_B \frac{\boldsymbol{L}}{\hbar} \tag{7.10}$$

with, **L** the *orbital angular momentum vector*. The negative sign again indicates that the magnetic moment is opposite to the angular momentum direction.

When an electron is part of an atom, its *total magnetic moment,* $\boldsymbol{\mu}_{\mathbf{total}}$, is the sum of the orbital and spin components (Eqs. 7.8 and 7.10):

$$\boldsymbol{\mu}_{\mathbf{total}} = \boldsymbol{\mu}_L + \boldsymbol{\mu}_S \tag{7.11}$$

It gives rise to *fine structure splitting* in atomic spectra due to *spin–orbit coupling*.

When an external magnetic field is applied, the electron's magnetic moment interacts with it, causing the energy levels to split. This is the *Zeeman effect* that we have previously described. The energy of an electron in a magnetic field **B** is given by: $E = -\boldsymbol{\mu}_s \cdot \mathbf{B}$. This interaction results in a shift in the energy levels proportional to the magnetic field strength and the orientation of the spin relative to the field.

For a free electron, the theoretical value of the *g-factor* is exactly 2 according to Dirac's theory. However, experiments show that the g-factor is slightly higher: $g_s \approx 2.002319$. This small difference is known as the *anomalous magnetic moment*, which arises from *quantum electrodynamics* (QED) corrections due to interactions with the quantum vacuum, virtual photons, and particles.

The *g-factor* is a fundamental parameter correlated to the intrinsic quantum nature of the electron and its interactions with magnetic fields. Besides the *spin* g_s*-factor*, it is possible to define also an *orbital g-factor,* g_L, that indicates the g-factor associated with the orbital angular momentum of an electron in an atom. For orbital motion, the g-factor is exactly $g_L = 1$. The *total g-factor,* g_J, refers to the g-factor for the total angular momentum $J = L + S$, which is a combination of both spin and orbital contributions.

Again, it is better to underline that the *electron magnetic moment* is not a result of physical rotation or any internal structure. Instead, it is a fundamental quantum property associated with the electron's intrinsic spin, as the Dirac equation describes. This intrinsic spin gives rise to a magnetic moment like a classical spinning charge would generate a magnetic field, but without requiring the electron to have any size or spatial extent. The presence of the magnetic moment for a point-like particle is thus a manifestation of the underlying quantum mechanical and relativistic nature of the electron.

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Chapter 8
Strange Relatives, Positrons, Muons, and Tau



The periodic table of elementary particles, represented by the *Standard Model*, includes, as we saw in Chap. 2, six particular elements, the *leptons*. Three of these are particles endowed with mass and charge, the *electron* (e), the *muon* (or *mu*) (μ), and the *tau* (τ); three others are their respective neutrinos, ν_e, ν_μ, ν_τ, without charge, with virtually negligible mass and spin 1/2. These particles have their equivalent in antimatter, so in total between *leptons* and *antileptons* there are twelve elements in the *Standard Model*. The electron in the table is placed in the first column and is thus a first-generation particle; the *muon* and *tau*, on the other hand, belong to the second and third generations, respectively. Each generation is completed by a pair of *quarks*. There are six *quarks* in all, plus the six *antiquarks*.

What the actual role of second- and third-generation particles is unclear. In fact, the entire universe consists of the first-generation ones, including the electron and the up and down quarks, which are the constituents of protons and neutrons. *Muons* are products of cosmic rays striking the upper part of the earth's atmosphere, while *tau* is observable only through particle accelerators. The electron compared to its *muon* and *tau* siblings is the lightest element and like the other two has a negative charge but has a characteristic that makes it unique, it is stable. The lifetime of *muon* and *tau* particles is very limited and we can observe them only for a few moments, especially the *tau* lepton. The electron is, therefore, the only possible choice as the constituent element of matter that is stable. However, the question remains as to why there are *muon* and *tau* siblings and what role they may play or have played in the formation of the universe.

This chapter is devoted to what can be defined as the "electron family." The *muon* and *tau*, as well as its antimatter twin, the *positron*. These elements can give rise to unexpected phenomena or generate strange atoms whose properties are still largely to be explored.

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P. Innocenzi, *What is an Electron?*, https://doi.org/10.1007/978-3-031-81984-1_8

8.1 The Discovery of the Electron's Antimatter Twin, the Positron

The *Standard Model,* as seen, includes twelve fermions, the six leptons and six quarks, and the equivalent antiparticles that are the constituents of antimatter. The idea of antimatter, formed by elementary particles that are absolutely analogous in mass but opposite in charge to those that form the matter, emerged early in the development of quantum mechanics in the 1920s. The existence of the positron, that is, the antimatter counterpart of the electron, was first hypothesized by physicist Paul Dirac in 1928. Dirac was deepening the quantum theory of electrons by trying to include relativistic behavior in the Schrödinger wave equation [1]. Dirac found a puzzling solution to his equations: an electron moving with negative energy, which is impossible for classical physics: "*The true relativistic wave equation should therefore be such that its solutions are divided into two non-combinable sets, referring to charge -e and charge e, respectively.*" Therefore, his relativistic wave equation included two solutions, one positive and one negative, which precisely implied the existence of a positively charged electron.

Experimental proof of the existence of the electron's antimatter twin came a few years later, in 1932, almost by chance [2]. The theoretical "discovery" of the positron had preceded the experimental one.

Particle physics has been largely driven by theory, as happened for example with the discovery of the Higgs Boson whose existence was predicted by Peter W. Higgs (1929–2024) theoretically as early as 1964 [3]. In the case of the positron, the process by which Paul Dirac came to determine its existence was described by him personally in a lecture he gave at the Accademia dei Lincei in 1975 [4]: "*The story of the positron is a story of a physical discovery that was made by theoretical arguments. Now you may wonder in the first place how is it possible to make a discovery by theoretical arguments. The theoretical physicist is just sitting in an armchair and thinking things over. How he can be led to a discovery. The answer is that he works from the mathematics. He takes the mathematics of the existing theory, there is a formalism a number of equations which have to be interpreted in a certain way. He studies this formalism and notices an imperfection in it and then he proceeds to try to remove this imperfection. If he succeeds in doing so, he will be led to some new equations. Then he looks for the physical meaning of these new equations. The thing that I want to emphasize is that mathematical development comes before physical development, which just follows along behind the mathematics.*"

However, Dirac's method and the "mathematical" discovery of the positron anticipating the experimental one will have methodological consequences that are not always entirely positive. Theoretical physics has developed this approach almost to the point of forgetting that without experimental proof, theories are only hypotheses to be tested. Nature does not follow the supposed "beauty" of equations, a belief held by some theoretical physicists for reasons that remain unclear, and the bold theoretical models used to explain complex phenomena are often not confirmed by

experimental evidence. Fortunately, this was not the case with the discovery of the positron, which followed Paul Dirac's predictions by a small margin.

The discovery of the positron was made by Carl Anderson (1905–1991), then a physicist at the California Institute of Technology (Caltech) in the United States. Anderson had studied physics also at Caltech and had Robert Millikan as his doctoral supervisor. Millikan himself insisted that he should remain at Caltech to study cosmic rays. For this purpose Anderson had built a large experimental chamber placed inside an intense magnetic field [6]. This experimental chamber is called *cloud chamber*, and it allows a relatively simple and direct study of cosmic rays, which are high-energy particles coming from outer space. The cloud chamber functions as a particle detector that enables the visualization of traces of charged particles, such as electrons, protons, and alpha particles. From the analysis of the trajectories described by the particles and observed in the cloud chamber, it is possible to trace the type of particle itself and possibly discover new ones. The detection system consists of a sealed chamber filled with a supersaturated vapor of water; Anderson added alcohol to the vapor to improve image quality. When a charged particle passes through the cloud chamber it ionizes the atoms in the vapor, creating a trail of charged particles that can be detected by a photographic plate or other suitable imaging device. These traces can be used to study the properties of particles and their interactions with matter.

Anderson's experimental apparatus was relatively simple and included a large magnet surrounding the chamber that served to bend the paths of the ionized particles and a lead plate in the center to slow them down. The trajectory described by the particles passing through the cloud chamber allowed him to determine whether they were positively or negatively charged (Fig. 8.1). Anderson took hundreds of photographs of the traces left by cosmic ray particles, but some of these traces he could not explain. In particular, in one image he noticed one particle whose trajectory seemed to suggest that it was positively charged but with a mass much smaller than that of a proton. This image, which is now reported in all the books and articles describing the discovery of the positron, was recorded on August 2, 1932, and Anderson carefully annotated it in his lab notebook: "*Track 75. Thru Pb showing energy change or double ejection.*" What he recorded was the first experimental trace of a positron.

The image in Fig. 8.1 shows the famous Anderson's picture where one can see the trace left by the particle as it passes through the cloud chamber. It can be observed that the radius of curvature of the particle's path above the plate is smaller than that at the bottom. This difference indicates that the particle entered from the bottom, hit the lead plate, and stopped soon after in the area above the plate due to energy loss. The energy loss and the direction of curvature of the path were useful elements in identifying that the charge was positive. The length of the track left by the particle after the slowdown induced by the lead slab suggested that it was an antielectron and not a proton because the upper track was much longer than expected for a proton. A proton, in fact, because of its mass much larger than that of an electron should have stopped in a shorter distance. The observed trajectory was that of a particle that must have been lighter than the proton.

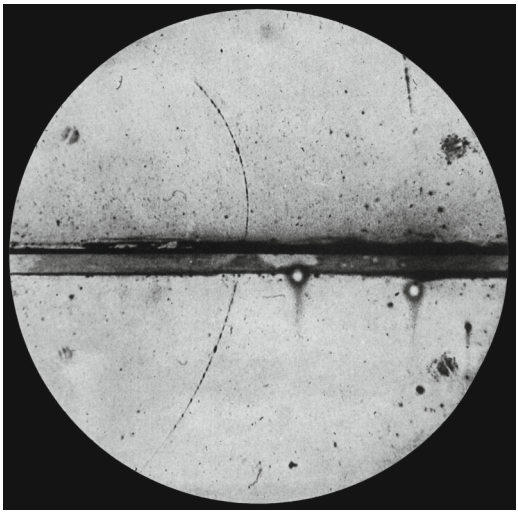


Fig. 8.1 Cloud chamber photograph of a positron passing through a lead plate, taken in 1932 by Carl D. Anderson. The thick horizontal line is a lead plane. The positron entered the cloud chamber at the lower left, was slowed down by the lead plane, and then curved upward to the left. The curvature of the path is caused by an applied magnetic field acting perpendicularly to the image plane. The higher energy of the incoming positron (lower part) resulted in less curvature of its path [5]

The name *positron* was suggested by the editor of the journal *Physical Review* to which Anderson had submitted his paper for publication. The similarities of this very particle to the electron suggested precisely that it was an entirely similar particle but of opposite charge. This was a hunch that later proved to be correct, the *positron* was precisely the antimatter twin of the electron. Like all elementary particles, positrons have dual properties of wave and particle, can collide with other particles, and can be diffracted like light.

At this point we can recall what Feynman had said about Wheeler's theory of the unique electron (see Chap. 6), namely that he had stolen it. In a 1949 paper specifically on the positron (*The theory of positrons*), Feynman proposed considering the antiparticle as an electron moving backward in time, a possible solution of the negative energy value that emerged from the Dirac equation. The metaphor used by Feynman is very evocative: "*It is as though a bombardier flying low over a road suddenly sees three roads and it is only when two of them come together and disappear again that he realizes that he has simply passed a long switchback in a single road*."[7].

This idea also appealed greatly to another Nobel laureate in physics, the Japanese Yoichiro Nambu (1921–2015), who again in 1949 in one of his papers wrote: "*The time itself loses sense as the indicator of the development of phenomena; there are particles which flow down as well as up the stream of time; the- eventual creation and annihilation of pairs that may occur now and then, is no creation nor annihilation, but only a change of directions of moving particles, from past to future, or from future to past; a virtual pair, which, according to the ordinary view, is foredoomed to exist only for a limited interval of time, may also be regarded as a single particle that is circulating round a closed orbit in the four-dimensional theatre; a real particle is then a particle whose orbit is not closed but reaches to infinity …*" [8].

Compared with other antimatter particles, the positron is relatively easy to produce and study. Positron production is now a well-established technological achievement, and medical diagnostic equipment, such as *Positron Emission Tomography* (PET), which uses positrons itself, has also been developed. Positrons are also produced naturally by cosmic rays and β^+ decays.

8.2 From Positron to Antihydrogen

Let us now return to atoms and consider the simplest of all, the hydrogen atom, which we know consists of an electron and a proton. What kind of atom could form if a positron took the electron's place? It is relatively easy to imagine an antiparticle that has an opposite charge to matter particles, but is it possible for antiparticles to form an atom of antimatter, i.e. antihydrogen? The answer is yes, and the formation of this atom albeit with great difficulty and in relatively recent times has been observed experimentally. Antihydrogen is the antimatter counterpart of hydrogen, which is composed of a positron and an antiproton, while the common hydrogen atom is composed of an electron and a proton (Fig. 8.2). The antiproton, in turn, consists of three antiquarks, two antiquark up, $\overline{u}$, and one antiquark down, $\overline{d}$. To have succeeded in producing antimatter atoms, such as antihydrogen, has important consequences experimentally and theoretically.

Antimatter is a puzzle because it should have been produced in quantities equal to normal matter during the Big Bang from which the universe should have been generated 13.6 billion years ago. Today, however, at least for our actual knowledge, there is no evidence of galaxies or clouds of antimatter and, on the other hand, antimatter is observed rarely and only for very short times, such as during certain types of radioactive decay before annihilating in a collision with normal matter. By measuring the properties of antiatoms, it is possible to determine whether their interactions are identical to those of normal matter.

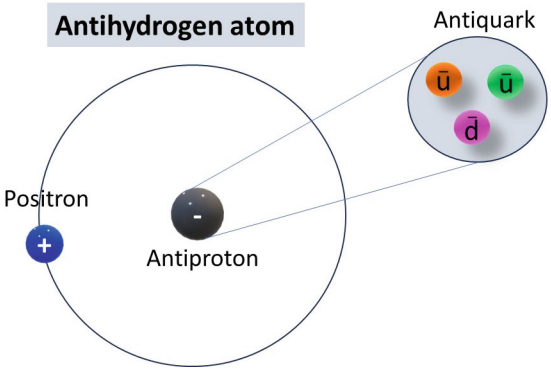


Fig. 8.2 Antihydrogen atom consists of a positron and an antiproton, which in turn consists of three antiquarks, two up antiquarks and one down antiquark

One of the first goals is to verify whether antiatoms respect *charge-parity-time* (CPT) *symmetry*, like matter atoms. For this reason, the properties of antihydrogen have been compared with those of hydrogen, and so far no significant differences have been found, which is good news. CPT symmetry implies that a particle would behave the same way in a mirror universe if it had opposite charge and moved backward in time. The CPT theorem is the basis of the *Standard Model*; the three letters stand for:

Charge. **C**harge conjugation: positive charge can be exchanged for negative charge (e.g., electron–positron).

Parity. **P**arity operation: a mirror image can be constructed.

Time. **T**ime reversal: you can replace t with -t.

If these three conditions of the CPT theorem were met, the laws of physics should remain the same, that is, be symmetrical. A CPT transformation applied to matter should, therefore, generate antimatter. Any hint of CPT symmetry breaking would cast serious doubt on our understanding of the laws of physics. However, the universe is made of matter and it seems that the equivalent part of antimatter has disappeared, the big hunt for dark matter that might be there but we do not see is still open. It could even, give the current state of knowledge, that dark matter does not exist and is only the result of faulty theoretical models and our limited knowledge of the universe.

So far, only antihydrogen atoms have been produced with great effort and a very short time; not yet antideuterium since in addition to an antiproton and a positron, it should have an antineutron making the system unstable [9].

8.3 Birth and Death of an Electron

In one of the previous chapters we saw that the decay of a neutron gives rise to the formation of an electron, as well as an electron antineutrino and a proton. The process governed by the weak nuclear force takes place under the laws of conservation of charge and mass. Intuitively the process is understandable, as we have observed, in fact, the neutron is heavier than the three particles into which it decays. But what if instead of a neutron having a definite mass we take a photon, would it still be able to generate electrons, even though the photon has no mass? As surprising as it may seem at first glance (but if the mass-energy correspondence were clear, it should not be completely surprising) the answer is yes, an electron–positron pair can be generated from a photon in what is referred to as an electron–positron pair production or pair creation process (*pair production*) (Fig. 8.3).

The process occurs through the conversion of a high-energy photon into a particle-antiparticle pair. This conversion can take place as long as the conservation of total relativistic energy, momentum, and charge are observed. When a photon loses all of its high energy during interaction with an atomic nucleus, an electron and a positron are formed. In *pair production*, a photon with sufficiently high energy can, therefore, convert spontaneously into an electron–positron pair consisting of a particle and its antiparticle. The photon in the interaction with the nucleus loses all its energy, while

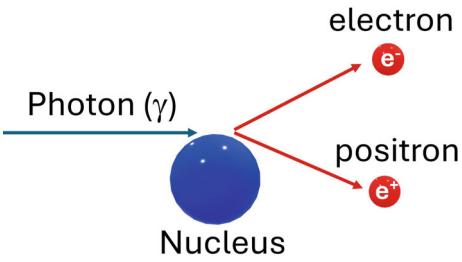


Fig. 8.3
Electron-antielectron pair production process generated by the interaction of a high-energy photon with an atomic nucleus

the nucleus, which has a very large mass, is not perturbed. An important condition is that the energy of the photon is greater than the rest mass of the pair of particles being formed; therefore, the process occurs especially at high energies such as that of cosmic rays and in the presence of intense electric or magnetic fields.

Pair production is an important process in many areas of physics, including particle physics and nuclear physics. It is also studied in astrophysics, as it can occur in the intense electromagnetic fields found in some astronomical objects, such as neutron stars and black holes. The process of pair production is not limited to the formation of an electron and a positron. It is a general physical phenomenon that occurs through the conversion of a neutrally charged boson (so no W^+ or W^-) into a subatomic particle and its antiparticle, so for example:

$$\textit{Neutral boson} \rightarrow \textit{Electron} + \textit{Positron}$$

$$\textit{Neutral boson} \rightarrow \textit{Muon} + \textit{Antimuon}$$

$$\textit{Neutral boson} \rightarrow \textit{Proton} + \textit{Antiproton}$$

How is it possible, however, that particles such as an electron and its positively charged twin can be formed from a particle that has no mass? Conservation principles help, and in physics they are fundamental in providing explanations of very complex processes that are difficult to interpret. Let us try to apply these principles to pair production, starting with the *Principle of Conservation of Charge*. The photon has no electric charge so $q = 0$, while the electron will have charge $q = -1e$ while the positron has opposite charge, i.e., $q = +1e$:

$$\text{photon}(0) \rightarrow \text{electron}(-1) + \text{positron}(+1) \rightarrow 0 = 0$$

So before and after the creation of an electron–positron pair the charge is conserved. Before checking whether another law, the *Law of Conservation of Momentum,* is obeyed in *pair production*, it is important to explain other details of the process. *Pair production* cannot take place in vacuum; the presence of an atomic nucleus with which the photon interacts is necessary. This interaction is necessary precisely so that momentum is conserved. The nucleus actually plays no active role in

pair creation, it serves only to ensure the conservation of momentum. For this reason, when a pair is produced, the atomic nucleus receives a certain backlash thereby going to satisfy the law of conservation of momentum of the system.

Another conservation law that the process must satisfy is the *Law of Conservation of Energy*. We should recall before the definition of rest mass, the minimum mass of an object in stationary conditions, and rest energy mass, the energy equivalent to the mass of a stationary object. The energy rest mass of an electron and positron (m_0) correspond to definite energy values, which in the case of electron and positron are identical:

$$\text{Electron:}m_0 = 0.511\ \text{MeV} \quad \text{Positron:}m_0 = 0.511\ \text{MeV}$$

Therefore, the pair formed by the photon must have total energy:

$$E_{\text{Photon}} = m_0 + m_0 = 2 \quad m_0 = 1.022\ \text{MeV}$$

Therefore, the photon to create the electron–positron pair must possess a minimum threshold energy of 1.022 MeV. This condition is dictated by the *Law of Conservation of Energy*. However, the energy required is quite high so gamma radiation (>100 keV) is needed for the process; visible light, for example, does not have enough energy to trigger pair production.

Finally, we can also check whether the *Principle of lepton number conservation is* satisfied. According to the principle the sum of lepton numbers before and after the interaction must be the same. The photon is a boson and has lepton number zero, the electron is a lepton and will have +1; the antielectron is an antilepton with lepton number −1, so $0 = (-1) + (+1) = 0$, and the *conservation of lepton number* is also satisfied.

The photon-matter interaction, because of the properties of the photon, can then result in the production of an electron without violating the fundamental laws. If the photon has a high energy, it can generate electron–positron pairs, at medium energy it will produce photoelectrons coming from atomic nuclei, while at low energies the Compton effect, in which a photon transfers part of its energy to an electron, will be observed instead.

Let us now consider the process opposite to pair production, annihilation. An electron and a positron when sufficiently close interact in a manner exactly opposite to pair production and annihilate by emitting two photons, which remain the only trace of their existence (Fig. 8.4). Matter, consisting of the electron and its antiparticle, the positron, disappears to transform into radiant energy. Again, all conservation laws are obeyed; the photons emitted in annihilation will have the energy corresponding to the masses and kinetic energies of the electron–positron pair. It is precisely the laws of conservation of energy and momentum that prohibit the creation of a single photon.

Different types of particles can also emerge from electron–positron annihilation, depending on the energy of the starting pair (Fig. 8.5). It is also possible, in the right

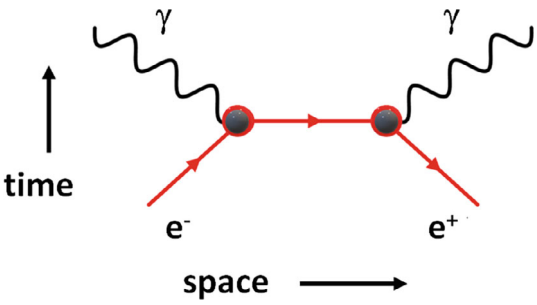


Fig. 8.4 Feynman diagram of the annihilation process between an electron and positron

range of energies, to observe the production of other leptons, for example, the muon/antimuon pair, tau/antitau or bosons W^+ – W^-.

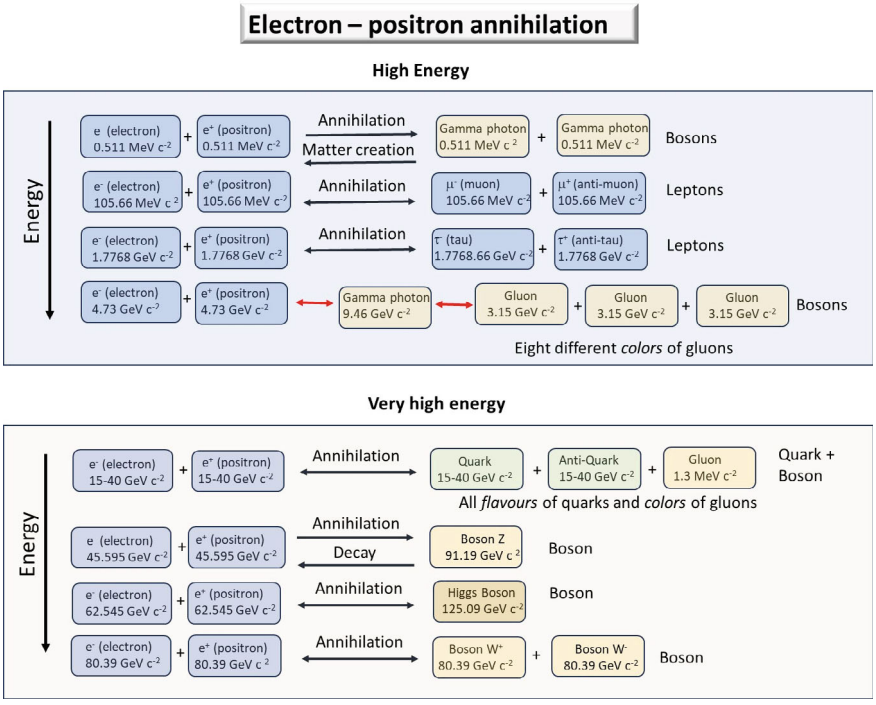


Fig. 8.5 Electron–positron annihilation at various energies

8.4 Electron-Antielectrons, Joined for a Few Moments, the Positronium

We have seen that the encounter between an electron and a positron is an event that leads to the annihilation of the two particles. It is a meeting of two worlds that of matter and antimatter that fortunately occurs only under special physical circumstances otherwise the whole universe of matter could not exist. However, the strangeness of the world of subatomic particles produces another truly special system called *positronium*. This system consists of an electron and an antielectron, that is, its antimatter twin, the positron (Fig. 8.6). How is this possible if the two oppositely charged particles should immediately annihilate each other? The *positronium* must then form before annihilation of the two particles occurs. Obviously this is an extremely unstable system and has a very short lifetime. Always considering the issue of spin, an electron–positron system could have two possible configurations, *parapositronium* (p-Ps) in which the positron and electron spins are antiparallel with an average lifetime of 125 ps in vacuum, and *orthopositronium* (o-Ps) in which, instead, the positron and electron spins are parallel and have a somewhat longer lifetime than p-Ps, of 142 ns in vacuum.

Positronium is thus an unstable system formed by an electron and a positron just before they annihilate. Somehow they form a bond, highly unstable and short-lived, finding a kind of balance between the charges. How this is possible can be guessed by thinking of the equilibrium that is established between electron and proton; similarly in *positronium* a state of bonding between the two particles is achieved, albeit for a very short time, similar to the proton-electron state of the hydrogen atom, which has only one electron and one proton that have opposite charges.

After their short life together, the two particles annihilate, and in the case of *parapositronium,* two photons are emitted, each with energy of 0.5108 eV, equivalent to the mass of the electron, while in the case of *orthopositronium,* three photons are formed whose energy and momentum follow their respective conservation laws.

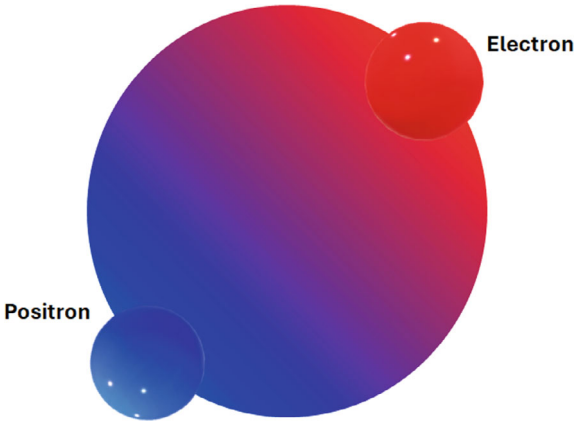


Fig. 8.6 Positronium consists of an electron and a positron in a quantum bound state

If *positronium* is a kind of hydrogen atom, defined as *hydrogen-like*, then it could theoretically also form molecules however ephemeral. This is exactly the case, and in 2007 the *di-positronium* molecule [10] consisting of two *positronium* atoms has been discovered [11].

8.5 The Heavy Brother of the Electron, the Muon

Around the mid-1930s, physicists thought they had already discovered virtually everything there was to know about subatomic particles. The proton, electron, and neutron were enough to explain atomic structure and its properties. In 1936, however, another strange particle was discovered, similar to an electron and with the same charge and spin, but larger mass. The discovery was so surprising that physicist Isidor Isaac Rabi (1898–1988), who would later win the Nobel Prize in 1944, when informed of the discovery blurted out, asking: *Who ordered that*?.

In 1936, again Carl David Anderson, the physicist who had discovered the positron, and his student Seth Neddermeyer were still studying cosmic rays with the fog chamber. During these observations they noticed that some particles deflected from their normal path as they passed through a magnetic field. In particular, the deflection was occurring with a smaller curvature than for electrons, but greater than for protons. This particular trajectory of the particle indicated that the charge must be the same as that of the electron even though its mass was larger while still being smaller than that of a proton. The intermediate behavior of this unknown particle led to its being called the mesotron. The particles with intermediate masses that had been discovered, however, did not end with it, and the term mesons was coined for them. This specific particle of the meson group was then given the name mu meson, using the Greek letter mu, μ, to distinguish it from other mesons. After some time, however, when more experimental details were added, it became clear that the mu has properties different from the actual mesons; in short, it was an infiltrator. Mesons are in fact heavy particles made up of quarks, defined as hadrons, while muons are elementary particles, in fact, in the *Standard Model* they are pigeonholed into the lepton family, along with the electron, the slimmer younger sibling, and the *tau* particle, the even more robust older sibling. Moreover, the decay of mesons occurs by emitting either a neutrino or an antineutrino, whereas in the case of the muon both are emitted, as we have already seen in Chap. 5. So it was clear that the *mu* meson was not a meson at all, and the particle was renamed a *muon*.

The muon is thus one of the elementary particles of the lepton family and in the *Standard Model* we find it enmeshed among the second-generation leptons, those of intermediate mass. Its identity card shows that it has negative electric charge, spin 1/2, a mass of 105.7 meV c^{-2} so it is about 207 heavier than an electron and about nine times lighter than a proton. However, it has a very short half-life that is exhausted in about 2.2×10^{-6} s. Its *antiparticle* is the *antimuon*, μ^+, which, as we shall see, can form a very exotic atom with the electron. If the muon decays into an electron, which has smaller mass and into a pair of neutrinos, a muon neutrino and an antielectron

neutrino, the antimuon will specularly decay into a positron, an electron neutrino and a muon antineutrino:

$$\mu^- \rightarrow e^- + \overline{\nu}_e + \nu_\mu$$

$$\mu^+ \rightarrow e^+ + \overline{\nu}_\mu + \nu_e$$

How are muons formed and where do they come from? As strange as the process may seem, muons are produced by cosmic radiation when it reaches the upper layers of the atmosphere and collides with particles it finds in its path. This is no small effect and it has been estimated that about 10,000 muons reaching one square meter of the earth's surface every minute.

8.6 Positronium Also Has a Sibling, Muonium

If two particles identical in all but charge, the electron and the antielectron, can find the conditions to form a bond with each other, albeit a very brief one, why shouldn't electrons and muons be able to do so as well? This is possible if an electron "falls in love" with the muon, or rather its antiparticle, the antimuon. The result of this strange union is an exotic atom, *muonium*, ($\mu^+ e^-$). The advantage of this marriage is that the muon, and thus the antimuon, has a mass more than two hundred times that of the electron and opposite charge which, should ensure relative stability. In fact, the *muonium* atom is quite similar in some ways to a hydrogen atom in which the proton is replaced by an antimuon. This exotic atom is thus more similar to a true hydrogen atom ($p^+ e^-$) than to positronium ($e^+ e^-$). *Muonium* was discovered in 1960 by U.S. physicist Vernon H. Hughes and has the chemical symbol *Mu* (Fig. 8.7).

This exotic atom is given by the combination of two leptons of different charge, but in fact the atom thus formed is not really the twin of positronium. If an electron and an antielectron are able to form an atom, however extremely unstable, why shouldn't the muon and antimuon be able to do so? If they could, they would go on to form a *true muonium*, ($\mu^+\mu^-$) the twin brother of positronium. This hypothetical exotic atom is referred to precisely as *true muonium* (*True Muonium*) and for now is only

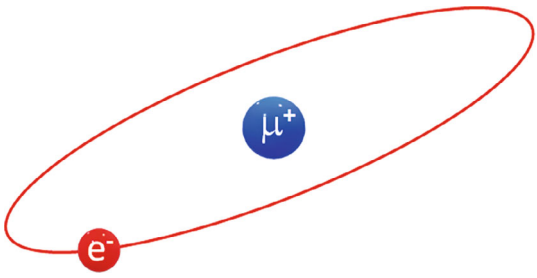


Fig. 8.7 Schematic diagram of a muonium atom formed by an electron and an antimuon, μ^+

a virtual particle even though its existence has been predicted within the *Standard Model*. A strange aspect of this exotic atom is that since it consists of an electron and a positively charged antiparticle the antimuon in theory should annihilate. However, the antimuon has a positive charge as the proton, so if the electron is able to orbit around a proton so can an electron rotate around the antimuon. The electron is not the antimatter counterpart of antimuon so the atom formed by this strange combination behaves similarly to the hydrogen atom.

8.7 The Most Unstable of All, the Tau Lepton

The family of elementary particles of negative charge and spin ½ is not exhausted by the electron and muon and includes a third particle, the heaviest of all, the *tau.* It has a mass as much as 3478 times that of the electron (~1.777 GeV) but a very short lifetime, 3.10^{-13} s, which explains why it was the last of the three particles to be revealed in the 1970s by Martin Perl who was awarded the Nobel Prize in Physics in 1995 for his discovery.

Tau, because of its large mass, is the only lepton capable of forming hadrons when it decays; the masses of the other leptons are in fact too small. What role does *tau* and more generally the third generation of particles play in the *Standard Model*? This is a question that has been asked many times before. Without a third generation, the Model would have been incomplete and because of the larger mass, they should have a strong coupling to the Higgs boson and likely hide the secret to understanding why the universe has more matter than antimatter. The new lepton was designated with the Greek letter *tau*, using the first letter of the word *tritos*, third in Greek and then represented as a triton.

One last question, but if electrons and muons are able to form strange exotic atoms, such as positronium and muonium, won't *tau* also be able to be part of a leptonic pair held together by the electromagnetic force? Probably so, but attempts to experimentally find the corresponding exotic atom, *tauonium* ($\tau^+\tau^-$), have so far been unsuccessful. However, ongoing experiments suggest that it should also be possible to find *tauonium* experimentally soon [12] (Fig. 8.8). The *tauonium* would be extremely small, the smallest and heaviest between the point-like lepton pairs. The *tauonium* Bohr radius is only 30.4 femtometers, i.e., approximately 1/1.741 the Bohr radius of a hydrogen atom. A very small particle that would allow to test quantum mechanics to its limits.

8.8 A Brief Schematic Summary

To close this chapter, let us briefly summarize the properties of the three leptons, e^-, μ^-, τ^-. Figure 8.9 shows the "identity card" of the three particles in terms of mass in MeV, mass compared to that of the electron, and lifetime in seconds.

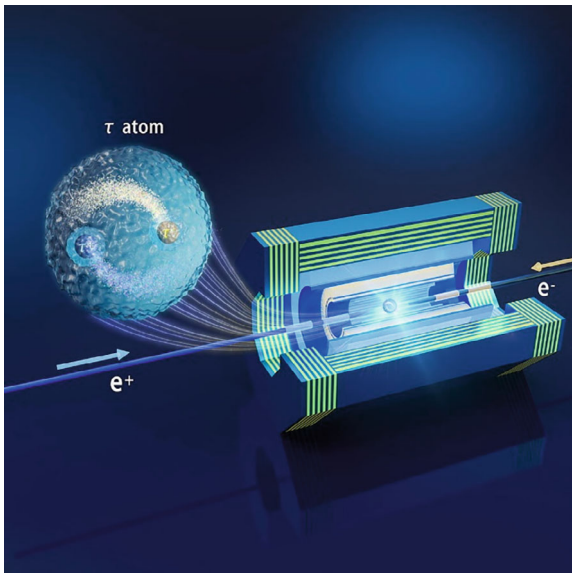


Fig. 8.8 Artistic image illustrating the formation of a tauonium. Reproduced with permission from ref. [12]

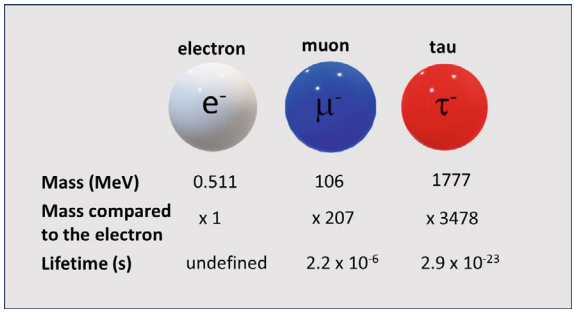


Fig. 8.9 Mass and lifetime of electron, muon, and tau

Table 8.1 Lepton pairs that form *quantum electrodynamics atoms*, with the year of their discovery

Pair	Year	QED atoms
Electron–positron	1951	Positronium (e^-e^+)
Electron–antimuon	1960	Muonium ($e^-\mu^+$)
Tau–antitau	Coming soon	Tauonium ($\tau^-\tau^+$)

Table 8.1 summarizes the different leptonic pairs held together by electromagnetic forces, the *positronium*, *muon,* and *tauonium*. These pairs are generally identified as *quantum electrodynamics (QED) atoms*.

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Chapter 9
Virtual Particles, Quasiparticles, and Quantum Fields



The electron, as we have seen, is one of the six leptons of the *Standard Model,* and using the Schrödinger equation and thus all the tools of quantum mechanics, it is always possible, albeit in probabilistic terms, to define its properties always keeping in mind its dual wave-particle nature. However, the electron remains something elusive and incomprehensible; after all, the question from which we started, i.e., what is an electron, still has no definitive answer. The ultimate ambition of physics is to arrive at a theory that describes everything, if that were even possible. The theory of everything, the one that would integrate gravity, i.e., general relativity and quantum mechanics, is not yet there, but *Quantum Field Theory* (QFT) is a sophisticated and fascinating theoretical model that accounts for the nature of fundamental particles. The QFT can be applied in different cases; the theory that describes the interaction between charged particles (such as electrons and protons) and the electromagnetic field is called *quantum electrodynamics*, (QED), while the theory that explains the strong force is the *quantum chromodynamics*, (QCD).

9.1 The Last Frontier, Quantum Field Theory

Quantum mechanics works well and does not rise interpretation problems as long as the shut up and calculate rule is applied. However, in the formulation expressed by the Schrödinger equation, quantum mechanics has an inherent limitation because it does not consider special relativity. If a quantum object is moving at the speed of light, the calculations using the semiclassical equations lead to wrong results. If one considers the atomic model in which the proton and electron interact within an electromagnetic field, this is, in fact, still described in classical terms (see Chaps. 1 and 2). The electromagnetic field varies continuously and not discretely as it should if this met the criteria of quantum mechanics instead. We have seen that Dirac wrote the extended formulation of the Schrödinger equation and that it had two solutions, one for matter

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P. Innocenzi, *What is an Electron?*, https://doi.org/10.1007/978-3-031-81984-1_9

and one for antimatter. Schrödinger's original equation did not incorporate Einstein's theory of relativity, which comes into play when the velocities of quantum-sized objects are close to those of light.

This discrepancy posed quite a problem because unless a limiting velocity, that of light, is introduced into quantum mechanics, we would even end up violating the cause-and-effect principle. If two measurements influenced each other at a speed faster than the speed of light, we could imagine that there is a reference system in which the future could influence the past, violating the principle of causality. That is not all; in fact, we have seen what happens to an electron when it encounters a positron, its antiparticle counterpart, it annihilates. *Beta decay*, in which a neutron, thanks to the weak nuclear force, transforms into a proton, an electron, and an antineutrino, would also find no explanation. One particle transforms into three new particles, and no quantum mechanical equation can explain why. Moreover, the famous question remains open: How come electrons are all the same?

This is where QFT comes in, to explain phenomena that cannot be understood within the theoretical frame of quantum mechanics that does not take into account the interactions of fields. Dirac himself found an early quantum formulation of the electromagnetic field surrounding a proton. This equation is so important that it is the basis of the *second quantum revolution.* The concept of the quantum field was later expanded to include the other force fields, the strong nuclear field, and the weak nuclear field.

How does QFT work? The theory associates each particle in the Standard Model with a field that extends throughout space–time. Fields are ubiquitously present and exist everywhere; they are filled with virtual particle pairs, that is, particle-antiparticle pairs that are constantly being created and destroyed. In the quark-up field, for example, there will always be a spontaneous creation of quark up and antiquark up that form and annihilate almost instantaneously. These virtual particles cannot be measured, and they form by borrowing energy from the vacuum, as long as the energy is put back into the same place from where it was taken, precisely the vacuum. This process happens so quickly that it is as if nature does not register them as real, measurable particles.

These fields can be visualized as a kind of infinite sea in which ripples are continuously and randomly produced even when the fields are in their lowest energy state. This turbulence is due to the creation and annihilation of virtual particles; if adequate energy is available, a large enough wave can be created such that it can be measured, and this wave is nothing more than a particle.

Therefore, a quark up will be created in the quark field if there is an energy of 2.3 MeV, corresponding to the mass of a quark up. A particle is nothing more than an excitation in this field, and these excitations are quantized and not continuous. Using the background energy, one can create only whole particles and not their fractions, one electron, two electrons, but not one and a half electrons.

Quantum field theory also explains why electrons are all identical, answering Wheeler's question (see Chap. 3). Electrons are the same because they are an expression of the same field that fills all space in the universe. We can imagine a field as a mathematical entity that fills space–time as if it were fluid, and each of these

points has scalar, vector, or tensor properties or even other more exotic properties. However, relativity imposes a number of constraints on the fields, in particular, that certain symmetries that coincide with the symmetries of space–time could be observed; thus symmetries of translations, rotations, or change of reference system. This is where specific mathematical objects come into play, which are the spins that we have seen before. If we use numbers, we will have a scalar field, such as that of the Higgs Boson, and each point will have zero spins since when a rotation of space around a point in the field occurs, the spin does not change. Vectors, on the other hand, have a specific orientation in the space and will have spin one because when a complete rotation is performed, the vector returns to describe exactly the same spin. Then there are *spinors*, and here, as we know, it takes as many as two revolutions to return to the initial configuration (always with the caveat that a revolution has not a physical meaning, but it is defined by appropriate mathematical operators). Each of these symmetries implies compliance with conservation laws.

According to QFT, energy can be transferred between two fields through a field of mediating bosons, since matter particles cannot interact on their own without a boson mediating the interaction. Thus, while in quantum mechanics there is no way for the neutron to split its energy into three different particles, a proton, an electron and an antineutrino, in *Quantum Field Theory* this can be done through the exchange of energy between fields.

For example, the transformation in *beta decay* is mediated by the weak nuclear force. The neutron, which is not an elementary particle, consists of two quarks down and one quark up. The weak force governs the decay of the neutron through the exchange of bosons. During *beta decay,* a quark down in the neutron changes into a quark up through the emission of a virtual W^- boson from the weak interaction's quantum field. As the quark down changes to an quark up, the energy associated with this transition is transferred from the quark fields (via their interaction with the weak interaction field) to the W^- boson field. The quantum field corresponding to the W^- boson gains energy and momentum, causing a virtual W^- boson to be emitted. The virtual W^- boson is unstable and decays almost instantaneously into an electron and an electron antineutrino. In this decay process, the energy from the W^- boson field is transferred to the electron and neutrino fields. This results in the production of a real electron and an electron antineutrino. The interaction is explained by the *Lagrangian* of the *Standard Model*, which encompasses terms that depict the interactions among the quark fields, the W^- boson field, and the lepton fields.

The transition of a quark down to an quark up is accompanied by the emission of a virtual W^- boson. This process is represented mathematically by interaction vertices in the Feynman diagram for *beta decay*.

So, starting from a single particle in the quark down field, one arrives at as many as three particles precisely because the fields are able to exchange energy. Although the number of particles has changed, the total energy is conserved if we add all the masses and kinetic energy at the beginning and end of the process. Therefore, the energy of the fields is the same before and after. One can mathematically explain and make a model for the creation and annihilation of particles and the mechanism

of force exchange. Moreover, the theory is entirely relativistic, i.e., free of causality violations.

Even this advanced model of quantum mechanics is not perfect; for example, gravity is still missing. A quantum field theory that takes gravity into account has not yet been defined; if we had it, one could also consider a field for gravitons, the hypothetical messengers of the gravitational force, and gravity would be mediated between matter particles through the exchange of this gravitational boson. QFT also provides an interpretive model for defining what an elementary particle is, that is, the fundamental particles of the *Standard Model*: *an elementary particle is the smallest possible vibration of a quantum field*.

Therefore, QFT can be considered still an incomplete model as well, but its great appeal is related to the idea of the intimate connection existing in the space–time that we call the Universe. QFT gives us an interpretative model according to which the Universe and everything in it, including all living things, planet earth, every star, galaxies, are all composed of excitations in the same quantum fields, and all these fields extend not only to all space but also to all time.

9.2 But What Happens When Two Electrons Meet?

With the introduction of QFT, it is possible to describe more accurately what happens when two electrons meet. The two particles have the same charge and repel each other; so far, nothing is unusual. Nevertheless, describing this kind of interaction is more complex than it might seem. The two particles are identical, cannot see each other, and do not come into contact, yet they feel their reciprocal presence enough to repel themselves.

In the brief description of QFT, a term that is very subtle to define has been used: the *virtual particle*. This definition, like many others we have encountered, for example, that of spin, is somewhat misleading from a semantic point of view. In fact, the *virtual particle* is not a particle at all despite the name used to define it. So, how can we define this mysterious entity that is so important in the transmission of information in a field but is not a particle? Probably a good definition is *field perturbation*, a disturbance caused by the presence of other particles or fields. A real particle can move by itself in space, but a *virtual particle* will disappear once the perturbation that created it ceases. That is why using the word particle to define an object that has not a physical nature is rather misleading.

In contrast, the electron is a real particle, and we are able to create electron currents and electron beams, just as with photons, which are real particles of light. When two electrons approach to each other, their electric charge will cause a perturbation of the electromagnetic field (Fig. 9.1). This perturbation is not a photon or a wave traveling at the speed of light; it is what is called a *virtual particle*.

This perturbation is crucial because it generates the repulsive electric force between two particles of the same electric charge, which is the force that the two

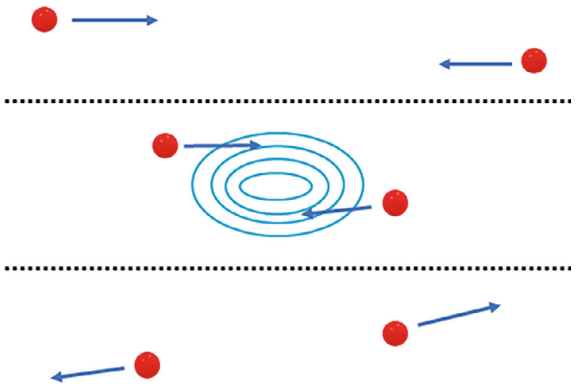


Fig. 9.1 When two electrons approach each other they generate a perturbation in the electromagnetic field (represented by the blue light ellipses) that corresponds in the QFT to the photon field. This perturbation will tend to separate them by moving them away from each other, and their trajectories are directed outward. The process is described as a "virtual photon exchange" even if it is not an exact definition of the process

electrons exert on each other. The world consists of "real particles" such as electrons, which for QFT would be nothing more than the excitations of the underlying quantum fields. This is also true of virtual particles that are themselves excitations of the underlying fields but only "temporarily."

To visualize the *virtual photo exchange* process, we now use Feynman diagrams. These diagrams are the most common way physicists use to graphically represent interactions between particles. Figure 9.2 shows the Feynman diagram of the electron–electron interaction according to QFT.

The location where lines intersect is referred to as a vertex. At each vertex, one can analyze the conservation laws that regulate particle interactions. It is required that at every vertex, charge, baryon number, and lepton number are conserved. Feynman diagrams are actually space–time diagrams, although arrows are usually omitted. The y-axis corresponds to ct, and the x-axis corresponds to space. Particles travel in space–time following the direction given by the arrows; for antiparticles, the arrow is reversed. The two vertices in Fig. 9.2 are connected by a wavy line with no arrows. This line corresponds to the virtual particle, in this case a photon, that is exchanged in the interaction.

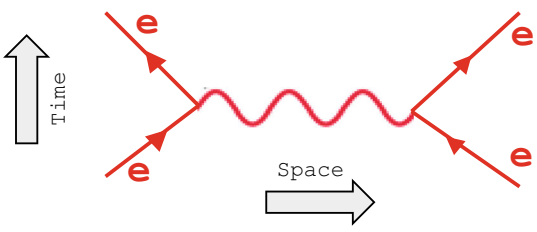


Fig. 9.2 Feynman diagram of the electron–electron interaction in QFT

9.3 Virtual Particles, What Are They?

In describing the interaction between two electrons, we have just resorted to the concept of *virtual particles*. Let us now delve deeper into the concept and see that perhaps these particles are not so virtual as we have defined and described them. To begin, we should underline that the connection between virtual particles and the uncertainty principle is something real and can explain the formation of such strange entities. Because the *virtual particles* have been defined as temporary fluctuations that occur in quantum fields, they are not directly observable and do not obey the usual energy–momentum relationship of real particles. The concept of virtual particles is closely linked to the energy-time version of the uncertainty principle:

$$\Delta E \cdot \Delta t \geq \hbar/2.$$

The uncertainty principle allows for short-lived violations of energy conservation, provided they occur within a short time interval. Specifically, if there is a large uncertainty in energy (ΔE), the uncertainty in time (Δt) can be correspondingly small, allowing for processes that seemingly violate energy conservation as long as they occur within a time frame dictated by the uncertainty relation (Fig. 9.3).

If a large amount of energy emerges with respect to the average zero because of the fluctuations in the quantum field, this event can last only a very short time (Fig. 9.3). Otherwise, if this energy is small, it can persist for longer. These temporary fluctuations can be thought of as "borrowing" energy from the vacuum to create particle-antiparticle pairs or other quantum states. These particles exist for a short time (determined by Δt) before annihilating or decaying. Since they do not last long enough to be directly observed, they are defined as "virtual." The energy required to create them is "borrowed" and must be returned quickly, following the energy-time uncertainty principle.

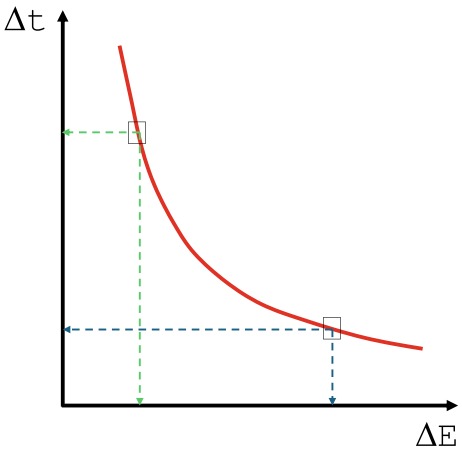


Fig. 9.3 Graph shows the correlation between the change in energy, ΔE, and the change in time, Δt

9.3.1 The Casimir Effect

We have defined the virtual particles as not physical entities, a possible derivation of the mathematics used to model the QFT. However, some experiments indicate that virtual particles may be more real than we imagine. One example is the so-called *Casimir effect*, first predicted by Dutch physicist Hendrik Casimir (1909–2000) in 1948 [1]. The *Casimir effect* is a physical phenomenon that arises due to quantum fluctuations of the vacuum, and it is intimately connected to the concept of virtual particles in QFT [2]. The *Casimir effect* describes the attractive force that occurs between two uncharged, parallel conducting plates placed very close to each other (the distance between the plates is in the nanometric scale) in a vacuum. The attractive force arises not because of any classical interaction (like electromagnetic forces), but due to quantum effects associated with the vacuum between the plates. The vacuum, according to QFT, is not completely empty. Fluctuations happen because energy is taken from the vacuum to produce pairs of particles and antiparticles, which quickly appear and disappear in the field.

When two conducting plates are close together, they impose boundary conditions on the quantum fields between them (Fig. 9.4). Specifically, they restrict the wavelengths of the virtual particles that can exist between the plates. Only certain modes (wavelengths) of the quantum field can exist between the plates, while the field is less constrained outside the plates. Between the plates the energy density of the vacuum is slightly lower than outside the plates due to these boundary conditions. This difference in vacuum energy leads to a net force that pushes the plates together.

The presence of the virtual particles becomes essential in mediating the *Casimir effect*, even if they are not directly observed. The *Casimir effect* provides an experimental example of how virtual particles, though not directly observable, may have measurable consequences in the physical world. However, the true nature of these perturbations, defined as *virtual particles*, which are otherwise well described and

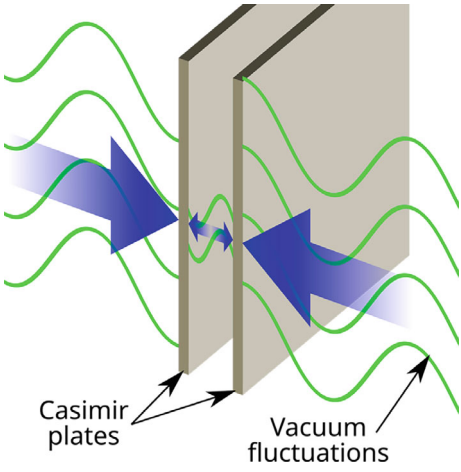


Fig. 9.4 Illustration of the Casimir effect with parallel plates mutually attracted to each other by the Casmir force. Inside the plates the wavelength of the virtual particles is restricted while outside the plates the field is not constrained. The energy density of the vacuum between the plates is slightly lower than the energy density outside the plates and this difference creates a net force that pushes the plates closer [3]

established by mathematical calculations, is still far from having found a comprehensive explanation [4, 5]. Their introduction is perhaps only a mathematical artifice that for the time being finds its coherence within the mathematics used in *Quantum Field Theory*. Then again, if *virtual particles* represented a physical reality, there should be no other way to do *QFT* calculations without considering them. Instead, there is a particular version of *QFT* that does not make use of virtual particles; it is the family of *Lattice Field Theories* in which space–time itself is defined using a discrete grid [6, 7]. It is not based on perturbation theory and, therefore, does not use virtual particles even though the same results are ultimately obtained. This represents a reasonable doubt that virtual particles may be merely a mathematical artifact.

9.4 Electrons Split, Spin-Charge Separation. But then Is the Electron Not so Elementary?

In the absence of experimental evidence for now, the electron remains a fundamental particle; therefore, no smaller elementary subatomic structures, such as the quarks, make it up, as is the case with protons and neutrons. However, the surprises with the electron never end…. Let us try to imagine a situation in which electrons are pushed to come into close contact with each other, how much can this process be forced? The electrons, since they all have the same negative charge, must repel each other, but what can happen if they are forced together by the physical conditions of the system? It can be assumed, for example, that electrons, to avoid getting too close to each other, may exhibit some exotic properties. An extreme condition is confinement at a very low temperature in a very thin wire of atomic size, for example, in a nanowire. In this state, it might be reasonable to expect to observe something strange because the movement of electrons is restricted to one dimension and slowed down by the low temperature. What might happen if, being in this situation, they were excited?

As could be expected, theoretical physicists have indulged in exploring this extreme physical state by trying to predict the properties of electrons. Sin-Itiro Tomonaga in 1950 and then Joaquin Mazdak Luttinger in 1963 were the first to develop a theory about quantum systems interacting in one dimension. In 1981, the physicist Duncan Haldane generalized the earlier theories by proposing the *Tomonaga-Luttinger Liquid Model* [8], by which he theoretically hypothesized that electrons, when spatially confined at low temperatures, would always change their properties to separate magnetism and charge into two new types of particles, the *spinons* and *holons*. The use of the term particle is not properly the appropriate one, and for this type of entity physicists have coined a special term, which is that of *quasiparticle*, on which we shall dwell a little later.

The electron should then exhibit under specific conditions, which are in any case extreme, temperature close to absolute zero and spatial confinement, a phenomenon entirely unexpected, at least on the basis of the elementary particle concept, the separation between two intrinsic properties, spin and charge (*spin-charge separation*)

[9, 10]. The consequence of this separation is the formation of *quasiparticles*, the two new entities defined as *spinons* and *holons*. The term *holon* indicates the nature of the *quasiparticle*, which is an electronic vacancy, a *hole* or electronic hole that will move through the one-dimensional system as if it were a real charged particle. Information about the spin of an electron is carried by a *spinon*, and information about charge by a *holon*. The process occurs separately and independently; that is, a *spinon* carries the spin, but not the charge, and conversely, a *holon* carries the charge but not the spin. The idea behind the separation of spin and charge is that electrons behave differently when their motion is restricted to one dimension, so in one-dimensional systems such as those consisting of atomic wires, they move with different velocities. If they move in one dimension, the repulsive force between their negative electric charges becomes dominant, and this generates collective effects that are strong enough to disrupt the flow of spin and charge information of a single electron. Thus in one-dimensional solids, the collective excitation of a system of electrons can lead to the appearance of new particles, or rather as we have seen *quasiparticles*. Unlike in three-dimensional metals whose low-energy excitations are fermionic *quasiparticles*, the low-energy excitations of fermions moving in a material, often an atom, are collective bosonic waves of spin and charge density (*spinons* and *holons*) that disperse linearly.

The surprises did not end, however. Theoretical models actually predicted the possible formation of another entity in a one-dimensional solid: the *orbiton* [11]. The *orbiton* can also be considered as a collective excitation of electrons in a one-dimensional solid that behaves just like an electronic *quasiparticle*. As an electron, it would be quite an odd one since it has orbital angular momentum but without spin or electric charge. This *quasiparticle* completes the series of the three theoretically predicted, and thus *spinons, holons*, and *orbitons* are three possible entities with which an electron can manifest itself when spatially confined at temperatures close to absolute zero. The three *quasiparticles* can move in the material not only with different velocities but also in different directions. The wave-like nature of the electron means that when confined in a one-dimensional material, the wave splits into three, and each of these waves carries, as we have seen, a different property that nevertheless cannot exist independently outside the material. From a theoretical point of view, one could think of an electron as the combination of the three quasiparticles, with the *spinon* carrying the spin of the electron, the *orbiton* carrying the angular momentum, and the *holon* carrying the charge.

However, these are *quasiparticles*, not physical particles in the strict sense; waves form in the electronic structure of the one-dimensional material, oscillations of electrons having the same spin (*spinons*), oscillations of electronic vacancies (*holons*), and finally of orbital angular momentum (*orbitons*). The electron does not change; only the specific physical confinement conditions generate these *quasiparticles*, which we can treat as if they were real particles, but they are only collective oscillation modes. To be more rigorous, the phenomenon cannot properly be defined as separation of spin and charge, because these are only collective oscillations of electrons that preferentially carry spin or charge information under one-dimensional conditions. It is enough, however, to imagine new applications in which information transport can occur in entirely new ways.

Thus, the particular conditions that restrict the motion of the electrons in one dimension cause these three *quasiparticles* to behave as if they were independent particles. Figure 9.5 shows how *spinons* form and move in a one-dimensional system.

(1) An electron is excited and jumps from the *b* band to the *a* band. This excitation gives rise to the formation of an exciton (the pair formed by the electron being excited in the *a* band and the gap formed as a result in the *b* band). The exciton with finite momentum (blue arrow) is created at the instant t = 0 when the electron is excited from the full *b* orbital to the empty *a* orbital.
(2) The electron in the *a* orbital jumps to the left, while an electron with opposite spin that is in the *b* orbital jumps to the right going to occupy the place left by the electron that jumped into the *a* band. A pair of electrons with the same spin is formed; their collective movement can be treated as a quasiparticle that will move leftward as a spin wave. The hole in turn then moves to the left, creating a *spinon* in the *b* orbital.
(3) The *orbiton* and the *hole* a move in a bound state to the left, while the *spinon* moves freely to the right.

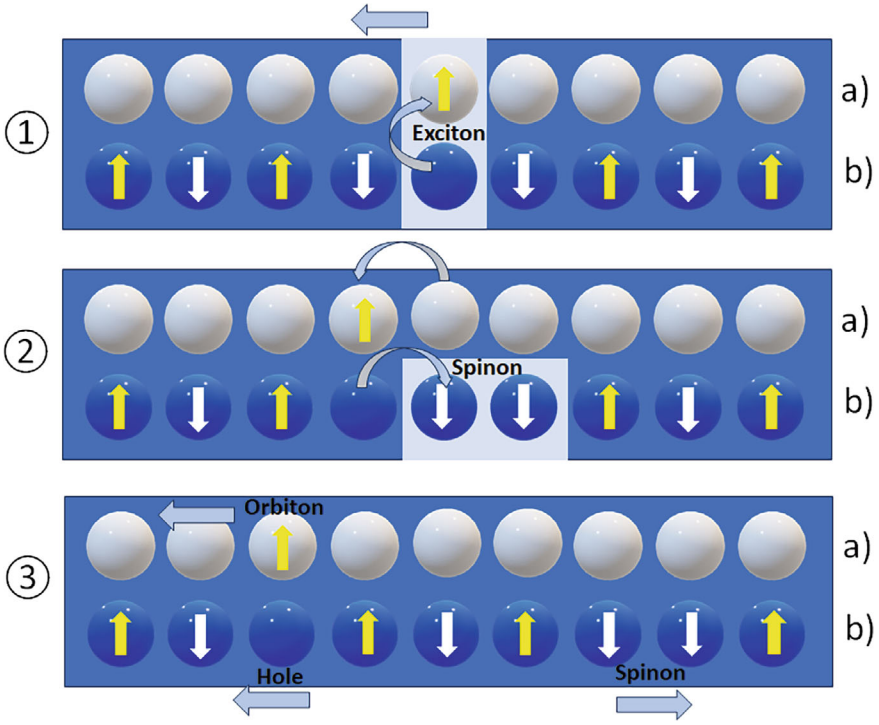


Fig. 9.5 Illustration of spin–orbit and spin-charge separation in a one-dimensional two-orbital chain (*a* and *b*) [12]

Clearly, the restriction in one dimension is crucial for *spinons* and *holons* to form; in a two- and three-dimensional solid, the information would be lost immediately because the holes would move randomly. Measuring the properties of an electron at extremely low temperatures and in a one-dimensional wire is not a small technological challenge, so it took until 1996 to find experimental evidence of *spinon* and *holon* formation [13, 14]. In 2012, another experiment finally demonstrated the formation of the quasiparticle pair, *spinon* and *orbiton* [15]. The effect of spin-charge separation in quasiparticles then came to life.

9.5 Quasiparticles, But What Are They?

The introduction of the term *quasiparticle* in physics dates back to the studies of the great Russian theoretical physicist Lev Landau (1908–1968) published in 1956, when he devoted himself to solving the problem of superfluidity of liquid helium [16]. A macroscopic system consists of an enormous number of particles, and it is practically impossible to describe the motion of every single particle. The example that is usually given to account for the inherent difficulty in describing an interacting system of particles in the solid state is that of a grain of sand of the size of 0.1 mm. The total number of nuclei composing the grain reaches the astronomical number of 10^{17} while the electrons are even more. The number of particles is far too large a number to be described by Schrödinger's second-order partial derivative equation. Indeed, the Schrödinger equation would, in principle, make it possible to calculate the exact properties of this system. However, one would have to use a partial differential equation on a vector space of 3×10^{18} dimensions, one dimension for each particle's coordinate (x, y, z), an impossible task.

One possible solution to this *impasse* is the one used by Landau, who introduced the *quasiparticle* concept. Instead of considering all the individual particles in a solid, electrons and protons, whose interactions are governed by Coulomb's law, he mathematically transformed their complex motion into that of imaginary particles, the *quasiparticles*. Landau postulated that *quasiparticles* with charge and spin 1/2 could be treated as virtually free electrons; the interaction effects are incorporated using a renormalized mass m*. The *quasiparticles* behave more like a non-interacting classical particle that would then move in a straight line and constant velocity and whose motion can be described in terms of classical physics. The description of *quasiparticles* motion is much simpler even though, for all intents and purposes, they can be regarded as *fake* particles. In general, the equations describing the dynamics of *quasiparticles* turn out to be simpler than those of the underlying interacting particles.

One of the most important examples of applying the *quasiparticle* concept is the description of the motion of electrons in a semiconductor. Semiconductors are used to create junctions that enable the realization of fundamental electronic devices such as diodes and transistors. Semiconductors such as silicon, when excited, promote an electron from the conduction band to the valence band. Band hopping by the

electron leaves an empty space, the "hole," which behaves virtually like a positively charged particle that can move through the semiconductor lattice. By doping the semiconductor, so that it has an excess of electrons (n-type semiconductors, doped by example with phosphorus) or an excess of holes (p-type semiconductors, doped by example with boron), p-n junctions can be made. The application of a potential difference makes it possible to control the collective movement of electrons and positively charged *quasiparticles* given by the presence of the holes.

Similar to other quantum systems, there is a fundamental state and several excited states, of which only those with energy close to the fundamental state are relevant: the Maxwell–Boltzmann distribution establishes that, given the same temperature, the highest thermal fluctuations are the least likely. Excited states of the lowest levels can contain an arbitrary number of elementary excitations (phonons, quasiparticles, and collective excitations), and *quasiparticles* are a type of excited state of the lowest energy levels.

The list of *quasiparticles* is quite extensive, and among them, we find some, such as *phonons* and *plasmons,* that are fundamental in describing some properties of condensed matter. *Phonons* are collective excited states that represent the vibrational motion of atoms, or ions, in a crystalline material. They are considered quantized packets of vibrational energy, analogous to the way photons are quantized packets of light. Phonon can be described as a quantum of a sound wave and are responsible for thermal conduction. The *plasmon* is the quantum of collective plasma oscillations in which all electrons oscillate simultaneously with respect to all ions.

The number of *quasiparticles* is continually expanding due to the possibility of studying collective phenomena in solid-state matter under increasingly extreme conditions of spatial confinement, such as in one-dimensional and two-dimensional materials and at temperatures close to absolute zero.

9.5.1 An Example of Quasiparticle by Electrons, the Polarons

One of the quasiparticles involving electrons is the *polaron.* A *polaron* is a quasiparticle that represents an electron (or hole) interacting with its surrounding lattice in a solid-state material [17]. When an electron moves through an ionic or polarizable lattice, its electric field can distort or polarize the nearby atoms or ions in the lattice (Fig. 9.6).

This distortion creates a local "cloud" of phonons (quanta of lattice vibrations) that accompanies the electron as it moves through the material. The combined system of the electron and the surrounding lattice distortion is defined as a *polaron.* This concept helps explain how an electron's effective mass and property can be altered due to its interaction with the lattice. *Polaron* theory is very useful in describing the properties of the electrons in some materials, especially semiconductors and ionic crystals. There are two main types of polarons: *Small polaron* that form when the electron is strongly coupled to the lattice, and the distortion it creates is highly localized, of the order of the lattice constants. The electron essentially becomes

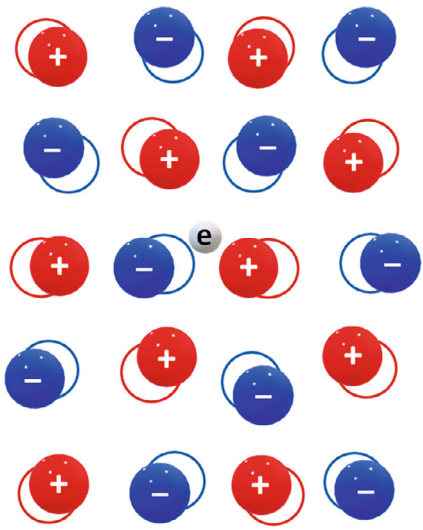


Fig. 9.6 Artist view of a polaron. A conduction electron in an ionic crystal or a polar semiconductor repels the negative ions and attracts the positive ions. A self-induced potential arises, which acts back on the electron and modifies its physical properties [18]

"trapped" in the lattice deformation and moves via hopping between sites. The other type is the *Large polaron* that occurs when the electron is weakly coupled to the lattice, resulting in a more extended distortion. The spatial extension of a polaron is large compared to the lattice parameter of the solid, the electron can move more freely and the lattice distortion spreads over a larger region.

9.5.2 *Other Electron-Based Quasiparticles, Plasmons, and Polaritons*

Let's make another example of quasiparticles involving electrons that is very much used in several applications, the *plasmon.* In a system with free electrons, metal, or semiconductor, applying an external stimulus, such as an electromagnetic field, directly affects the electrons. The free electrons in the material can collectively oscillate, forming a quantum mechanical wave named as a *plasmon.* Plasmon formation can occur in the bulk of the material, typically a metal or semiconductor (*bulk plasmon*), or at the surface (*surface plasmon*). This phenomenon is particularly interesting if it occurs at the interface between a metal and a dielectric because the surface plasmons are very sensitive to any change of the state of the surface.

Bulk and surface plasmons differ in the location of collective electron oscillations within a material and their interaction with the environment. *Bulk plasmons* are collective oscillations of the electron gas that occur throughout the volume of the material. The conduction electrons oscillate within the material relatively to the positively charged ions. These oscillations are not confined to the surface. The frequency of the oscillations is the bulk plasmon frequency and depends on the density of

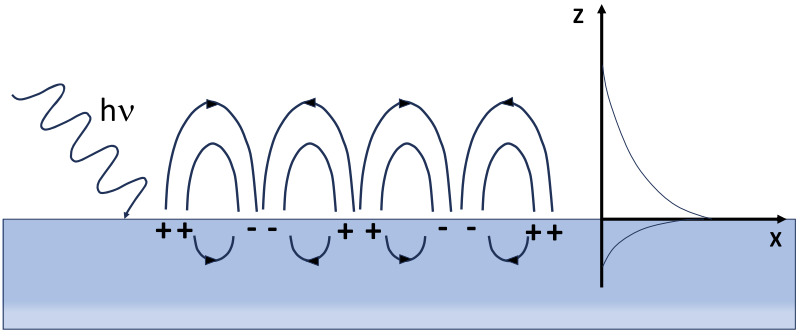


Fig. 9.7 Diagram illustrating how an electron density wave moves along a metal–dielectric interface. These waves, known as surface plasmon-polariton waves, involve oscillations in charge density and associated electromagnetic fields. The graphic on the right displays the exponential decrease in electromagnetic field intensity as the distance from the interface increases. These waves can be easily activated by visible light

conduction electrons in the material. It is typically higher than the surface plasmon frequency.

Instead, *surface plasmons* are confined to the interface between a conductor and a dielectric, such as air, glass, or vacuum; the electron oscillations travel along the material surface and decay exponentially into the metal and the surrounding dielectric. The collective oscillation motion of electrons in a surface plasmon also creates an electromagnetic field in the dielectric at the interface with the metal.

The surface plasmons are generated by light excitation and can strongly couple with the associated electromagnetic field to form *Surface Plasmon Polaritons* (Fig. 9.7). When light interacts with the metal–dielectric interface at a specific frequency, it can excite the electron oscillations, and the two systems become coupled, forming a polariton, a mixed state of photon and plasmon. These hybrid light-matter waves propagate along the surface.

Plasmons in nanoparticles or thin metallic films are a special case because they can localize at specific points (*Localized Surface Plasmons*), leading to enhanced electromagnetic fields used for sensing or enhancing chemical reactions.

9.5.3 Strange Demons Can Appear

Now suppose we have a metal that is a little stranger than conventional ones, in which there can be, at least theoretically, multiple electron bands. In this case multiple electron populations would be observed, like conduction electrons in different energy bands, which oscillate relative to each other. In 1956, U.S. physicist David Pines [19] hypothesized that in such a metal, there could be a collective oscillation of electrons from different bands that would not produce a net electric field.

Unlike *plasmons*, where the conduction electrons in a metal oscillate collectively against the positively charged ion background, producing charge density waves, the demon involves oscillations between different groups of electrons, and these oscillations cancel out overall, resulting in no net charge density fluctuation. A demon corresponds to counter-propagating currents of electrons from different bands. They are like ghost oscillations, but they are real, even if difficult to detect. In fact, because it is charge-neutral, the Pines' demon does not couple with electromagnetic radiation, which makes it harder to detect experimentally using optical methods like those used for plasmons. By the way, the strange name has been chosen to metaphorize Maxwell's demon.

The demons have been so elusive to detect that they were finally experimentally observed only in 2023. Evidence of Pines' demon has been observed in strontium ruthenate (Sr_2RuO_4) [20], a superconducting layered ceramics with a multiband electronic structure.

9.6 Electron Crystals: The Wigner Crystals

As we have observed in various examples, the surprises seem never-ending when a phenomenon involving electrons is shifted in scale, approaching atomic or at least nanometer scale, and at very low temperatures. One of these weird physical phenomena is the formation of the so-called *Wigner Crystals*, named after the Hungarian physicist Eugene Paul Wigner (1902–1995), one of the founders of the theory of symmetries in quantum mechanics, who predicted their existence in 1934 [21].

According to Wigner, when Coulomb interactions between electrons become stronger than their kinetic energy, true electron crystals can be observed in which electrons form tightly packed structures. At extremely low temperatures, near absolute zero, electrons possess minimal kinetic energy and can arrange themselves in an organized manner. At low densities, potential energy overrides kinetic energy, and the spatial organization of electrons becomes important because it helps minimize the system's energy. Electrons arrange themselves spatially in an orderly manner, forming a body-centered cubic "crystal" structure in a three-dimensional material, 3D, a triangular lattice in two-dimensional structures such as graphene or other 2D materials, and an equispaced lattice also in one-dimensional structures, 1D [22].

A group of Princeton physicists obtained the first direct visualization showing the formation of Wigner crystals in 2024 (Fig. 9.8) [23]. The experiment used a scanning tunneling effect microscope and a defect-free graphene sheet as the two-dimensional material. The sample was cooled to very low temperatures while applying a magnetic field to form a two-dimensional electron gas. The images confirmed the extraordinary ability of electrons to form "crystal-like" structures. These crystals form despite the fact that electrons, by their charge, tend to repel each other. However, because of the finite density imposed by the dimensional restrictions of the 2D material, they

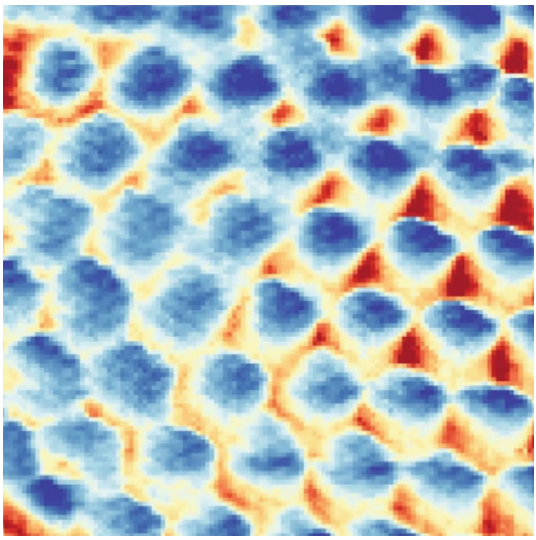


Fig. 9.8 Image of a triangular Wigner crystal taken by scanning tunneling microscope. Credit: Yen-Chen Tsui and team, Princeton University

cannot separate beyond a specific limit, and to minimize the energy of the system, they arrange themselves in an orderly manner.

9.7 When Electrons Couple, Cooper Pairs

In the previous chapters, we have seen that under "extreme" conditions of temperature and spatial confinement, electrons exhibit totally unexpected properties. According to classical physics theory, the closer the temperature approaches absolute zero, the more the internal motions of the atoms of a solid will slow down until they stop when absolute zero is reached, at 0 K. However, this is not quite the case, and when, overcoming technical difficulties, it was possible to approach absolute zero, a strange behavior of condensed matter was observed. Dutch physicist Heike Kamerlingh Onnes (1853–1926) of Leiden University was almost obsessed with low-temperature phenomena and devoted virtually his entire life of study to them. While studying mercury he observed something amazing, at 4.2 K the electric current seemed to flow without any resistance, i.e., without loss of energy, Onnes had stumbled upon the phenomenon of superconductivity [24, 25]. Later it became clear that this phenomenon was not limited only to the case of mercury but other alloys and metals had similar property.

Before the experiments on conducting materials at low temperatures, several assumptions had been made about how electrons might behave near absolute zero. It was reasonable to predict that the movement of electrons would stop and resistivity would become infinite at absolute zero. Otherwise, according to other hypotheses, resistivity should have steadily decreased until it reached zero. Electrons flowing

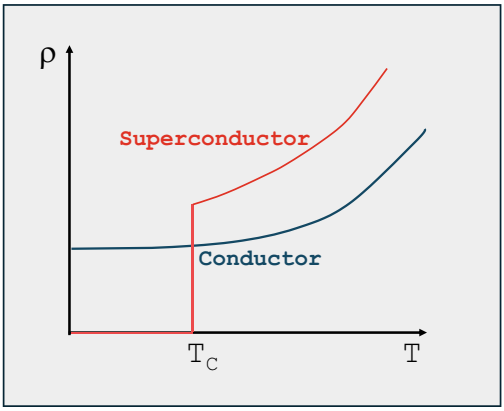


Fig. 9.9 Temperature-dependent change in resistivity, *r*, near absolute zero. In a superconductor, the resistivity goes to zero at the Critical Temperature, T_C

in a normal conductor, such as copper wire, lose some of their energy when they "bump" the atoms in the lattice of the material. In a superconductor, on the other hand, the current appears to flow as if the atoms in the solid were not present. The value at which the drop in resistivity occurs is defined as the *Critical Temperature*, T_C (Fig. 9.9). Another strange phenomenon observed in a superconductor is related to magnetism. When external magnetic fields are applied, as long as they are not too strong, they are unable to penetrate the superconductor but remain confined to its surface. This phenomenon is known as the *Meissner effect,* after the German physicist Fritz Walther Meissner (1882–1974), a student of Max Planck, who first observed it in 1933.

Having discovered the fascinating phenomenon, it remained to explain why, but the physics behind superconductivity was very complicated because it is closely related to quantum mechanics. Therefore, it was necessary to wait another forty-six years before the phenomenon was finally explained. In 1957, three U.S. physicists, John Bardeen (1908–1991), Leon Cooper (1930), and John Robert Schrieffer (1931–2019) proposed a theoretical explanation, the BCS theory from the initials of their last names, that finally allowed light to be shed on the strange phenomenon of electronic superconductivity [26]. In 1972, the BCS theory earned them the Nobel Prize.

The BCS explanation is intrinsically linked to the nature of the electron, which, as we now know well, is a fermion in that it has spin ½ as opposed to bosons, which have integer spin. We also know that a boson has no energy restriction and, therefore, multiple bosons can be in the same energy state, as opposed to electrons, which are limited by Pauli Exclusion Principle. An electron moving in the lattice of a conducting material interacts both with the other electrons from which it is repelled by having the same negative charge and with the positive ions of the lattice in the metal from which they are attracted. The attraction of the electron to the positive ion causes, in turn, a distortion of the average position of the ion toward the electron itself, causing a local increase in the positive charge density. This positive charge density will attract other electrons until such attraction can lead to the creation of electron "pairs," the famous *Cooper Pairs* (Fig. 9.10).

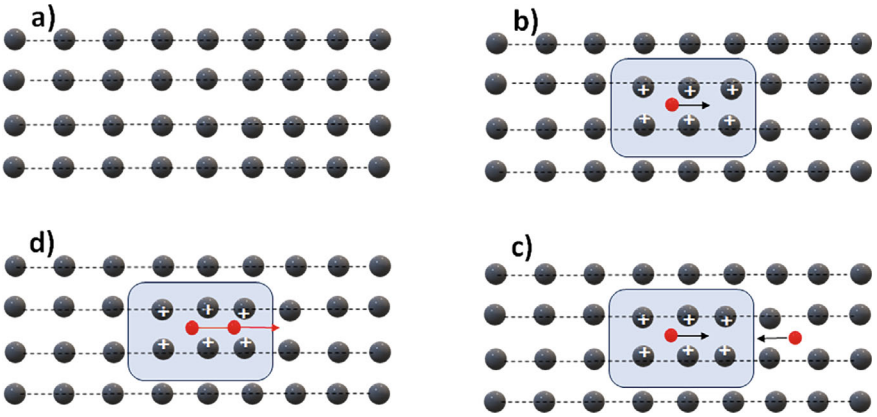


Fig. 9.10 Visual model of the formation of *Cooper Pairs*. **a** Undistorted crystal lattice. **b** An electron moves within the lattice and causes a slight distortion toward its trajectory. **c** Another electron moving in the opposite direction is attracted by the positive charge density formed as a result of the attraction created on the lattice by the first electron. **d** The two electrons couple, forming a *Cooper Pair* within the lattice

This pair of electrons, as long as the temperature is low enough, stays together, and the half-spins interact in such a way as to form a whole spin, so the pair of electron begins to have the properties of a boson. Swapping the electrons of the pair, it will also change the sign of the wave function twice, and therefore, nothing will change because, precisely, the pair is a boson. The bosonic nature frees the pairs from the constraints imposed by *Pauli's Exclusion Principle*, so that at low temperatures they occupy the lowest quantum state. However, this state consists of negatively charged particles which, even when coupled, can freely conduct an electric current because they do not interact with the atoms in the conductor. As temperature increases, the decoupling causes individual electrons, now fermions that can only singly occupy one energy state, to interact with the lattice atoms and the resulting resistivity of the material to electron flow. For this reason, the phenomenon of superconductivity is observed only at temperatures near absolute zero; as the temperature increases, the vibrations of the lattice ions will increase the interactions with the electrons, causing the Cooper electron pairs to decouple. The interaction energy of the pairs is relatively weak, around 10^{-3} eV, so they can easily separate with a small increase in the thermal energy of the system.

There is another experimental element to support the BCS model: the critical temperature at which the material becomes a superconductor depends on the isotopic mass. Materials with heavier ions exhibit a higher critical temperature at which superconductivity is observed because heavier ions are more difficult for electrons to attract. This result can be explained if an electron-lattice interaction is assumed; by changing the mass of the lattice atoms that varies with isotopes, the electron-induced distortion will vary accordingly. Beyond Cooper's semiclassical explanation, the quantum mechanical model shows that phonon-electron interactions underlie the

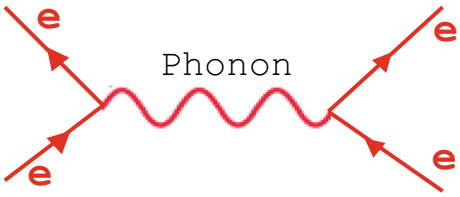


Fig. 9.11 Feynman diagram of the electron–phonon interaction that generates *Cooper Pairs*

phenomenon. Phonons (the excitations in the crystal lattice that act as quasiparticles) generated by the collective motion of lattice ions are thus involved in the process. A Feynman diagram well represents again the phonon-mediated electron coupling (Fig. 9.11). In this case the wavy line represents the phonon exchange, instead than photon such as shown in Fig. 9.2.

It is precisely the electron–phonon interaction that is the mechanism considered to underlie superconductivity in the quantum mechanical model.

The bosonic nature of *Cooper Pairs* makes the phenomenon of superconductivity particularly intriguing because, at low temperatures, the phenomenon of Bose–Einstein condensation can be observed for these pairs. The formation of numerous electron pairs in the superconductor leads to the formation of a condensed state. The superconductor then becomes a condensate of *Cooper Pairs*. The Bose–Einstein condensate is considered a particular state of matter in which bosons, when cooled to temperatures near absolute zero, behave like a single entity instead of separate particles. Particles found in the Bose–Einstein condensate have quantum properties that are detectable even at the macroscopic scale.

Before concluding this chapter we can reflect further on the nature of *Cooper Pairs*, and in this regard we quote Feynman's remarks: "*I don't wish you to imagine that the pairs are really held together very closely like a point particle. As a matter of fact, one of the great difficulties of understanding this phenomenon originally was that that is not the way things are. The two electrons which form the pair are really spread over a considerable distance; and the mean distance between pairs is relatively smaller than the size of a single pair. Several pairs are occupying the same space at the same time.*" [27].

BCS theory with all its fascinating quantum implications well explains the phenomenon of superconductivity at low temperatures. However, the phenomenon of superconductivity observed at temperatures much higher than absolute zero, from 80 K (−193 K), where other mechanisms underlie the phenomenon remains to be elucidated.

9.8 The Electron in the Box and Strange Nanoparticles, the Quantum Dots

Let us now ask a scientific and philosophical question: where does the frontier of scientific knowledge in physics stand now? Certainly the great cosmological questions are still the focus of interest of astrophysicists and theoretical physicists. Perhaps there are even more questions than answers. However, let us shift the focus from the big to the small; that is, let us once again bring the electron back to the center of the village. The world of cosmology, governed by general relativity, has its opposite in the world of elementary particles, where physical phenomena are described using quantum mechanics instead. The two descriptions of the universe are currently irreconcilable, no matter how much effort has been put into it a theory of quantum gravity still appears to be the holy grail of physics. There is. However, as Feynmann had once again guessed, much space is yet to be explored, the space of nanometer dimensions. We have seen that spatially confined electrons have exotic properties. In the nanoworld, electrons are still quantum entities, and nanomaterials, therefore, exhibit properties that depend on their shape and size. This fascinating world of nanoscience appears as a Pandora's box of endless possible discoveries. The examples would be almost endless, but in order not to have to write a second volume as an appendix to this one let us take one example, that of quantum dots.

To describe quantum dots, we start with the theoretical model of the electron in the box, which considers an electron confined in a potential well with infinitely high walls. The barriers outside the one-dimensional box have infinitely large potential. In contrast, the interior of the box has a constant, zero potential (Fig. 9.12a). Inside the box, no forces act upon the particle, which means that the part of the wave function inside the box oscillates through space and time with the same form as a free particle. The infinite barrier confines the electron inside the well and the wave function that gives the position of the electron must be zero at the walls. The electron can only occupy discrete energy levels due to the boundary conditions imposed by the walls of the box (Fig. 9.12b). Since the potential is infinite at $x = 0$ and $x = L$, the wave function must vanish at the boundaries: $\psi(0) = \psi(L) = 0$. These boundary conditions restrict the allowed wave functions to specific forms (standing waves) inside the box. Considering the boundary conditions, the time-independent Schrödinger equation for the electron in the box gives the wave function that represents the general solution and the values of the allowed energy levels inside the box:

$$E_n = \frac{n^2 h^2}{8\, mL^2} \tag{9.1}$$

with E_n = the energy of the electron in the n-th state, n the quantum number ($n = 1, 2, 3, \ldots$, positive integer numbers), h the Planck's constant, m the mass of the electron and finally L the length of the box.

Equation (9.1) shows that the energy levels of an electron confined in the box can increase with the square of n. The energy does not grow linearly with n but grows

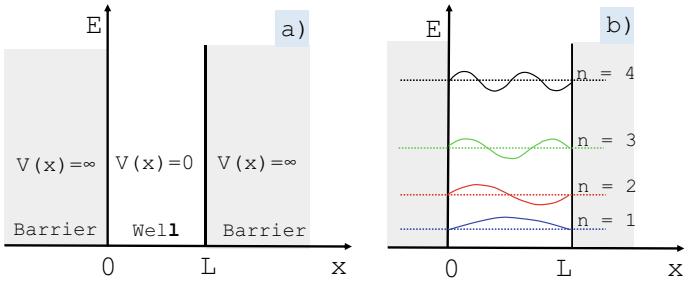


Fig. 9.12 **a** Model of an electron in a box. The barriers outside the box have infinitely large potential, inside the box there is a constant zero potential. **b** Allowed energy levels for an electron in a box

as the square of n. The lowest energy state ($n = 1$) is the ground state, while the higher values of n correspond to excited states. If the quantum number n doubles, the energy increases by a factor of $2^2 = 4$. If n triples, the energy increases by a factor of $3^2 = 9$, and so on. As the quantum number n increases, the energy gap between consecutive energy levels increases. For instance, the energy difference between E_1 and E_2 is smaller than the difference between E_2 and E_3. At higher quantum numbers, therefore, the electron requires significantly more energy for a transition to the next energy level compared to lower energy states. To visualize the energy levels, it is possible to imagine a series of "steps" where each step represents an energy level. The height of each step corresponds to the energy of that level. Moving up to higher quantum numbers ($n = 2, 3, 4\ldots$) the steps become steeper, reflecting the quadratic nature of the energy increase.

Because the electron is confined within a finite region (the box), its momentum has inherent uncertainty. The smaller the box, which means a smaller L, the greater will be the uncertainty in momentum, which leads to higher possible energy states.

If we move from the theoretical ground to the real world, fabricating a semiconductor particle of nanosize means confining the electrons in very specific energy levels that depend on the particle dimensions. Quantum dots are defined as semiconductor nanocrystals typically ranging from 2 to 10 nm in size. Quantum dots are like quantum wells, but electrons are confined in all three spatial dimensions, not just vertically, which means they can be tuned to only emit energy at certain frequencies, as colors. It is important that the nanoparticles are semiconductors to behave as quantum dots. The energy gap of the semiconductor is, in fact, modulated by the reduced dimensions (Fig. 9.13).

The emission wavelength (color) of quantum dots depends directly on their size due to the quantum confinement effect. Larger quantum dots emit-absorb light toward the red region of the UV–Vis spectrum (longer wavelengths) because have a smaller bandgap. This means that less energy is necessary to excite the electrons from the valence to the conduction band. Smaller quantum dots have instead a larger bandgap and emit-absorb light in the blue region of the UV–visible spectrum (shorter wavelengths). As the quantum dot size decreases, the energy levels become more discrete,

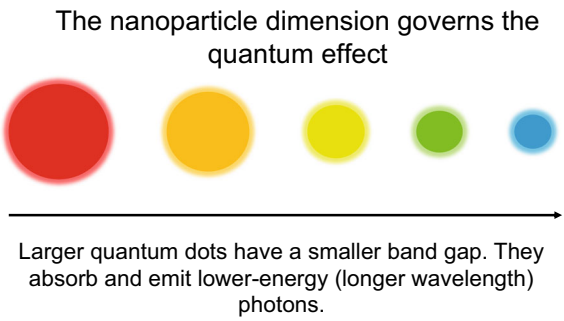


Fig. 9.13 Size of the nanoparticle affects the quantum properties and the emission–absorption of photons

resembling the behavior of individual atoms rather than bulk materials. In bulk materials, energy levels form continuous bands, but in quantum dots, these bands break up into distinct, quantized energy levels. This size-dependent shift in the energy levels allows tuning that the optical and electronic properties of the quantum dots by modulating their size in the nanoscale.

Quantum dots, thanks to their ability to modulate the absorption and emission of light based on their size and shape, are extraordinary objects whose properties have been used in many different fields. For the discovery and development of quantum dots, Moungi Bawendi, Louis E. Brus, and Aleksey Yekimov were awarded the 2023 Nobel Prize in Chemistry.

9.9 The End of the Story. What If the Electron Is Not an Elementary Particle?

Having reached the end of this journey to discover an elementary particle that underlies virtually all our daily activities, more than a few doubts are still legitimate. We can easily guess that we are not at the end of the story, that much remains to be understood, and that only some of the pieces are in place. The *Standard Model* in which the electron is a fundamental particle explains the world of subatomic particles quite well. However, it fails to account for Dark Matter, assuming it exists and is not just an incorrect theoretical prediction. The dual wave-particle nature of the electron and many experiments, such as the double slit experiment, still leave room for different interpretations, such as quantum mechanics itself. The *Standard Model* shows what the constituent elements of matter and thus of the cosmos are, leptons and quarks, but are we sure that these are really the ultimate elements, or is there not something even smaller that constitutes the ultimate building block with which the universe is constructed? This question is part of the baggage of doubts and uncertainties in science. But the problem is that if these particles really existed, they would be so small that we would practically be doomed never to observe them experimentally. If leptons and quarks were composed of smaller particles, then they should have a measurable size. However, experiments carried out so far using the most powerful

accelerators have failed to obtain any measure of their size experimentally. A lepton has no size; if it has, it is smaller than what we can measure now with our instruments, which is about 1/10 thousandth of the size of a proton. So the question is still open.

Meanwhile, a name has been given to these hypothetical particles smaller than a quark, the *prequarks*. Physicists, with their usual imagination, had proposed many evocative names for these hypothetical particles, *prequarks, subquarks, maons, alphons, quinks, rishons, tweedles, helons, haplons, Y-particles*, and *primons*, but in the end, the winner was the *preon*. The first hypothesis about the possible existence of these particles was made by Jogesh Pati and Abdus Salam in 1974. However, after some initial interest, it was almost abandoned until more recent times when new hypotheses about the existence of *preons* have been advanced. The idea of *preons* arose in an attempt to understand the seemingly complex structure of quarks and leptons and to provide a more fundamental explanation of their properties. However, the concept of *preons* is now nothing more than a speculative idea and is not accepted by the scientific community without solid empirical evidence.

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Conclusions

The end of this voyage to the discovery of the electron and its strange properties brings us somewhat back to the starting point: What is the electron? We started with the electron and then got tangled up in the meanderings of the oddities of quantum mechanics. In the end, however, the accounts add up, and almost magically, wave or particle that is the electron, everything works if we just use mathematics. The electron is an elementary particle, a point on which the structure of matter and the universe as we know it is founded. A theoretical castle made up of theorems, principles, theories, and models, such as the Standard Model and Quantum Field Theory, has been built around the electron. It is a castle that constitutes the fortress within which the secrets of the electron are well kept.

One last question. Does the electron really exist? A rather paradoxical question, but for some theoretical physicists, not too much of one. However, the very concept of "existence" is highly complicated for a particle to define. Whether an electron "truly" exists is a profound philosophical question and a matter of scientific interpretation. It touches on fundamental concepts of quantum mechanics, the philosophy of science, and the nature of reality itself. To answer this question, it is important to consider what we mean by "existence" and how this concept might be applied in modern physics.

In classical physics, defining an object is relatively simple; whether large or small, it will have a defined position, size, and mass and occupy a specific point in space at a specific time. However, quantum mechanics greatly complicates this picture. Particles like electrons do not behave according to the laws of classical physics; they are not very small, solid spheres that, in motion, can describe well-defined trajectories in space and time. Instead, they are governed by the principles of quantum mechanics, which consider them as dual objects in nature, wave-like or particle-like, described by a wave function that gives as a solution a density of probability. Therefore, from a quantum mechanical perspective, an electron is represented by a wave function that contains all the information about the electron, such as its probability distribution in space and time. This description has the well-known implication, widely described in this book, that the electron doesn't have a definite position until it is measured. Before the measure, the electron is in a superposition of all possible states, defined

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P. Innocenzi, *What is an Electron?*, https://doi.org/10.1007/978-3-031-81984-1

as a cloud of probabilities rather than as a point-like particle. On the other hand, the principle of complementarity introduces an additional element that defines the quantum nature of the electron. It cannot be a wave or particle at the same time; the two descriptions are complementary. Whether one description or the other is used depends on the experimental context, but they are mutually exclusive. Therefore, according to Bohr, asking "Where is the electron?" or "Does it exist at a particular point?" is not a valid question until a measure is made, as the electron does not have a definite position in the absence of measurement.

Furthermore, according to Heisenberg's Uncertainty Principle, the position and momentum of an electron cannot be simultaneously known to arbitrary precision. This uncertainty means the electron doesn't have a well-defined position in the classical sense. In the book, we have discussed and described these quantum properties in several after ways. It remains to connect these pillars of quantum mechanics with the abstract terms that define reality and "existence". If we accept that quantum particles "exist," it is probably necessary to extend this definition to something that cannot be known in terms of uniquely determined physical properties. However, using probability to describe a particle raises a philosophical conflict between realism, for which quantum entities have an independent "existence" even when they are not measured, and anti-realism, the quantum properties only come into being when measured. The opposition between realism and anti-realism also extends to the interpretation of the nature of the wave function. Indeed, one can regard the wave function as real and physical, although this physical reality is still subject to various possible interpretations. Conversely, the wave function is a pure mathematical tool that works very well but remains confined to an abstract conceptual realm.

In Quantum Field Theory, the electron is not considered a "particle" in the traditional sense but rather a quantized excitation of an underlying electron field. In this view, all electrons are manifestations of a single, pervasive field that permeates all of space. This means that an electron is not an independent object, but rather a *localized excitation* of the electron field. The QFT perspective suggests that what we think of as particles (like electrons) are really just discrete *units of energy* and *momentum* within a field. Thus, the electron's "existence" is tied to the presence of an excitation in the field.

Still it can be that "Reality is not what it seems." The theoretical physicist Carlo Rovelli thinks that the electron does not always "exist:" "*Electrons don't always exist. They exist when they interact. They materialise in a place when they could lead with something else. The quantum leaps from one orbit to another constitute their way of being real: An electron is a combination of leaps from one interaction to another*." [1]. It is a fascinating concept that opens up a new perspective on the concept of "existence."

Although the philosophical question of the electron's existence is complex, there is overwhelming experimental evidence that electrons are real enough to be at the ground of post-industrial revolution technologies. The electron's negative charge is a well-defined property that we can measure accurately. The electron's spin and magnetic moment have been observed with great precision. Electrons absorb and emit photons, leading to observable phenomena like the photoelectric effect, which

helped establish the quantum theory itself. These properties suggest the electron is "real" because it has measurable effects and interacts with other entities. We may not be able to pinpoint its exact position, but we can measure its properties and predict its behavior statistically. At the end of the day, when switching on and off the light, we never think about whether the electrons and, therefore, electricity exist. Classical physics works extraordinarily well for making any type of practical calculations involving electrons.

Ultimately, whether the electron "exists" depends on what we mean by existence. The electron clearly exists if we define existence based on measurable effects and empirical evidence. If we require a well-defined position and trajectory, its existence becomes more ambiguous. The electron defies classical definitions and forces us to adopt a more abstract, probabilistic view of reality. Therefore again, does an electron truly exist? From the perspective of classical physics, the electron exists because it has measurable properties and interacts with other particles and fields in predictable ways. From a philosophical and quantum point of view, it depends very much on how we interpret the nature of quantum mechanics. However, it is also important not to fall into the interpretative traps typical of theoretical physics when it loses its connection with experimental reality. Not only is our daily life based on the existence and use of electrons, but we have also been able to photograph the transition from one atomic orbital to another using attosecond physics.

The nature of the electrons fundamentally differs from classical objects and challenges our capability even to find a proper language to describe it. Transferring the perception of reality as we understand it on the basis of our daily experience on an atomic scale is undoubtedly extremely challenging. Do we need to know more about electrons? The readers should now have the instruments to give a reasonable answer.

Reference

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